

HEM / HEMK / PCSM / PCSK GEN3

UTILITY SCALE INVERTERS

Faults, warnings and troubleshooting manual

Power Electronics' staff
SW versions 2.2.0 and later

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ABOUT THIS DOCUMENT

PURPOSE AND AUDIENCE

This document contains the necessary information to carry out the diagnosis of HEM_K and PCSM_K GEN3 inverters once they are in fault mode. It includes all the different types of faults that may be triggered and provides information so that a first remote diagnosis is possible as long as the safety and integrity conditions of the users and the inverters allow it and, secondly, a physical diagnosis process in those cases where it is necessary.

This manual is intended for Power Electronics' personnel and qualified customers approved by Power Electronics who will be operating Power Electronics' solar and battery GEN3 inverters.

Power Electronics reserves the right to modify product features.



NOTICE

DISCLAIMER

The points described in the diagnosis process for each single failure must be read completely, from beginning to end, before starting any type of intervention on the inverter.

REFERENCE MANUALS

The following reference documents are available for the solar and battery GEN3 inverters:

- Programming and Software Manual.
- Hardware and Installation Manual.
- Accessories Manual.
- Maintenance Manual.



NOTICE

WARRANTY DISCLAIMER

The manufacturer is not liable for damages, losses, costs or expenses incurred by any user of the product if such damages, losses, costs or expenses result from a failure to comply with the applicable safety instructions or general instructions or operating instructions given by the manufacturer in any of the documents and manuals of the product, including, but not limited to hardware installation, programming and operation, maintenance instructions, handling, or any other. Any damages, losses, costs or expenses resulting from the improper handling, manipulation, modification or operation of the product will be subject to the company's warranty terms. **Do not modify the product in any way not authorized by Power Electronics. If you do so, the manufacturer will not assume any liability and the product warranty will be voided.**

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REVISIONS CONTROL

DATE (DD / MM / YYYY)	REVISION	DESCRIPTION
17 / 11 / 2023	E1	First release: F41, F45, F63.
28 / 01 / 2024	E2	Faults released: F204-207, F228, F230-233, F118, F31, F32, F51, F63. Misprint corrections in F45.
09 / 02 / 2024	E3	Faults released: F157, F192, F158, F150, F3, F203, F40, F55, F103, F27, F112. Misprint corrections. Added the list of warnings. New coding is added, and statuses are updated.
28 / 02 / 2024	E4	Faults released: F25, F48, F130, F58, F13, F83, F84, F71-82, F119-124. Added more information: F150, F45, F204-F207, F228, F230-F233, F51, F158
01 / 03 / 2024	E5	Faults release: F159, F164, F227, F49, F3, F203, F40, F55, F103, F27, F112, F25, F48, F130, F58, F13, F83, F84, F71-F82, F119-F124 Added more information: F5, F6, F7, F8, F14, F44, F66, F69, F70, F85, F106, F108, F109, F114, F115, F116, F152, F153, F155, F165, F170, F197, F198, F208, F217, F220, F221, F222, F223, F90, F132, F101
08/03/2024	E6	Faults release: F5, F6, F7, F8, F14, F44, F66, F69, F70, F85, F106, F108, F109, F114, F115, F116, F152, F153, F155, F165, F170, F197, F198, F208, F217, F220, F222, F223, F90, F224, F225, 132, F101, F35 Added more information: F2, F4, F11, F23, F43, F54, F56, F65, F86, F98, F99, F107, F128, F161, F162, F163, F166, F167, F169, F174, F213, F214, F215, F216, F229, F240, F253, F15, F33, F104, F171, F172, F211, F212, F28, F95, F238, F252, F134, F200, F219, F221
15/03/2024	E7	Released faults: F173, F255, F102 Added more information: F2, F4, F11, F23, F43, F54, F56, F65, F86, F98, F99, F107, F128, F161, F162, F163, F166, F167, F169, F174, F213, F214, F215, F216, F229, F240, F253, F15, F33, F104, F171, F172, F211, F212, F28, F95, F238, F252, F134, F200, F219 Contact with PE: F10, F59, F64, F113, F125, F129, F131, F190, F210, F226, F235, F236.

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TABLE OF CONTENTS

ABOUT THIS DOCUMENT	2
ACRONYMS	8
IMPORTANT SAFETY INSTRUCTIONS.....	10
SAFETY SYMBOLS	18
1. INTRODUCTION	19
2. LIST OF ALL FAULTS AND WARNINGS.....	20
2.1. Faults.....	20
2.2. Warnings	22
3. NECESSARY INSTRUMENTATION	24
4. FAULTS TO BE REPORTED TO POWER ELECTRONICS	26
4.1. Fault code F1: Watchdog	26
4.2. Fault code F46: Data Fault	26
4.3. Fault code F47: Watchdog uP	26
4.4. Fault code F10: Comms SCI FPGA-DSP	26
4.5. Fault code F59: Module SW Mismatch.....	26
4.6. Fault code F64: Invalid MAC	26
4.7. Fault code F113: CRC	26
4.8. Fault code F125:Diff lac mod run	27
4.9. Fault code F129:Module Isolation.....	27
4.10. Fault code F131:Faults limit exceeded	27
4.11. Fault code F190:Combined F-V.....	27
4.12. Fault code F210: uC-FPGA Comms.....	27
4.13. Fault code F226:IMI Hardware	27
4.14. Fault code F235: T/H sensors comms.....	27
4.15. Fault code F236: DU sensor comms lost.....	27
5. FAULTS ANALYSIS AND TROUBLESHOOTING.....	28
5.1. Fault code F41: GFDI	28
5.2. Fault code F45: General Isolation.....	33
5.3. Fault code F63: IMD Earth Connection	36
5.4. Fault code F204-F207: communications loss with fans	39
5.5. Fault code F228: Fan setpoint.....	50
5.6. Fault code F230-F233: fan type not correct.....	52
5.7. Fault code F118: Module Disabled	54
5.8. Fault code F31: Selfdiagnosis fault.....	57
5.9. Fault code F32: Error Mod selfdiagnosis	59
5.10. Fault code F51: PPC Comms	60
5.11. Fault code F62: DU Comms	63
5.12. Fault code F157: Feedback AC breaker	70
5.13. Fault code F158: Low pressure SF6.....	85
5.14. Fault code F150: MV cell wrong state	88

5.15. Fault code F159: Overtemperature DU	108
5.16. Fault code F164: Excessive Manoeuvres	110
5.17. Fault code F227: Fault Unknown.....	111
5.18. Fault code F49: IMD Selfdiagnosis Error.....	112
5.19. Fault code F3: SoftCharge.....	113
5.20. Fault code F35: MT Premag.....	121
5.21. Fault code F203: MV overtemperature	190
5.22. Fault code F40: Internal overtemperature	192
5.23. Fault code F55: External fault/OVGR	194
5.24. Fault code F103: IGBTs temperature	197
5.25. Fault code F27: Modules start fault	200
5.26. Fault code F112: OCAC emergency.....	201
5.27. Fault code F25: Low Vdc.....	205
5.28. Fault code F48: Internal comms	210
5.29. Fault code F130: Softcharge Feedback.....	211
5.30. Fault code F58: Control SW mismatch	217
5.31. Fault code F13: No modules.....	218
5.32. Fault code F83: Multiple desaturation.....	219
5.33. Fault code F84: Communications	224
5.34. Fault code F71-F82: Desaturation (<i>outers</i>).....	230
5.35. Fault code F119-F124: Desaturation Inners	235
5.36. Fault code F5-F8: Grid failures	239
5.37. Fault code: F14: Drive-Select DSP and F44: Drive-Select MCU	246
5.38. Fault code F66: Module VDC Unbalance	247
5.39. Fault code F69 – F70: Overvoltage DC+/-.....	253
5.40. Fault code F85: Timeout soft charge	258
5.41. Fault code F106: Mod. Vdc unbalanced	266
5.42. Fault code F108: Mod. High Vdc	272
5.43. Fault code F109: Mod. Low Vdc	278
5.44. Fault code F114-F116: Ir, Is, It module current sensing.	285
5.45. Fault code F152: DGPT2 Temp failure	291
5.46. Fault code F153: DGPT2 Pressure failure.....	298
5.47. Fault code F155: DGPT2 oil or gas level.....	304
5.48. Fault code F165: Improper Modulation	309
5.49. Fault code F170: Thermal L Feedback	310
5.50. Fault code F197: HW Overvoltage	314
5.51. Fault code F198: HW Undervoltage	316
5.52. Fault code F208: MV cri overtemperature	323
5.53. Fault code F217: Critical temperature filter LC	327
5.54. Fault code F220: PT100 Inductance no connected	329
5.55. Fault code F221: PT100 MV Transformer no connected.	337
5.56. Fault code F222: NTC admission temperature no connected.....	348
5.57. Fault code F223: NTC LC filter no connected.....	350
5.58. Fault code F90: T ^a IGBTs HF.....	352
5.59. Fault code F224: Ind overtemperature	356
5.60. Fault code F225: Ind cri overtemperature.....	366
5.61. Fault code F234: Thermal Bus Plus	377
5.62. Fault code F132: High voltage DC module	380
5.63. Fault code F101: Mod. measure DC.....	381
5.64. Fault code F2: HW Vbus	390
5.65. Fault code F4: Discharge	392
5.66. Fault code F11: Active Anti-island	394
5.67. Fault code F23: Unbalanced Vac	395

5.68. Fault code F43: Emergency stop.....	403
5.69. Fault code F54: DU Overcurrent.....	406
5.70. Fault code F56: Remote emergency stop.....	409
5.71. Fault code F65: Overcurrent AC.....	410
5.72. Fault code F86: Feedback breaker.....	419
5.73. Fault code F98: Idc derivation	428
5.74. Fault code F99: Mod. Iac derivation	439
5.75. Fault code F107: Iac not reached	450
5.76. Fault code F128: Spring loaded.....	453
5.77. Fault code F161: LOTO AC	463
5.78. Fault code F162: LOTO DC.....	467
5.79. Fault code F163: WD MV Cell	472
5.80. Fault code F166: E-STOP	500
5.81. Fault code F167: Open DU.....	505
5.82. Fault code F169: LCL Feedback	506
5.83. Fault code F174: AC circuit breaker fb	519
5.84. Fault code F213: Negative DU current	537
5.85. Fault code F214: IMD Comms.....	540
5.86. Fault code F215: Timeout GFDI	542
5.87. Fault code F216: Timeout IMI Meas	556
5.88. Fault code F229: Timeout IMI Channel	561
5.89. Fault code F240: Du BP overcurrent	564
5.90. Fault code F253: Imbalance V.....	568
5.91. Fault code F255: BAT open detection	574
5.92. Fault code F15: Synchronization	581
5.93. Fault code F33: Unable reconnect.....	582
5.94. Fault code F104: PCB temperature	584
5.95. Fault code F171: Filter LC Current	589
5.96. Fault code F172: C LC Filter.....	595
5.97. Fault code F211: Intern Isolation	601
5.98. Fault code F212: Extern Isolation	604
5.99. Fault code F28: Passive Anti-island	613
5.100. Fault code F95: Source PCB	614
5.101. Fault code F238: AEMO comms lost	619
5.102. Fault code F252: RFI OverCur Persistent.....	622
5.103. Fault code F102: Unbalanced current.....	628
5.104. Fault code F134: Module sensor comms.....	630
5.105. Fault code F200: Overvoltage Lim DC.....	631
5.106. Fault code F219: Overvoltage Lim AC.....	640
5.107. Fault code F173: CTs measure	647

ACRONYMS

The terms commonly used in the documentation of Power Electronics' products are listed in the table below.

Please notice this is a general series of terms and it encompasses all our product divisions (industrial, solar, storage, and electric mobility), thus, some of the following expressions may not apply to this particular manual.

ACRONYM	MEANING
AASS	Auxiliary Services
AC	Alternating Current
AI	Analogue Input
AO	Analogue Output
BESS	Battery Energy Storage System
BMS	Battery Manager Solution
CCID	Charge circuit interrupting device
CCL	Charge Current Limit.
CCS	Combined charging system – charging and communications protocol following the standard IEC 61851-23 Annex CC
CHAdemo	Charging and communications protocol following the standard IEC 61851-23 Annex AA
CPU	Central Processing Unit
DC	Direct Current
DCL	Discharge Current Limit
DI	Digital Input
DSP	Digital Signal Processor
DO	Digital Output
EMS	Energy Management System
EV	Electric Vehicle
FPGA	Programmable device (Field-Programmable Gate Array)
FRU	Field Replaceable Unit
GFDI	Ground Fault Detector Interrupter
GPRS	General Packet Radio Services, a data transmission system
HVAC	Heating, Ventilation, and Air Conditioning
IGBT	Insulated Gate Bipolar Transistor
IMI	Insulation monitoring device
INV3	Power Electronics' GEN3 inverters (HEM, HEMK, PCSM and PCSK)
IT	Grid system where the power supply is kept isolated and the electrical equipment system is grounded.
LOTOTO	Lock Out – Tag Out – Try Out
MCB	Miniature Circuit Breaker
MCCB	Moulded Case Circuit Breaker
MPCS	Multi Power Conversion System
MID	Measuring Instrument Directive
MV	Medium Voltage. This term is used to refer to high voltage in general
PE	Ground connection
PI	Proportional and Integral
POI	Point Of Interconnection

EN

ACRONYM	MEANING
PPE	Personal Protection Equipment
PV	Photovoltaic energy
RCD	Residual Current Device
RCM	Residual Current Monitor
RFID	Radio Frequency Identification
SOC	State Of Charge – referred to battery
SOH	State Of Health – referred to battery. It compares the actual state of the battery to its initial conditions. It is measured in percentage
STO	Safe Torque Off
TN	Grid system where the power supply is grounded, and the electrical equipment system is brought to the same ground through the neutral connector.
TT	Grid system where both the power supply and the electrical devices are connected to the ground via separate connections
UPS	Uninterruptible Power Supply
VSD / VFD	Variable Speed Drive, Variable Frequency Drive. Both terms are used

IMPORTANT SAFETY INSTRUCTIONS

This manual contains important instructions for **HEM_K** and **PCSM_K GEN3** inverters that shall be followed during diagnosis after a fault and troubleshooting.

Read carefully all documentation before manipulating the product and pay special attention to safety recommendations to maximize the performance of this product and to ensure its safe use.

It is the user's responsibility to follow the instructions provided on this manual, to follow good electrical practices and to identify all mentioned warnings and recommendations before performing any interventions on the inverter.

During the diagnosis procedure of each of the fault codes described in this manual, several steps and sequences of different nature are described to identify and isolate the root causes for such faults.

In order to perform the described actions, both remote and on-site interventions may be required on the products within the scope of this manual.

It is absolutely mandatory to follow the safety instructions described in the [Safety Instructions for Operation, Troubleshooting and Maintenance](#) intended for that purpose and which define the statuses of the products and their meaning, the areas of the equipment to which access is allowed and under what conditions; all this always accompanied by the appropriate PPE and the necessary tools for each intervention.

The following table explains the status codification:

EXAMPLE: INVMG3_ST3B – Status 3B, only applies to HEM / PCSM inverters			
PRODUCT		STATUS	NUMBER
INVG3	All GEN3 inverters	ST	1A
INVKG3	HEMK / PCSK inverters		1B
INVMG3	HEM / PCSM inverters		2
			3A
			3B
			4
			5A
			5B
			6



WARNING

If the instructions on this manual are not followed, the products may present a SHOCK HAZARD. Read and retain the *Hardware and Installation Manual* for future reference. Before installing the inverter, read all instructions and cautionary markings. Failure to follow the safety instructions could result in severe shock or even death. Exercise extreme caution always to prevent possible accidents.

EN



WARNING

Ensure the inverter is completely disconnected from the power supply and connected to ground before manipulating it or performing any maintenance tasks. Otherwise, electric shock hazard exists.

To prevent electrical hazards, disconnect input supply, connect the product to ground, remove all control voltages before carrying out any task and ensure busbars are completely discharged.

Warning and safety labels must be properly placed in terminals, enclosures and control panels according to local normative in force.

In addition to the recommendations on this manual, local and site-specific safety procedures must be considered regarding electrical installation, buildings, fire and other regulations or network requirements applicable to the installation and/or product. Also, national and local electrical codes must be followed in order to avoid personal injuries and/or product damage.

Failure to comply safety instructions and electrical codes may void the warranty.

When working in electrical installations always remember to apply the FIVE GOLDEN RULES:

1. Visible cutout of all sources in voltage.
2. Mechanical locking of all cutting elements.
3. Check the absence of voltage.
4. Ground and short-circuit all possible voltage sources.
5. Define and mark the work area.

Do not modify the product in any way not authorized by Power Electronics.



NOTICE

Consult the *Safety Instructions for Operation, Troubleshooting and Maintenance* of all the products included in the project.



WARNING

A solar or battery inverter is a dual power source. There may be voltage present, from the photovoltaic generator and / or the product. It is especially important to consider this when disconnecting parts of the system.

The input configuration and the fuse values must be selected according to the installation characteristics and meet all code requirements. It is the responsibility of the installer to select the proper overcurrent values.

Up to 1500 Vdc may be present in the solar array / BESS. **Use LOTOTO procedures to ensure the system is safely deenergized prior to making connections.**

Once LOTOTO procedures are complete, ensure AC are also cut off before making any actions on the product. There could be hazardous voltages present in the busbars and you may get an electric shock.

Do not connect any supply to the product while the doors are open.
Otherwise, you could get an electric shock.

Do not apply power to a damaged inverter or to an inverter with missing parts even if the installation is complete.
Otherwise, you could get an electric shock.

Keep all doors closed during product operation.
Otherwise, you could get an electric shock

Do not open the doors except for periodic inspections or wiring, even if the input power is not applied. Otherwise, you could contact the charged circuits and get an electric shock.

Always operate switches with dry hands and use Personal Protective Equipment (PPE). Otherwise, you could get an electric shock.

Any tasks performed must be carried out with, at least, two operators. The inverter should never be operated by a single operator.

Do not use cables with damaged insulation. Do not subject the cables to abrasions, excessive stress, heavy loads or pinching. Otherwise, you could get an electric shock.

Ensure compliance of all safety regulations for medium voltage operations that are in force at the place where the inverter is installed. Otherwise, you could get an electric shock.

In case the product stops due to a power loss, do not perform any works on it. Automatic restart function might be enabled, and you could get an electric shock.

Any tests performed on the products must be previously validated and approved by Power Electronics.

In the case of HEM / PCSM products, under no circumstances shall any modifications be made to the transformer taps. If a transformer ratio test is required, it must be performed using the factory tap.

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WARNING



5 min

The product contains capacitors on the AC and DC sides. The discharge time of the capacitors is 5 minutes. Wait until the capacitors have been discharged before carrying out any maintenance task.

Work at heights shall be carried out under the appropriate safety conditions and equipment and always in accordance with the regulations of the country where the operation is carried out.

In the cabinets where high voltage is present, the installer must add a warning signal “DANGER – HIGH VOLTAGE”.

PRODUCTS WHICH INCLUDE GFDI PROTECTION:

When a ground fault is indicated, normally grounded conductors may be ungrounded and energized or normally ungrounded conductors may be grounded

PRODUCTS WITH UNGROUNDED POLES CONFIGURATION:

When a ground fault is indicated, normally ungrounded conductors may be grounded.

A solar or battery inverter is dual power source. There may be voltage present from the PV generator or BESS and/or inverter. It is particularly important to consider this when disconnecting parts of the system.

The input configuration and fuse ratings must be selected according to the characteristics of the installation and comply with regulatory requirements. It is the responsibility of the installer to select the appropriate overcurrent values.

Up to 1500Vdc may be present in the solar field or batteries. **Use LOTOTO procedures to ensure that the system has been safely de-energized before making connections.**

ISOLATION GUIDELINES HEM / PCSM



To ensure the safety of the personnel who will be working on the inverter, it is imperative to open the input circuit breaker and earth the MV switchgear supplying voltage to the product. Besides this, the customer must remove voltage from the MV facility network and discharge it to ground by keeping it in ground position while the product is open (unless a task specific Health and Safety study is completed, and task specific procedure is available).

In addition, it is necessary to completely disconnect the DC power supply from the PV array or BESS system to the product, unless a Health and Safety study has been performed and a task-specific procedure is available. In this case, please note that the DC cabinet cannot be opened, as there is voltage present from the PV field or BESS.

EN



WARNING

To perform any maintenance or activity inside the products, the user must verify that no voltage is present on the product, as well as carry out the LOTOTO procedure. Always apply the five golden rules to ensure that no dangerous voltages are present.

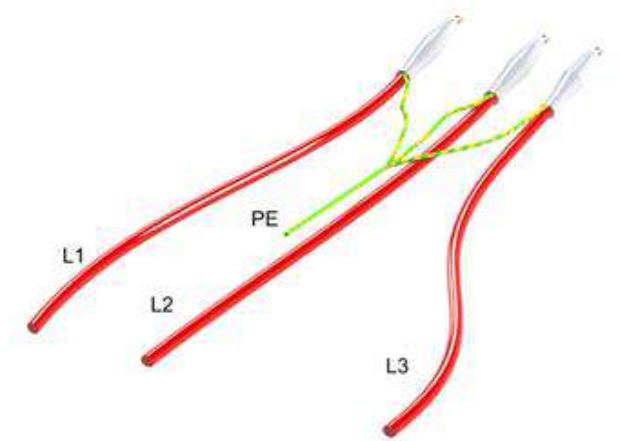
These manoeuvres are beyond the scope of Power Electronics personnel. The customer will be responsible for ensuring the safety of Power Electronics personnel working on the product.

GROUND CONNECTION HEM / PCSM

Electric shock prevention:

- The product chassis must be properly grounded to prevent a possible electric shock if a leakage current flows through the enclosure. Disconnect all the voltage supplies before proceeding with maintenance operations within the product. The product contains capacitors which may have a hazardous voltage even when the input voltage is disconnected. Make sure the product is discharged before performing any operation.
- Only connect the earth grounding to the product's earth plate. Do not use the case or the chassis screw for grounding.
- The earth protection cable must be connected first and disconnected last.
- The product must be connected to ground by using the ground busbar located in the medium voltage switchgear.

AC output cables are medium voltage cables. As shown in the figure below, they are coaxial cables, each one including earth. To carry out a correct connection, it is necessary to bond the earth of each phase and connect it to the ground busbar of the product (located in the MV voltage switchgear).



**CAUTION**

Install the product on a solid and levelled surface. Follow the recommendations on how to build the foundation. Otherwise, there is a risk of malfunction and even permanent damage.



Disconnect the input power in case the inverter gets damaged.

Otherwise, it could result in a secondary accident or a fire.



Do not allow lint, paper, wood chips, dust, metallic chips or other foreign matter into the inverter. Otherwise, a fire or an accident could occur.

After the input power is applied or removed, the inverter will remain hot for a few minutes. Touching internal hot parts could result in skin burns.

**CAUTION**

According to IEC 62920 type A inverters are not designed to be used in a residential environment. This inverter may cause radio frequency interference, in which case the user may be required to take additional mitigation measures against electromagnetic interference.

Observe all minimum clearance requirements for electrical code, thermal and safety requirements. Refer to NEC 110.34 for additional clearance requirements for product above 600 Volts

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NOTICE

RECYCLING

Product packaging must be recycled. Separate all different materials (plastic, paper, cardboard, wood...) and place them in the corresponding containers. Ensure waste collection is properly managed with a Non-Hazardous Waste Agent.



To guarantee health and natural environmental sources protection, the European Union has adopted the WEEE directive concerning discarded electric and electronic product (SEEA).

Waste of electrical and electronic product (WEEE) must be collected selectively for proper environmental management.



Our products contain electronic cards, capacitors and other electronic devices that should be separated when they are no longer functional. These WEEEs should be managed accordingly with a Hazardous Waste Agent.

Power Electronics promotes good environmental practices and recommends that all its products sold outside of the European Union, once they reach the end of their life, are separated and the WEEE managed according to the particular country applicable legislation (especially electronic cards, capacitors and other electronic devices).

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REQUIRED PERSONAL PROTECTION EQUIPMENT (PPE from now on)

PPE is required according to standards for repairing and maintaining Freesun inverters. Always Follow the site operative instructions to fulfil the compliance of national or local normative.

In case of actions with voltage presence, it is compulsory to use an Arc Flash Safety Kit (gloves, clothes and face shield).

A detailed example of PPE used will be shown below. Customer must specify in his safety instructions (risk assessment and method statements) which are the PPE required and when and how they should be used according to his electric arc studies, the characteristics of site, inverters, installation and country.

Power Electronics accepts no responsibility or liability for any damages resulting from inappropriate use of the product or failure to comply national or local normative.

The following table includes just an example of commonly used PPE description. **Always follow local regulations / NFPA-70E standard about Safety and Health.**

ITEM	DESCRIPTION
Safety glasses	Eye protection according EN 166.
Electric gloves	Gloves with mechanical, dielectric and against arc flash. Class according to voltage. EN 60903; ASTM D120 specifications and NFPA 70E standards.
Safety footwear	S3 class complying with BS EN ISO 20345.
Insulation carpet	Isolation carpet according to IEC 61111 Class according to voltage.
Safety kit arc flash	Arc flash personal protective equipment kit (including arc flash protective face shield & hard hat), fire resistant 40 cal/cm ² .
Lock Out / Tag Out elements set	Padlock and auxiliary elements set to lock out dangerous product.
HI-VIS vest	Fr VIS vest 9 cal/cm ² .

PPE FOR INSTALLATION

Mechanical gloves



Safety boots



Safety helmet



Safety glasses

Additional PPE for commissioning and maintenance tasks

Safety clothes according to NFPA-70E and safety labels

The following table shows **the protection class type**, depending on the working voltage.

EN**ELECTRICAL INSULATED GLOVES**

Class	AC (V _{AC})	DC (V _{DC})
00	500	750
0	1000	1500
1	7500	11250
2	17000	25500
3	26500	39750
4	36000	54000

ELECTRICAL SAFETY MATTING

Class	AC (V _{AC})	DC (V _{DC})
0	1000	1500
1	7500	11250
2	17000	25500
3	26500	39750
4	36000	54000

SAFETY SYMBOLS

Always follow safety instructions to prevent accidents and potential hazards from occurring.

In this manual, safety messages are classified as follows:

	WARNING	Identifies potentially hazardous situations where dangerous voltage may be present, which if not avoided, could result in minor personal injury, serious injury or death. Be extremely careful and follow the instructions to avoid the risk of electrical shocks.
	CAUTION	Identifies potentially hazardous situations, which if not avoided, could result in product damage, or minor or moderate personal injury. Read the message and follow the instructions carefully.
	NOTICE	Identifies important measures to take in order to prevent damage product and warranty lost, as well as encouraging good use and environmental practices.

Other symbols used in this manual for safety messages are the following:

	Hot surface. Be careful and follow the instructions to avoid burns and personal injuries.
	Risk of fire. Be careful and follow the instructions to prevent causing an unintentional fire.
	Caution, risk of electric shock. Energy storage timed discharge. Wait for the indicated time to avoid electrical hazards.
	Caution, risk of hearing damage. Wear hearing protection.

EN

1. INTRODUCTION

1

This document contains the necessary information to carry out the diagnosis of HEM_K and MPCSM_K GEN3 inverters once they are in fault mode. It includes all the different types of faults that may be triggered and provides information so that a first remote diagnosis is possible as long as the safety and integrity conditions of the users and the inverters allow it and, secondly, a physical diagnosis process in those cases where it is necessary.

Throughout the diagnosis, the state in which the inverter must be in will be indicated.

Please, consult the detailed information about states in document [Safety Instructions for Operation, Troubleshooting and Maintenance](#).

EN

2. LIST OF ALL FAULTS AND WARNINGS

2

2.1. Faults

Fault	Fault	Fault
F1: Watchdog	F51: PPC Comms	F95:Source PCB
F2: HW Vbus	F54: DU Overcurrent	F98:Idc derivation
F3: SoftCharge	F55: External fault/OVGR	F99: Mod. lac derivation
F4: Discharge	F56: Remote emergency stop	F101:Mod. measure DC
F5: High Vac	F58: Control SW mismatch	F102:Unbalanced current
F6: Low Vac	F59: Module SW Mismatch	F103:IGBTs temperature
F7: High frequency	F62: DU Comms	F104:PCB temperature
F8: Low frequency	F63: IMD Earth Connection	F106: Mod. Vdc unbalanced
F10: Comms SCI FPGA-DSP	F64: Invalid MAC	F107: lac not reached
F11: Active Anti-island	F65:Overcurrent AC	F108: Mod. High Vdc
F13: No modules	F66:Module VDC Unbalance	F109: Mod. Low Vdc
F14: Drive-Select DSP	F69: Overvoltage DC+	F112: OCAC emergency
F15: Synchronization	F70: Overvoltage DC-	F113: CRC
F23: Unbalanced Vac	F71:Desaturation R1(H)	F114: Measure Ir
F25: Low Vdc	F72:Desaturation R2(H)	F115: Measure Is
F27: Modules start fault	F73:Desaturation R2(L)	F116: Measure It
F28:Passive Anti-island	F74:Desaturation R3(L)	F118: Module Disabled
F31: Selfdiagnosis fault	F75:Desaturation S1(H)	F119:Desaturation R1(L)
F32: Error Mod selfdiagnosis	F76:Desaturation S2(H)	F120:Desaturation R3(H)
F33: Unable reconnect	F77:Desaturation S2(L)	F121:Desaturation S1(L)
F35: MT Premag	F78:Desaturation S3(L)	F122:Desaturation S3(H)
F40:Internal overtemperature	F79:Desaturation T1(H)	F123:Desaturation T1(L)
F41: GFDI	F80:Desaturation T2(H)	F124:Desaturation T3(H)
F43: Emergency stop	F81:Desaturation T2(L)	F125:Diff lac mod run
F44: Drive-Select MCU	F82:Desaturation T3(L)	F128:Spring loaded
F45: General Isolation	F83:Multiple desaturation	F129:Module Isolation
F46: Data Fault	F84:Communications	F130:Softcharge Feedback
F47: Watchdog uP	F85:Timeout soft charge	F131:Faults limit exceeded
F48: Internal comms	F86:Feedback breaker	F133: Softcharge board
F49: IMD Selfdiagnosis Error	F90:T ^a IGBTs HF	F134: Module sensor comms

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Fault
F150: MV cell wrong state
F152: DGPT2 Temp failure
F153: DGPT2 Pressure failure
F155: DGPT2 oil or gas level
F157: Feedback AC breaker
F158: Low pressure SF6
F159: Overtemperature DU
F161: LOTO AC
F162: LOTO DC
F163: WD MV Cell
F164: Excessive Maneuvers
F165: Improper Modulation
F166: E-STOP
F167: Open DU
F169: LCL Feedback
F170: Thermal L Feedback
F171: Filter LC Current
F172: C LC Filter
F173: CTs measures
F174: AC circuit breaker fb
F190: Combined F-V

Fault
F197: HW Overvoltage
F198: HW Undervoltage
F200: Overvoltage Lim DC
F203: MV overtemperature
F204: Fan impulsion MT
F205: Fan DU zone
F206: Fan zone filter LC
F207: Fan zone modules
F208: MV cri overtemperature
F210: uC-FPGA Comms
F211: Intern Isolation
F212: Extern Isolation
F213: Negative DU current
F214: IMD Comms
F215: Timeout GFDI
F216: Timeout IMI Meas
F217: Critical Temp LC filter
F219: Overvoltage Lim AC
F220: PT100 induc no connected
F221: PT100 traf not connected
F222: NTC adm temp not connect

Fault
F223: NTC LC filter not connec
F224: Ind overtemperature
F225: Ind cri overtemperature
F226: IMI Hardware
F227: Fault Unknown
F228: Fan setpoint
F229: Timeout IMI Channel
F230: Fan type not correct DU
F231: Fan type not correct MT
F232: Fan type not correct MOD
F234: Thermal Bus Plus
F235: T/H sensors comms
F236: DU sensor comms lost
F240: Du BP overcurrent
F252: RFI OverCur Persistent
F253: Imbalance V
F255: Reserved

2.2. Warnings

Warning
W1:Input high voltage
W2:Low voltage input
W3:High frequency
W4:Low frequency
W5:No start conditions
W6:DU communications
W7:P limit
W8:Q limit
W9:Is limit
W10:Idc limit
W11:Derating Vdc
W12:Derating thermal IGBT
W13:Derating thermal adm
W14:Low P
W15:Low Vdc
W16:Delta therm PCB Admission
W17:Delta therm IGBT teorik
W18:Tracker output low Vac
W19:LVRT
W20:HVRT
W21:Curve P(f)
W22:Curve Q(V)
W23:Curve P(V)
W24:DU close night
W25:Close AC without premag.
W26:Overvoltage AC
W27:Overvoltage limit AC
W28:Derating freq temp IGBT
W29:DGPT2 Temperature alarm
W30:Overtemperature DU
W31:MV Autocalibration
W32:MV Premagnetization
W33:ECON Mode
W34:Night Conditions
W35:Autonight Night Day
W36:Autonight Day Night
W37:Night Mode
W38:Opening DU
W39:DU Open
W40:Opening AC
W41:AC Open
W42:AC Disconnect
W43: Air conditioning
W44:Active heating enabled

Warning
W45:No modules active heating
W46:Filter LC Current Unbal
W47: C LC Filter
W61: Act Balancing Triggering
W65:Overcurrent AC
W66:Mod. Bus DC Unbalanced
W67:Overcurrent DC
W69:Overvoltage DC (P)
W70:Overvoltage DC (N)
W71:Desaturation R1(H)
W72:Desaturation R2(H)
W73:Desaturation R2(L)
W74:Desaturation R3(L)
W75:Desaturation S1(H)
W76:Desaturation S2(H)
W77:Desaturation S2(L)
W78:Desaturation S3(L)
W79:Desaturation T1(H)
W80:Desaturation T2(H)
W81:Desaturation T2(L)
W82:Desaturation T3(L)
W83:Multiple desaturation
W84:Communications
W85:Time out soft charge
W86:Feedback breaker
W87:Softcharge target
W90:T ^A IGBTs HW
W91:Thermal L
W93:Thermal inductance LCL
W94: DC SC Module
W95:Source 15V
W96:T ^A PCB module
W98:Idc derivation
W99:Differential lac stop
W101:Mod. measure DC
W102:Unbalanced current
W103:IGBTs temperature
W104:PCB temperature
W106:Unbal/Max Vbus
W107:lac not reached
W108:Overcurrent Idc
W109:Vdc module low
W111:Fan module
W112:Emergency OCAC

Warning
W113:Checksum comms
W114:Sensed current R
W115:Sensed current S
W116:Sensed current T
W117:Sensed current DC
W118:Disable mod. SC
W119:Desaturation R1(L)
W120:Desaturation R3(H)
W121:Desaturation S1(L)
W122:Desaturation S3(H)
W123:Desaturation T1(L)
W124:Desaturation T3(H)
W125:Diff lac mod run
W126:Desv. T ^A IGBTs
W127:Mod. IGBT Delta T ^A est
W128:Spring loaded
W129:Isolation Module
W130:Softcharge Feedback
W131:Exceeded Faults limit
W133: Softcharge board
W151:Lockable module
W152:PPC comms
W156:High resistor isolation
W158: General Isolation
W159: Intern Isolation
W160: Extern Isolation
W161:Fan impulsione MT
W162:Fan DU zone
W163:Fan zone filter LC
W164:Fan zone modules
W165:Fan motor status
W166:Temp sensor read
W167:AntiPID ON
W170:Overvoltage DC
W171:No RTC
W172:LC filter Temperature
W173: IMI Disabled Channels
W178:Blown fuse
W179:FTP_OPEN
W180:SD problem
W181: Aux transf overvoltage
W182: Therma Bus Plus
W183: Internal comms
W184: MV transformer Temp

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Warning
W185: IMI not calibrated
W186: Supply low
W187: DU Delta Temp Threshold
W188: DU sensor lost comms
W189: Cold start ongoing
W191:Antipid wrong V
W192:Antipid V not reached
W193:Antipid retries exceed

Warning
W252:RFI OverCur Persistent
W253:Imbalance V
W254:VDC not reached
W255:BAT open detection
W303:RFI OverCurrent
W304:Imbalance P
W305:Limit VDC BESS
W306: CCL Limit

Warning
W307: DCL Limit
W308: HW Limit
W309: Busbar Disable
W310: Module Disable
W311: Low Vdc
W312: High Vdc
W313: Unbalanced Vdc

3. NECESSARY INSTRUMENTATION

3

The table below shows the devices and tools required to perform the various troubleshooting tasks. Consult PowerElectronics.

Description PE	BRAND AND REFERENCE EXAMPLE	Image V_1
Endoscopic camera with led	Pancellent Inc. 4331891352	
RJ45 Tester/Crimper	SGILE SE120485	
Ethernet cable tester	PROSTER PST204	
1500V Multimeter	AMPROBE HD110C	
Test probes DSP4	Test probes DSP4 ELECTRODE i	
Multimeter with clamp ammeter	FLUKE 376 FC	
Multimeter with insulation measurement (Megohmmeter)	FLUKE 1587 FC	

EN

Description PE	BRAND AND REFERENCE EXAMPLE	Image V_1
Antistatic bracelet	LUPO S-ETP-0301	
Power Inverter 1000W DC 12V to 230V	Yinleader 1000W12V-P-EU	
Power Inverter 2000W DC 12V to 230V	CANTONAPE DX-GAC2000W	
NETWORK ANALYZER	PURE ELSPEC SPG-531A-0090	
THERMOGRAPHIC CAMERA	FLUKE TIS+20 FLUKE TIS+60 BOSCH GTC400C	
RESISTANCE METER (MILLIOHMMETER)	CROPICO DO4000	
ISOLATION METER	KYORITSU 3125A (Megger)	
CAPACITANCE METER	PEAKTECH 2180 DMM/LCT METER	
MEDIDOR LCR GEN LCR GEN3 METER3	CEM DT-9935	
Mini contact thermometer with high accuracy and anti-drop, choice of thermocouple probe K/J	UNIT UT320A/UT320D	
2-CHANNEL OSCILLOSCOPE	PICO TECHNOLOGY PICOSCOPE 2205A	
4-CHANNEL OSCILLOSCOPE	PICO TECHNOLOGY PICOSCOPE 3403D	
TELUROMETER	PEAKTECH 2700	
CURRENT PROBE	FLUKE I3000s Flex-24	
THREE-PHASE CURRENT PROBE	ELSPEC SOA-3003-1271	
DIFFERENTIAL PROBE	TESTEC TT-SI 9010	

4. FAULTS TO BE REPORTED TO POWER ELECTRONICS

4

This section comprises the identified faults that will require no specific action but contacting Power Electronics.

Please, before starting any support request, use PowerCommsV with role Developer and download the following data files; they will be required to analyze the causes of the fault:

- SD card and black box of each module.
- Central black box.
- Inverter parameters in.xlsx and plant configuration in .dcfv format.

Then start the support request. Do not forget to attach the downloaded files.

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4.1. Fault code F1: Watchdog

This fault is normally related to an unsuccessful process of software upgrade, or in those cases when the response to one of the chips is out of time.

4.2. Fault code F46: Data Fault

The system has detected parameters out of range.

4.3. Fault code F47: Watchdog uP

Similar to F1, this fault indicates there was process that did not end on time, or with an error, in the microprocessor. Please, besides downloading the files listed at the beginning of this section, download also the F47 report.

4.4. Fault code F10: Comms SCI FPGA-DSP

The FPGA has not been configured correctly. This is due to a communication problem between DSP and central FPGA.

4.5. Fault code F59: Module SW Mismatch

This fault occurs when not all modules have the same software version.

4.6. Fault code F64: Invalid MAC

This fault occurs when the MAC address entered is invalid.

4.7. Fault code F113: CRC

This fault occurs when four consecutive frames with invalid CRC have been detected.

4.8. Fault code F125:Diff lac mod run

This fault appears when the zero sequence current calculated by the module is higher than the value set from PowerCommsV for more than 500us.

4.9. Fault code F129:Module Isolation

During diagnosis, an insulation fault was detected by switching one of the modules with open DU and after no insulation fault was obtained by measuring AC without switching.

4.10. Fault code F131:Faults limit exceeded

This fault appears when the module has given more than N faults in the configured time and the DSP has locked it. The number of faults can be configured by PowerCommsV.

4.11. Fault code F190:Combined F-V

This fault occurs when one of the set protection thresholds has been exceeded.

4.12. Fault code F210: uC-FPGA Comms

Communication between the MCU and the FPGA on the central control board has failed.

4.13. Fault code F226:IMI Hardware

The hardware of the insulation monitoring device is not compatible with the number of modules in the inverter.

4.14. Fault code F235: T/H sensors comms

This fault appears when the communications with the temperature and humidity sensors of all the inverter boards fail: central control, FPGA module and DU.

4.15. Fault code F236: DU sensor comms lost

This fault appears when the communications with the temperature and humidity sensors of the two DUs of the inverter fail: master - slave, if the inverter has two DUs, or only one DU of the inverter (master), if the inverter only has 1 DU, for example, MPCS.

5. FAULTS ANALYSIS AND TROUBLESHOOTING

5

5.1. Fault code F41: GFDI

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The GFDI circuit breaker is in TRIP status. This fault only applies to solar inverters with the negative pole connected to ground.	
Applicable to	☒ HEM	☐ PCSM/MPCSM
	☒ HEMK	☐ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

This fault is caused either because there is a derivation to ground on the positive pole or in any of the low voltage AC phases, or because there is a parasitic capacitance that has discharged through the GFDI Q12.

1. Use PowerCommsV to verify that voltages inside the product are stable and are all within range. Any abnormal value indicates that the derivation is inside the inverter. **INV3_ST1A**.
2. Check if there is any grounded point in the photovoltaic field. For this, it is recommended to open the GFDI and use the manual maneuver of the IMD, with which we can check the insulation of both the photovoltaic field and the low voltage side of the inverter. **INV3_ST1A**.
3. If no ground points are detected, use the GFDI maneuver to check its correct operation (opens and closes without problems). If the TRIP status appears when the device tries to close, there is probably an excessively large parasitic capacitance connected to ground. In this case it will be necessary to analyze why, please contact Power Electronics. **INV3_ST1A**.
4. To verify whether the problem is a parasite capacitance or a defect in the GFDI itself, disconnect the GFDI from earth and the negative pole **INVMG3_ST5A**, **INVKG3_ST5A** then repeat the GFDI maneuver. **INV3_ST1A**.



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- 4.1. If the GFDI does not trip, then there is a parasite capacitance. Search for possible sources of parasite capacitance (for example any system or device that is connected in DC; such as DCDC from strings, or tracker systems connected to the PV field) or, as a last resort, try to measure the parasitic capacitance to ground in case it is a problem of the installation itself.
- 4.2. If the GFDI trips, the GFDI has a problem. Replace it (refer to document *INVG3IOP85376*).
5. If everything went well, it is possible that there is a derivation inside the inverter (it can be in AC or DC). In this case, perform an insulation measurement with a megohmmeter (without voltage) between each of the phases (maximum at 1500V) and ground and between DC+ and DC- to ground (Measure on the busbars BBEP.75, BBEP.77, BBEP.79, BBEP.81, BBEN.60, BBEN.62, BBEN.64, BBEN.66)
INVMG3_ST5B, INVKG3_ST5B.

**DC (+) – GND****DC (-) – GND**

- 5.1. If the measurement is greater than 100 MΩ at all points, then everything is ok. Also, on the AC side the measurements should be similar, to verify that everything is ok.



AC (L1) - GND

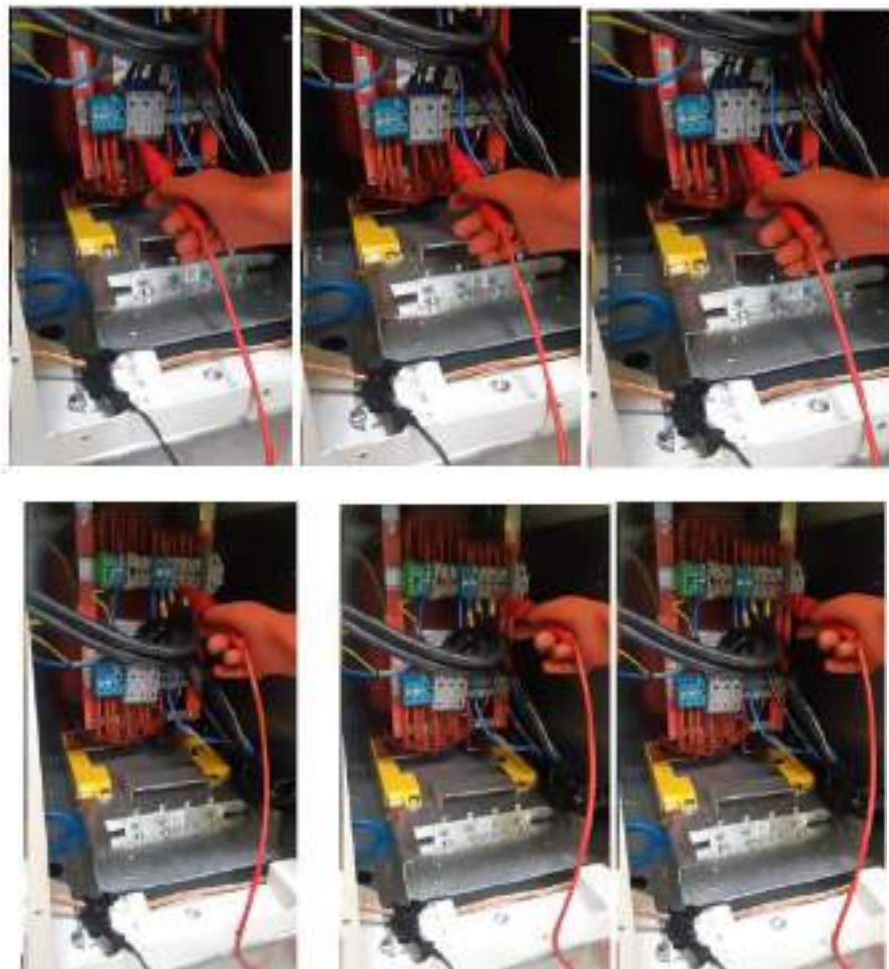


AC (L2) - GND



AC (L3) - GND

- 5.2. If the measurement fails, start disconnecting parts of the inverter and repeat the measure until the cause of the failure is found.
- 5.3. Make a visual inspection of: C filter, inductance L01, power stages, AASS transformer, LV busbar of the MV transformer (in HEM inverters). Verify there are no loose cables or foreign objects. If anything is found, fix it.
- 5.4. Open the disconnecter of the SSAA transformer T02, perform an insulation resistance test IR with (insulation tester test). If it fails, replace the transformer (refer to document *INVG3/OP85378*).



- 5.5. Disconnect the AC plates of the power stages. Perform an insulation resistance test IR (insulation tester test) at the inductance L01 and at each power stage. If it fails, replace the damaged

component. Refer to documents *INVG3/OP85375* (inductance) or *INVG3/OP85369* (power stages).



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- 5.6. Disconnect the plates of the C filter and perform an insulation resistance test. If it fails, repair or replace it; refer to document *INVG3/OP85367*.



- 5.7. In HEM inverters, disconnect the MV transformer and perform an insulation resistance test. If it fails, isolate and replace the transformer (refer to document *INVG3/OP85379*).
- 5.8. If nothing has been found, move on to the next step.
6. If the measurement went ok, enable DC and AC selectors (AC selector is only available in HEMK inverters) and repeat the manual maneuver of the IMD without closing the GFDI. If the IMD detects a resistance lower than 1 k Ω , visually inspect the inverter looking for the insulation fault point. [INVG3_ST1A](#).

7. If the GFDI is in TRIP status, it is probably because there is an earth leakage somewhere in the inverter. It will be necessary to make a technical inspection all over the inverter trying to find out any possible grounded point. **INVKG3_ST5A, INVMG3_ST5A.**

If everything has worked properly, re-start the inverter (get inverter back up again), configure the SD custom log with the specific parameters of the GFDI and (put the inverter in an operating point) generate an operating point similar to when the fault occurred. If the inverter includes the Bus Plus option and has DCDC converters, download the parameters and logs and contact Power Electronics.

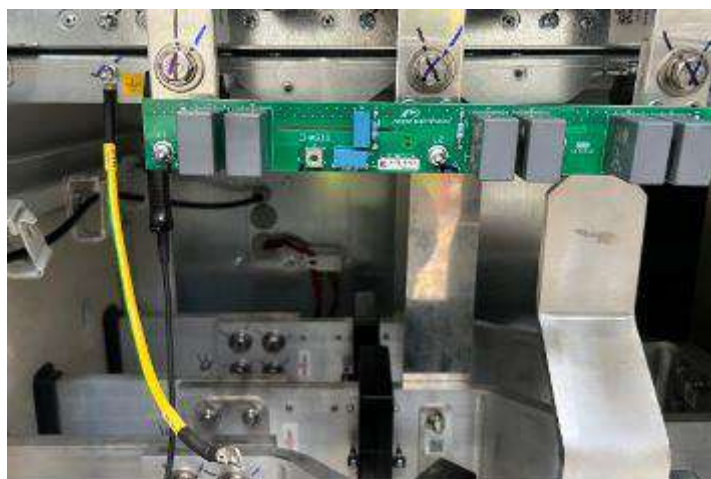
EN

5.2. Fault code F45: General Isolation

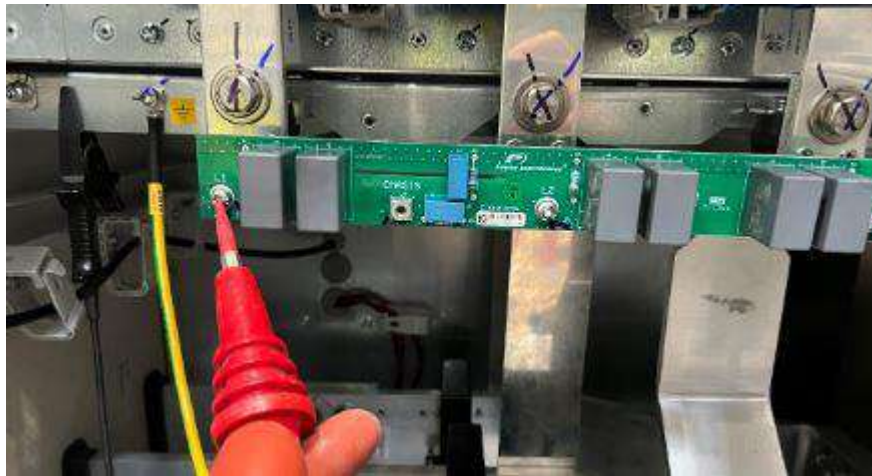
Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The inverter has measured a ground resistance lower than the fault threshold set in ([G9: IMD], subgroup [G9.2: Channels]) in any of the configured channels. When an insulation failure (temporary or permanent) has occurred somewhere in the system and when we cannot distinguish whether the origin of the insulation fault is inside or outside the inverter, this fault will be triggered.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	In some cases, warning [W155: Low resistor isolation] could appear before the fault.	

Analysis procedure

- In inverters with grounded pole: if the fault was detected first thing in the morning, it could be due to humidity presence. Depending on the weather conditions of the plant and the day, it is recommended to re-start the inverter and repeat the measurement 15 minutes to 2 hours later, when the humidity level has decreased depending on the environmental conditions of the plant. [INV G3_ST1A](#).
- If the fault occurs without humidity conditions, then we must measure the resistance in each channel. To do so:
 - Verify the IMD is calibrated. Consult the *Software and Programming Manual*. [INV G3_ST1A](#).
 - With the AC and DC open (all selectors in OFF position), enable the manual maneuver of the IMD in group [\[G9.3: Maneuver\]](#) and check the resistance of each channel in parameter [Resistor] within group [SV9.2: Channels]. It is recommended to do this test with at least 100V on all channels.
If the fault is triggered in this case, check which IMD channel is triggering the fault and:
 - If the fault was detected in AC and the inverter is not connected to the grid, proceed to measure insulation. [INVKG3_ST5A](#), [INVMG3_ST5A](#). In this case:
 - Use a high voltage insulation measuring instrument (1500V Multimeter). If there is continuity and the correct insulation is not present, go to the end of this point. The insulation between each of the phases of the RFI board (E16) and earth must be checked. To make the check it is necessary to access the LC filter cabinet where the RFI board (E16) is located.
Disconnect the ground cable from the RFI board (E16).



Connect the ground wire (black) of the Insulation tester to the chassis of the inverter and the line probe (red) of the Insulation tester to phase L1 of the RFI board (E16). **INVKG3_ST5A, INVMG3_ST2.**



Set the Insulation tester to provide a voltage of 1000Vdc (maximum 1500Vdc) with a measurement time of 1 minute and press the test button to take the measurement. Repeat this step for each of the phases L1, L2 and L3 of the RFI board (E16).



If the measurement value obtained is less than 100MΩ, it is not a correct insulation value. Therefore, it is necessary to progressively disconnect parts of the inverter that are connected to the AC busbar in order to find the point of failure. If a faulty component is detected, it must be replaced.

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Below is the list of most common components where this fault could have occurred, along with the related replacement procedure (should it be necessary):

- Inductance L01. Replacement procedure: *INVG3/OP85375*.
- C filter. Replacement procedure: *INVG3/OP85367*.
- Power stages. Replacement procedure: *INVG3/OP88076*.
- MV transformer T01. Replacement procedure: *INVG3/OP85378*.
- AASS transformer T02. Replacement procedure: *INVG3/OP85379*.

2.2.3. If the fault was triggered in DC, contact the plant staff and isolate the channels or measure ground potentials with two equal testers (positive-ground, negative-ground) to detect the channel causing the defect. *INVG3K_ST3A, INVG3M_ST3A*.

3. Activate the DC selector (close DC circuit breakers) and repeat the manual manoeuvre of the IMD with the inverter in DC READY status. *INVG3_ST1A*.
 - 3.1. If a fault is detected in this case, there is probably an insulation fault inside the inverter (power stages area or filter inductance). To find it, start closing the DC circuit breakers one by one and check the IMD measure. *INVG3_ST1A*.
 - 3.2. If the inverter includes the Bus Plus option, besides the previous step, start closing one by one the PV circuit breakers and check the IMD measure. *INVG3_ST1A*.
4. In HEMK and MPCSK inverters: activate the AC selector (close the AC circuit breaker) and with the inverter in READY status, repeat the insulation measurement. If a fault is detected, there is probably a fault in the low voltage zone of the inverter. *INVG3_ST1A*. Should this happen, check the previous actions for faults detected in an AC or DC channel.

If no fault was detected in any step, it is possible that it is an occasional fault. Configure the SD custom log with the IMD general and advanced parameters (both subgroups are within [SV9-IMD]) and contact Power Electronics.

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5.3. Fault code F63: IMD Earth Connection

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The inverter has detected that it is not grounded, when it should be.	
Applicable to	☒ HEM ☒ HEMK	☐ PCSM/MPCSM ☐ PCSK/MPCSK
Associated faults or warnings	N/A	
Probable cause	The ground resistance measured after finishing the GFDI maneuver (grounded) is greater than the ground fault resistance value (parameter [Fault Threshold], within [G9.2: Channels]).	

Analysis procedure

Remote actions:

1. Verify the IMD is calibrated. Consult the *Software and Programming Manual*. **INVG3_ST1A**.
2. Check the value of the insulation resistance measurement by forcing a manual maneuver of the IMD when the GFDI is closed (subgroup [Channel AC], within group [SV9.2-Channels]). Typical values are between [0, 800] Ω , ideally 0 Ω . **INVG3_ST1A**.

In case the insulation resistance measured remotely is higher than the configured threshold, perform the following steps. It is essential to remove all voltage from the inverter. Before removing the voltage, ensure the GFDI is closed (parameter [GFDI Output], within group [SV9.3-Maneuver]).

3. Steps that can be carried out with the inverter short-circuited: **INVMG3_ST5A, INVMG3_ST5A**.
 - 3.1. Disconnect completely the resistance boards (E71-E75) associated with each IMD channel and measure. Check that the resistance in each branch is 240 k Ω $\pm$ 1%. Otherwise, the board must be replaced.



- 3.2. Check continuity with 1500V multimeter between the GFDI and the PV negative pole.



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- 3.3. Check continuity with 1500V multimeter between all the jumpers that connect the GFDI poles in series.



- 3.4. Check continuity with 1500V multimeter between GFDI (at the circuit breaker output) and ground.



- 3.5. Check continuity with 1500V multimeter between the IMD and ground.



- 3.6. Check continuity with 1500V multimeter between the metal plate of the IMD panel and the main ground terminal of the inverter.

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4. Release the grounding cables: **INVMG3_ST5B**, **INVKG3_ST5B**.

Remove the motor from the GFDI, close the GFDI by hand and check continuity with 1500V multimeter between the PV negative pole and ground.

5.4. Fault code F204-F207: communications loss with fans

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

When communications between fans and the central control fail, one of the following faults will be triggered, depending on where the fan is located:

- F204¹: Fan impulsion MT: MV fans (cyclones, located in the MV transformer cabinet).
- F205: Fan DU zone: DU zone fans (located in the DC cabinet).
- F206: Fan zone filter LC: fan at the LC filter cabinet.
- F207: Fan zone modules: fans located in the modules (above the power stages).

There are several **probable causes** for this fault:

- Fan configuration has been lost.
- Physical breakage of the communication channel.
- Inverter misconfiguration.
- Loss of power supply to the fan.
- Physical failure of the fan

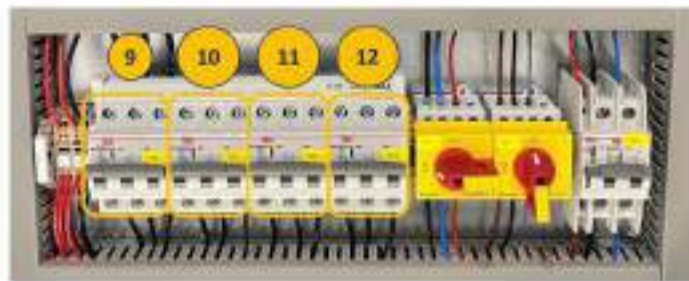
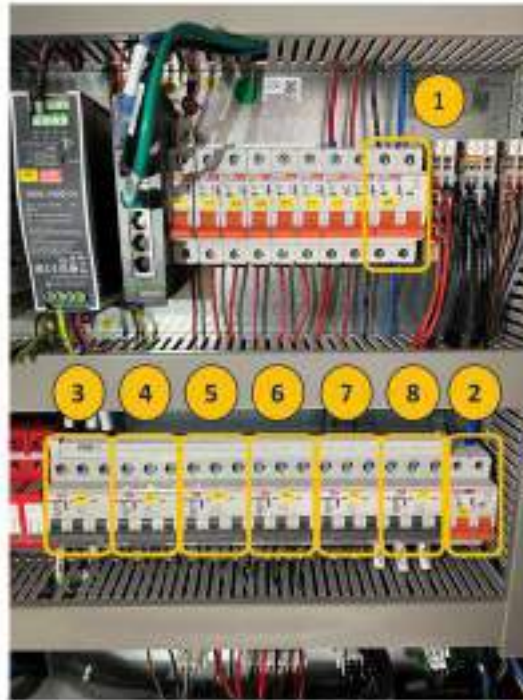
The following table lists the warnings that may appear before the fault, as well as the specific components (fans and circuit breakers) that must be verified during the analysis procedure which will be detailed below. In the event that during the analysis procedure it is determined that any fan needs to be replaced, please consult document *INVG3/OP85370*.

Fault	Associated warning	Main components that must be verified
F204	W161: Fan impulsion MT.	Fans: PMV8, PMV9, PMV10 and PMV11. Circuit breakers: MV cabinet (Q20, Q21, Q22 and Q23) and general MV circuit breaker (Q11). Power fan connector: XC761, XC764, XC767 and XC770. Communications connector: XC760, XC763, XC766 and XC769.
F205	W162: Fan DU zone	Fans: PMV1 and PMV2. Circuit breakers: Q04 and Q05. Power fan connector: XC551 and XC554. Communications connector: XC550 and XC553.
F206	W163: Fan zone filter LC	Fan: PMV7. Circuit breaker: Q10. Power fan connector: XC251. Communications connector: XC250.
F207	W164: Fan zone modules.	Fans: PMV3, PMV4, PMV5 and PMV6. Circuit breakers: Q06, Q07, Q08 and Q09. Power fan connector: XC201, XC204, XC207 and XC210. Communications connector: XC200, XC203, XC206 and XC209.

¹ Fault F204 only applies to HEM and MPCSM inverters. Faults F205-F207 apply to all GEN3 inverters.

Analysis procedure

1. Identify which fan has triggered the fault and verify it is assigned to the correct zone. Check parameters [State comms] and [Fan zone], within group [SV11: Fans]. **INVKG3_ST1B**. If the settings are correct, go to the next point; otherwise, adjust them so that the fan type matches the location where it is installed.
2. Run the search process (in [G11.1: Detection]) for the fan that triggered the fault. **INVKG3_ST3A**, **INVMG3_ST3A**.



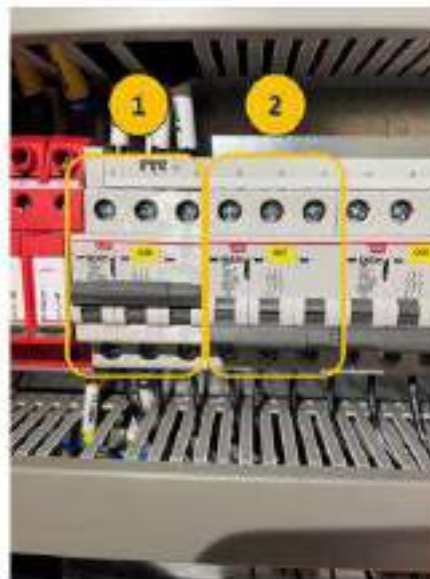
Breaker	Reference	Fan
8 - 12	Q11 (General MV Breaker), Q20, Q21, Q22 & Q23	PMV8, PMV9, PMV10 and PMV11
1 - 2	Q04 & Q05	PMV1 and PMV2.
7	Q10	PMV7
3 - 6	Q06 (Module 4), Q07 (Module 3), Q08 (Module 2) & Q09 (Module 1)	PMV3, PMV4, PMV5 & PMV6

3. Check the RS485 fan communication group [G6.1: Modbus].
 - 3.1. Verify whether there is communication with the other fans. If so, go to the next step.
 - 3.2. Otherwise, if there is no communication with any fan, verify all configuration values (speed, parity, stop bit...) within group [G6.1: Modbus] are set to their default value. **INVKG3_ST1B**.
 - 3.3. Ensure that the RS485 communication of the fans is enabled (parameter [Fans 485 Enable], within group [G11: Fans]). **INVKG3_ST1B**.

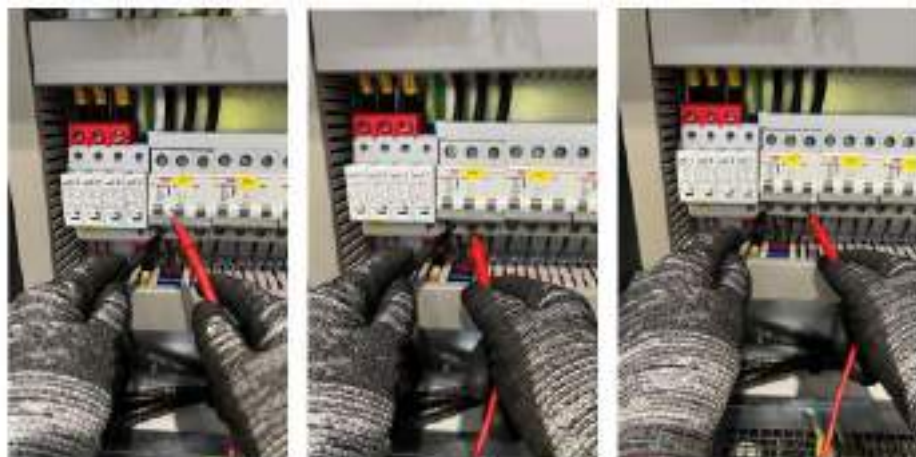
- 3.4. Check the ratio of correct frames [Modbus OK frames] and frames with error [Modbus error CRC], it must be lower than 1%. Higher values mean frames are being lost. **INVKG3_ST1B**.

If the above steps have been carried out and the source of the fault has not yet been found, it will be necessary to check physically the inverter (circuit breakers, wiring, etc.). Continue with the following steps and notice the status of the inverter will vary depending on the test to be performed.

4. Visually check the protection circuit breakers of the fan zone (see the table above to find out which circuit breakers correspond to each zone). **INVKG3_ST5A, INVMG3_ST5A**.
- 4.1. Verify all circuit breakers of the fan zone are closed (in ON position). **INVKG3_ST5A, INVMG3_ST5A**. Otherwise, if any circuit breaker is not closed:
- 4.1.1. Check on the fan side (bottom of the magnetic switch, points 2, 4 and 6 on the appropriate thermistor depending on the fan) that there is no short circuit between the phases. Check with 1500V multimeter that the resistance is greater than 0. If the resistance is 0Ω or close to 0Ω, disconnect the corresponding fan connector and repeat the measurement. If the measurement is no longer 0Ω, replace the corresponding fan. If the measurement is still 0Ω, replace the fan wiring. **INVKG3_ST5A, INVMG3_ST5A**. In case the fans in the power modules area need to be checked, the status for HEMK/PCSK is **INVKG3_ST2**.



Breaker	Reference
1	Breaker cerrado (ON)
2	Breaker abierto (OFF)



- 4.1.2. If no short circuit was found (INVKG3_ST5A, INVMG3_ST5A), prepare to close the circuit breaker. Check the *Safety Instructions for Operating, Troubleshooting and Maintenance* to safely close it. Please notice the status of the inverter and the required PPE vary depending on the area of the inverter in which the circuit breakers to be checked are located. INVKG3_ST3A, INVMG3_ST6.
5. After previous step has been performed, rearm the circuit breaker, and see if the fault is triggered again.
- 5.1. If the circuit breaker is triggered immediately after re-starting: there is a short circuit in the wiring or in the fan itself. In this case, carry out the following checks. INVKG3_ST5A, INVMG3_ST5A (in case the fans located in the power stage area need to be checked, the status for HEMK/PCSK is INVKG3_ST2):
- 5.1.1. Disconnect the corresponding fan connector (INVMG3_ST5A, INVKG3_ST5A). Repeat the test for resetting the circuit breaker. To do this, refer to the *Safety Instructions for Operation, Troubleshooting and Maintenance* for safe locking. Please note that the condition of the inverter and the required PPE varies depending on the area of the inverter where the circuit breakers to be checked are located. INVKG3_ST3A, INVMG3_ST6. If the circuit breaker no longer trips, replace the fan. INVMG3_ST5A, INVKG3_ST5A. If a fan needs to be replaced, refer to document INVKG3/OP85370. In case the fans located in the area of the power stages need to be replaced, the status for HEMK/PCSK is INVKG3_ST2.



Module fan connector



C Filter fan connector

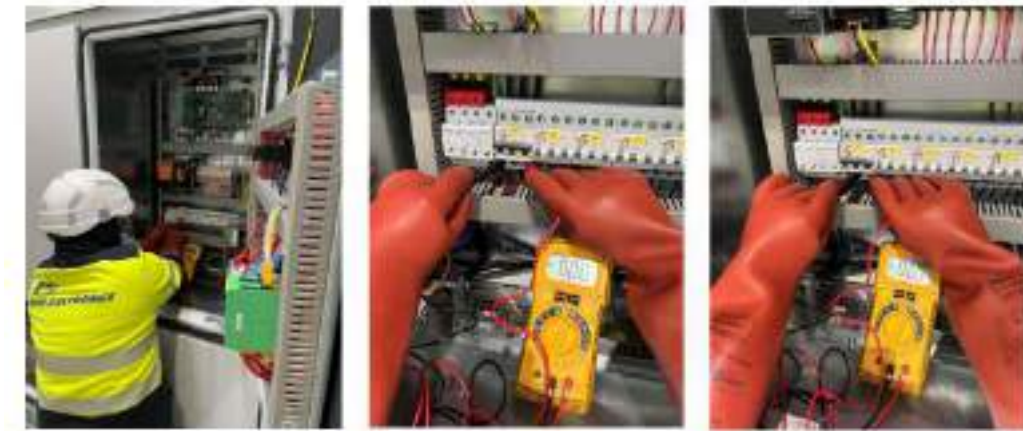


DC fan connector



MV fan connector

- 5.1.2. If the circuit breaker keeps tripping, there is a point in the wiring that is short-circuited or has an insulation fault. Inspect the wiring from the circuit breaker to the fan connector for shorted or stripped points. If the fault is not found, replace the circuit breaker and fan wiring. **INVG3_ST5A.**
- 5.2. If the fault does not appear immediately, send the start command to the fan through PowerCommsV (**parámetro [Modo] en grupo [G11: Ventiladores]**) and see if the fault is triggered when the fan starts. In this case, there is a problem in the fan. Replace the fan (refer to document *INVG3IOP85370*). Please notice the status of the inverter will vary depending on the fan to be working in:
- **INVMG3_ST3B** for MV fans.
 - **INVKG3_ST3B, INVMG3_ST3B** for C filter and module fans.
 - **INVKG3_ST5A, INVMG3_ST5A** for DC cabinet fans.
- 5.3. If the fault is not triggered after sending the start command and the fan has started up well, configure the SD custom log with fan speed and communication status parameters. Contact Power Electronics to investigate why was the fault triggered in the first place. **INVG3_ST1B.**
- 5.4. If the fault persists when the circuit breaker is raised, measure the voltage with 1500V multimeter upstream of the circuit breaker (between points 2-4, 4-6 and 6-2 of the circuit breaker depending on the fan).the fault has not been triggered, measure the voltage of the circuit breaker upstream the circuit breaker and check if when closing it there is a voltage unbalance (i.e. if any of the voltages is < 400V). Check that in all phases there is 400V. If there is an unbalance, carry out a technical inspection of the wiring. Follow schematics and refer to the *Safety Instructions for Operating, Troubleshooting and Maintenance* to see the status that applies for this action. **INVKG3_ST3A, INVMG3_ST6.** If voltage unbalance or no voltage is measured at any or all points, perform the following checks:



- 5.4.1. Check via PowerCommsV (group SV3: Network display) whether the input voltage unbalance measured by the circuit breaker is reflected in the input voltage of the device. If this is the case, it corresponds to a connection problem of the equipment or of the customer's network. Check with the customer if he is performing any task that causes this unbalance. If the customer mains voltages are OK, check that the fuses of the auxiliary transformer (F100) are in good condition. If they are not in good condition, replace the fuses of the auxiliary transformer.



F100



Feedback to check F100 state (Example OK)

5.4.2. If the fuses and the mains are balanced, measure voltage with 1500V multimeter on the primary of the auxiliary transformer, between points L1-L2, L2-L3 and L3-L1. In all cases there should be a balanced voltage between 600 and 700Vac. If an unbalance or missing phase is measured, check continuity with 1500V multimeter between the following points:

- F100:5 with L3 of the primary of the SSAA transformer.
- F100:3 with L2 of the primary of the SSAA transformer.
- F100:1 with L1 of the primary of the SSAA transformer.
- If there is no continuity at any of the points, replace the wiring.



Homepage PowerCommsV

SV1.2.1-Grid voltage RS LV	701.7	V
SV1.2.2-Grid voltage ST LV	697.9	V
SV1.2.3-Grid voltage TH LV	702.6	V

PowerCommsV parameter

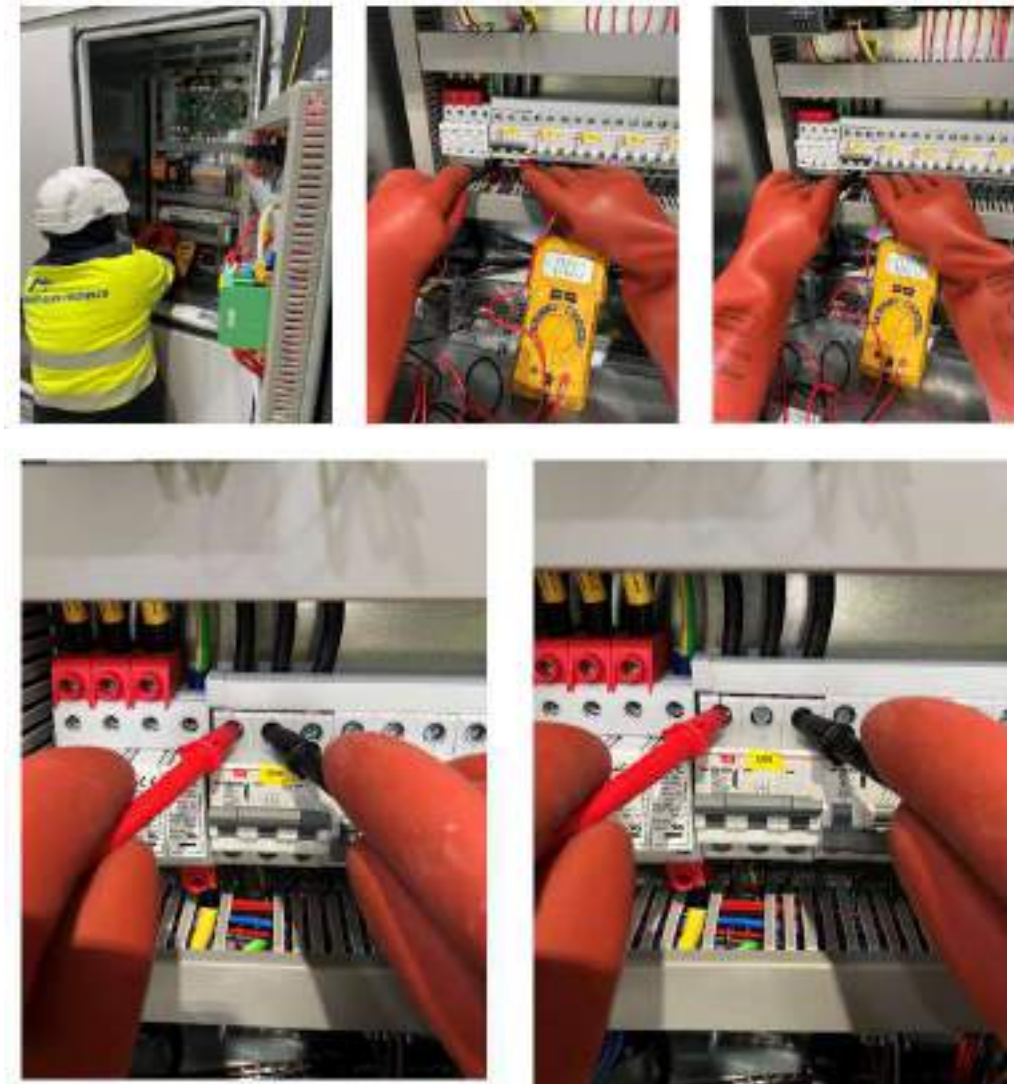
5.4.3. If the SSAA transformer primary voltages are correct, measure voltage with 1500V multimeter on the 430Vac secondary of the auxiliary transformer between points L1-L2, L2-L3 and L3-L1. The measured voltage on the auxiliary transformer should be approximately 400Vac and balanced. If the unbalance does not appear, check continuity with 1500V multimeter between points L1, L2 and L3 of the 430Vac secondary of the auxiliary transformer and points 1, 3 and 5 of the circuit breaker corresponding to the fan (in the case of the DU fans, also check the neutral).



6. Check visually that circuit breakers are not burnt. **INVKG3_ST3B, INVMG3_ST3B**. If any of them was, refer to the corresponding document and replace it. **INVKG3_ST2, INVMG3_ST2**.

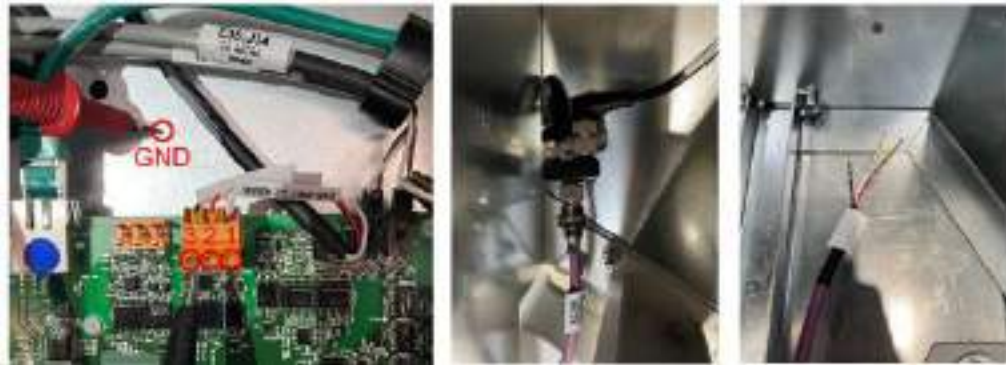
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7. If the technician has tools to measure temperature without making contact (contact block), check the circuit breakers and see if there is one for which the temperature draws the attention (at least 20 degrees above the rest of the circuit breakers). It should not exceed 70°C. **INVKG3_ST3A, INVMG3_ST6**. If any circuit breaker is overheated, replace it. **INVKG3_ST2, INVMG3_ST2**.
8. Measure voltage with 1500V multimeter downstream and upstream the circuit breaker, between all phases. **INVKG3_ST3A, INVMG3_ST6**.
 - 8.1. The voltage RS upstream and downstream the circuit breaker must be the same (also for ST and RT phases). Otherwise, replace the circuit breaker. **INVKG3_ST2, INVMG3_ST2**.



9. Check that the fan power connector is in good condition (no pins are broken, bent, displaced, oxidized, nor do they have any other defect that could cause a bad contact). If any defects are observed, replace the connector. **INVKG3_ST5A, INVMG3_ST5A**.
10. Check that the RS485 communications connectors are in good condition (no pins are broken, bent, displaced, oxidized, nor do they have any other defect that could cause a bad contact). If any defects are observed, replace the connector. **INVKG3_ST5A, INVMG3_ST5A**.
11. Check the continuity with 1500V multimeter of the RS485 communications bus up to the fan that has triggered the fault at the following points:
 - Pin 1 of connector J36 on the E35 board with pin 4 of the communications connector on the failed fan.
 - Pin 2 of connector J36 on the E35 board to pin 2 of the communications connector on the failed fan.

- Pin 3 of connector J36 on the E35 board with pin 3 of the communications connector on the failed fan.



- If any of the checks do not give continuity, repeat the above measurements with 1500V multimeter but between the E35 board and a connector close to the E35 board to find the fault. Once the fault has been pinpointed, replace the corresponding cable or T section and check the communications. Narrow down the sections until the problem is detected and replace the faulty component. [INVKG3_ST5A](#), [INVMG3_ST5A](#).



- Once continuity has been checked, measure the resistance with 1500V multimeter between the pins (A-B, A-GND, B-GND) of the RS485 connector of the failed fan. The measuring equipment must have a range greater than 10 MΩ. If any measurement gives infinity, the internal wires of the fan are likely broken. Refer to document [INVG3IOP85370](#) to repair or replace the fan. [INVKG3_ST5A](#), [INVMG3_ST5A](#).



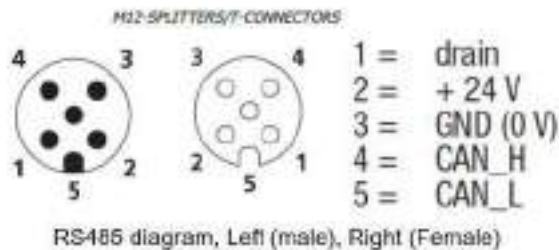
Connector



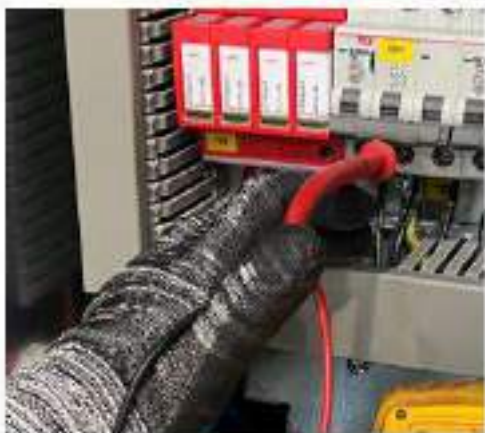
Measure example



Zoom measure



13. Check the physical condition of the fan: installed properly, without scuff marks, can rotate smoothly, etc. If any problem is detected, fix it. If fan replacement is necessary, refer to document *INVG3IOP85370*. [INVKG3_ST5A](#), [INVMG3_ST5A](#).
14. Measure continuity with 1500V multimeter in the fan power supply cables. [INVKG3_ST2](#), [INVMG3_ST2](#). The points are given below:
 - 14.1. Between the failed fan MCB and the failed fan power connector.
 - Point 6 of the fan circuit breaker and point 3 of the fan connector.
 - Point 4 of the failed fan circuit breaker and point 2 of the fan connector.
 - Point 2 of the failed fan circuit breaker and point 1 of the fan connector.
 - If any of the points do not give continuity, replace the wiring.



- 14.2. With the circuit breaker open, measure the resistance between phases with a 1500V multimeter, measuring between points 2-4, 4-6 and 6-2 of the faulty fan circuit breaker. It should be greater than 1k Ω . If any of these points measure a permanently lower resistance, replace the fan (see document INVG3IOP85370).



- 14.3. Check the resistance between phase and earth. Check the resistance between points 2, 4 and 6 of the failed circuit breaker and earth with the 1500V multimeter in the resistance position. The expected measurement should be greater than 10MOhms (ideally the multimeter should give infinity). If this is not the case, disconnect the fan power connector and repeat the measurement. If the low measurement remains, replace the fan wiring. If the fault disappears, replace the fan concerned (see document INVG3IOP85370).



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5.5. Fault code F228: Fan setpoint

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	One of the fans is not operating at the configured speed setpoint. When the speed reference is changed, the microcontroller checks if all fans are rotating at the configured setpoint (a tolerance, called "setpoint deviation error", is defined in group [G11-Fans]).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	W165: Fan motor status.	

Analysis procedure

There are several reasons why the fan may be slowing: excess inrush current, temperature derating, a physical problem in the fan itself, etc. Follow the steps below to identify the possible source of the fault and resolve it:

1. Ensure the software conditions to trigger the fault are configured correctly: parameters [Setpoint deviation err] and [Setpoint error delay], within group [G11: Fans], must be set to their default value. **INV3_ST1B.**
2. If any fan of the inverter has been replaced recently, check if all parameters are configured correctly. **INV3_ST1B.**
 - 2.1. Go to groups [G11.3: DU config], [G11.4: MT config], [G11.5: Config Modules] and [G11.6: Config LC filter].
 - 2.2. Check the values of parameters [Setpoint], [Minimum speed], [Maximum speed]. Make sure they fit fan manufacturer and model and that they are set to their default values.
3. Check whether the fan has reported any faults. See parameters [Motor Status] and [Warnings] in group [SV11: Fans]. If there are any faults reported, take note and check the manufacturer's manual for instructions on how to proceed. **INV3_ST1B.**
4. Fan faults are usually very fast, and their status and fault variables could change rapidly during the process, making it nearly impossible to diagnose the problem only by looking at the PowerCommsV screen. Thus, follow the next actions: **INV3_ST1B.**
 - 4.1. Set up the SD data log (custom log) with all the visualization group of the corresponding fan and also with the status of the inverter, to check the fan startup time.
 - 4.2. Reset the inverter, and make it run as usual. The aim is trying to reproduce the fan's fault again.
 - 4.3. If the fan is slow to start but eventually reaches the setpoint, extract logs and contact Power Electronics. Please indicate fan manufacturer and model.

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If the fan starts and stops, take note of the fan faults and warnings and refer to the fan's manual to analyze them. If repair or replacement was necessary, please refer to the document *INV3/OP85370* and notice that in this case the inverter must be in status **INVKG3_ST5A**, **INVMG3_ST5A**. If repair or replacement was necessary, please refer to document *INV3/OP85370* and notice that in this case the inverter must be in status **INVKG3_ST5A**, **INVMG3_ST5A**. In case the fans in the area of the power modules need to be replaced, the status for HEMK / PCSK inverter is **INVKG3_ST2**.

5. Start the fan at fixed speed (in [G11: Fans], [Mode] = Fixed) and check if there are any strange noises which could be caused by the fan rubbing against any nearby inverter's part. Whether this is the case, then check fan out in order to replace or repair any damaged or loose part from the fan or near parts. Refer to document *INV3/OP85370*. **INVKG3_ST5A**, **INVMG3_ST5A**. In case the fans in the area of the power modules need to be replaced, the status for HEMK / PCSK inverter is **INVKG3_ST2**.

If after all the steps have been done, no fan failure has been found, it will be necessary to replace the fan. Refer to document *INV3/OP85370*. **INVKG3_ST5A**, **INVMG3_ST5A**. In case the fans in the area of the power modules need to be replaced, the status for HEMK / PCSK inverter is **INVKG3_ST2**.

5.6. Fault code F230-F233: fan type not correct

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	N/A.			

All fans installed in each fan zone must be from the same supplier. If there are any fans from different manufacturers, one of the following faults will be triggered, depending on where the fan is located:

- F230: DU zone fans (located in the DC cabinet).
- F231²: MV fans (cyclones, located in the MV transformer cabinet).
- F232: fans located in the modules (above the power stages).
- F233: fan at the LC filter cabinet.

The **analysis procedure** when any of these faults appear is:

1. Verify the manufacturer **configured in group G11 or SV11** and confirm it matches with the fans. If they do not match, configure them properly. If configuration is correct, move on to the next step. **INVKG3_ST1B**.
2. If not all the fans of the zone are from the same manufacturer, install fans from the same supplier in the cabinet and run again the search process from the subgroup **[G11.1-Detection]** to configure them correctly. **INVKG3_ST5A, INVMG3_ST5A**.

Images of the different fan suppliers are shown below.



² Fault F231 only applies to HEM and MPCSM inverters. Faults F230, F232 and F233 apply to all GEN3 inverters.



- 2.1. For MV fans (HEM and MPCSM inverters), if the fans are from the Delta manufacturer but different model, install the same model. **INVG3M_ST5A**.

The following table lists associated warnings on software versions prior to 2.2.0, as well as the specific fans that must be verified during the **analysis procedure**.

Fault	Associated warning (in versions prior to 2.2.0)	Fans that must be verified
F230	Used to be warning W174.	DC cabinet fans: PMV1 and PMV2.
F231	Used to be warning W175.	MV transformer cabinet fans: PMV8, PMV9, PMV10, PMV11.
F232	Used to be warning W176.	Inverters cabinet fans: PMV3, PMV4, PMV5, PMV6.
F233	Used to be warning W177.	LC filter cabinet fan: PMV7.

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5.7. Fault code F118: Module Disabled

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".		
Description	This fault is triggered by the system to stop a module when it detects a situation that needs to be checked.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	W118: Disable mod. SC		

Analysis procedure

- Check the LOG SYS of the SD and check if the module that triggered the fault has been disabled during diagnosis. If the customer/SCADA has disabled the module, ask the cause of this action and inform that this is an improper action. **INV3_ST1A**.
- If the inverter includes the Bus Plus option: currently any fault that isolates DC on Bus Plus causes the rest of the modules to fail by triggering fault F118. **INV3_ST1A**. Thus:
 - Check the faults and warning history (groups [SV7.1-Faults], [SV7.2-Warnings] and [SV7.4-Module events]).
In case of encountering any fault that isolates DC (opens DU or special fault), read such fault within this document and fix the problem in the module on which the fault occurred. The module that has triggered the F118, a priori, should not have any problems.
- If another element is connected to the module bus (e.g., a DCDC on Bus Plus), verify that this element is not operating during module discharge (i.e., not injecting current) and is charging the module during discharge. If a problem is detected, the problem is outside the inverter, check that element. **INV3_ST1B**.
- Using PowerCommsV, activate the DC virtual selector and verify that the diagnosis process (module bus charging and discharging) meets the fault conditions (not dropping below 200V from start of discharge in less than 20 seconds or not dropping below 100V in 60 seconds). **INV3_ST1A**.
 - If the first fault condition is met, go to step 5.
 - If the second fault condition is met, go to step 6.
- If after the above steps the source of the fault has still not been detected, it will be necessary to physically check the inverter:
 - Check the physical status of the DC switch disconnectors: **INV3_ST5A**, **INV3_ST5A**.
 - Measure continuity with 1500V multimeter between the bus part of the module and the external DC part (BESS or PV). If there is continuity, make a technical inspection of the inverter and locate the problem. **INV3_ST2**, **INV3_ST5A**
 - Verify that when they are opened, they are actually open (check absence of continuity with 1500V multimeter between the input and output of the DC isolating switch). **INV3_ST2**, **INV3_ST5A**. If any of them is not open (there is continuity), replace it. Refer to document **INV3IOP85377**.



5.2. Check the status of the power stage:

- 5.2.1. Check the condition of the diodes of the power stage (see document *INVG3IOP85369*). If any problem is found, disassemble the power stage and check the diodes of each IGBT individually. If any of them are not OK, replace the IGBT or the power stage (depending on availability). **INVKG3_ST4, INVMG3_ST4.**



Check the capacity of the module's half buses. (Note: make sure that the 1500V multimeter can measure up to 3000 uF). **INVKG3_ST4, INVMG3_ST4.** If the inverter includes the Bus Plus option: disconnect the DC cables and measure capacitance. **INVKG3_ST5A, INVMG3_ST5A.** If the capacitance is not below 2700 uF, replace the capacitor stage or the capacitor bus (depending on availability). **INVKG3_ST4, INVMG3_ST4.** Refer to document *INVG3IOP85369*.

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BUS2 - BUS_N



BUS2 - BUS_P



BUS_P - BUS_N

EN

5.8. Fault code F31: Selfdiagnosis fault

Mandatory	The user must check the <i>Safety Instructions for Operating, Troubleshooting and Maintenance</i> before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	The DC or AC self-diagnosis has not finished successfully before the timeout (set internally) elapses.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

Analysis procedure

There are many causes that could trigger this fault. Thus, the first recommended step is to check the faults and warning history [SV7-Faults and warnings] using PowerCommsV to get an idea of the problem. Then, keep on reading to find out about the probable causes of the problem and how to proceed:

Wrong configuration of the insulation monitor, or an IMD badly calibrated, or excessive parasitic capacitance to ground or very high resistance to ground with some capacitance:

1. Using PowerCommsV, check that the settings of the insulation monitor (group [G9-IMD]) match with the inverter hardware. During self-diagnosis the inverter checks the insulation measurements; if any channel is not enabled or is not measuring correctly, the inverter ends up failing due to timeout. **INVG3_ST1A.**

2. Verify that the IMD is calibrated (parameter [Calibration status], within group SV9.4) **INVG3_ST1A.**

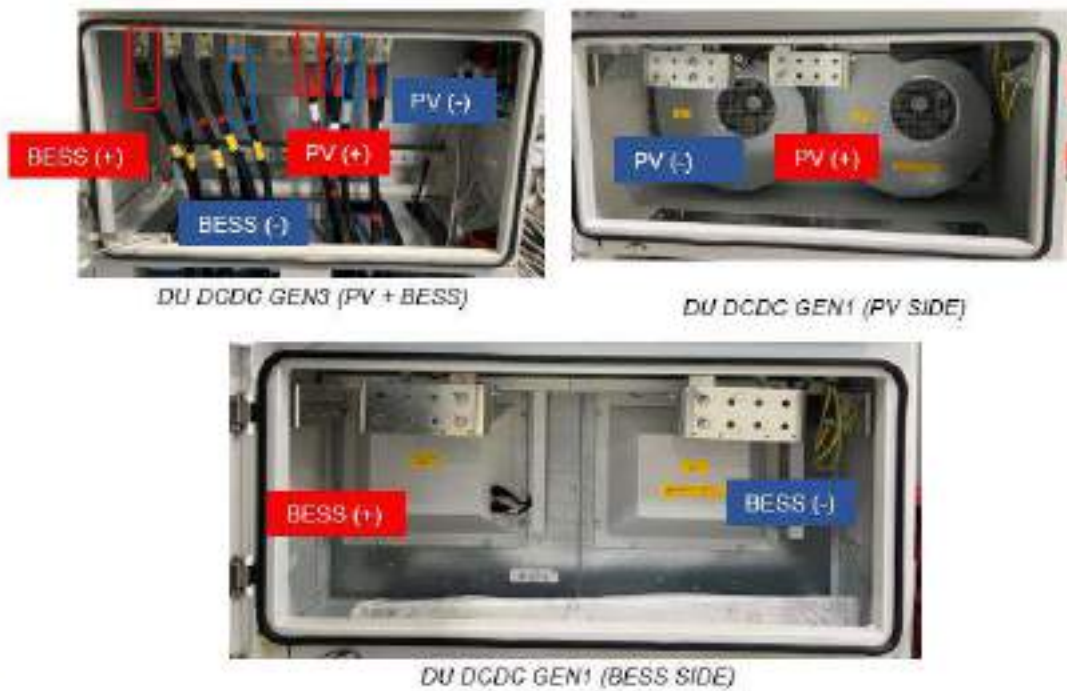
A soft charge board in bad condition, wrong soft charge resistors or a module that switches incorrectly.

3. Check in the main screen of PowerCommsV that the modules perform correctly the soft charge during self-diagnosis. That is, that all modules are able to increase bus voltage and then discharge it in a similar time (and less than the configured timeout). **INVG3_ST1A.**

In inverters with Bus Plus option, there could be a bad configuration of the connected bus capacitance.

4. Check that the correct number of DCDC converters is configured on the inverter (parameter [Number of DC/DC] in subgroup [SG10.1-General]). **INVG3_ST1A.**
5. Check the state of the DCDC converters to verify that they have not interfered in any way with soft charge:
 - 5.1. Through PowerCommsV, check that during self-diagnosis the DCDC modules are in the OFF (stopped) state. **INVG3_ST1A.**
 - 5.2. Through PowerCommsV, check that the DCDCs have not been charging batteries during the self-diagnosis. **INVG3_ST1A.**
 - 5.3. If the DCDCs were discharging batteries, they would cause the inverter bus voltage to rise in an uncontrolled manner (this would disrupt the normal operation of the self-diagnosis). **INVG3_ST1A.**

Check that the DCDCs do not have some kind of internal defect in their PV or BESS bus that would cause them to consume from the inverter bus. **INVKG3_ST3B, INVMG3_ST3B** for the inverter and the DCDC must be off, disconnected and with LOTOTO applied following the *Safety Instructions for Operating, Troubleshooting and Maintenance* of the product



EN

Some unknown external power supply charges the bus:

6. In case the installation includes DCDC converters which are not manufactured by Power Electronics, check if the voltage bus rises in an uncontrolled way in the main PowerCommsV screen (for example, being OFF that the bus rises as if soft charge is enabled); that would mean that there is an external power supply that is loading the product bus without our knowledge. [INVKG3_ST1A](#). Check what is connected to the inverter's internal bus or Bus Plus busbar (if applicable); check the project electrical schematics to make sure there is no soft charge from the field. [INVKG3_ST5A](#), [INVMG3_ST5A](#). If the inverter includes the Bus Plus option, the DCDC must be off, disconnected and with LOTOTO applied following the *Safety Instructions for Operating, Troubleshooting and Maintenance* of the product.
7. If nothing is connected, check the soft charge circuits to see that none are connected in an undesired manner. [INVKG3_ST2](#), [INVMG3_ST2](#).



GROUNDED POLE



ISOLATED POLES

5.9. Fault code F32: Error Mod selfdiagnosis

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .		
Description	One of the modules (or more) has triggered a fault while the self-diagnosis was running.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

Analysis procedure

1. Check through PowerCommsV the faults and warning history at the moment the fault occurred (check groups [SV7.1-Faults], [SV7.2-Warnings] and [SV7.4-Module events]) and proceed accordingly depending on the fault that is found. [INV G3_ST1A](#).
2. If no faults were found in the previous step, use PowerCommsV to reset the F32 fault and verify that all modules correctly perform the charge and discharge process during the self-diagnosis, reaching the target PV or BESS DC voltage. [INV G3_ST1A](#). Otherwise, check soft charge circuit (in solar inverters: check wiring of the soft charge board; in storage inverters: check wiring of the resistance boards and contactors) and the power stage. [INVKG3_ST2](#), [INVMG3_ST5A](#). Refer to document [INV G3/OP85369](#).
3. If the problem remains unresolved, contact Power Electronics.

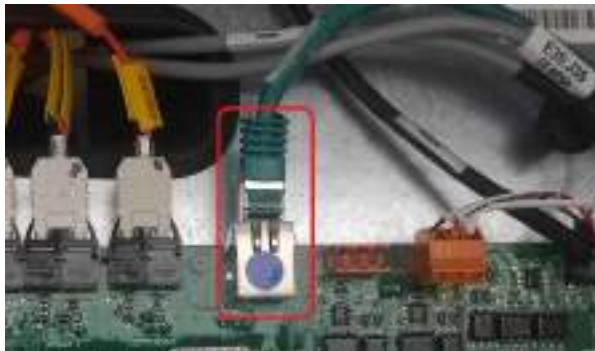
EN

5.10. Fault code F51: PPC Comms

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	This fault is triggered if the inverter stops receiving the heartbeat from the PPC or SCADA in the Heartbeat communications register.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	[W152: PPC comms] must appear before triggering F51.	

Analysis procedure

In order to troubleshoot this fault, the inverter must have a connection cable from the E35 central control board to the equipment switch (U01) and another cable from the switch (U01) to the equipment door (U02) (default configuration). If the project has a different configuration, it will be necessary to put the inverter in status **INVKG3_ST5A**, **INVMG3_ST5A** and connect such cable. Once connected, it is possible to change the PPE to those of the status indicated at each point.



The PPC or SCADA does not send any data to the Heartbeat, or it fails to reach the inverter due to a misconfigured IP:

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1. Using PowerCommsV, check the configuration of [G6.2: Ethernet] and ensure that the inverter's IP is correct according to the project configuration. If the IP is not configured correctly, update it according to the project/customer documentation. **INVG3_ST1B.**

2. Ensure that the counter [PPC Heartbeat Reg], within group [G6.5: PPC] is consistently increasing.

If it is increasing, it indicates that the inverter is properly configured at communications level and the issue is most likely a sporadic hardware failure.

If it does not increase, proceed to the next point. **INVG3_ST1B.**

Rule out any potential configuration problems with inverter access permissions:

3. If the counter is not increasing, ensure that IP filtering is disabled (parameter [IP filter enable], within group [G6.5: PPC]) and that the SCADA IP is not configured as "Black list" in the access configuration (parameter [User Level IP], within group [SG14-Access Settings]). **INVG3_ST1B.**

If the counter is not increasing, there may be an issue with the communications wiring at the client software level of the central control board.

4. To rule out a configuration problem, connect an external computer to the equipment via Ethernet and send different values from 0 to the Modbus address corresponding to the parameter [PPC Heartbeat Reg]. **INVG3_ST1B.**

4.1. If the Heartbeat counter increases, it is a problem with either the SCADA software or the communication installation. Inform the issue to the client, they must find a solution. **INVG3_ST1B.**

4.2. If the counter does not increase, ping the central control board. If it works properly, check the configuration of [G6.2: Ethernet] and the IP addresses. **INVG3_ST1B.**

If the ping fails, replace the central control board card (E35) and if the problem persists contact Power Electronics. **INVKG3_ST5A, INVMG3_ST5A.**

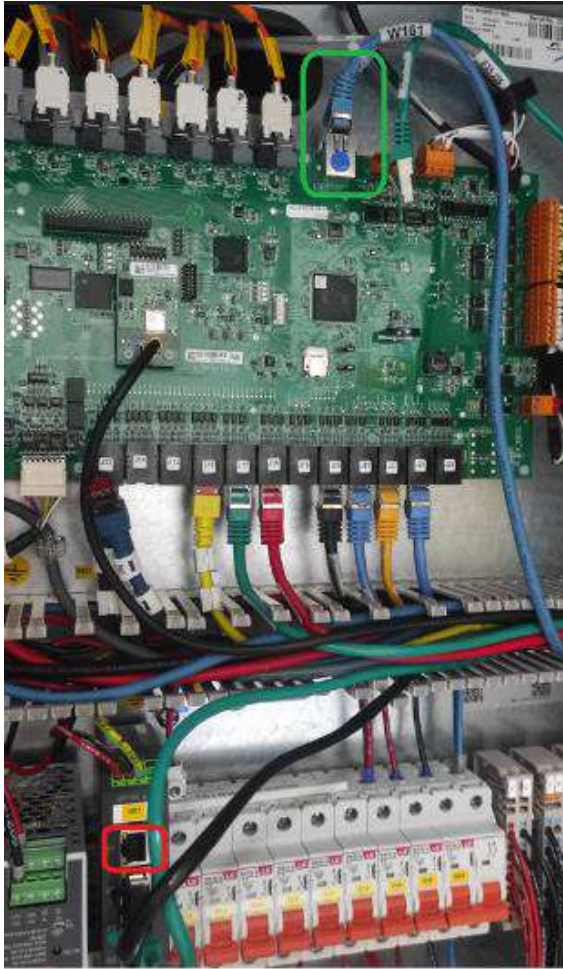
5. Check the RJ45 connectors of the client's network to ensure that they are properly crimped, not bent, and in good condition. **INVKG3_ST5A, INVMG3_ST5A.** If any defect is found, inform the client so they can fix it.

If the counter increases, but irregularly over time, there may be a hardware communication problem with the SCADA: connect a computer and regularly send different values from 0 to the Heartbeat register **INVG3_ST1B.** There could be two possible outcomes:

6. If it increases regularly, the problem is with the inverter's external communication network. Inform the client so they can fix it.
7. If communication continues to be erratic and heartbeats are lost, the problem is with the central control board (E35). Replace it. **INVKG3_ST5A, INVMG3_ST5A** (See document INVG3IOP88849). If the problem persists after replacing the board, contact Power Electronics.

If sporadic losses are detected, there may be a problem with the inverter's communication switch where the customer's cable is connected.

8. Disconnect from switch (U01) the customer cable and connect it directly to the E35 central control board. **INVKG3_ST5A, INVMG3_ST5A.** Observe if communication is restored. **INVG3K_ST3A, INVG3M_ST3A.** If the heartbeat counter increases regularly, the problem is the inverter's switch; replace the switch U01 and see if the problem is solved. **INVKG3_ST5A, INVMG3_ST5A.**



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If the PPC Heartbeat Register increases, and it does so regularly, it indicates that communication is OK.

9. Check the communication timeout setting (parameter [HB lost time], within group [G6.5: PPC]). This time must be much longer (4 times or more) than the increment time of the PPC Heartbeat register (if in doubt, use the default timeout value). If the heartbeat increments regularly but very slowly, contact the customer to find out if this time is acceptable (there is no fault in the SCADA software) and if so, change the timeout setting according to the customer's requirements. **INVKG3_ST1B**.

There may be interference (noise) in the communication cables disrupting the communication.

10. Check that all customer cables are shielded with the FTP or STP mark. **INVKG3_ST5A**, **INVMG3_ST5A**. If they are not shielded, discuss the problem with the customer to try to find a solution.

If the problem is still not detected after all the above checks:

11. The customer must check the continuity of his RJ45 communication cables with a RJ45 tester. If there are any problems, should solve them. **INVKG3_ST5A**, **INVMG3_ST5A**.

5.11. Fault code F62: DU Comms

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".		
Description	There is a communications problem between the central control board and the DU currents measurement board.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A		

Analysis procedure

- 1. If the error occurs after a software upgrade, there may be an incompatibility problem, or the upgrade process may have failed.
 - 1.1. Attempt to reload the software via Remote Upgrade. **INVG3_ST1B**.
 - 1.2. If it does not work, the board software must be uploaded manually using JTAG. **INVKG3_ST5A**, **INVMG3_ST5A** with external 24Vdc power supply.

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LOCATION OF MOTHER AND SLAVE BOARDS



MOTHER



SLAVE

- 1.3. Check that the DU board version is later than E21801204CA (to ensure compatibility with the software package). If not, contact Power Electronics. **INVKG3_ST5A, INVMG3_ST5A.**
- 1.4. If the fault still occurs, replace the board. **INVKG3_ST5A, INVMG3_ST5A.** If the fault persists, contact Power Electronics.

In any other case:

2. If the inverter is a PCSM/K of one busbar (only one battery), this device does not have a DU board. Therefore, it is a bad configuration. Check the Drive Select [SG10] configuration with PowerCommsV. If it is correct, contact Power Electronics. **INVG3_ST1B.**
3. Check if the DU board(s) have the LEDs on (which means the board has 24V power supply), as well as the power supplies in the central cabinet. **INVG3K_ST2 with an external alimentation of 120Vdc, INVG3M_ST3A** and only in this case, notice the use of PPE for electrical hazards is mandatory.



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LEDS SHOULD BE ON

4. If the LEDs on the board do not light up but those on the power supply do: check the wiring, the polarity and the magnetothermal protection. **INVKG3_ST5A, INVMG3_ST5A.**
- 4.1. Check continuity with 1500V multimeter between pin J15:1 on the E67 board (for solar, also on E66) and the negative pin of the 24Vdc output of the G01 power supply. If there is no continuity,



replace the wiring.

PIN J15:1 OF BOARD E67 WITH NEGATIVE PIN OF POWER SUPPLY OUTPUT G01



PIN J15:1 OF BOARD E66 WITH NEGATIVE PIN OF POWER SUPPLY OUTPUT G01

- 4.2. Check continuity with 1500V multimeter between the positive pin of the 24Vdc power supply G01 and pin1 of the circuit breaker Q17 and the thermal circuit breaker Q18. If there is no continuity, check the wiring and the comb that bridges the circuit breakers Q17 and Q18.



POSITIVE PIN OF THE POWER SUPPLY WITH PIN 1 OF THE CIRCUIT BREAKER Q18



POSITIVE PIN OF THE POWER SUPPLY WITH PIN 1 OF THE CIRCUIT BREAKER Q17

EN



LOCATION OF Q17 AND Q18

- 4.3. Check continuity with 1500V multimeter between pin J15:2 of card E67 and pin 2 of circuit breaker Q18 (for solar check also between board E66 and circuit breaker Q17). If there is no continuity, replace the wiring.



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PIN J15:2 OF E67 BOARD WITH PIN 2 Q18, ON PIN J15:2 USE BLACK MULTIMETER CLAMP



PIN J15:2 OF E66 BOARD WITH PIN 2 Q17, ON PIN J15:2 USE BLACK MULTIMETER CLAMP

- 4.4. If the wiring is not OK, correct it and if the wiring is OK, replace the board. Check the document [INVKG3_ST5A](#), [INVMG3_ST5A](#).

5. If the power supplies in the central cabinet do not have the LEDs lit, make the following: **INVG3K_ST2, IN VMG3_ST2.**
 - 5.1. Check continuity with 1500V multimeter between terminal block X100:1:3 and pin L(+) of power supply input G01. If there is no continuity, replace the wiring.



EN

- 5.2. Check continuity with 1500V multimeter between terminal block X100:1:4 and pin N(-) of the power supply input G01. If there is no continuity, replace the wiring.



If the continuity checks have gone well and the equipment has communication (the central control has been turned on before and it has been able to communicate), it means that the power supply is broken. Replace the G01 power supply.

6. Check that the fiber optic connectors are correctly mounted (Rx of the central control board (E35) with Tx of the DU board (E66) and vice versa). If not, correct them. **INVKG3_ST5A, INVMG3_ST5A.**



E67 FIBER CONNECTION



E66 FIBER CONNECTION

7. Check the condition of the fiber optic cables:
 - 7.1. Verify that the fibers are connected to the appropriate connector ("DU fibers") on the currents measurement board (E66). If they are not, change the fiber to the corresponding connector of the DU board. **INVKG3_ST5A, INVMG3_ST5A.**



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- 7.2. Check that the fibers are not excessively bent, nor damaged (there are no cuts or burnt zones). If they are not in perfect conditions, replace the fiber that is defective. **INVKG3_ST5A, INVMG3_ST5A.**



OK



NO OK

- 7.3. Check the fiber connector, the fibers should be aligned with the connector and well-polished. If they are not, replace the connector or the fiber (depending on availability). **INVKG3_ST5A, INVMG3_ST5A.**



GOOD CONDITION



BAD CONDITION

5.12. Fault code F157: Feedback AC breaker

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	Inconsistency between the AC circuit breaker switching order and the feedbacks.	
Applicable to	☐ HEM	☐ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

The circuit breaker trips a protection and opens without the inverter detecting it. It could be a problem in the circuit breaker relay, a problem with the internal current measurement in the circuit breaker or a HW problem of the equipment (short circuit, phase loss, etc.).

To rule out whether it is an AC circuit breaker problem or a feedback wiring problem.

1. Visually check if there is any trip alarm activated in the circuit breaker relay (any led of the circuit breaker relay is on). [INVKG3_ST6](#).



- 1.1. If any LED is on, it indicates that the circuit breaker has tripped autonomously in operation for some reason. Go to [INVKG3_ST2](#) status and search for the cause.
- 1.2. If no LEDs are on, it is probably a backfeed wiring problem; go to the next step.
2. If the circuit breaker relay has tripped due to overcurrent, check that there is no short circuit between boards in the inductance zone, measure continuity with 1500V multimeter between phases with the circuit breaker open. [INVKG3_ST2](#).



- 2.1. If there is no continuity between phases and there are no boards with a safety distance of less than 11 millimetres, check the relay and the circuit breaker toroids.

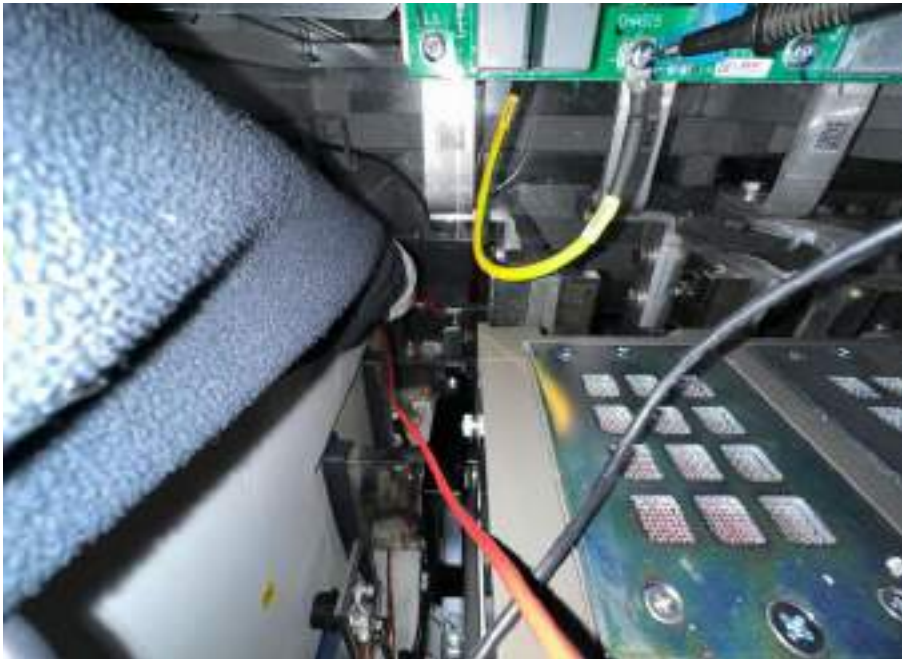


If a part is found to be damaged or in poor condition, replace the circuit breaker or relay (depending on availability). **INVKG3_ST2.**

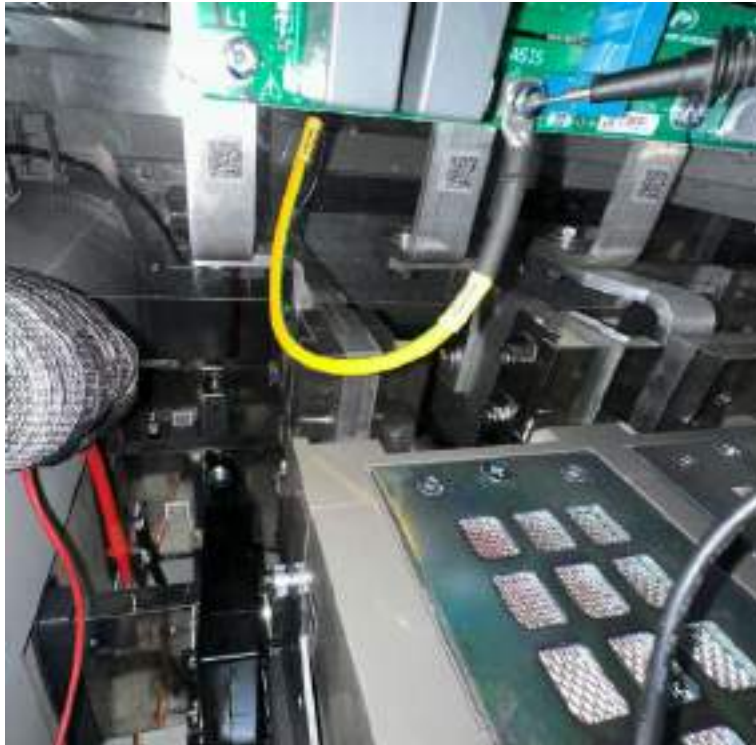
3. Check the integrity of the circuit breaker for the presence of hot spots or signs of degradation of the circuit breaker. **INVKG3_ST2.**

If the circuit breaker's Ig protection is tripping, it is because the circuit breaker is sensing a very high phase-to-phase unbalance that could be closing to ground.

4. Check that all PCBs (BBI1.03, BBI2.03, BBI3.03) are connected and that no phase is missing. **INVKG3_ST2.**
5. Check that there is no point in the power circuit connected to earth. Measure continuity with the 1500V multimeter between the AC terminals (BBI1.03, BBI2.03, BBI3.03) and ground. **INVKG3_ST2.**



EN



EN

- 5.1. If there is continuity (resistance < 1kOhm). Make a visual inspection of the inverter and search elements that can shunt a phase to ground. **INVKG3_ST2**
- 5.2. If no fault has been detected in the previous point, set the inverter to "READY" status and measure with the IMD (enable the manual maneuver of the IMD in group [G9.3: Maneuver] and check the resistance of each channel in parameter [Resistor] within group [SV9.2: Channels]). **INVKG3_ST1B.**
- 5.3. If everything is OK, this indicates that the problem is in the relay; replace it. **INVKG3_ST2.**

If the protection continues to trip, there is probably a problem with the internal current measurement in the circuit breaker. To rule out problems with the internal current measurement:

6. Measure the impedance of the internal toroidals of each phase of the circuit breaker, the correct impedance value must be 83Ω. To do this, contact Power Electronics. **INVKG3_ST2.**

Check that the circuit breaker opens and closes correctly.

7. Activate and deactivate the AC selector (in "DC Ready" state). Reset the circuit breaker and check that it opens and closes correctly (you should hear it close).



If it physically opens and closes correctly, but fails to do so, check the feedback status. If it does not open and close correctly, the circuit breaker coils are most likely not supplied with power or are damaged. Go to the next item. **INVKG3_ST1B.**

Check that the circuit breaker has 120V power supply: **INVKG3_ST6 with an external alimentation of 120V.**

8. Check that 120Vdc power is supplied to the main terminal block of the C filter cabinet. With a 1500V multimeter configured to measure DC voltage, measure between X22:2 (Positive 120V) and X22:1 (Negative 120V). To do so, before you need to remove the polycarbonate. **INVKG3_ST6.**



- 8.1. If it is supplied, check with 1500V multimeter step by step that voltage arrives to each of the following points:

- 8.1.1. Between positive pole of terminal X22:2 of the filter cabinet and the negative pole of the closing coil (XC100.1 A2), opening coil (XC100.1 C2), undervoltage coil (XC100.1 D2) and spring motor (XC100.1 U2). **INVKG3_ST2 with an external alimentation of 120 V.**

EN



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If there is no 120Vdc at all of these points, the negative wire leading from the main terminal block (X22:1) to the point where we do not measure 120Vdc voltage is broken, check and replace. **INVKG3_ST4**.



EN

- 8.1.2. Between the negative pole of the filter board terminal block (X22:1) and the positive of each of the activation relays of the inputs/outputs board (E17, J5:1, J6:1, J7:1), and the positive of the UVT undervoltage coil (terminal D1 of connector XC100 of the circuit breaker itself). **INVKG3_ST2 with an external alimentation of 120Vdc**. If at any of the points there are not 120Vdc, check the wiring and replace. **INVKG3_ST4**.





EN





EN

8.1.3. Check continuity with 1500V multimeter between the relay outputs of the inputs/outputs board and the connection point to the AC circuit breaker. If there is no continuity at any of the points, check and replace the wiring. Also check if the intermediate connector has any changed pins. [INVKG3_ST4](#).



Board E17 where continuity is checked



First point to check continuity of the E17 board



U1 point to check continuity with the first point of the E17 board



Second point to check continuity of the E17 board

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C1 point to check continuity with the second point of the E17 board



Third point to check continuity of the E17 board



A1 point to check continuity with the third point of the E17 board

- 8.1.4. If all the checks have been good, verify with a 1500V multimeter that the motor resistance (U1 and U2) is approximately 30Ω (with the spring unloaded), and replace the coil that is not acting according to the diagnosis made at the beginning. Check document [INVKG3_ST4](#).



Backfeed wiring problems: the corresponding backfeed is not activated, or there is an undefined status (both backfeeds are activated or neither is activated).

9. Using PowerCommsV check the status of the backfeed for the AC circuit breaker: **INVKG3_ST1B**.
 - 9.1. If both backfeeds are activated or deactivated at the same time, it will be necessary to visualize the actual status of the circuit breaker to know which backfeed is failing. **INVKG3_ST6**.
 - 9.2. If the backfeeds are correct at this time, it will be necessary to identify at what time the fault has been triggered to know the expected status at that time to determine if it is a circuit breaker status problem. Download in PowercommsV the SD and attempt to reproduce the fault to identify the problem. **INVKG3_ST1B**.

Once the faulty backfeed has been found, proceed as follows:

- 9.3. If the backfeed appears open when it should be closed, bypass the backfeed on the inputs/outputs board and check if it changes on the central control board. **INVKG3_ST4**.
- 9.4. If it does not change, the problem is the wiring to the central control or the digital inputs/outputs board.
 - 9.4.1. Check that the RJ45 cable is properly connected on both boards. **INVKG3_ST2**.



- 9.4.2. Check continuity with RJ45 tester of the RJ45 cable to the central control board. If not OK, replace. **INVKG3_ST2**.



EN



9.4.3. If the RJ45 cable is OK, replace the inputs/outputs board. **INVKG3_ST2**. Re-energize and check if the backfeed is still open. **INVKG3_ST4**. If this does not work, replace the central control board. **INVKG3_ST2**.

9.4.4. If none of these points work, contact Power Electronics.

5.13. Fault code F158: Low pressure SF6

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The pressure switch indicating the SF6 gas pressure in the high-voltage switchgear has been activated.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

This fault occurs either because the SF6 pressure in the high-voltage switchgear has been lost or because the wiring is in poor condition.

1. Check the pressure status of the SF6, which is located in the high voltage switchgear cabinet. **INVG3_ST1B with electrical hazard PPE**. It should have the needle in the green zone of the indicator, if it is not in the correct values, it will be necessary to consult Power Electronics.



2. Check that the connectors of the digital inputs and outputs board are correctly connected:
 - 2.1. HEM / PCSM: E82 inputs and outputs board. **INVG3M_ST3A**
 - 2.2. HEMK / PCSK: E101 inputs and outputs board, located in the LV panel or cabinet of the MV Skid.
In the case of TWIN SKID, the inputs and outputs boards are: **SKIDG3_ST3A**.
 - 2.2.1. E101 board (corresponding to inverter n°1)
 - 2.2.2. E105 board (corresponding to inverter n°2).
3. Disconnect the SF6 connector of the digital inputs and outputs board:
 - 3.1. HEM / PCSM: J3-1 connector of the E82 inputs and outputs board. **INVG3M_ST2**
 - 3.2. HEMK / PCSK: J3-2 connector of the E101 inputs and outputs board. In the case of TWIN SKID, it corresponds to connector J3-1 on the E105 inputs and outputs board. **SKIDG3_ST2**.

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EN

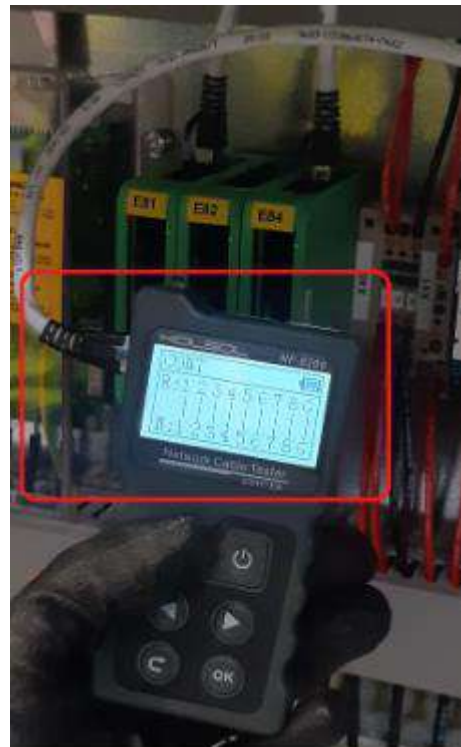
4. Check continuity with 1500V multimeter between the SF6 connector, which has been disconnected before, and the 24Vdc connector (J2-1 of the E02, E01, E05 inputs and outputs board, as appropriate). **INVG3M_ST2, SKIDG3_ST2.**



5. If there is no continuity, it is a wiring fault from the digital inputs and outputs board until the E35 central control board. To check this wiring, measure continuity with a network cable RJ45 tester between J1 connector of the inputs and outputs board (E02, E01, E05, as appropriate) and J14 connector of the E35 central control board. **INVG3M_ST2, SKIDG3_ST2.**



J14 CONNECTOR

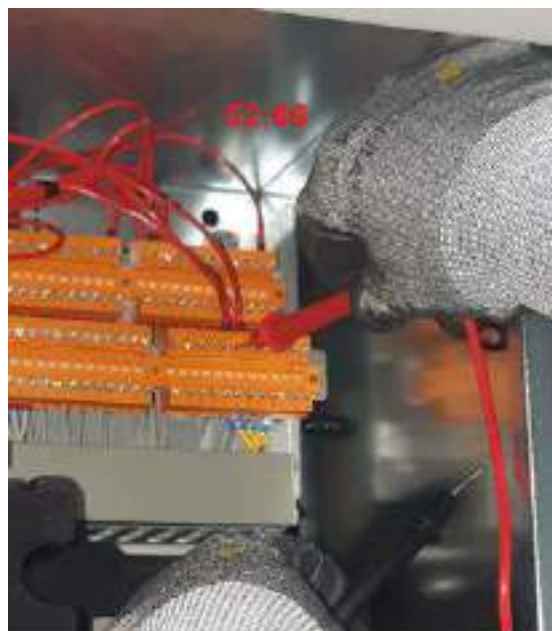


J1 CONNECTOR

- 5.1. If there is no continuity, replace the RJ45 cable.
- 5.2. If there is continuity and the switchgear is OK (SF6 measure equipment in the green zone), check the wiring. Measure continuity with 1500V multimeter from the J3-1 connector of the digital inputs and outputs board (E82, E101, E105, as appropriate) to the 52:66 terminal of the Q02 switchgear.
INVG3M_ST2, SKIDG3_ST2.



J3-1 CONNECTOR



52:66 TERMINAL OF THE Q02 SWITCHGEAR

NOTE: Due to the distance between connectors inside the equipment, continuity is measured with 1500V multimeter through the equipment chassis.

5.14. Fault code F150: MV cell wrong state

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	Inconsistency between the status of the MV switchgear and the inverter command.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

The switchgear relay has tripped autonomously for some reason. Check the status of the relay on the MV switchgear display (Q02).

- 1. Open the switchgear door and check the status of the relay. If a fault has occurred during commissioning, check that the relay parameterisation matches the nominal data of the inverter. For this, refer to your project specific information. [INVKG3_ST3A](#), [INVKG3_ST3A](#), [SKIDG3_ST3A](#).



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2. Check the fault of the relay. **INVKG3_ST3A, INVMG3_ST3A, SKIDG3_ST3A.**
 - 2.1. For the model Ekor RPA serie 021 (Ormazabal manufacturer), push button ↓ in the main menu until reach the historial menu.



- 2.2. Push the → button to see the 1 historic (H1). Pushing the ↑ and ↓ buttons it is possible to navigate through the last 10 faults registered.



The fault codes are:

- 50: Instantaneous phase overcurrent
- 51: Inverse time phase overcurrent
- 50N: Instantaneous neutral overcurrent
- 51N: Inverse time neutral overcurrent
- EXT: Fallo externo

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3. In the case of HEM/PCSM inverters check if the relay indicator is "External fault" ("EXT.TRIP" led on). If so, check that it has not tripped for any of the following causes:
 - 3.1. Emergency button (S02). Carry out a technical check of the contact block and all associated wiring **INVMG3_ST5A, INVKG3_ST2**. To do this, check that there is no continuity with a 1500V multimeter between pins 63 and 64 of the emergency button. Note that the emergency button must not be pulsed. If continuity is found between the pins, replace the contact block of the emergency button sensor (NO contact). **INVMG3_ST5A, INVKG3_ST2**.



NO contact of emergency stop button S02

- 3.2. Check that relay KA08 is activated (relay pilot light on). If the pilot light does not come on, check with 1500V multimeter that 24Vdc voltage is present between terminals A1 and A2 of relay KA08. If 24Vdc voltage is present and not activated, replace relay KA08. **INVMG3_ST3A, INVKG3_ST3A, SKIDG3_ST3A**



3.3. If there is no 24Vdc voltage, carry out the following checks with 1500V multimeter:

- Check for 24Vdc voltage between X40:1 and X40:3 **INVMG3_ST3A**. If there is, replace diode X41. **INVMG3_ST5A, INVKG3_ST2**. **Note:** Due to the distance between the connectors within the equipment, it will be needed to measure with 1500V multimeter the continuity through the equipment chassis.

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- Check continuity with 1500V multimeter between connector X40:1 and Q19:2. **INVMG3_ST2, INVKG3_ST2, SKIDG3_ST2**



- Check continuity with 1500V multimeter between Q19:2 and the positive terminal of the 24Vdc power supply (G02). **INVMG3_ST5A, INVKG3_ST2**.



- Check continuity with 1500V multimeter between connectors X40:3 with X11:3. **INVMG3_ST5A, INVKG3_ST2**



- Check continuity with 1500V multimeter between connector X11:3 and the negative terminal of the 24Vdc power supply (G02). **INVMG3_ST5A, INVKG3_ST2.**



- If any of the above does not give continuity, replace the corresponding wiring or connector. **INVMG3_ST5A, INVKG3_ST2.**
- Test if in the documentation of the project it has the option of shot of extern cell. If this was the case, speak with the client in order to know why it happened. **INVMG3_ST3A.**



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4. In case of the inverters **HEMK/PCSK with MV Skid motorized**, check if the indicator of the relay is “External failure” (led of “EXT.TRIP” on). If it appears, it is because the client has triggered a shot of external cell. Check with the client in order to find out the reason. **INVKG3_ST3A, SKIDG3_ST3A**

4.1. If another failure in the display cell appears, follow the next steps:

4.1.1. Overcurrent (P TRIP) (led if “P TRIP” on). In this case:

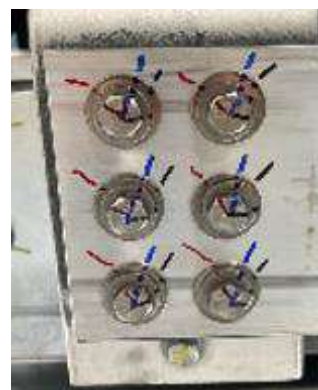
- Check in the display of the relay to which current has the failure happened. If the current is similar to the nominal current (check the documentation of the project), check the relay parameterization. **INVMG3_ST3A, INVKG3_ST3A, SKIDG3_ST3A.**
- In case of a very high current revise that has not tripped immediately after pre-magnetizing the MV transformer and closing the cubicle (in case of HEMK/PCSK applies when motorized MV SKID option is available). If so, check the actions described in fault F35. **INVMG3_ST3A, INVKG3_ST3A.**
- Check on the relay display which phase(s) has given the overcurrent, and check if this overcurrent is visible on the SD card (Log_1). **INVG3_ST1.** If the SD card shows anomalous behavior, carry out a technical check of the equipment. Also check the toroids of the MV cell. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**

4.1.2. Ground fault (N TRIP) (led of “N TRIP” on). In this case:

- Check in the display of the relay of any phase has read 0A at time of failure. **INVKG3_ST3A, INVMG3_ST3A, SKIDG3_ST3A.** If it is like that, check the toroids of the MV switchgear and its wiring.
- If this is not the case, check physically the equipment, zone of the MV mainly. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.** If it always jumps at the same power, check the SD card (Log_1), **INVG3_ST1**, the measures of the current of the relay just before the failure in order to detect posibles las medidas de corriente del relé justo antes del fallo para detectar posibles failures in the Reading of the toroids of the MV cell. **INVMG3_ST3A, SKIDG3_ST3A.**

4.2. Make a technical inspection of the equipment in order to verify that there is not any short-circuit or derivation and that all the phases are correctly connected. **INVKG3_ST2, INVMG3_ST2.** For it:

4.2.1. Check the torque marks on the connections of the MV transformer, the inductance and the capacitor filter. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**



Note: if any of the points are not tightened properly or do not have the torque mark properly, inform Power Electronics. Check torque and mark when tightened properly. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

- 4.2.2. Verify the ground connections of the medium voltage cables (W408, W409, W410 and the equipment input cables). INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.



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Note: If any of the medium voltage cable points are not grounded, inform Power Electronics and connect correctly. If the problem is in the external connections of the equipment, inform the customer to solve the incidence. INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

- 4.2.3. Check the MV transformer (T01) for cracks or defects. INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

Note: if damage to the MV transformer is found, inform Power Electronics and do not start the equipment.

- 4.2.4. Inspect the main inductance (L01) for hot spots or points where arc flash may have occurred. INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

Note: if any points in bad condition are found, inform Power Electronics and do not start the equipment.

- 4.2.5. Perform a visual inspection of the capacitor filter (C25 to C48). Verify that all capacitors are in good condition (not swollen, defective, or leaking acid). INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

Note: If capacitors in bad condition are found, replace the capacitor filter or the damaged capacitor according to availability. See document INVKG3/OP85367. If the fault is repetitive, consult Power Electronics. INVKG3_ST2, INVMG3_ST2.

- 4.2.6. Check that the contactors are not fused (KM01 to KM04). Manually check that they open and close correctly. INVKG3_ST2, INVMG3_ST2. To do this, manually operate the contactor by pressing the orange button.



If there is continuity in the contactor coil, the contactor has fused contacts, it is necessary to replace the contactor and inform Power Electronics. **INVG3K_ST2, INVMG3_ST2.**

- 4.2.7. Check that the notch is not earthed (L07, L08 and L09). Check that there is no continuity with 1500V multimeter between the screws that hold the notch to the core and that the notch plates maintain their insulation.



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If any of the measured points give continuity, replace the notch filter and inform Power Electronics. See document *INVG3IOP85367*.

- 4.2.8. Check that the RFI board (E16) is in good condition and grounded. **INVG3K_ST2, INVMG3_ST2.**



If the board is not in good condition or has broken turrets, replace it. See document *INVG3IOP88849*.

- 4.2.9. Check the capacitance of the filter capacitors. Measure with the 1500V multimeter directly between terminals L1-L2, L2-L3 y L3-L1 of the boards E20, E21, E22 y E23 that there is 180uF (acceptable range 170-190uF). It is not necessary to disconnect anything. Set the multimeter to measure capacitance. There is no polarity in this measurement. **INVG3K_ST2, INVMG3_ST2.**



EJEMPLO MEDIDA ENTRE FASE L1-L2

If any of the capacitor bank measurements are not within the acceptable measurement range, inform Power Electronics.

If the measurement is outside the range 203 to 166uF, replace the capacitor filter. Refer to document *INV3/OP85367*. **INV3K_ST2, INVM3_ST2.**

4.2.10. Check that the inverter module is in good condition. See document *INV3/OP85369*.

- 4.3. If all the above checks have been successful, reassemble. If the fault reappears, contact Power Electronics.
5. Check that the SF6 gas level in the MV switchgear is within acceptable levels. If not, contact Power Electronics or the switchgear manufacturer. **INVM3_ST3A, SKIDG3_ST3A.**

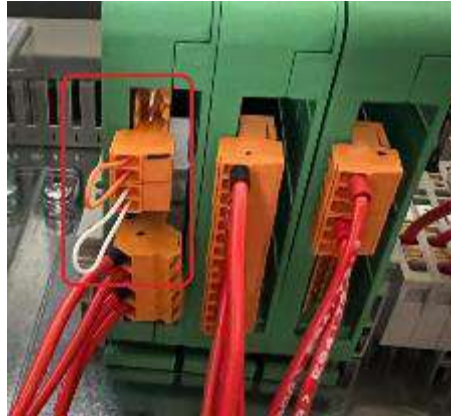


Fault in the switchgear feedbacks. The fault shall trip when:

- The state of the circuit breaker is undefined: both feedbacks activated or deactivated at the same time.
- The feedback does not match the status being sent by the equipment:
 - Feedback open, activated with the close command.
 - Feedback closed, activated with opening command.

In this case, carry out the following checks:

6. Put a bridge the feedbacks on the input/output board E81. Disconnect connector J4. Bridge pins J4:3 and J4:4 (circuit breaker closed feedback) and pins J4:5 and J4:6 (circuit breaker open feedback).
INVKG3_ST2, INVKG3_ST2.



Check in PowerCommsV to see that the circuit breaker closed and circuit breaker open signals are activated ([DSP Status RJ13] in group [SV1.5-Digital Inputs and Outputs Status]).

If the status of the feedback is not reflected in PowerCommsV, skip to step 6.

If the feedbacks are visible on the central control board, skip to the next step.

7. Check continuity of the switchgear feedbacks with the input/output board:

-HEM / PCSM: E81 board.

-HEMK / PCSK: E100 and E104 boards.

To do this, carry out the following checks: **INVKG3_ST2, INVKG3_ST2, SKIDG3_ST2.**

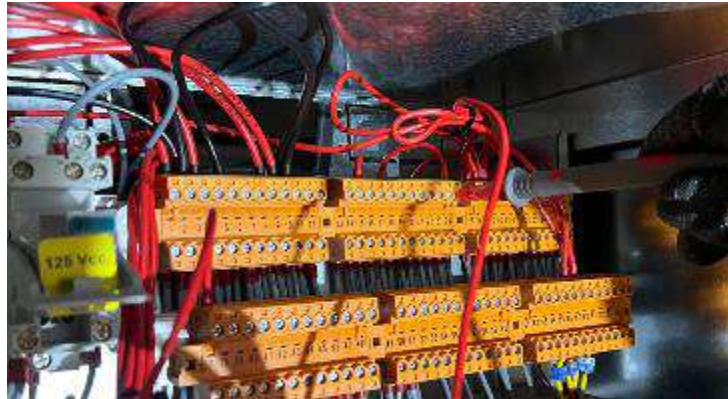
- 7.1. Check the closed switchgear feedback, verifying continuity with 1500V multimeter between the following points:

NOTA: Due to the distance between connectors inside the equipment, it may be necessary to measure continuity with 1500V multimeter through the equipment chassis.

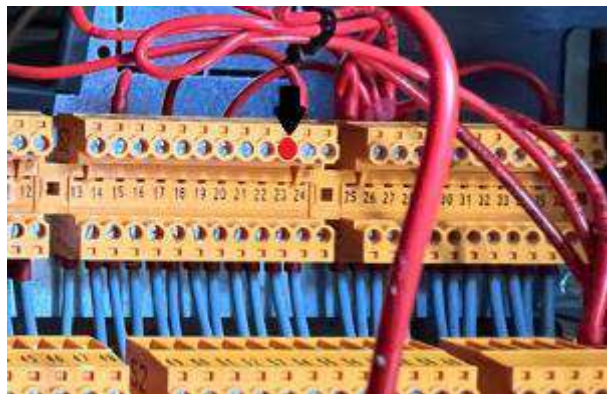
-HEM / PCSM: between connector J4:3 of the E81 board and 52:25 of the Q02 switchgear.
INVKG3_ST2.



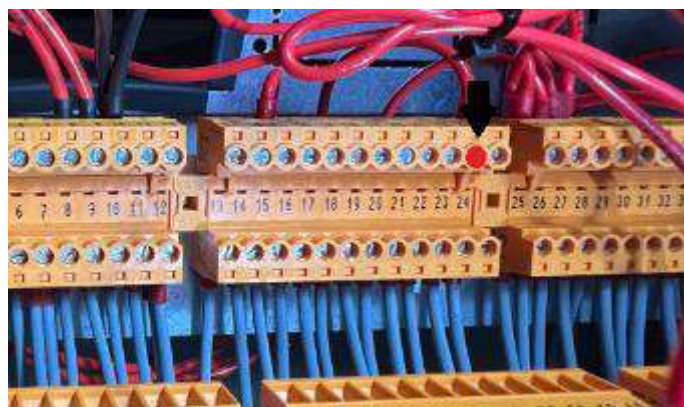
-HEM / PCSM: between connector J4:4 of the E81 board and 52:24 of the Q02 switchgear. **INVMG3_ST2.**



-HEMK / PCSK with MV SKID: between connector J4:3 of the E100 board and 52:22 of the Q02 switchgear. **INVKG3_ST2, SKIDG3_ST2.**



- HEMK / PCSK with MV SKID: between connector J4:4 of the E100 board and 52:23 of the Q02 switchgear. **INVKG3_ST2, SKIDG3_ST2.**

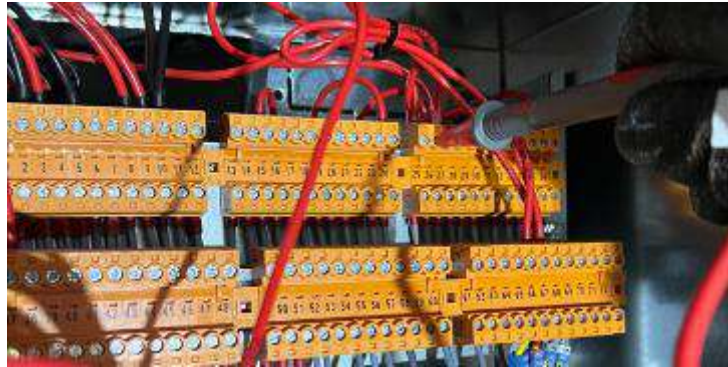


- If there is continuity in all the above points, consult Power Electronics about the possibility of changing the medium voltage switchgear. See document *INVG3IOP87922*. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

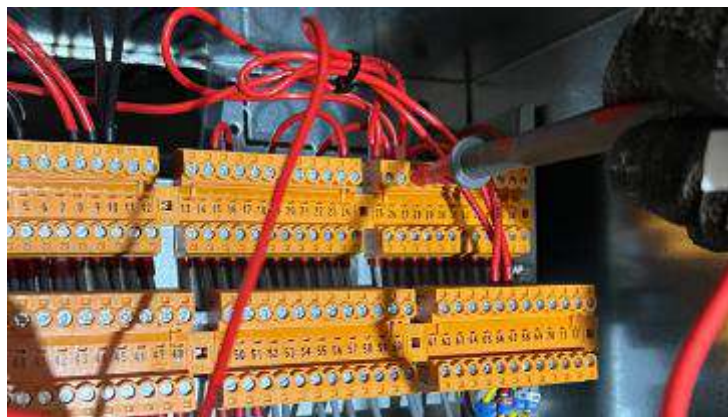
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- 7.2. Check the open switchgear feedback by measuring continuity with 1500V multimeter between the following points:

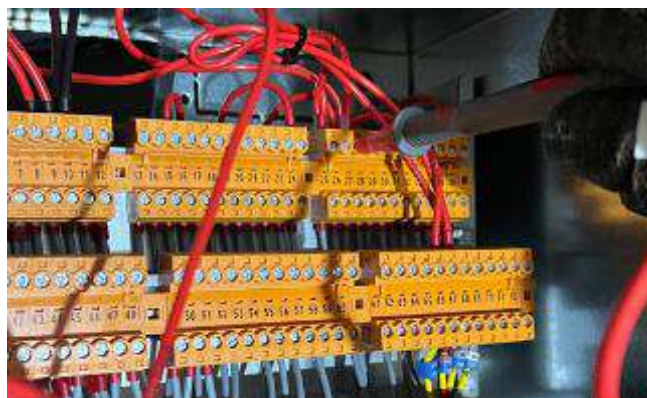
-HEM / PCSM: between connector J4:6 of the E81 board and 52:26 of the Q02 switchgear.
INVMG3_ST2.



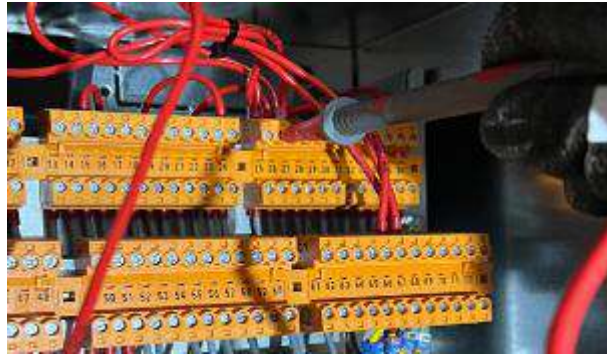
-HEM / PCSM: between connector J4:5 of the E81 board and 52:27 of the Q02 switchgear.
INVMG3_ST2.



- HEMK / PCSK with MV SKID: between connector J4:6 of the E100 board and 52:27 of the Q02 switchgear. **INVKG3_ST2, SKIDG3_ST2.**



- HEMK / PCSK con MV SKID: between connector J4:5 of the E100 board and 52:26 of the Q02 switchgear. **INVKG3_ST2, SKIDG3_ST2.**



- If there is continuity in all the above points, consult Power Electronics about the possibility of changing the medium voltage switchgear. See document *INVG3IOP87922*. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

8. Check the continuity of the RJ45 cable connecting the input and output board:

-HEM / PCSM: E81 board.

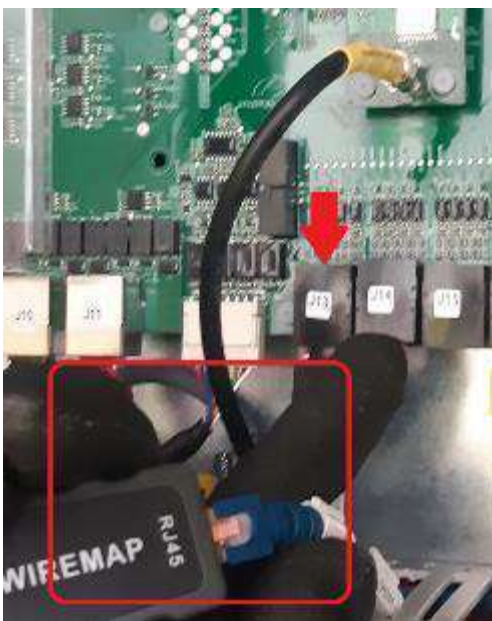
-HEMK / PCSK: E100 (SKID) and E104 board

and connector J13 of the central control board E35 with a RJ45 tester. **INVKG3_ST2, INVMG3_ST2.**

If the cable is not in good condition, repeat the test by exchanging it with the intermediate Harting connector. (XC702.1B).

If any of the cable sections have continuity faults, replace it.

If the cable has no continuity fault, replace the input and output board according to the equipment. See document *INVG3IOP88849*. **INVKG3_ST2, INVMG3_ST2**



CONNECTOR J13



CONNECTOR J1

9. If the fault continues to trip after replacing the input/output board, replace the central control board (E35).
See document *INVG3/OP88849* [INVKG3_ST2](#), [INVMG3_ST2](#).

Failure in the switchgear drives. If the feedbacks are correct, it is possible that the drive is damaged.

10. Check with 1500V multimeter that 120Vdcc voltage reaches the medium voltage cabinet:
[INVKG3_ST3A](#), [INVKG3_ST3A](#)

-HEM / PCSM: between X31:1 y X31:4

-HEMK / PCSK: between X112:1 y X112:4.

11. If that not happens check continuity with 1500V multimeter between the following points: [INVKG3_ST2](#), [INVMG3_ST2](#), [SKIDG3_ST2](#).

Note: Due to the distance between connectors within the cabinet, it will be needed to measure continuity with 1500V multimeter through the equipment chassis.

-between connectors X31:4 (negative) y X100:2:1.

In case of the MV SKID, connector X112 instead of X31.



-between connectors X31:1 (positive) and X101:2:2.

In case of the MV SKID, connector X112 instead of X31.



-between connector X101:2:2 and S04:6.



-between S04:5 and S02:42.

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-between S02:42 and KA01:21.

**EN**

-between KA01:21 and KA01:12.



-between KA01:12 and X100:3:2.



-if there is no continuity in the points above, change the wiring or damaged connector.
INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

12. Check the state of the opening and closing coil.

NOTE: It is advisable to do this checking without high voltage, as this could accidentally close the switchgear and damage the equipment. In order to do this checking:

Dismantle the opening and closing coils, and power them externally with 120Vdccc in a safe way (powering the coils with an external 120Vdccc source and previously enclosing the cil in an isolated enclosure with overcurrent protections. If the coils do not operate, replace the damaged coils.
INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.

13. Check that the drive wiring runs correctly to the high voltage switchgear. To do this, check continuity with 1500V multimeter between the following parts: **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

NOTE: Due to the distance between connectors withing the cabinet, it will be needed to measure continuity with 1500V multimeter through the equipment chassis.

13.1. Switchgear closing drive.

-check continuity with 1500V multimeter between the connector J6:1 of the card E81 (E100 for the MV SKID) and terminal block X31:1 (X112:1 for MV SKID).



CONNECTOR J6:1



TERMINAL BLOCK X31:1

-check the continuity with 1500V multimeter between the connector J6:2 of the card E81 (E100 for the MV SKID) and the pin 52:3 of the high voltage switchgear (Q02).



CONNECTOR J6:2



PIN 52:3

EN

-check the continuity with 1500V multimeter between the connector X31:4 (X112 for SKID) and the pin 52:5 of the high voltage switchgear (Q02).



CONNECTOR X31:4



PIN 52:5

-if there is no continuity at any of the points above, change the wiring. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

13.2. Switchgear opening drive. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

-check the continuity with 1500V multimeter between the connector J5:1 of the card E81 (E100 for MV SKID) and the terminal block X31:1 (X112:1 for MV SKID).



CONNECTOR J5:1

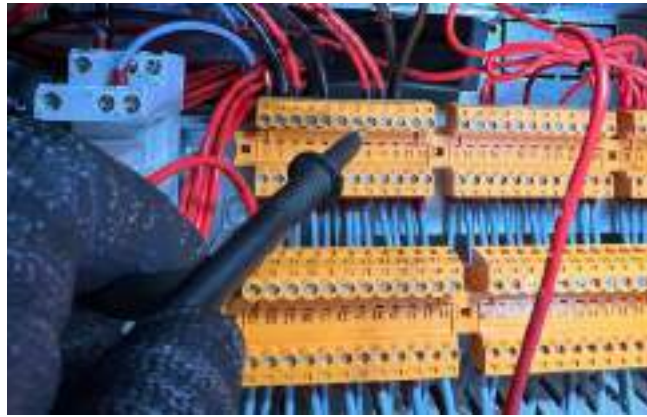


TERMINAL BLOCK X31:1

-check the continuity with 1500V multimeter between the connector J5:2 of the card E81 (E100 for MV SKID) and the pin 52:8 of the high voltage switchgear(Q02).



CONNECTOR J5:2

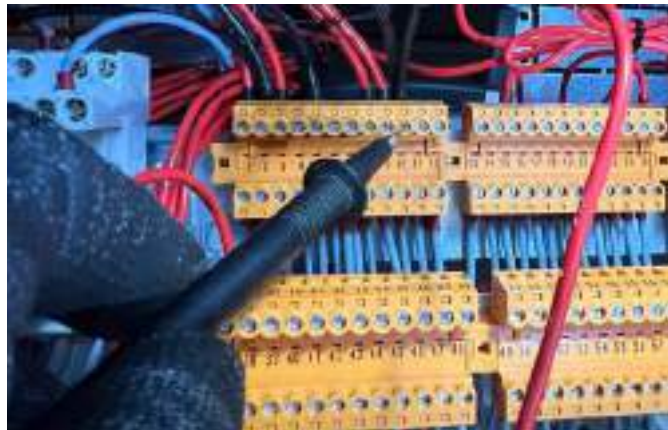


PIN 52:8

-check the continuity with 1500V multimeter between the connector X31:4 (X112 for MV SKID) and the pin 52:10 of the high voltage switchgear (Q02).



CONNECTOR X31:4



PIN 52:10

-if there is no continuity at any of the points above, change the wiring. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

14. Check the continuity of the wire RJ45 that joins the central control board:

-HEM / PCSM: card E81

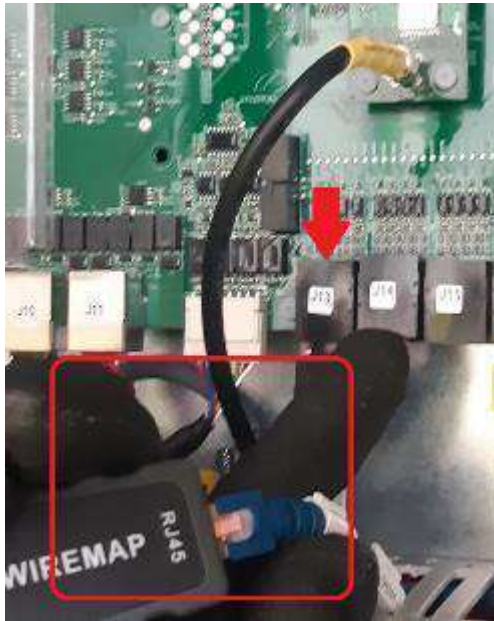
-HEMK / PCSK: card E100 (SKID) AND E104

and the connector J13 of the central control card E35 with a RJ45 tester. **INVKG3_ST2, INVMG3_ST2.**

If the wire is in bad conditions, repeat the test exchanging the intermediate harthing connector (XC702.1B).

If one of the wiring sections has a continuity failure, change it.

If the wire has not a continuity failure, change the central control board according the equipment. Check the following file *INVKG3/OP88849*. **INVKG3_ST2, INVMG3_ST2**



CONNECTOR J13



CONNECTOR J1

The switchgear has been manually operated.

15. If the switchgear relay is correct and there is no problems with the feedbacks and drives, talk to the plant manager and ask if the switchgear has been manually operated for some reason. If this is the case, indicate that this is not the correct way to operate the inverter, and check that *ECO7554* has been implemented, which blocks manual operation of the switchgear.



5.15. Fault code F159: Overtemperature DU

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The temperature read by the DU board has overcome the threshold configured internally (group [G5.4: Temperature]).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	In HEM and HEMK, the warning related to DU temperature derating may have been triggered (W30:Overtemperature DU).	

Analysis procedure

This fault indicates that there is a cooling problem in the DU or there is a component that is generating more heat than it should.

Remotely download the inverter data from the SD and check at the time of the fault if the DU temperature was at the value corresponding to the fault. [INVKG3_ST1A](#).

1. If it was not within the value corresponding to the fault, replace the DU board (E67 for PCSM/PCSK, E66 and E67 for HEM/HEMK). Refer to document [INVKG3/OP88849](#). [INVKG3_ST5A](#), [INVMG3_ST5A](#). If the fault persists, consult Power Electronics.
2. If the temperature was equal to or higher than the fault temperature, check the temperature protection set in group [G5.4: Temperature]. If it is not equal to the default value consult with the customer, the reason for the change and proceed to return to the default value if applicable. [INVKG3_ST1A](#).
3. If the temperature read by the board is higher than 90°C, replace the DU current measuring board (E67 for PCSM/PCSK and E66 or E67 for HEM/HEMK). Refer to document [INVKG3/OP88849](#). [INVKG3_ST5A](#), [INVMG3_ST5A](#).

If the temperature read was at its default value or higher, most probably there is a problem in the ventilation circuit of the inverter.

4. Check through PowerCommsV that the DU and filter fans are operating correctly. Configure in fixed reference mode at 100% speed reference (parameter [Mode], in group [G11.3: DU config]) and check that they rotate at the configured revolutions. [INVKG3_ST1A](#). If they do not, check the fans visualization parameters [SV11], check the fan fault / warning and follow the manufacture fan manual. [INVKG3_ST2](#), [INVMG3_ST5A](#).
5. Check using PowerCommsV that there is no fan warning during the fault. [INVKG3_ST1A](#). If there is a warning, check the fans for damage, check that they do not rub against any part of the equipment and that they are securely anchored. [INVKG3_ST2](#), [INVMG3_ST5A](#).
6. Check the inverter air inlet grids. If they are saturated or clogged, clean them. [INVKG3_ST2](#), [INVMG3_ST5A](#).



Bad conditions



Good conditions

EN

7. Check the inlet filters of the air exchanger. If saturated or dirty, replace them. [INVKG3_ST2](#), [INVMG3_ST5A](#).



Bad conditions



Good conditions

8. Check the DU for hot spots (burned or discoloured plates, burned choke coils, burned circuit breakers, etc.). If present, check connections of the affected plates or replace damaged components [INVKG3_ST2](#), [INVMG3_ST5A](#).
9. Check fans, check that the fan blades are not broken or worn, and that the fan, power and communication cables do not rub against the moving parts of the fan. If any defect is found, replace them. Refer to document [INVG3IOP85370](#). [INVKG3_ST2](#), [INVMG3_ST5A](#).

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5.16. Fault code F164: Excessive Manoeuvres

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The DSP has detected 6 soft charge, self-diagnosis or DC breaker maneuvers in 50 minutes, or 15 C filter contactors maneuvers in 30 minutes	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

Remotely check the SD. See DSP and module statuses to check which of the manoeuvres has triggered the F164. **INV3_ST1A**.

1. If the inverter has entered AC DIAG or SOFTCHARGE DC status: check in the SD the status of the selectors to verify that the customer has not continuously activated and deactivated their status, if he is constantly activating and deactivating the selectors, indicate them that this is an incorrect way of maneuvering the inverter. **INV3_ST1A**.
2. Check if the inverter is stopping and reconnecting module or busbar due to voltage conditions within [G2: Running conditions]. In this case, there may be a wrong setting of the stop and start parameters; or, in MPCSK/M inverters, there may be a problem in the battery rack associated to the inverter that is causing this unbalance between busbars. **INV3_ST1A**.
3. Check if modules are being enabled or disabled in [SG1.5: Modules configuration] or [G2.8: Busbars Configuration] (the latter, for MPCSK/M). If modules are being enabled and disabled, indicate to the customer that this is an incorrect way to operate the inverter. **INV3_ST1A**.
4. Check that there are no consecutive starts and stops of the inverter which only involve C filter maneuvering. In this case, check if they are being commanded by the client (inform that it is an incorrect way of maneuvering the inverter) or by the software due to a wrong adjustment of the start and stop parameters in group [G2: Running conditions] (if this is the case, adjust the parameters according to the plant). **INV3_ST1A**

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5.17. Fault code F227: Fault Unknown

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The DSP or the FPGA have reported a fault unknown to the microcontroller.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

One of the inverter devices has triggered a fault that is not recognized by the rest of devices.

1. Check through PowerCommsV that the software versions of all devices are correct. **INV3_ST1A.**
2. Load the software again to make sure the package is OK. **INV3_ST1A.**
3. If the fault persists after these actions, download logs and contact Power Electronics. **INV3_ST1A.**

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5.18. Fault code F49: IMD Selfdiagnosis Error

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The IMD's periodic hardware self-diagnosis has reported an error	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

The IMD board (E70) is in bad condition.

- 1. Attempt to reset the fault and run again the self-diagnosis. INVKG3_ST1B.
- 2. If the fault persists, replace the board. Refer to document INVKG3/OP88849. INVKG3_ST5A, INVKG3_ST5A.



- 3. Check for water marks or signs of condensation on any part of the inverter.
 - 3.1. If water marks are present, find the source of the water ingress and seal it. INVKG3_ST5A, INVKG3_ST5A.
 - 3.2. If there are signs of condensation, contact Power Electronics. INVKG3_ST5A, INVKG3_ST5A.



If the fault appears sometime after replacing the board, it is possible that the board is being damaged by an improper operation. Contact Power Electronics.

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5.19. Fault code F3: SoftCharge

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".		
Description	The device has exceeded the soft charge device timeout while in soft charge. This timeout depends on the configured product (300 seconds for solar and battery inverters; 400 seconds for Bus Plus). Timeout of one of the soft charge processes.		
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK	
Associated faults or warnings	W85: Time out soft charge W94: DC SC Module W130: Softcharge Feedback W133: Softcharge board		

Analysis procedure

This fault will not be triggered if any of the modules has been able to make the soft charge (closing the DC circuit breaker). This fault may also appear during the diagnosis process if the module takes longer than expected to make the soft charge.

Product misconfiguration.

1. Check that the device is configured correctly so that the corresponding timeout (grupo [SG10: Drive select]) is applied. If it is not configured properly, change the configuration. **INVG3_ST1B.**

The soft charge process has taken longer than expected.

2. Check the SD at Log2 the Vdc voltage of each module or repeat the soft charge process to reproduce the failure and check which module(s) are taking longer than expected to make the soft charge (visual check on PowerCommsV main screen). See which module is failing and the status it reports. **INVG3_ST1A.**
3. Check the modules that show problems and check the soft charge circuit of the modules.

The checks are carried out according to the component references of module 1. If the fault is in a different module, the references vary as shown below:

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
Softcharge board	E60	E61	E62	E63
RJ45 cables	W013	W014	W015	W016
Contactors	KM10-KM11	KM12-KM13	KM14-KM14	KM16-KM17
Terminal blocks	X25	X26	X27	X28
Sofcharger resistors	R30-R31	R32-R33	R34-R35	R36-R37
Sofcharger fuses	F101	F102	F103	F104
Sofcharger positive plates	BBIP.07	BBIP.05	BBIP.03	BBIP.01

Note that the checks to be performed on the softcharge circuit vary depending on the solar inverter or battery inverter.

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3.1. In solar inverters:

3.1.1. Check the wiring between the soft charge boards (from E60 to E63 depending on the module) and module PCB (E02) **INVKG3_ST2, INVMG3_ST5A**.

- Check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W013), first disconnect both RJ45 connectors (E60 board, J2 port; E02 board, J11 port) and connect them to pin tester. All number must match. **INVKG3_ST2, INVMG3_ST5A**



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If there is not continuity, replace de RJ45 cable. **INVKG3_ST2, INVMG3_ST5A**.

- Check the power supply of softcharger board (E60). **INVKG3_ST2, INVMG3_ST5A**.
- Check continuity between J1 connector on the softcharger board (E60) to X100 terminal block. Use a 1500V multimeter at the following points:



Positive pole
From J1 pin 1 (E60) to X125:4:2



Negative pole
From J1 pin 3 (E60) to X125:4:4

From X125:4 (as same point of previous step) to X100:3



Positive pole
From X125:4:2 to X100:3:2



Negative pole
From X125:4:4 to X100:3:4

- Check continuity with 1500V multimeter in bus+ and bus- of the softcharge board (E60). INVKG3_ST2, INVMG3_ST5A. Following steps must be performed:



Positive pole
From J3 pin (E60) to BBIP.07 plate



Negative pole
From J4 pin (E60) to BBIN.07 plate



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3.2. In battery inverters, the following steps must be performed:

3.2.1. Disconnect the modules power supply through the selectors. **INVKG3_ST2_**
INVMG3_ST5A.

3.2.2. Proceed to make measurements on the softcharge board (E60) by externally power supply. Disconnect the wires connected to the (+) and (-) terminals from the power supply output (G01), and power supply these wires with the external 24Vdc power supply. **INVKG3_ST2**
INVG3_ST5A.



G01 without external power supply



G01 with external power supply

3.2.3. Check the wiring between the soft charge boards (from E60 to E63 depending on the module) and module PCB (E02) **INVKG3_ST2, INVMG3_ST5A.**

- Check continuity with a RJ45 tester on all pins of the RJ45 cable (W013), first disconnect both RJ45 connectors (E60 board, J2 port; E02 board, J11 port) and connect them to pin tester. All number must match. **INVKG3_ST2, INVMG3_ST5A**



If there is not continuity, replace de RJ45 cable. **INVKG3_ST2, INVMG3_ST5A**.

3.2.4. Check the softcharge board (E60).

- Make a bridge between pin 1 and 2 of the J7 connector of the softcharge board (E60). **INVKG3_ST2_ INVMG3_ST5A**. Check the KM10 and KM11 are closed.

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- Measure using a 1500V multimeter (continuity mode) between A1 and A2 in both contactors. INVKG3_ST2_ INVMG3_ST5A. If there is not continuity, replace the contactor. INVKG3_ST2_ INVMG3_ST5A.



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3.2.5. Check that contactors coils KM10 and KM11 are supplied.

- Measure using a 1500V multimeter 1500V (Vdc mode) between the terminals X25:1:2 and X25:1:1 of the X25 terminal block. The measurement must be 24 Vdc. INVKG3_ST2_ INVMG3_ST5A with External power supply



- If it is not 24Vdc, check the wiring since the power supply (G01). INVKG3_ST2_ INVMG3_ST5A:
- Using a 1500V multimeter in continuity mode, measure between X25:5:1 and frame. INVKG3_ST2_ INVMG3_ST5A



3.2.6. Once the contactors have been checked for correct operation, relay R3 on the E60 soft-load charge must be checked. For this purpose

- Disconnect the wires from connector J7 of the E60 soft charger board, and by bridging pins 1 and 2 of connector J3 of the E60 soft charger board, measure with a 1500V multimeter in continuity mode between pins 1 and 2 of connector J7.

INVKG3_ST2_ INVMG3_ST5A



- If there is no continuity, the E60 soft charger board must be replaced.. See document NVG3/OP88849. INVKG3_ST2_ INVMG3_ST5A.

3.2.7. Check softcharge resistors R30-R31

- Measure the value of resistor R30 and R31. INVKG3_ST2_ INVMG3_ST5A. Use a 1500V multimeter in resistance mode between pin A2 of the KM10 contactor and the BBIP.07 board INVKG3_ST2_ INVMG3_ST5A. This value must be 1200 Ohms $\pm 5\%$



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3.2.8. Check F101 fuse

- Remove the fuse and measure with a 1500V multimeter in continuity mode. This value should be 0.2 ohms. INVKG3_ST2_ INVMG3_ST5A. If the value is different, replace the fuse.

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Also check if there is any external load on the module that is interfering with the soft charge of the module, if there is, disconnect the external load (e.g. in Bus Plus equipment disconnect the DCDC converters connected to the inverter). It is advisable to measure resistance with 1500V multimeter between the positive and negative DC connections of the failing module to detect a possible short circuit. **INVKG3_ST5A, INVMG3_ST5A.**

5.20. Fault code F35: MT Premag

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	A problem occurred during premag of the MV transformer.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

Check if any fault has occurred in the SD card (SYSLOG) or in the history during the Blackstart state, in which case the corresponding actions indicated in the Troubleshooting manual must be applied to solve it.

Note: The Blackstart state is the state in which the device is during the entire pre-magnetisation sequence.

If no fault has occurred, it will be necessary to identify the part of the pre-magnetisation sequence in which the equipment has failed.

The normal pre-magnetisation sequence is:

- Check the conditions for starting the pre-magnetisation.
- Close the AC circuit breaker (only on HEMK and PCSK units with motorised SKID).
- Close the capacitor filter.
- Start the modules to generate AC voltage.
- Raise the voltage of the modules and synchronise it with the external voltage measurement.
- Check the DC conditions on the modules.
- Close the medium voltage switchgear.

Bear in mind that once the pre-magnetisation has started and the conditions to start it have been checked, the maximum time to finish the manoeuvre is 100s, otherwise the equipment will fail.

Failure will also occur if the equipment loses pre-magnetisation conditions during the manoeuvre.

If none of the above points have been identified, it is necessary to check if the customer has carried out any manoeuvre on the equipment during the pre-magnetisation.

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and solve it.

AC selector disabled during pre-magnetisation.

1. Check the SD card (SYSLOG) to see if the AC selector has been deactivated during the pre-magnetisation sequence. If the virtual AC selector has been deactivated, it is due to incorrect operation of the equipment and inform the customer. **INV3_ST1B.**

Activation of F161: LOTO AC during pre-magnetisation.

2. Check the SD card (SYSLOG) to check if the F161 has been activated, if so, inform the customer about an incorrect manoeuvre of the equipment. **INV3_ST1B.**

Non-optimal conditions of the equipment for pre-magnetisation start-up.

In this case the fault will appear very quickly, for this we are going to carry out the following checks:

3. Check the SD card (SYSLOG) to check if there is any module and/or busbar disabled before starting the manoeuvre. **INV3_ST1B.**
4. Check that the AC voltage conditions are correct either in PowerCommsV in group [SV3: Grid display] or on the SD board (LOG1). Check with the customer that the AC voltage is also correct, otherwise go to the point where the AC voltage measurements are checked. **INV3_ST1B.**

5. Check that the MV switchgear is visually configured correctly, to do this check that the line disconnecter is closed and the earthing switch and circuit breaker are open. **INVMG3_ST3A, INVKG3_ST3A.**



NOK

If not, ask the PV plant manager of the customer's installation if there is any reason to have that configuration and if not, ask the customer to operate it and leave it configured correctly. **INVMG3_ST3A, INVKG3_ST3A.**

6. Check that the MT cell feedbacks in PowerCommsV within the group [SV1.4-Pairing] correspond to the previous point. **INVMG3_ST3A, INVKG3_ST3A.**
If the MV cell states are not displayed correctly in PowerCommsV, the following steps must be followed:
 - 6.1. Check the line disconnecter feedback, to do so:
 - 6.1.1. Make a jumper between connector pins J2:1 and J6:1 of the input/output board.



-HEM / PCSM E82 input/output board. **INVG3M_ST3A.**

-HEMK / PCSK E101 input/output board, located in the LV cabinet of the MV SKID. In the case of TWIN SKID, the input and output boards are:

INVKG3_ST3A, SKIDG3_ST3A.

-E101 corresponding to inverter 1

-E105 corresponding to inverter 2



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Check in PowerCommsV in the group [SV1.5 Digital I/O status] that the feedback disappears.

If not, the problem is between the input/output board, as applicable, and the central control board E35.

6.1.1.1. check the continuity of the RJ45 cable with a RJ45 tester.

INVKG3_ST2, INVMG3_ST5A. Note that if the cable has Harting connectors in between, each cable section must be checked with the 1500V multimeter.

Replace the faulty cable section. **INVKG3_ST2. INVMG3_ST2.**



- 6.1.1.2. if the RJ45 cable does not fail, replace the input/output board as appropriate. **INVKG3_ST2, INVMG3_ST2**. (See document INVG3IOP88849).
- 6.1.1.3. if the fault persists, replace the central control board E35. (See document INVG3IOP88849). **INVKG3_ST2, INVMG3_ST2**.
- 6.1.2. If the feedback disappears, the problem is in the feedback wiring of the closed line disconnector. Check for continuity with a 1500V multimeter at the following points:
- 6.1.2.1. Between connector J2:1 on the input/output board, as applicable, and J8 of Q02 in the MV switchgear. **INVKG3_ST2, INVMG3_ST2**.



Q02:J8 - CHASSIS



Q02:J8



J2:1 - CHASSIS

- 6.1.2.2. Between connector J6:1 of the input/output board, as applicable, and J9 of Q02 on the MV switchgear **INVKG3_ST2, INVMG3_ST2**.

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Q02:J9 - CHASSIS

EN



J6:1 - CHASSIS

6.1.2.3. If there is a continuity fault in any of the above points, replace the corresponding cable. **INVKG3_ST2. INVMG3_ST2.**

6.1.2.4. If there is no continuity fault, consult Power Electronics to replace the MV switchgear (Q02). **INVKG3_ST2. INVMG3_ST2.** (See document INVG3IOP87922).

- 6.2. Check the grounding switch feedback for this purpose:
- 6.2.1. jumper the connector pins J2:1 and J4:2 on the input/output board, as appropriate. **INVMG3_ST3A, INVKG3_ST3A, SKIDG3_ST3A.**



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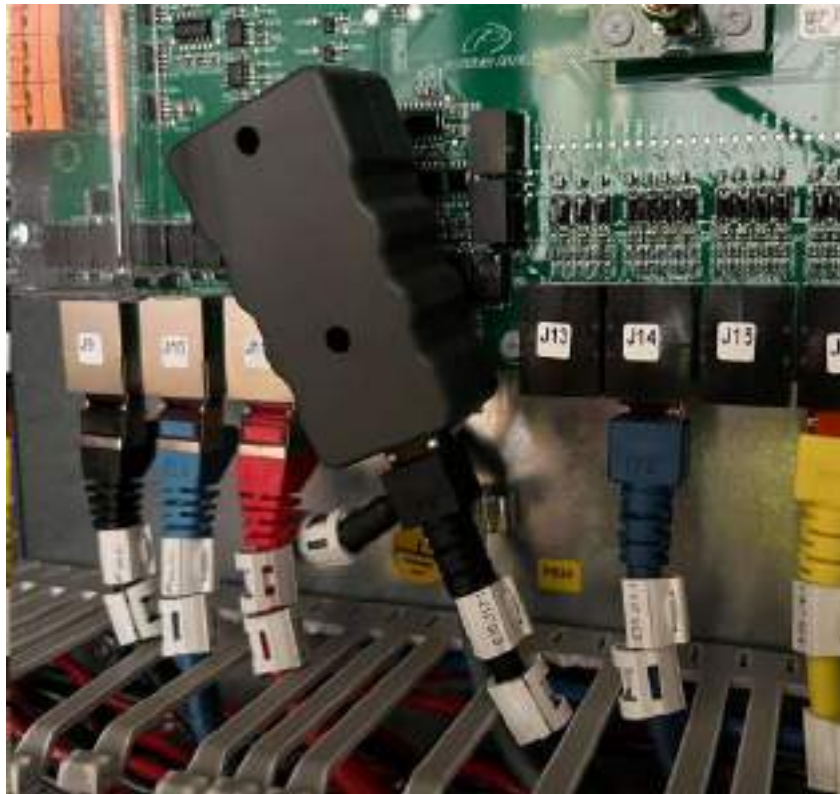
Check in PowerCommsV in the group [SV1.5 Digital I/O status] that the feedback disappears.

If not, the problem is between the input/output board, as applicable, and the central control board E35.

6.2.1.1. check the continuity of the RJ45 cable with a RJ45 tester.

INVKG3_ST2, INVMG3_ST5A. Note that if the cable has Harting connectors in between, the test must be carried out on each cable section. Replace any cable section that fails. **INVKG3_ST2. INVMG3_ST2.**





6.2.1.2. if the RJ45 cable does not fail, replace the input/output board as appropriate. **INVKG3_ST2, INVMG3_ST2**. (See document INVG3IOP88849).

6.2.1.3. if the fault persists, replace the central control board E35. (See document INVG3IOP88849). **INVKG3_ST2, INVMG3_ST2**.

6.2.2. If the feedback disappears, the problem is in the feedback wiring of the open earthing switch. Check for continuity with 1500V multimeter at the following points:

6.2.2.1. between connector J2:1 and J16 of Q02 in the MV switchgear.

INVKG3_ST2, INVMG3_ST2



J2:1 - CHASSIS

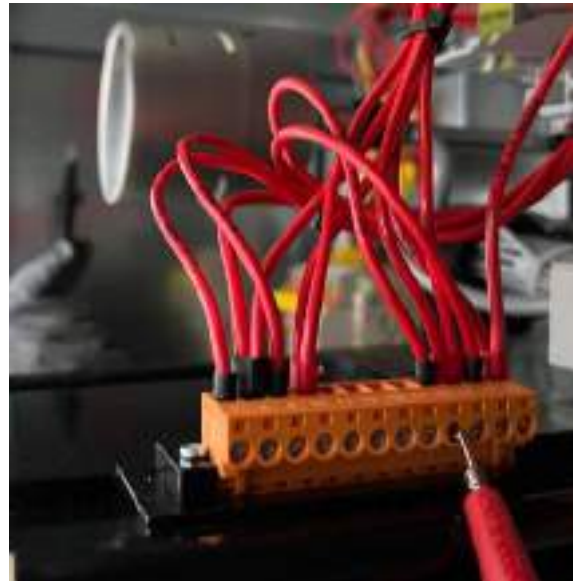


Q02:J16 - CHASSIS

6.2.2.2. Between connector J4:2 and J15 of Q02 in the MV switchgear.
INVKG3_ST2. INVMG3_ST2.



J4:1 - CHASSIS



Q02:J15 - CHASSIS

6.2.2.3. If there is a continuity fault at any of the above points, replace the corresponding cable. **INVKG3_ST2. INVMG3_ST2**

6.2.2.4. If there is no continuity fault, consult Power Electronics for replacement of the MV switchgear. (Q02). **INVKG3_ST2. INVMG3_ST2.** (See document INVG3IOP87922).

- 6.3. Check the feedback of the circuit breaker of the MV switchgear, for this purpose:
- 6.3.1. jumper the pins of connector J4:5 and J4:6 on the input/output board.



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-HEM / PCSM E81 input/output board. **INV3M_ST3A**.
 -HEMK / PCSK E101 input/output board, located in the LV cabinet of the MV SKID. In the case of TWIN SKID, the input and output boards are **INVKG3_ST3A**, **SKIDG3_ST3A**.



Check in PowerCommsV in the group [SV1.5 Digital I/O status] that the feedback disappears.
 If this is not the case, the problem lies between the input/output board and the central control board E35.

6.3.1.1. Check the continuity of the RJ45 cable with a RJ45 tester **INVKG3_ST2**, **INVMG3_ST5A**. Note that if the cable has Harting connectors in

between, the test must be carried out on each cable section. Replace the failed cable section. **INVKG3_ST2, INVMG3_ST2.**



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6.3.1.2. If the RJ45 cable does not fail, replace the input/output board accordingly. **INVKG3_ST2, INVMG3_ST2.** (See document INVG3IOP88849).

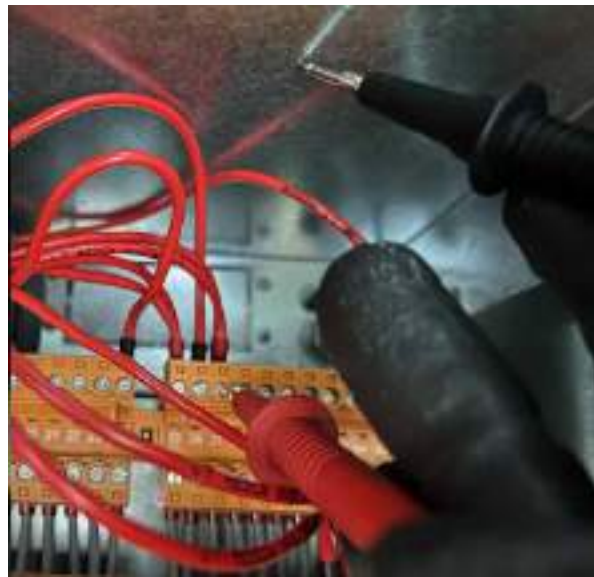
6.3.1.3. If the fault persists, replace the central control board E35. (See document INVG3IOP88849). **INVKG3_ST2, INVMG3_ST2.**

6.3.2. If the feedback disappears, the problem is in the feedback wiring of the open circuit breaker in the MV switchgear. Check that there is continuity with 1500V multimeter at the following points:

6.3.2.1. between connector J4:5 and 52:27 of Q02 on the MV switchgear.
INVKG3_ST2, INVMG3_ST2



J4:5 - CHASSIS

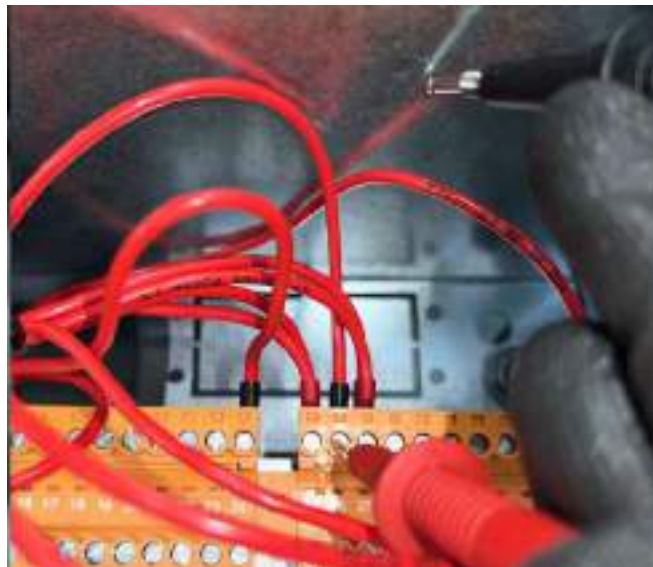


Q02:52:27 - CHASSIS

6.3.2.2. Between connector J4:6 and 52:26 of Q02 of the MV switchgear.
INVKG3_ST2. INVMG3_ST2



J4:6 - CHASSIS



Q02:52:26 - CHASSIS

6.3.2.3. If there is a continuity fault in any of the above points, replace the corresponding cable. **INVKG3_ST2. INVMG3_ST2**

6.3.2.4. If there is no continuity fault, consult Power Electronics to replace the MV switchgear (Q02). **INVKG3_ST2. INVMG3_ST2**. (See document INVG3IOP87922).

If the fault still appears quickly when starting the manoeuvre, contact PowerElectronics.

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AC circuit breaker malfunction (HEMK and PCSK with motorised SKID).

In case the AC circuit breaker has not been closed or no feedback has been received, consult PowerCommsV in group [SV1.5 Digital I/O status], it will be necessary:

7. Check that there is voltage at terminal X22 of the filter cabinet C, to do this measure with the 1500V multimeter a voltage of 120Vdc (acceptable measuring range 105-140 Vdc) between pins X22.5.3 and X22.5.4. **INVKG3_ST3A.**



8. If there is no voltage, measure continuity with 1500V multimeter at the following points:
 - 8.1. Between the negative of connector X22.5.4 and connector X100:2:1. **INVKG3_ST2.**

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8.2. Between the positive of connector X22.5.3 and connector X101:1:2. **INVKG3_ST2.**



8.3. Between connector X101:1:2 and disconnecter S04:4. **INVKG3_ST2.**



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8.4. Between disconnecter pins S04:3 and emergency mushroom S02:32 **INVKG3_ST2**

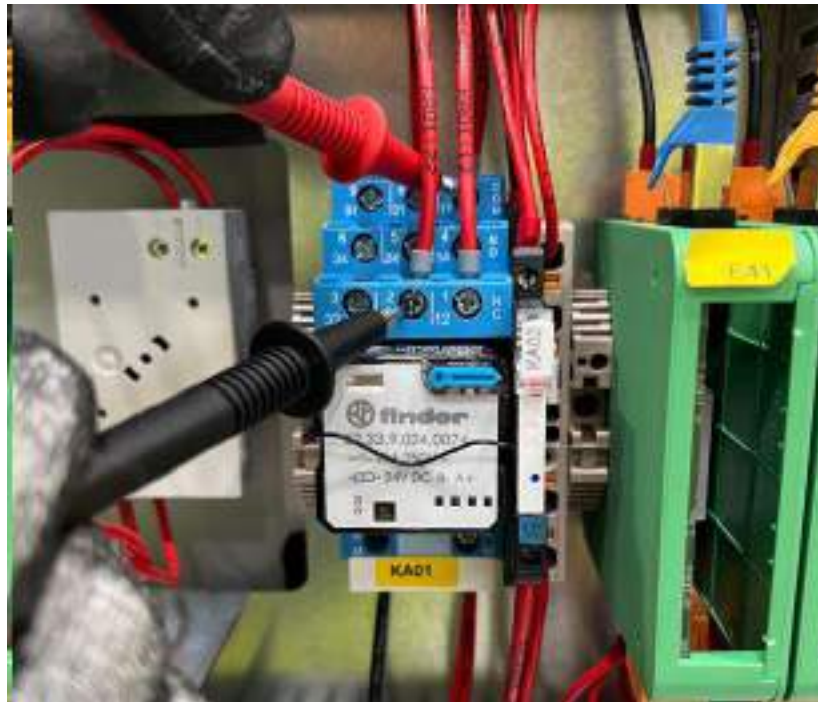


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- 8.5. Between E-stop button pin S02:31 and relay KA01:21 **INVKG3_ST2**



- 8.6. Between the relay pins KA01:22 and KA01:11 **INVKG3_ST2**



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8.7. Between relay pin KA01:12 and connector X100:3:2 **INVKG3_ST2**





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If there is no continuity at any of the above points, replace the corresponding wiring or connector.

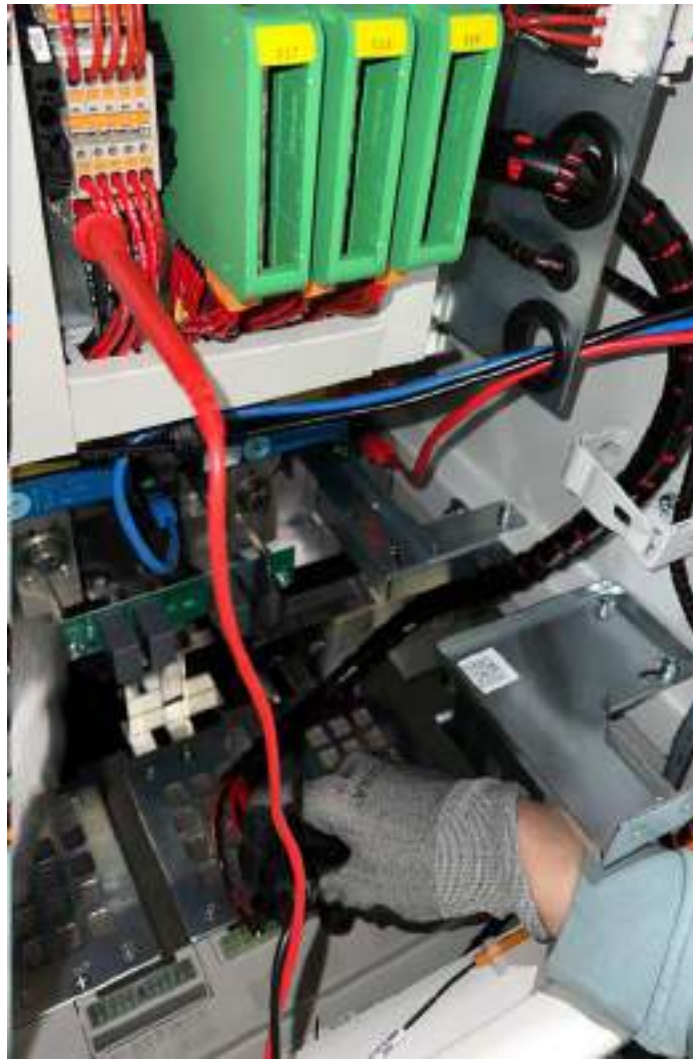
INVKG3_ST2.

9. Check audibly that the AC circuit breaker minimum coil is activated and deactivated when we activate and deactivate the AC LOTO switch (S04), so we can verify that the coil is powered.

INVG3_ST1B.

If not, measure continuity with 1500V multimeter at the following points: **INVKG3_ST2.**

- 9.1. Between the negative of connector X22 and terminal D2 of the AC circuit breaker.



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- 9.2. Between the positive of connector X22 and terminal D1 of the AC circuit breaker.



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If there is no continuity at any of the above points, replace the corresponding wiring or connector. **INVKG3_ST2.**

If there is no continuity fault, replace the AC circuit breaker undervoltage coil. **INVKG3_ST2.** (See document INVG3IOP85374).

10. Check the power supply to the AC circuit breaker closing coil. With no AC voltage at the equipment terminals, jumper the pins of connector J3:6 and J3:1 and check that the AC circuit breaker closes. **INVKG3_ST2.** To do this, supply terminal X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand.



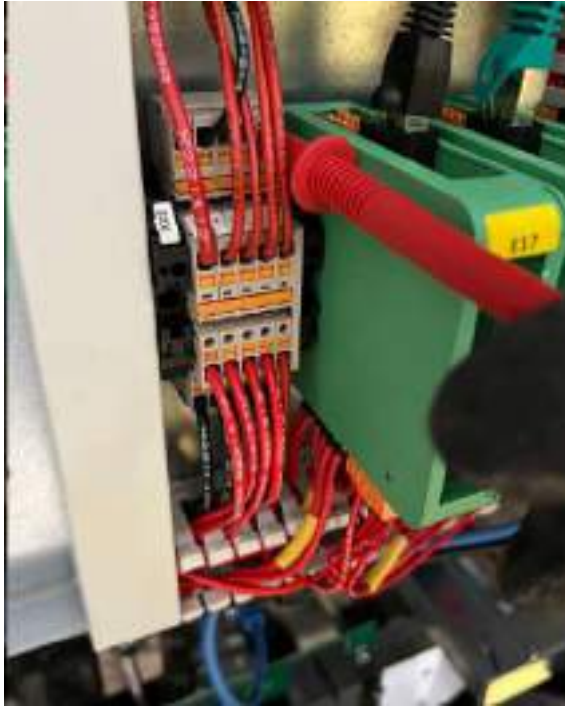
External power supply



If not, measure between pins A1 and A2 of the AC circuit breaker to check for 120Vdc. **INVKG3_ST2**. To do this, supply terminal X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand. If voltage is present, replace the closing coil. **INVKG3_ST2**. (See document INVKG3IOP85374).



If there is no voltage, measure continuity with 1500V multimeter at the following points:
INVKG3_ST2.
 10.1. Between the negative of connector X22 and terminal A2 of the AC circuit breaker.



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- 10.2. Between the positive of connector X22 and connector J7:1 of the digital inputs and outputs board E17.



- 10.3. Connect connector J7:2 of the digital inputs and outputs board E17 to terminal A1 of the AC circuit breaker.



If there is no continuity at any of the above points, replace the corresponding wiring or connector.

INVKG3_ST2.

If there is no continuity fault, check the continuity of the RJ45 cable with the RJ45 tester.

INVKG3_ST2. If faulty, replace the RJ45 cable. **INVKG3_ST2.**

If not, replace the E17 input/output board. **INVKG3_ST2.** (See document INVG3IOP88849).



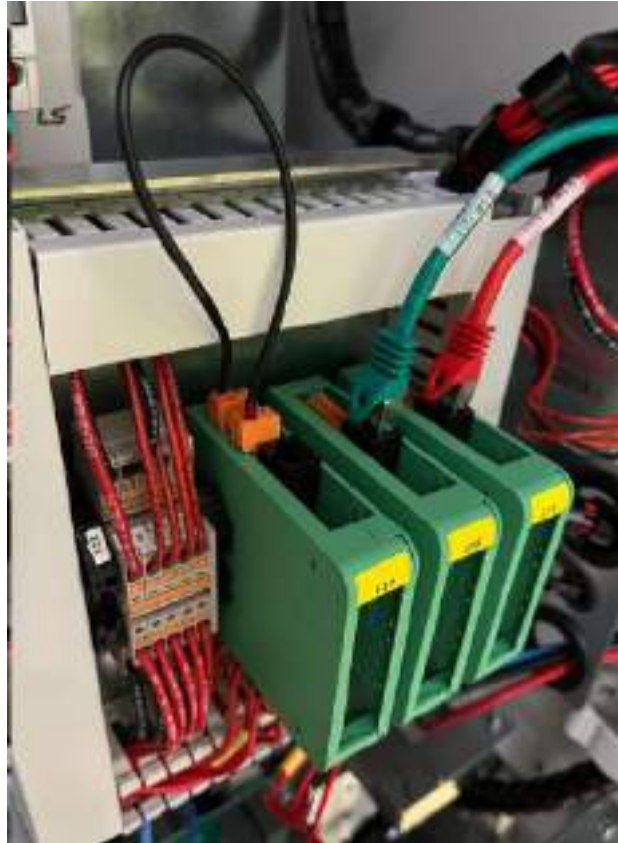
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If the fault persists, replace the central control board E35, **INVKG3_ST2**. (See document INVG3IOP88849).

11. Check the power supply to the spring-loaded motor. To do this, without AC voltage at the equipment terminals, jumper the pins of connector J3:6 and J3:5 and check whether the spring of the AC circuit breaker is loaded. **INVKG3_ST2**. To do this, supply terminal X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand.



External power supply



12. If not, measure between pins U1 and U2 of the AC circuit breaker to check for 120Vdc. **INVKG3_ST2**. To do this, supply terminals X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand.



External power supply

If voltage is present, replace the spring-loaded motor of the AC circuit breaker. **INVKG3_ST2**. (See document INVG3IOP85374).

If there is no voltage, measure continuity with 1500V multimeter at the following points:
INVKG3_ST2.

- 12.1. Between the negative of connector X22 and terminal U2 of the AC circuit breaker.



- 12.2. Between the positive of connector X22 and connector J5:1 of the board digital inputs and outputs E17.



- 12.3. Connect connector J5:2 of the digital inputs and outputs board E17 to terminal U1 of the AC circuit breaker.

EN



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If there is no continuity at any of the above points, replace the corresponding wiring or connector.

INVKG3_ST2.

If there is no continuity fault, check the continuity of the RJ45 cable with the RJ45 tester **INVKG3_ST2.** If faulty, replace the RJ45 cable. **INVKG3_ST2.**

If not, replace the input/output board E17. **INVKG3_ST2.** (See document INVG3IOP88849).



If the fault persists, replace the central control board E35. **INVKG3_ST2**. (See document INVG3IOP88849).

13. Check the closing, opening and spring loaded feedback by bridging the following points: **INVKG3_ST2**. To do this, supply G01 and G02 from an external 24Vdc power supply. Disconnect the cables from both sources beforehand.



G01 sin alimentación externa



G01 con alimentación externa



G02 sin alimentación externa



G02 con alimentación externa

- 13.1. Between connector pins J4:1 and J4:2 of the E17 digital inputs and outputs board. Check via PowerCommsV e of group [SV1.5-Digital I/O status]) the feedback of circuit breaker closed (bit 5: Retro Circuit Breaker Closed). **INVKG3_ST3A**.



EN

- 13.2. Between the pins of connector J4:3 and J4:4 of the board digital inputs and outputs E17. Check via PowerCommsV e of group [SV1.5-Digital I/O status]) the feedback of circuit breaker open (bit 4: Retro Circuit Breaker Open). **INVKG3_ST3A.**



- 13.3. Between connector pins J4:5 and J4:6 of the board digital inputs and outputs E17. Check via PowerCommsV e of group [SV1.5-Digital I/O status]) the feedback of the circuit breaker spring loaded (bit 3: Circuit Breaker loaded). **INVKG3_ST3A.**



If any of the above signals are not activated, the digital input circuit between the digital input/output board E17 and the central control board E35 is damaged.

Therefore, check the continuity of the RJ45 cable with the RJ45 tester. **INVKG3_ST2**. If it fails, replace the RJ45 cable. **INVKG3_ST2**.



If not, replace the input/output board E17. **INVKG3_ST2**. (See document INVG3IOP88849).

If the fault persists, replace the central control board E35. **INVKG3_ST2**. (See document INVG3IOP88849).

If all signals activate and deactivate correctly, measure continuity with 1500V multimeter at the following points: **INVKG3_ST2**.

- 13.4. Between connector J4:1 on the E17 digital input/output board and pin 13 of the AC circuit breaker.



EN



- 13.5. Between connector J4:2 of the digital input/output board E17 and pin 14 of the AC circuit breaker.



EN

- 13.6. Between connector J4:3 of the digital input/output board E17 and pin 11 of the AC circuit breaker.



- 13.7. Between connector J4:4 of the digital input/output board E17 and pin 12 of the AC circuit breaker.



EN

- 13.8. Between connector J4:5 of the digital inputs and outputs board E17 and pin 413 of the AC circuit breaker.



- 13.9. Between connector J4:6 of the digital input/output board E17 and pin 414 of the AC circuit breaker.



EN

If all checks are correct, contact Power Electronics to replace the AC circuit breaker.
INVKG3_ST2. (See document INVKG3IOP85374).

One or more contactors of the LC filter have not been closed.

If feedback has not been received from one or more LC filter contactors during pre-magnetisation, check in PowerCommsV ([DSP Status RJ17] and [DSP Status RJ18] within group [SV1.5-Digital I/O status]) which contactor is not functioning correctly, then this is necessary:



14. Measure with the 1500V multimeter between the connector pins X22.5.3 and X22.5.4. of the filter cabinet C that there is 120Vdc (acceptable measuring range 105 -140Vdc). **INVKG3_ST3A**, **INVMG3_ST3A**.



EN

15. If there is no voltage, use the 1500V multimeter to measure continuity between the following points: **INVKG3_ST2, INVMG3_ST5A.**
15.1. Between negative of connector X22 and connector X100:2:1



- 15.2. Between the positive of connector X22 and connector X101:1:2.



EN

15.3. Between the pins of connector X101:1:2 and selector S04:4.





EN

15.4. Between selector S04:3 and emergency pushbutton S02:32.

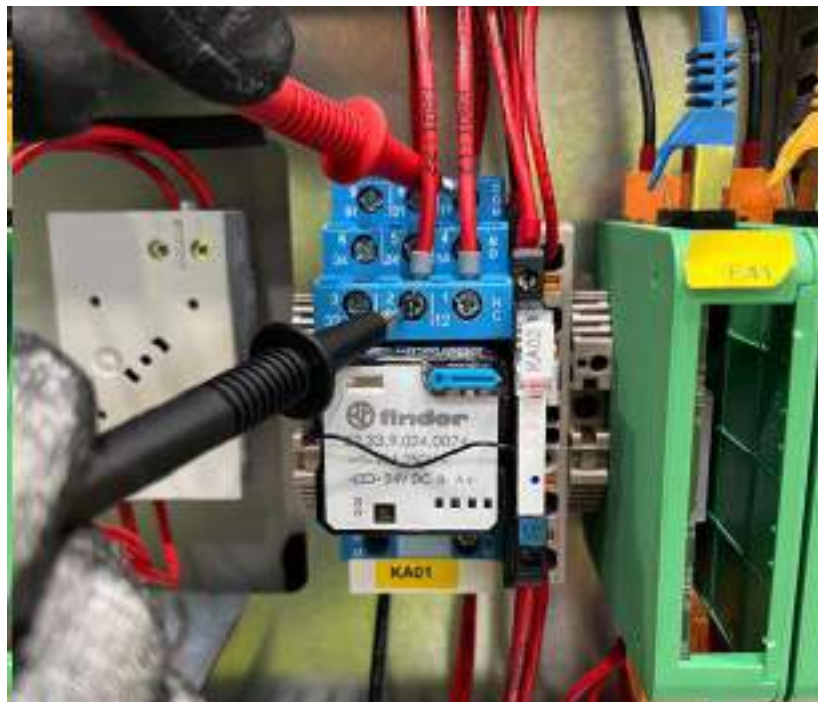


15.5. Between E-stop button S02:31 and relay KA01:21.



EN

15.6. Between relay KA01:22 and KA01:11.



15.7. Between relay KA01:12 and terminal block X100:3:2.



EN

If there is no continuity at any of these points, replace the damaged wiring or connector.
INVKG3_ST2, INVMG3_ST5A.

16. Check that the contactor is activated correctly. To do this, we are going to supply the contactor coil that is not working correctly with 120Vdc. Make a bridge from the connector pin J3.6 of the inputs and outputs board E18 with the pin J3.X that corresponds to the contactor. **INVKG3_ST2, INVMG3_ST5A.** To do this, supply terminals X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand. Supply G01 and G02 from an external 24Vdc power supply. Disconnect the wires from both sources beforehand.

The following table shows the pin corresponding to each contactor.

Contactor	Verification
1	Jumper between J3:6 and J3:5 of the E18 board
2	Jumper between J3:6 and J3:3 of the E18 board
3	Jumper between J3:6 and J3:1 of the E18 board
4	Jumper between J3:6 and J3:5 of the E19 board



G01 without external power supply



G01 with external power supply

EN



G02 sin alimentación externa



G02 con alimentación externa



J3:6 - J3:5 (E18)



J3:6 - J3:3 (E18)



J3:6 - J3:1 (E18)



J3:6 - J3:5 (E19)

17. If the contactor is functioning correctly, check the feedbacks in PowerCommsV. ([DSP Status RJ17] and [DSP Status RJ18] within group [SV1.5-Digital I/O status]). **INVKG3_ST2**, **INVMG3_ST5A**. To do this, supply terminal X22.1.2 (positive) and X22.1.1 (negative) from an external 120Vdc power supply. Disconnect the cables beforehand. Supply G01 and G02 from an external 24Vdc power supply. Disconnect the wires from both sources beforehand.



External power supply

EN



G01 without external power supply



G01 with external power supply

18. If the contactor does not trip, check for continuity with 1500V multimeter at the following points:
INVKG3_ST2, INVMG3_STA5.
 - 18.1. Between the negative of connector X22 and pin A2 of the coil of the non-functioning contactor.



EN

- 18.2. Between the positive of connector X22 and the pin of the input/output board E18 corresponding to the contactor.



J5:2 (E18)



J6:2 (E18)



J7:2 (E18)



J5:2 (E19)

Contactor	Verification
1	J5:2 from board E18
2	J6:2 from board E18
3	J7:2 from board E18
4	J5:2 from board E19

- 18.3. Connect pin A1 of the non-functioning contactor to its corresponding pin on the input/output board E18:



A1 (KM0X)



J5:1 (E18)



J6:1 (E18)



J7:1 (E18)



J5:1 (E19)

Contactor	Verification
1	J5:1 from board E18
2	J6:1 from board E18
3	J7:1 from board E18
4	J5:1 from board E19

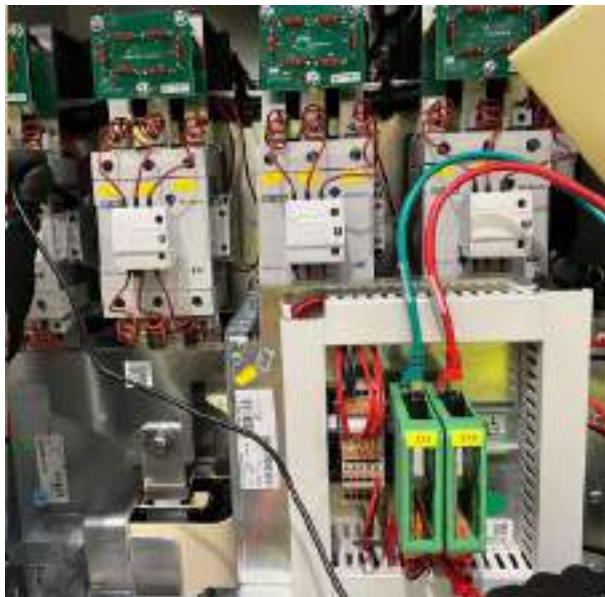
19. If all the above checks have been satisfactory, the contactor coil is damaged or has something that is preventing it from operating correctly.

Check that the capacitor precharge is fitted correctly. If not, adjust. **INVKG3_ST2**, **INVMG3_STA5**.



If set correctly, the contactor coil is damaged, replace the filter contactor. **INVKG3_ST2, INVMG3_STA5.**

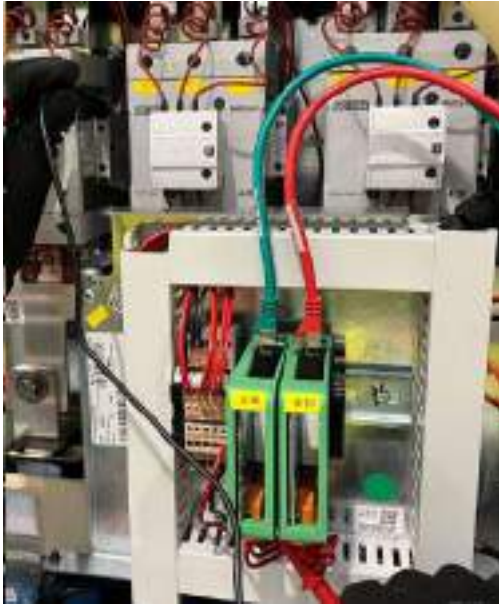
20. If the contactor closes and opens correctly, but does not receive feedback (Verification step 19), check the wiring of the feedback.
 - 20.1. Check that there is continuity with 1500V multimeter between the following points depending on which contactor is faulty: **INVKG3_ST2, INVMG3_ST5A.**



KM01



PIN 13 CONTACTOR



KM02



KM03



KM04

Contactor	Verification
1, 2, 3 y 4	Between pin 13 of the contactor and J4:5 of the E18 board
1	Between pin 14 of the contactor and J4:6 of the E18 board
2	Between pin 14 of the contactor and J4:4 of the E18 board
3	Between pin 14 of the contactor and J4:2 of the E18 board
4	Between pin 14 of the contactor and J4:6 of the E19 board

If there is no continuity in any of the above, replace the affected wiring. **INVKG3_ST2**, **INVMG3_ST5A**.

- 20.2. If there is no continuity fault, check the continuity of the RJ45 cable with the network cable continuity tester. **INVKG3_ST2, INVMG3_ST5A.**
If faulty, replace the RJ45 cable. **INVKG3_ST2, INVMG3_ST5A.**



- 20.3. If all the above checks have been satisfactory, replace the input and output board as appropriate (contactors 1, 2 and 3 is E18 and for contactor 4 is E19). **INVKG3_ST2, INVMG3_ST5A.** (See document INVG3IOP88849).
20.4. If not, replace central control board E35. **INVKG3_ST2, INVMG3_ST5A.** (See document INVG3IOP88849).

No module has been started for pre-magnetisation.

21. In case no modules are ready or could not start up, check on the SD card (**SYSLOG**) if at least one module has started up and managed to generate AC voltage. **INVKG3_ST3A, INVMG3_ST3A.**
22. If not, check the warnings and module statuses in PowerCommsV in the groups [SV7.2: Warnings] and [SV7.4: Module events].

Act on the modules according to the module warnings/failures displayed in PowerCommsV. **INVG3_ST1B.**

The required voltage reference has not been reached or could not be synchronised with the grid.

23. Check in PowerCommsV ([**V MV Calibration**] in group [**SG6-Calibrations**]) that the equipment has calibrated the MV voltage. **INVG3_ST1B.**

If it is not calibrated, do the closing procedure without pre-magnetisation (load the DC bus of the modules beforehand) and calibrate the measurement. **INVG3_ST1B.**

24. If the measurement is calibrated, check in PowerCommsV ([**V MV Calibration**] in group [**SG6-Calibrations**]) if the MV voltage measurement is balanced and has normal values (depending on the plant), and also the frequency is read correctly. **INVG3_ST1B.**
25. If the measurements are not good, check in PowerCommsV ([**MV network**] in group [SV3: Network display]) the measurements coming to the MV voltage measurement board (HEM: E80, SKID: E89).

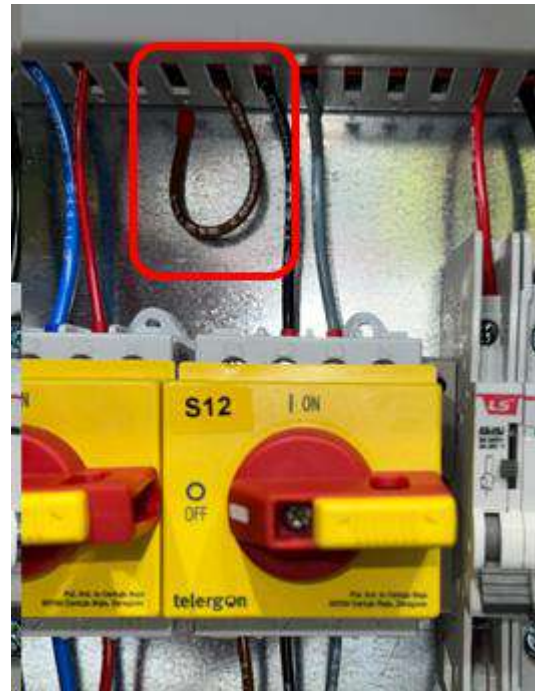
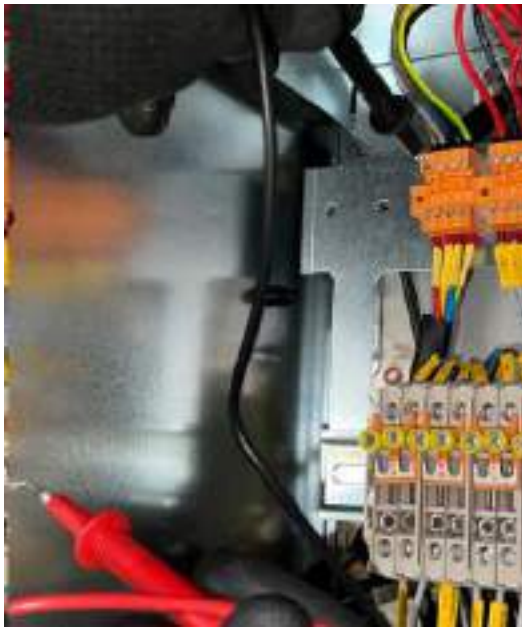
To do this, measure $\approx 160\text{Vac}$ (TWIN SKID $\approx 120\text{Vac}$) at voltage disconnector terminals (HEM: S12, SKID: S13) of the MV voltage measurement board (HEM: E80, SKID: E89). **INVKG3_ST3A, SKIDG3_ST3A, INVMG3_ST3A.**



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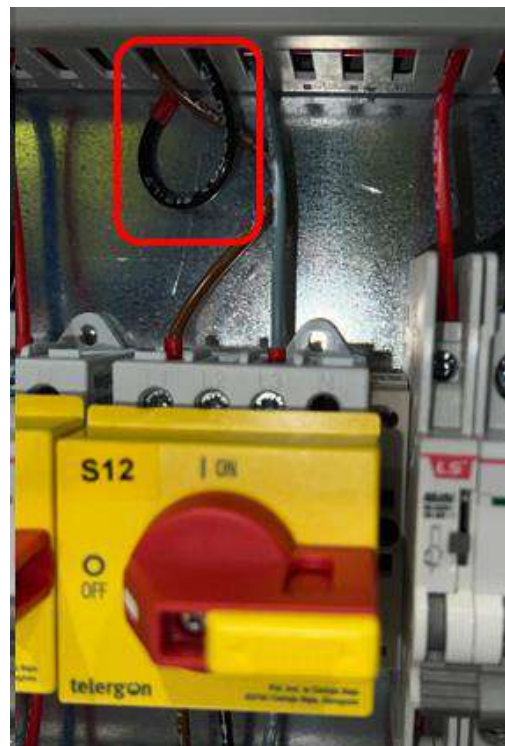
If there is no voltage to any of the phases or the voltage is very low, carry out the following checks:
25.1. Check continuity with 1500V multimeter between the following points: **INVKG3_ST2**, **SKIDG3_ST2**, **INVMG3_ST2**.

- between pin XU:1:L1 of Q2 and pin L1 of S12 or S13.

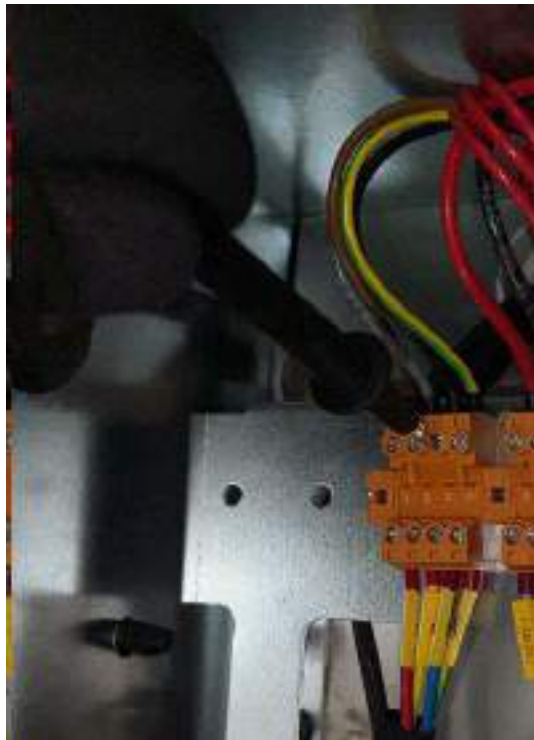


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- between pin XU:2:L2 of Q2 and pin L2 of S12 or S13.



- between pin XU:3:L3 of Q2 and pin L3 of S12 or S13.



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NOTE: the order of the connections described may not be the same, but it is not a problem for the equipment or for carrying out the measurements.

- Check with 1500V multimeter that the MV voltage measurement board (HEM: E80, SKID: E89) is correctly connected to ground and verify that there is continuity between this ground, the equipment ground and the MV cubicle ground.



EN

If there is no continuity, correct the wiring. **INVKG3_ST2**, **SKIDG3_ST2**, **INVMG3_ST2**.

- 25.2. If there is no continuity in any of the above, correct the wiring. **INVKG3_ST2**, **SKIDG3_ST2**, **INVMG3_ST2**.
- 25.3. Check again in PowerCommsV ([MV Network] in group [SV3: Network Display]) if the measurements are still bad, then contact Power Electronics.
26. If the medium voltage measurements are OK, repeat the pre-magnetisation manoeuvre and observe how the low voltage voltages change, check in PowerCommsV ([LV network] in group [SV3: Network display]). If they do not rise balanced or some do not rise at all, carry out the following checks:
 - 26.1. Check that the auxiliary fuse disconnecter (F100) is closed.



- 26.2. If this is not the case, close it securely (no AC voltage). **INVKG3_ST2, INVMG3_ST5A.**
 Check that the auxiliary fuses (F100) are in good condition by measuring continuity with a 1500V multimeter:



- If not, replace them. **INVKG3_ST2, INVMG3_ST5A.**
 Check continuity with 1500V multimeter between the following points: **INVKG3_ST2, INVMG3_ST5A.**
 HEM / PCSM:
 - between terminal 1 of F100 and J3 of central control board E35.



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- between terminal 3 of F100 and J4 of central control board E35.



- between terminal 5 of F100 and J5 of central control board E35.



EN

HEMK / PCSK:

- between terminal 1 of F100 and J3 of central control board E35.



- between terminal 3 of F100 and J4 of central control board E35.



EN

- between terminal 5 of F100 and J5 of central control board E35.



NOTE: To carry out this measurement correctly, first disconnect the measurement cables that go to terminals 1, 3 and 5 of the F100 fuse holder.

If there is no continuity in any of the cables, replace it. **INVKG3_ST2, INVMG3_ST5A.**

The modules do not have DC conditions to close the medium voltage switchgear. After synchronisation with the grid, if the modules do not have DC voltage above the rectified grid voltage, the MV switchgear cannot be closed.

27. Check on the SD card (Log_2) that the DC voltage of the modules has not suddenly dropped during pre-magnetisation. **INVKG3_ST1B**.

In case of PCS, check with the customer why he has opened batteries and report that it is an improper manoeuvre for the inverter.

In case of solar, it is possible that the starting DC voltage is a bit low, correct the starting voltage and retry the pre-magnetisation when there is more irradiation.

28. If only one module had voltage below the rectified voltage, and was disabled and/or without voltage on its busbar during the manoeuvre, check that the AC boards of the module are correctly connected. **INVKG3_ST2, INVMG3_ST5A**.



If they are not connected, ask the plant manager if there is any reason why they are disconnected, and if there is no reason, connect the module to AC. Also check for loose or missing screws, if screws are loose or missing, correct. **INVKG3_ST2, INVMG3_ST5A**

If the boards are not connected and the plant manager indicates that there is a cause for not connecting the module, contact PowerElectronics.

The medium voltage switchgear has tripped on closing.

29. If the equipment has reached the point of closing the switchgear, but has failed, check if the switchgear relay has tripped due to a fault. **INVKG3_ST3A, INVMG3_ST3A**.



30. If a fault has tripped, check the cubicle parameters. If the fault is unbalance or current to ground:

For the model Ekor RPA serie 021 (Ormazabal manufacturer), push button ↓ in the main menu until reach the historial menu.



Push the → button to see the 1 historic (H1). Pushing the ↑ and ↓ buttons it is possible to navigate through the last 10 faults registered.



EN

The fault codes are:

- 50: Instantaneous phase overcurrent
- 51: Inverse time phase overcurrent
- 50N: Instantaneous neutral overcurrent
- 51N: Inverse time neutral overcurrent
- EXT: External fault

- Check the connections of the measuring toroids of the medium-voltage switchgear. If any are disconnected, replace. [INVKG3_ST2](#), [INVMG3_ST2](#).
- Check if there are any points on the medium voltage part of the equipment that are earthed. If there is any point, repair and correct. [INVKG3_ST2](#), [INVMG3_ST2](#).

The medium voltage switchgear could not be closed.

The medium-voltage switchgear has not been closed or the feedback received is not as expected, carry out the following checks:

31. Measure 120Vdc between terminal block pins X31:1 (positive) and X31:4 (negative), (SKID terminal block X112 instead of X31). [INVKG3_ST3A](#), [INVMG3_ST3A](#).
32. If there is no voltage, check for continuity with 1500V multimeter between the following points (only for HEM / PCSM): [INVMG3_ST2](#).

- between the connector X31:4 (negative) and X100:2:1.



- between X31:1 (positive) and X101:2:2



- between X101:2:2 and S04:6



- between S04:5 and S02:42

EN



- between S02:42 and KA01:21



- between KA01:21 and KA01:12

EN



- between KA01:12 and X100:3:2



If there is no continuity at any of the measured points, correct the damaged wiring or connector.
INVMG3_ST2.

33. Check that relay KA08 is activated (only for HEM / PCSM). Verify that the relay light is illuminated.
INVMG3_ST3A.
- 33.1. If the relay lamp does not light, measure with a 1500V multimeter in Vdc mode between terminals A1 and A2 of the KA08 relay. **INVMG3_ST3A.**



If voltage is present and not activated, replace relay KA08. **INVMG3_ST5A.**

33.2. If there is no voltage, carry out the following checks:

33.2.1. measure with a 1500V multimeter in Vdc mode between connector pins X40:1 and X40:3, **INVMG3_ST3A.**



If voltage is present, replace diode X41. **INVMG3_ST5A.**

33.2.2. check continuity with 1500V multimeter between connector pins X40:1 and Q19:2. **INVMG3_ST5A.**



33.3.3. check continuity with 1500V multimeter between terminal Q19:2 and the positive terminal of the 24Vdc power supply G02. **INVMG3_ST5A.**



33.3.4. check continuity with 1500V multimeter between the X40:3 connector and the X11:3 connector. **INVMG3_ST5A.**

EN



33.3.5. check continuity with 1500V multimeter between connector X11:3 and the negative terminal of the 24Vdc power supply G02. **INVMG3_ST5A.**



EN

If there is no continuity at any of the measured points, repair the corresponding wiring or connector. **INVMG3_ST5A.**

34. Failure in the switchgear feedbacks. The closed cubicle feedback has not been removed and the open cubicle feedback (indeterminate status by SW) or the feedback has been activated. Perform the following checks:

34.1. jumper the pins of connector J4:3 to connector J4:4 of the board E81 (SKID: E100)
INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.



Check in PowerCommsV in the group [SV1.5 Digital I/O status] that the digital switch signal of the closed switchgear is closed. **INVKG3_ST3A, INVMG3_ST3A, SKIDG3_ST3A.**

34.2. disconnect pins J4:5 and J4:6 of the board E81 (SKID: E100). **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**



Check in PowerCommsV in group [SV1.5 Digital I/O status] that the digital open switchgear signal is activated. **INVKG3_ST3A, INVMG3_ST3A, SKIDG3_ST3A.**

34.3. if any of the above two actions did not work, continue with the following checks:

34.3.1. check the continuity of the RJ45 cable between connector J2 on the input/output board E81 (SKID: E100) and connector J13 on the central control board E35 with RJ45 tester. **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST2.**



Note that if the cable has Harting connectors in between, the Verification must be performed on each cable section with the 1500V multimeter (XC702.2:B1 for HEM and XC1026.2:1B for SKID).

Replace the failed cable section. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

EN



EN

34.3.2. if the RJ45 cable does not fail, replace the input/output board E81 (SKID: E100). **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**. (See document INVG3IOP88849).

34.3.3. if the fault persists, replace the central control board E35. (See document INVG3IOP88849). **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**

34.4. if the previous checks have been successful, repeat the pre-magnetization maneuver and check audibly if the unit is able to close the cubicle. If you hear the cubicle close, but it opens immediately, this means that a cubicle feedback has failed, continue with point 34.5. If it has not closed, this means that the cubicle actuators are faulty, continue with point 34.6. **INVG3_ST1B**.

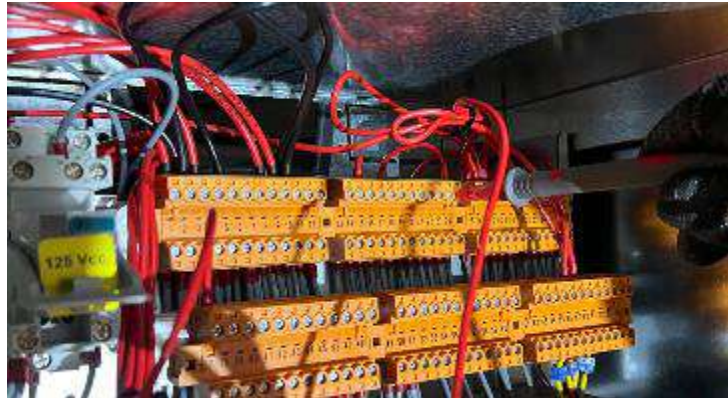
34.5. fault in the switchgear backfeed, to do this check that there is continuity in the switchgear closing backfeed to the medium voltage switchgear. Check continuity with 1500V multimeter between the following points: **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**.

HEM / PCSM:

- between connector J4:3 of the E81 board and 52:25 of the Q02 switchgear.



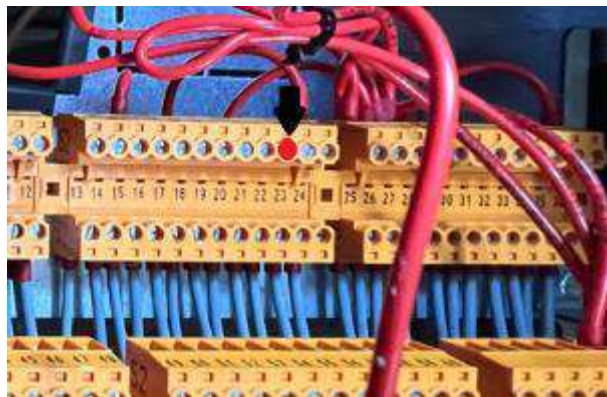
- between connector J4:4 of the E81 board and 52:24 of the Q02 switchgear.



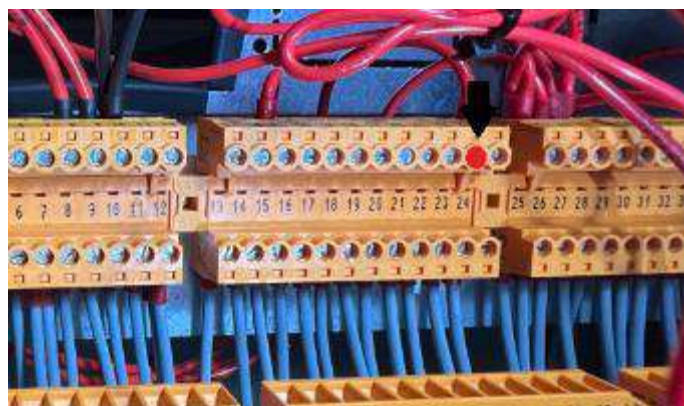
SKID:

- between connector J4:3 of the E100 board and 52:22 of the Q02 switchgear.

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- between connector J4:4 of the E100 board and 52:23 of the Q02 switchgear.



If there is continuity at all measured points, consult Power Electronics about the possibility of replacing the MV switchgear (Q02). **INVKG3_ST2**. **INVMG3_ST2**, **SKIDG3_ST2**. (See document INVG3IOP87922).

34.6. Switchgear operating fault.

34.6.1. Repeat the pre-magnetization operation.

Measure 120Vdc with an oscilloscope and a differential voltage probe between pins J6:2 of the E81 board (SKID: E100) and terminal block X31:4 (SKID: X112:4). **INVKG3_ST2**. **INVMG3_ST2**, **SKIDG3_ST2**.



E81:J6:2



X31:4



X112 (SKID)

EN

Once the instrumentation equipment is connected, close the equipment and perform the pre-magnetization maneuver. **INVKG3_ST1B**.

34.6.2. If at the moment of closing the cubicle no 120Vdc voltage is measured on the oscilloscope, it means that there is no voltage in the drive coil.

34.6.2.1. Check continuity with 1500V multimeter between points J6:1 of E81 (SKID: E100) and terminal block X31:1 (SKID: X112:1).



If there is no continuity, correct the wiring. **INVKG3_ST2**. **INVMG3_ST2**, **SKIDG3_ST2**.

34.6.2.2. if there is continuity in the above point, check the continuity of the RJ45 cable between connector J2 on the input/output board E81 (SKID: E100) and connector J13 on the central control board E35 with RJ45 tester. **INVKG3_ST2**, **INVMG3_ST5A**, **SKIDG3_ST2**.



Note that if the cable has Harting connectors in between, the verification must be performed on each cable section with the 1500V multimeter (XC702.2:B1 for HEM and XC1026.2:1B for SKID). Replace the failed cable section. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**



34.6.2.3. if the RJ45 cable does not fail, replace the input/output board E81 (SKID: E100). **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.** (See document INVG3IOP88849).

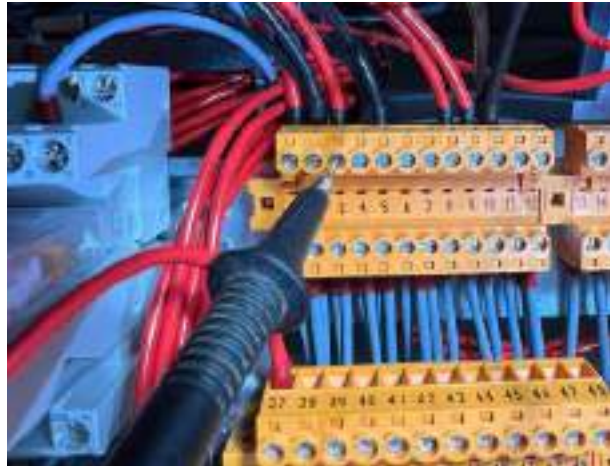
34.6.2.4. If the fault persists, replace the central control board E35. (See document INVG3IOP88849). **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**

34.6.3. If there is 120Vdc voltage on the oscilloscope verification, it means that the wiring to the switchgear or the switchgear closing coil is bad. Perform the following checks:

34.6.3.1. Check continuity with 1500V multimeter between connector J6:2 of the E81 board (SKID: E100) and pin 52:3 of the high voltage switchgear Q02. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2**



CONNECTOR J6:2



PIN 52:3

34.6.3.2. Check continuity with 1500V multimeter between connector J6:1 of the E81 board (SKID: E100) and pin 52:5 of the high voltage switchgear Q02. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**



CONNECTOR J6:1



PIN 52:5

If there is no continuity at any of the measured points, replace the necessary wiring. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

34.6.3.3. measure the resistance of the closing coil of the medium voltage switchgear with the 1500V multimeter. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

If the measured value is not correct, replace the closing coil of the medium voltage switchgear. **INVKG3_ST2, INVMG3_ST2, SKIDG3_ST2.**

Loss of AC voltage conditions in the maneuver.

35. During the whole maneuver, AC voltage conditions have been lost. If you do not see a specific cause of the failure, add the following Modbus addresses (4300, 4301, 4302) in PowerCommsV ([Custom Log] in group [G12- SD Card]) add the following Modbus addresses (4300, 4301, 4302) and repeat the procedure.
Then download the logs from the SD card and view on the SD card (Custom Log) the MV AC voltages.
If it fails again, check if the AC voltage drops during pre-magnetization.
If it drops, consult the customer to see if he is performing operations or maintenance on his medium voltage line that are influencing the equipment. **INV63_ST1B**.

The modules have not been stopped or the capacitor filter has not been opened.

36. If it is observed that the fault occurs after closing the MV switchgear and the modules have not stopped or the filter has not been opened after the maneuver, contact Power Electronics.

5.21. Fault code F203: MV overtemperature

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	One of the PT100 of the transformer has exceeded the temperature set in the overtemperature protection of the high voltage transformer.	
Applicable to	☒ HEM ☐ HEMK	☒ PCSM/MPCSM ☐ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this fault. Follow the steps below to identify the possible cause and resolve it.

Transformer overtemperature problem.

1. Check Log_3 of the SD ([G12: SD card]) of the inverter and check if the temperature increase is real or it was a noise problem in the temperature measurement. **INVG3_ST1B.**
 - 1.1. If the temperature has risen progressively and without ripple, there is a transformer overtemperature problem.
 - 1.2. If there is excessive ripple in the PT100 temperature measurement, it may be a wiring problem.

Wrong configuration of the temperature thresholds. **INVG3_ST1B.**

2. Check the fault configuration. If the fault temperature is below its default value, consult Power Electronics in case there is a special configuration of this plant.

In case there is an overtemperature problem:

3. Check the status of the transformer cabinet fans using PowerCommsV. Check that they are rotating at the correct speed [SV11: Ventilation]. If not, find the fan problem and fix it (see information on fan faults in this document). Also check the configuration of the protection group (grupo [G5.16: LC filter]). **INVG3_ST1B.**
4. Check the condition of the cyclone filters, check that the air inlet is not clogged with anything, if the filters are damaged or clogged in any way, replace them. **INVMG3_ST5A.**
5. Check the condition of the high voltage transformer, check that there are no breaks in the resin or hot spots on the transformer. If there are any, consult Power Electronics about the possibility of replacing the MV transformer. **INVMG3_ST5A.** (Check document *INVG3/OP86974*).
6. Check for hot spots or bad tightness in the low voltage plates and high voltage cables of the transformer. If there is a lack of tightness or missing screws, repair it. **INVMG3_ST5A.**
7. Check the ventilation channels of the high voltage transformer, check that they are not obstructed, nor the windings are displaced or obstructing the ventilation. If they are clogged, clean them and remove the obstruction. If they are displaced, contact Power Electronics to see what actions to take to correct the problem. **INVMG3_ST5A.**

There is a problem in the PT100 wiring. **INVMG3_ST5A.**

8. Check and measure continuity with 1500V multimeter between ground and the ground mesh, on all PT100 wires (W043, W044 and W045) of the high voltage transformer. If there is no continuity, ground the mesh.

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9. Check for loose or improperly tightened wires at the transformer and central control board terminals. Check that W405, W406 and W407 are connected to XPT1, XPT2 and XPT3. Check the E35 control board, J29, J28 and J29 connectors. If any, retighten and clean the wires if necessary.

The transformer temperature is adequate, but there has been an unexplained F203 fault. It is possible that there was a one-time erroneous measurement. In this case, follow the steps below:

10. Save in the custom log (group [G12: SD card]) the transformer temperature and attempt to reproduce the fault. **INVG3_ST1B.**
11. Analyze the SD by looking in Log_1 the exact time of the fault. Then look at the three temperatures at the column of the MT transformer (HEM) or the transformer temperature (HEMK / PCSK with temperature probe) at the time of the fault (parameters [MV PT100 (1)], [PT100 (2)] and [MV PT100 (3)], within group [SV1.6: Ambient]). **INVG3_ST1B.**
12. If the temperature has a very high value (approximately 65535) then either there has been a loose wiring problem (the value will be seen for several seconds in the custom log) or the software has not read the temperature at that time. **INVG3_ST1B.**
13. If the problem happens frequently, check connections of the wiring and ground mesh if any. **INVMG3_ST5A.**
14. If the wiring has been checked and secured, contact Power Electronics.

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5.22. Fault code F40: Internal overtemperature

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	The main control board has measured an own PCB temperature higher than the limit set by the overtemperature fault.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

Analysis procedure

There are different causes that can trigger this fault. Follow the steps below to identify the possible cause and resolve it.

Excessive temperature in the main control cabinet.

1. Check the logs of the SD and check the temperature displayed, if the temperature rises progressively and without ripple, it is a real overtemperature.
 - 1.1. Check if the temperatures of the rest of the electronic boards in the area (module PCBs and DU) have also an excessively high temperature. If so, it is a ventilation problem of the whole area. **INV3_ST1B.**
 - 1.2. Check in PowerCommsV that the DU and filter C fans have correctly configured their references, operating mode and fan type (within group [G11: Fans]). If they are not at their default value, consult the customer if it is a special configuration for some reason and correct. Ventilation should be at default unless otherwise justified. **INV3_ST1B.**
 - 1.3. Configure in PowerCommsV, within group [G11: Fans], DU and filter C fans to "Fixed" mode. **INV3_ST1B.**
 - 1.3.1. Check that the fans are rotating correctly. To do this, check in PowerCommsV the rotational speed and that there are no strange noises in the ventilation. **INV3_ST1B.** If they do not rotate correctly or there are strange noises, check the affected fan and correct the problem, i.e. check that there is no object obstructing the air inlet of the inverter and check the air inlet filters. If there is anything obstructing the air intake, remove it. **INV3_ST2, INV3_ST5A.**



If the filters are clogged, replace them. INVKG3_ST2, INVMG3_ST5A.

- 1.4. Check the DU exchanger air inlet on the inside. Check that it is not clogged. If it is clogged, clean it, and if it is damaged, consult Power Electronics about the possibility of replacing it. INVKG3_ST5A, INVMG3_ST5A.
2. If only the central control has an excess temperature:
 - 2.1. Check if in the customer cabinet or in the central control cabinet there are elements that can generate an excess of temperature. If there are any, perform a visual inspection to check that it is not operating improperly and is overheating the cabinet. INVG3K_ST3A, INVG3M_ST3A. If it is necessary to replace the component, contact Power Electronics.
 - 2.2. Check the ventilation ducts of the central control. Check that they are not obstructed; if they are, remove the object that is obstructing them. INVKG3_ST5A, INVMG3_ST5A



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Overtemperature fault threshold setting:

3. Check if the overtemperature threshold (parameter [Main control board], within group [G5.4: Temperature]) is at its default value. If it is not, consult Power Electronics for a special configuration. Unless there is good reason, the threshold value must be at its default value. If it is not, set it. INVG3_ST1B.

PCB sensor measurement error:

4. If a very noisy measurement is seen in SD Log_3 or the temperature is not seen to have reached the fault threshold, the sensor or the central control board may be in poor condition. INVG3_ST1B. Replace the central control board. Check document INVKG3IOP88849 INVKG3_ST5A, INVMG3_ST5A.
5. If the displayed temperature is excessively high, with a delta of +30°C above the inlet temperature with the unit running (subgroup [SV1.6: ambient]), it is possible that the sensor or the (E35) central board is damaged. INVG3_ST1B. In this case, replace the central control board. (Check document INVKG3IOP88849) INVKG3_ST5A, INVMG3_ST5A.

5.23. Fault code F55: External fault/OVGR

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The inverter has detected an emergency shutdown ordered by the customer.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

The customer does not use the external failure option:

1. If the customer does not wire an external fault signal to terminal X35:1 and X35:2, it will be necessary to jumper these connections to avoid the fault. **INV3K_ST3A, INV3M_ST3A.**



The customer has opened the digital input of the external emergency fault.

2. Consult the customer if he has triggered the external emergency fault. If it has been done intentionally, it is normal behaviour of the inverter to trigger this fault, if the customer has solved the problem, the inverter can be reset and returned to service. **INV3_ST1B.**
3. If the customer has not triggered the external emergency fault voluntarily, jumper the external emergency fault terminal block (X35:1 and X35:2). Then check whether the status of the corresponding digital signal changes (parameter [DSP Status RJ20] in group [SV1.5: Digital I/O status]). If the status of the digital signal does not change, it is a problem with the wiring of the device (next point). If it does change, then it is a problem with the customer's external emergency fault installation. If this is the case, inform the customer to solve the problem. **INV3K_ST3A, INV3M_ST3A.**



Wiring problem:

4. Jumper connectors J4:3 and J4:4 of the E44 board and check that the digital signal "external emergency fault" is set to 1 (parameter [DSP Status RJ20], in group [SV1.5: Digital I/O status]). If it is set to 1, there may be a wiring fault between the external emergency trip terminal block (X35) and the digital input/output board. To do this, ring and troubleshoot the problem. To do this, refer to the wiring diagrams. **INVKG3_ST2_ INVMG3_ST5A with External power supply**



5. If the digital signal at the central control does not change, ring the RJ45 cable between the central control board and the digital input/output board. **INVKG3_ST2_ INVMG3_ST5A with External power supply**



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6. If the digital signal still does not appear on the central control **INVG3K_ST3A**, **INVG3M_ST3A**, replace the digital input/output card E44. **INVKG3_ST2**, **INVMG3_ST5A**.
7. If the signal still does not appear, the central control board may be damaged. If so, replace the (E35) central control board. **INVKG3_ST5A**, **INVMG3_ST5A**.

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5.24. Fault code F103: IGBTs temperature

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The temperature of one or more IGBTs has exceeded the maximum permissible value according to the software configuration.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	W103: IGBTs temperature	

Analysis procedure

To resolve this fault, check whether the temperature readings of the IGBTs have reached the threshold configured in the temperature protection group ([\[G5.4: Temperature\]](#)).

The module ventilation is not working properly:

1. This can be caused by a problem in the fans, or in the intake of the inverter. If this occurs, the temperature of all IGBT in the module will be abnormally high with respect to adjacent modules. For this, check the state of the module ventilation:
 - 1.1 Check the fan settings for PowerCommsV. It should be at its default value unless otherwise justified ([group \[G11.5: Config Modules\]](#)), [INV G3_ST1A](#).
 - 1.2 Check that all fans are rotating at the configured setpoint and without unjustified speed variations (groups [SV11.3: Fan 3], [SV11.4: Fan 4], [SV11.5: Fan 5] and [SV11.6: Fan 6]). If this is not the case, check the fan warnings and correct according to the relevant warning. [INV G3_ST1A](#).
 - 1.3 Check that there are no strange noises in the fan and that they rotate correctly [INV G3_ST1A](#). If there are noises: check that the fan support is fixed and that the nozzle is in good condition without rubbing against the fan, that there is no object obstructing the air inlet and outlet of the inverter, check the fan bearing and also the air inlet filters. If there is anything obstructing the air inlet, remove it. [INVKG3_ST3B](#), [INVMG3_ST3B](#). If the problem persists:



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1.3.1. Check that there is no dirt or foreign matter on the top of the module heatsink. To access the air duct where the heatsink is located, the stage must be removed. **INVKG3_ST3B, INVMG3_ST3B**. (See document *INVG3/OP85369*).

1.3.2. Check the AC and DC power connections of the module for the presence of hot spots. **INVKG3_ST3B, INVMG3_ST3B**.

Excess current for that phase/module. When this happens, one more phase or module will normally be displayed with a higher temperature than another (parameters [Current R], [Current S] and [Current T], from group [SV12: Modules Information]). **INVG3_ST1A**. In this case:

2. Activate the reverse sequence of the device (parameter [Inverse sequence], in the [Advanced config] group) to balance the AC currents per phase. If the problem persists, contact Power Electronics. **INVG3_ST1B**.

There is a connection problem in the ribbon cables (W033 and W034) connecting the E01 board to the E02 module FPGA board. If there is a bad connection, the temperature of the IGBTs displayed will be approximately 203°C (group [SV12.x.3: Temperature Measurements]). **INVG3_ST1B**. If this occurs:

3. Check that the cable is well connected and not damaged. If it is in good condition but still fails, replace the flat cables **INVKG3_ST3B, INVMG3_ST3B**. If it still fails:
 - 3.1. Replace (E01) IGBT board. (Check document *INVG3/O988076*) **INVKG3_ST3B, INVMG3_ST3B**.
 - 3.2. If the problem persists, replace module control board (E02). (Check document *INVG3/O988076*) **INVKG3_ST3B, INVMG3_ST3B**.

There is a problem connecting the flat cables (W020 to W028) from the E01 distribution board to the GATE DRIVE board of the IGBT. (IGBT01-09). **INVKG3_ST3B, INVMG3_ST3B**. If this occurs, an IGBT at a temperature of approximately 201°C will be displayed. In this case:

4. Check that the IGBT cable is well connected and not damaged. If it is in good condition but still fails, replace the IGBT GATE DRIVE board (IGBT01-09) and ribbon cable. (Check document *INVG3/O988076*) **INVKG3_ST3B, INVMG3_ST3B**.
5. If the problem persists, replace the IGBT (E01) IGBT board (Check document *INVG3/O988076*) **INVKG3_ST3B, INVMG3_ST3B**.
6. If the problem persists, replace the module control board (E02). (Check document *INVG3/O988076*) **INVKG3_ST3B, INVMG3_ST3B**.

There is an IGBT in bad condition due to a bad tightening of its electrical connections, internal defect, overheated points or thermal interface (graphite) in bad condition. It is detected because the thermal behavior of the damaged IGBT is different from the rest, rising much faster than the others in the same position and reaching higher temperature than the rest for the same current. Likewise, the damaged IGBT will drop in temperature faster than the rest. In steady state with the same current it will remain about $\geq 5-10^{\circ}\text{C}$ higher than other IGBTs in same position. If this happens:

7. Visually check IGBT connections for: hot spots, discolored or degraded platens or screws, connections susceptible to retightening, or degraded adjacent parts **INVKG3_ST3B**, **INVMG3_ST3B**. If this is the case:
 - 7.1. Re-tighten connections and repair what has been detected. Check all power module hardware. **INVKG3_ST3B**, **INVMG3_ST3B**.
 - 7.2. Check temperature of the IGBT in operation and see its behavior by PowerCommsV (group [SV12: Modules Information]). **INVG3_ST1A**. In case the IGBT follows the same thermal behavior, disassemble the power module and replace the problematic IGBT. **INVKG3_ST3B**, **INVMG3_ST3B**. (See document *INVG3/OP88076*).
 - 7.3. If the problem persists then replace the power module. **INVKG3_ST3B**, **INVMG3_ST3B**. (See document *INVG3/OP88076*).
 - 7.4. If the problem still persists, please contact Power Electronics.

5.25. Fault code F27: Modules start fault

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The inverter has not been able to start any module during start-up.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

The modules have failed due to a fault at start-up.

1. Check the module statuses (group [SV7.4: Module events]), the warning history (group [SV7.2: Warnings]) and the module status in the SD (SYSLOG) to see which fault has triggered the modules and act according to the fault. By PowerCommsV INVG3_ST1B.

The modules have been disabled before start-up.

2. Check the warning history (group [SV7.2: Warnings]) and the SD (SYSLOG) to see what caused the modules to be disabled. If it was the client, comment that this is not a correct way of operating the inverter. By PowerCommsV INVG3_ST1B.
3. the reason for the failure does not appear in the SD or in the message history, set the [Debug logging level] parameter of group [SG2.1: uP debug] to "Debug log level" to review the events prior to the disabling of the module. After analysis, return this parameter to the default value. By PowerCommsV INVG3_ST1B.

EN

5.26. Fault code F112: OCAC emergency

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".			
Description	The module has detected an overcurrent that has been sustained for more than 500us. This fault usually triggers because the module has not been able to cut off the current within a certain time.			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	May be preceded by desaturations of IGBT, F95, F65, F106, F108, F69, F70.			

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect parameterisation of the protection AC current thresholds of the FRU module.

1. In case the overcurrent thresholds of the FRU stage are not the default ones, check if there is any justifiable cause for this, otherwise leave them as their default value (parameter [Overcurrent threshold] of group [SG10.2: Protections]). **INVG3_ST1B**.

Incorrect parameterisation of the inverter's control loops.

2. Check that the parameterisation is set to its default values (subgroups [Sync-measure freq], [Vdc loops], [Current loops] within the group [G8.2: Advanced config]). Check the project specifications as there may be some special parameterisation for these loops. **INVG3_ST1B**.

Incorrect measurement of hall current transformers in one of the phases.

3. It is possible that a current sensor has some kind of problem and does not read correctly on a permanent or punctual. In this case:
 - 3.1. Download the file corresponding to the Blackbox (group [G12.2: Blackbox]) and analyse it to check if the overcurrent is generalised or if it is present only in one phase **INVG3_ST1B**. If it is only one phase, check which phase is affected, remove the power module and visually check the status and connections of the faulty phase sensor. **INVKG3_ST3B**, **INVMG3_ST3B**. (For the removal of the power module see document *INVG3IOP85369*).



Connection detail: Clamp (+) - DC(-)



Clamp (-) – L1

EN



Clamp (-) – L2



Clamp (-) – L3



Connection detail: Clamp (-) – DC(+)



Clamp (+) – L1



Clamp (+) – L2



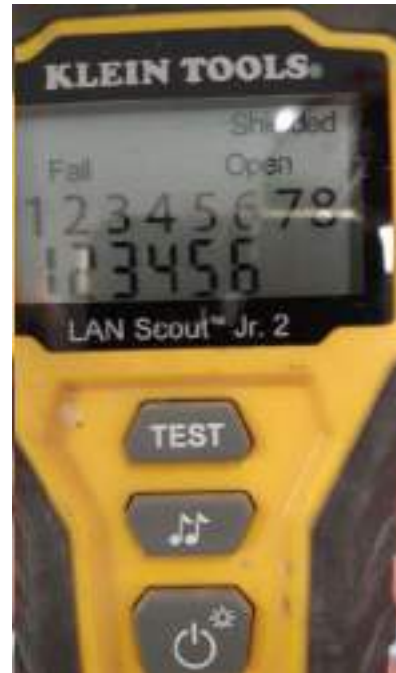
Clamp (+) – L3

Check continuity using a RJ45 tester of cables W030, W031 and W032, from connector U1 on the HALL board (E04, E05, E06) to connector U53, U54 and U55 on module board E02. If there is no continuity, replace RJ45 cable.

If the cable is in good condition, replace the HALL (HT02, HT03, HT04) with the board (E04, E05, E06). (Refer to document *INV3IOP88076*)



RJ45 tester OK



RJ45 tester NOK



RJ45 to check



Receiver RJ45 Tester

IGBT is damaged.

4. One of the IGBTs in the module is short-circuited, which has caused the module not to be able to cut off the current in time. If this occurs:
 - 4.1. Visualise the temperatures of the IGBTs in PowerCommsV to verify that there is no out-of-range or fixed temperature (203°C). To do this, consult the parameters corresponding to each of the phases R, S, T within the group [SV12.x.3: Temperature Measurements] of each module). **INVG3_ST1B**.
 - 4.2. Inspect the power module that has triggered the fault, measuring the diodes with 1500V multimeter of all the IGBTs individually and repair if necessary. **INVKG3_ST3B, INVMG3_ST3B**. (See document *INVG3IOP88076*).



Diode 1 measure



Diode 2 measure



Expected value

EN

5.27. Fault code F25: Low Vdc

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The inverter has read a DC voltage lower than the rectified mains voltage.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	It can be preceded by desaturations of IGBT. F95: Source PCB. F65: Overcurrent AC. F106: Mod. Vdc unbalanced. F108: Mod. Low Vdc. F69: Overvoltage DC+. F70: Overvoltage DC-.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Misconfiguration of the failure.

1. Check that the fault threshold (parameter [Low DSP Vdc] of group [G5.10: Internal protections]) is at its default value. If it is not, consult with the customer why it is at that value and if there is no justified cause, return to its default value by PowerCommsV. **INVG3_ST1B**.

DC bus short-circuit.

2. The module has a bus short-circuit and has caused a voltage drop which has caused the DC bus voltage to drop significantly. In this case:
 - 2.1. Check if any of the power modules or external buses of the inverter have any short circuit, hot spot, foreign body, point of the module with presence of water or humidity. **INVKG3_ST5A**, **INVMG3_ST5A**.



presence of water in the module area

EN

- 2.1.1. If the short circuit is in a power module, check the damaged stage by checking all the IGBT by measuring the voltage drop of each diode and replace the damaged module(s) or IGBT. **INVKG3_ST3B**. **INVMG3_ST3B**. (See document *INVG3/OP88076*).



Connection detail: Clamp (+) - DC(-)



Clamp (-) – L1



Clamp (-) – L2



Clamp (-) – L3



Connection detail: Clamp (-) – DC(+)



Clamp (+) – L1

EN



Clamp (+) – L2



Clamp (+) – L3

- 2.1.2. If it is in another part of the DU, repair the part that is causing the short circuit and check if the stages have been damaged during the event, checking each of the IGBT by measuring the voltage drop of each of the diodes. **INVKG3_ST5A, INVKG3_ST5A**. (See document *INVG3/OP88076*).
- 2.1.3. If the short circuit is external to the equipment, contact the customer to solve the problem. Otherwise, check connections on the Bus Plus side and on the PV side of the DCDC converters (for inverters that include the Bus Plus option). **INVKG3_ST5A, INVKG3_ST5A**.

Power module disconnected from AC.

3. There is a bad AC connection of a module, this causes its DC voltage to be lower than the rectified mains voltage, when it is disconnected from DC.
 - 3.1. Check that the AC connecting plates are well connected visually. They must not have any hot spots, overheated screws, damaged insulating parts or elements. They must also have the corresponding tightening torque. If not, retighten or replace if damaged or overheated parts are present or present. **INVKG3_ST3B, INVMG3_ST3B**.



Screws AC side (Modules)

EN



Screws AC output (Main inductance)



AC busbar screws to LC Filter

EN

Network event.

4. The inverter has not been able to overcome the grid event, consuming all the power of the PV array and causing the DC bus voltage to drop in very low irradiance situations (HEM /K solar inverters).
 - 4.1. Check Blackbox of the modules and SD of the device by PowerCommsV. **INVG3_ST1B**.
 - 4.2. The DC connection of the module has spontaneously failed and caused its DC voltage to drop below the rectified grid voltage. Check the SD and Blackbox by PowerCommsV **INVG3_ST1B** to see if the DC circuit breaker has opened. If so, check the DC connection and repair as necessary. **INVKG3_ST3B, INVMG3_ST3B**.

Internal bus measurement disconnected or faulty (applies to solar inverters with optional Bus Plus). In this case:

5. Check that the E36 board is connected to the J1 and J2 connectors. Subsequently, check that the DC measurement of the internal bus is correctly displayed in the SD Log_2. **INVG3_ST1B**.



EN

FPGA module board (E02).

6. In case a module is not read correctly, replace the board. (Check document *INVG3/OP88849*)
INVKG3_ST3B, INVMG3_ST3B.

5.28. Fault code F48: Internal comms

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Internal communication problem between MCU and DSP.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	W183: Internal comms	

Analysis procedure

This fault may occur during the software update process. If so, this is to be expected. If it occurs continuously, it may be due to the following causes:

The version of the uP and the DSP is not compatible.

1. Check that the software versions of each device correspond to those of the software package installed in PowerCommV. To do this, see the "versions" section within the platform itself. [INVG3_ST1B](#). In case it does not correspond:
 - 1.1. Install the correct version. [INVG3_ST1B](#).

Loss of communication frames between the uP and DSP.

2. If the software has already been checked for correctness and still the fault continues to appear, then:
 - 2.1. Reinstall the uP and DSP. Check if the fault has disappeared. If possible, reinstall both versions manually with the available loaders. [INVKG3_ST3A](#), [INVMG3_ST3A](#).
 - 2.2. If fault still occurs, replace main central control board (E35). (Check document [INVG3/OP88849](#)) [INVKG3_ST5A](#), [INVMG3_ST5A](#).
 - 2.3. If the problem persists, contact Power Electronics.

EN

5.29. Fault code F130: Softcharge Feedback

Mandatory

The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "[Introduction](#)".

Description	Soft load feedback not received (discrepancy in soft load resistor feedback).			
Applicable to	☐	HEM	☒	PCSM/MPCSM
	☐	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	W130: Softcharge Feedback			

Analysis procedure

The following checks are performed based on the component codes of module 1. If the fault is in a different module, the following table shows the code of each component equivalent to each module.

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
Soft-load card	E60	E61	E62	E63
RJ45 cable	W013	W014	W015	W016
Contactors	KM10-KM11	KM12-KM13	KM14-KM15	KM16-KM17
Terminals	X25	X26	X27	X28

24Vdc does not reach the terminal block (X25) corresponding to the soft-load contactors:

1. Check that 24Vdc is supplied by means of a 1500V multimeter in Vdc mode, at terminals X25:5:1 and X25:1:1. If the voltage is not 24Vdc, with no power supply, ring the wiring between terminals X25:5:1 and X25:1:1 and the G01 power supply.
 - a. Using a 1500V multimeter in continuity mode, measure between terminal X25:5:1 and the wire connected to the pin (+) of the power supply G01.



Terminal X25:5:1



Pin (+) of the power supply G01

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- b. Using a 1500V multimeter in continuity mode, measure between terminal X25.1:1 and the cable connected to the pin (-) of the power supply G01



Terminal X25.1:1



Pin (-) of the power supply G01

- i. If there is no continuity, replace the cable. **INVKG3_ST5A, INVMG3_ST5A** with external power supply 24Vdc.



G01 without external power supply



G01 with external power supply

Soft-load contactor is damaged.

2. Verify that the contactors open and close correctly by measuring continuity with a 1500V multimeter between the input and output. To do this, supply its coil with an external 24 Vdc power supply. **INVKG3_ST5A, INVMG3_ST5A**
3. To supply power externally, disconnect the positive connected to the pin (+) of the power supply G01 and connect it to the positive of the external power supply. In the same way, disconnect the negative connected to pin (-) of the power supply G01 and connect it to the negative of the external power supply. In this way, terminal block X25 is supplied with 24 Vdc. **INVKG3_ST5A, INVMG3_ST5A**



G01 without external power supply



G01 with external power supply

4. Once the power supply has been supplied, the order to close contactors KM10 and KM11 must be sent. To do this, pins 1 and 2 of connector J7 on the E60 soft-load card must be jumpered. **INVKG3_ST5A, INVMG3_ST5A**
5. Once powered and with the close command, check that they open and close, using a 1500V multimeter in continuity mode, measure between pins A1 and A2 of contactors KM10 and KM11. **INVKG3_ST5A, INVMG3_ST5A**



6. In the event that they do not open and/or close:
 - 6.1. Check if 24Vdc power is supplied when the soft load is activated by measuring with a 1500V multimeter in Vdc mode at points X25:1:3 and X25:1:4 for the KM10 contactor and points X25:2:3 and X25:2:4 for the KM11 contactor. The reading should be 24Vdc. **INVKG3_ST5A, INVMG3_ST5A**

EN



If any of them do not open and close correctly and 24Vdc power is supplied when the soft load is activated, replace the contactor. **INVKG3_ST5A**, **INVMG3_ST5A** with external 24Vdc power supply. **INVKG3_ST5A**, **INVMG3_ST5A**

EN



G01 without external power supply



G01 with external power supply

The soft-load digital input/output board (E60) or the contactor activation circuit of terminal block X25 is damaged or incorrectly connected.

Once the contactors have been checked for correct operation, relay R3 on the E60 soft-load board must be checked.

Disconnecting the wires from connector J7 of the E60 soft charger board, and jumpering pins 1 and 2 of connector J3 of the E60 soft charger board, measure with a 1500V multimeter in continuity mode between pins 1 and 2 of connector J7.

If there is no continuity, the E60 soft charge board must be replaced. (Refer to document *INVG3IOP88849*)

INVKG3_ST5A, **INVMG3_ST5A**

Feedback circuit in bad condition.

7. Feedback circuit is damaged or incorrectly connected. In this case:
8. Check using a 1500V multimeter in continuity mode. To do this, with contactors KM10 and KM11 closed, measure continuity between pins 5 and 6 of connector J4 on the E60 soft-load board.

If there is no continuity, with the contactors closed, measure with a 1500V multimeter in continuity mode between pins 1 and 2 of contactors KM10 and KM11. If there is no continuity, replace the contactors.

Check that the connections to the E60 board are correct:



Connection from pin 5 of connector J4 to X25:3:2 and from this to pin 1 of contactor KM10.
 Connection from pin 6 of connector J4 to X25:4:2 and from this to pin 1 of the KM11 contactor.
 If not, wire correctly.

INVKG3_ST5A, INVMG3_ST5A.

EN

RJ45 cable damaged.

9. Disconnect the RJ45 cable connected to J2 of the E60 soft load I/O board and check continuity with a RJ45 tester on all pins of the RJ45 cable (W013) connecting J2 of the soft load I/O board to J11 of the module PCB (E02). **INVKG3_ST2, INVMG3_ST5A.**

If there is no continuity on any of the pins, replace the RJ45 cable.

FPGA module board (E02) damaged in feedback signal.

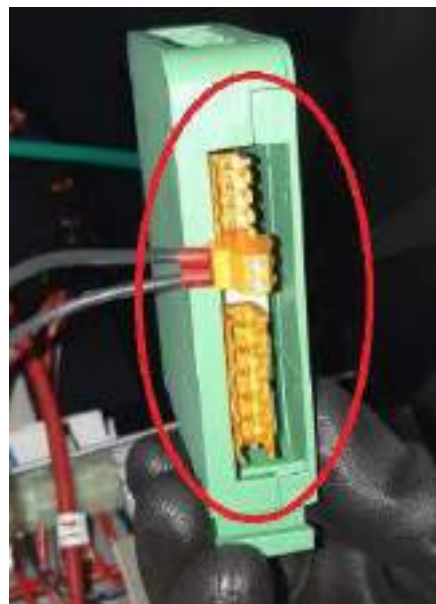
10. Activate the feedback with an external input/output card (**INVKG3_ST2, INVMG3_ST5A**). To do this, connect an external input and output card (same model as the E60), making a jumper between pins 5 and 6 of connector J4 of the external card, and connecting via an RJ45 cable from J2 of the external input and output card to J13 of the E02 module FPGA.

Verify by PowerCommsV (SV12: module inputs and outputs) that the soft load feedback is activated. **INVG3_ST1A**

11. If the feedback is not activated, replace the module FPGA board (E02). (Refer to document *INVKG3IOP88076*) **INVKG3_ST2, INVMG3_ST5A**

FPGA module board (E02) damaged

12. If all cables are in good condition, verify in PowerCommsV the signals corresponding to the soft load command (subgroup [FPGA module interf] within group [SV1.5: Digital I/O status]). **INVKG3_ST1A**. If not, replace the FPGA module board (E02). (Refer to document *INVKG3IOP88076*) **INVKG3_ST2, INVMG3_ST5A**.

EN

5.30. Fault code F58: Control SW mismatch

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	Incompatibility between MCU and DSP versions.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

The fault is due to the fact that the software version of uP and DSP are not compatible. In this case:

1. Check that the software versions of each device correspond to those of the software package installed in PowerCommV. To do this, see the "versions" section within the platform itself. **INVG3_ST1B**. In case it does not correspond to the software version:
 - 1.1. Install the correct version remotely. **INVG3_ST1B**.
2. If the inverter still fails, load the software package manually by connecting directly to the uP and DSP devices. **INVKG3_ST3A, INVMG3_ST3A**.
3. If the fault persists, contact Power Electronics.

EN

5.31. Fault code F13: No modules

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	All modules have stopped for some reason while the inverter was running.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

All modules have reported a warning or fault code. In this case:

1. Review via PowerCommsV the module fault and warning history (groups [SV7.1: Faults], [SV7.2: Warnings] and [[SV7.4: Module events](#)]) to detect the fault occurred **INVG3_ST1B**.

Note: Please note that before F13 is activated, other fault(s) shall be activated beforehand. The actions corresponding to the faults should be consulted throughout this document.

In the case of PCS inverters, the customer may have disabled all modules. If this is the case:

2. Consult with the customer about the reason for the disabling and proceed to re-enable the modules (parameter [Enable Bubars] in group [G2.8: Busbars Configuration]). Subsequently, reset the fault, if necessary, via PowerCommsV. **INVG3_ST1B**

EN

5.32. Fault code F83: Multiple desaturation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Several of the IGBT have reported desaturation.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	It can be preceded by any of these faults: F71-F82 y F119-F124: Desaturations F95: Source PCB	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Misconfiguration. The modules can give sporadic desaturations at operating points with a low DC bus voltage, which is lower than the rectified mains voltage ($V_{AC} \cdot 1.41$) + 50V. If this occurs:

1. Set the following parameter correctly ([Min Vdc] within group [G2.2-Stop]) y [Min Vdc] within group [G4.1-MPPT] according to the available DC power in order not to affect the DC bus voltage of the inverter.
2. Consult with the customer the standard parameters of the inverter. In addition, check if the current parameters of the inverter are correctly configured. **INVG3_ST1B**.
3. Consult Power Electronics in case the configuration needs to be adjusted.

One or more power stage IGBT are damaged.

4. Check the condition of the IGBT and if any are damaged, replace them. **INVMG3_ST3B**, **INVKG3_ST3B**. (See document *INVG3IOP88076*).





Connection detail: Clamp (+) - DC(-)



Clamp (-) – L1



Clamp (-) – L2



Clamp (-) – L3



Connection detail: Clamp (-) – DC(+)



Clamp (+) – L1



Clamp (+) – L2

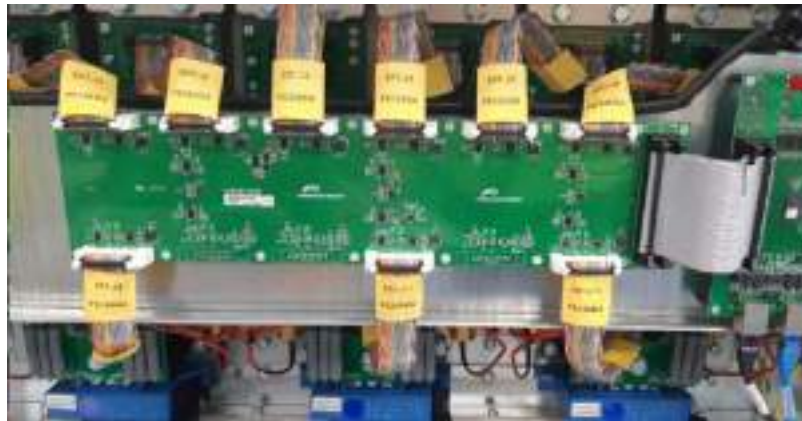


Clamp (+) – L3

Damaged flat cables or touching the snubbers capacitors.

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5. Check the condition of the flat cables and that they are properly connected to the E01 board and IGBT01-09 GATE DRIVE board. If they are not or if any of them is in contact with any of the snubber capacitors, replace the cable. **INVMG3_ST3B, INVKG3_ST3B**. (See document *INVG3IOP88076*).



6. If the fault continues and it is not due to any of the causes described above, contact Power Electronics to replace the power stage. (See document *INVG3IOP85369*).

In case there is no possibility to make the replacement, follow the next instructions:

7. Disassemble the power stage. See document *INVG3IOP85369*.
8. Connect the power supply and the optical fibers of the power stage to the power supply and the optical fibers of the inverter.
9. Supply with 24Vdc the output of the G01 and G02 power sources of the inverter to supply all boards.



10. Replace some components of the power stage gradually until the fault disappears. To replace a component, it is necessary to remove the 24Vdc power supply beforehand to avoid damaging the power stage.
11. The order of replacement should be as follows:
 - Flat cables.



- E01 board.



- IGBT01-09 GATE DRIVE board.



- E02 module board.

**EN**

5.33. Fault code F84: Communications

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	There is a communication frame lost between the FPGA of the main control board and the FPGA of the module board.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

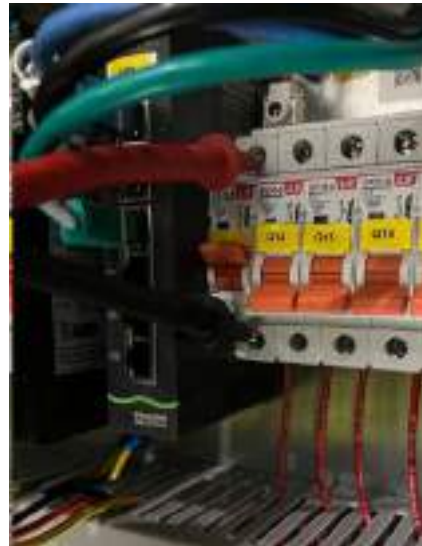
There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

FPGA module board (E02) power supply fault.

1. Check the 24Vdc power supply of the FPGA module board. Verify the correct polarity and the condition of the power supply connector. To do this, disconnect connector J15, and check with a 1500V multimeter in Vdc mode that the measurement is +24 Vdc between pins J15:2 and J15:1, bearing in mind that pin J15:2 corresponds to 24 Vdc and pin J15:1 to GND. If there is no power supply, check the circuit breaker Q13. To do this, check with a 1500V multimeter in continuity mode between pins 1 and 2 of Q13 with the circuit breaker closed. If the circuit breaker Q13 is correctly connected, use a 1500V multimeter in continuity mode to check that there is continuity between pin 2 of circuit breaker Q13 and connector J15:2 of the E02 board, and the cable that connects pin 1 of circuit breaker Q13 to pin (+) of the G01 power supply. To check the negative, continuity must be measured with 1500V multimeter in the same way between pin J15:1 and X11:1:3, and between X11:1:1 and pin (-) of power supply G01. **INVMG3_ST3A, INVKG3_ST3A.**



J15:2 – J15:1



Q13:1 – Q13:2

EN



E02:J15:2



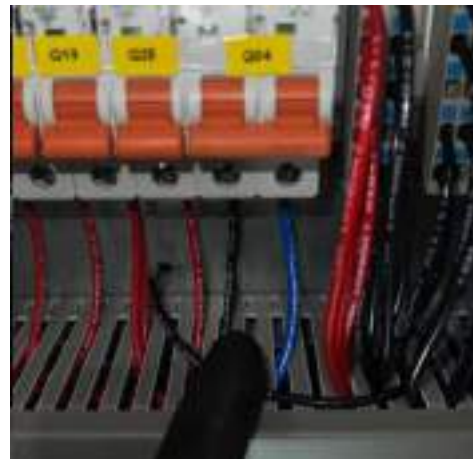
Q13:2



Q13:1 - G01(+)



J15:1



X11:1:3

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X11:1:1 – G01(-)

2. If there is continuity at all points replace the E02 module FPGA, otherwise the fault will persist. **INVMG3_ST5A, INVKG3_ST2.**

Damaged optical fiber cables.

3. Check that the optical fibers are not excessively bent or damaged (without cuts or burnt areas). Replace the damaged optical fiber. **INVMG3_ST5A, INVKG3_ST2.**



OPTIC FIBER OK



OPTIC FIBER NOK

4. Check in the optical fibers connector that they are aligned with the connector and well-polished. If they are not, replace the connector or the optical fiber (depending on availability). **INVMG3_ST5A, INVKG3_ST2.**

EN



OPTIC FIBER OK



OPTIC FIBER NOK: Reversed connector

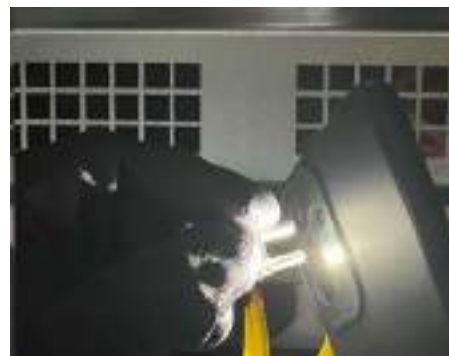


OPTIC FIBER OK: Flush burr-free connector

5. Check that the light is transmitted correctly from one optical fiber to another. Shine a light source, either natural or artificial, from one end of the optical fiber and check that light is received at the other end. If not, replace the damaged optical fiber (depending on availability). **INVMG3_ST5A, INVKG3_ST2.**



OPTIC FIBER OK: Light



Artificial light testing of optic fiber



OPTIC FIBER NOK: No light

6. If the fault still appears from time to time, check if there are losses (dB) in the communication between the FPGA of the E02 module board and the FPGA of the E35 central control board. Check in PowerCommsV [(FPGA Central-Module) within group [SV6.2-Central Board]] if the percentage of good frames in the communication is very lower than 100%, INVKG3_ST1B, in this case:
 - 6.1. Exchange the optical fiber connection of an E02 module board in this state for another optical fiber connection of an E02 module board with good optical connection. INVMG3_ST5A, INVKG3_ST2.

EN



- 6.2. If the percentage is around 100% and the fault no longer appears, **INVG3_ST1B**, replace the optical fiber between the E02 module board and the E35 central control board, **INVMG3_ST5A**, **INVKG3_ST2**. If not, check the following point.

[illegible]

Damaged optical fiber transducers.

7. Exchange the optical fiber connection of a failed E02 module board with the E35 central control board for a properly E02 module board. **INVMG3_ST5A, INVKG3_ST2.**



8. Check in PowerCommsV [(FPGA Central-Module] within group [SV6.2-Central Board]) **INVG3_ST1B**, if the fault still appears in the E02 module board, in this case replace the E35 central control board. **INVMG3_ST5A, INVKG3_ST2.**
9. If fault the E02 module board that has been exchanged faults, replace the E02 module board that initially failed. **INVMG3_ST5A, INVKG3_ST2.**

NOTE: Do not start or supply power with the optical fiber connections of the E02 module boards exchanged. Return all connections to their default state before starting.

EN

5.34. Fault code F71-F82: Desaturation (*outers*)

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.	
Description	A desaturation has been triggered in one of the IGBT (<i>outers</i>).	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

The following table lists the fault codes with the components to be checked during the following analysis procedure. If during the procedure it is determined that an IGBT needs to be replaced. (See document *INVG3IOP88076*).

Fault code	IGBT to be checked
F71	R1(P)
F72	R1(N)
F73	S1(P)
F74	S1(N)
F75	T1(P)
F76	T1(N)
F77	R3(P)
F78	R3(N)
F79	S3(P)
F80	S3(N)
F81	T3(P)
F82	T3(N)



EN

Misconfiguration. The modules may give sporadic desaturations at operating points with a low DC bus voltage, which is lower than the mains rectified voltage ($VAC \cdot 1.41$) + 50V. If this occurs:

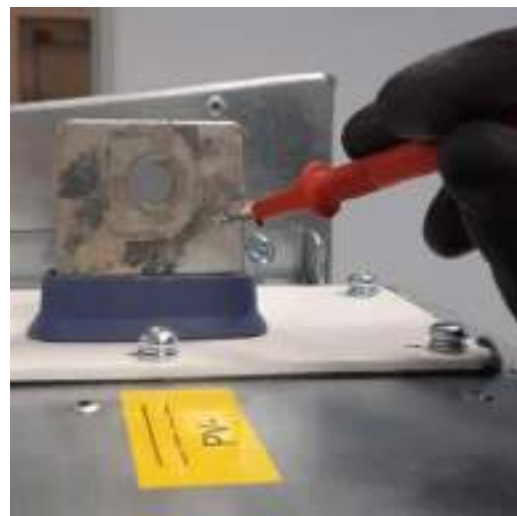
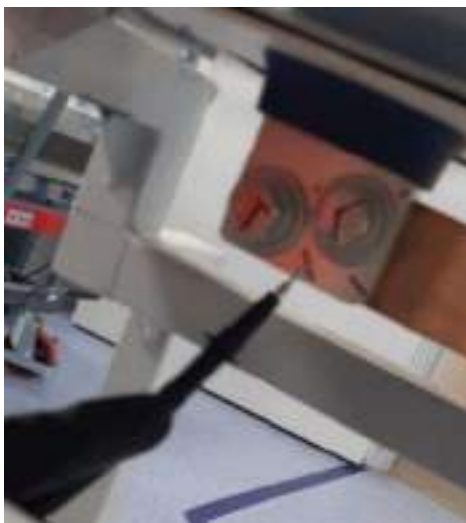
1. Adjust the minimum Vdc voltage and the minimum MPPT voltage (parameter ([Vdc mín] within group [G2.2-Stop]) y [Vdc min] within group [G4.1-MPPT]) must be set correctly according to the available DC power in order not to sink the DC bus voltage of the inverter. **INVKG3_ST3A**, **INVMG3_ST3A**.

One or more IGBT of the power stage are damaged.

2. Technically check with 1500V multimeter the IGBT and replace any damaged IGBTs. **INVKG3_ST2**, **INVMG3_ST5A**. (See document *INVG3IOP88076*).

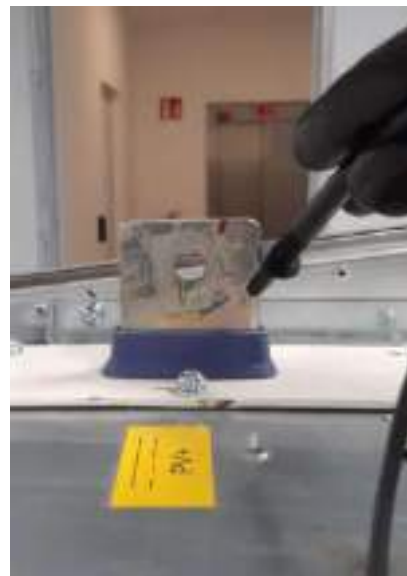


EN



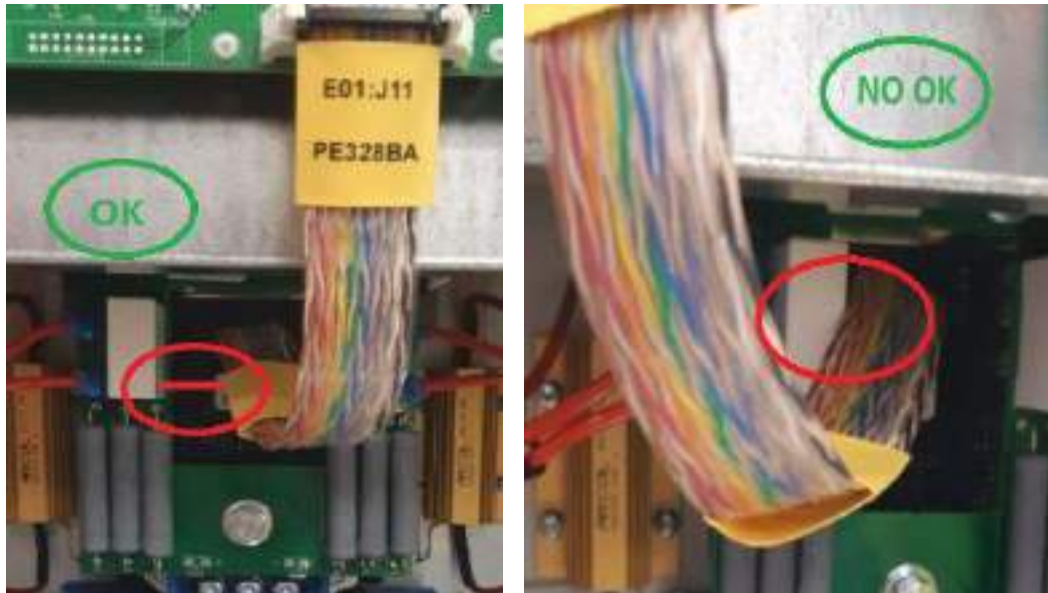


EN



Bad ribbon cables or touching Snubbers capacitors (see document HEMkPCSMkG3ECOID77720).
INVKG3_ST2, INVMG3_ST5A.

3. Check that all ribbon cables are in good condition and properly connected. If they are not in good condition or are touching any of the snubber capacitors, replace/repair the lead(s). INVKG3_ST2, INVMG3_ST5A.



EN

4. If the fault persists and is not due to any of the causes described above, contact Power Electronics to replace the power stage (see document INVKG3IOP85369). If replacement is not possible, the following procedure must be carried out:
 - 4.1. Disassemble the power stage, for this purpose refer to document INVKG3IOP85369. INVKG3_ST2, INVMG3_ST5A
 - 4.2. Connect the power supply and the fibres of the power stage to the power supply and the fibres of the inverter. INVKG3_ST2, INVMG3_ST5A
 - 4.3. Supply 24Vdc to the output of sources G01 and G02 of the inverter in order to have all the inverter cards powered. To do this, disconnect the cable connected to the (+) terminal of power supply G01 and connect it to the positive of the external power supply, and disconnect the cable from the (-) terminal and connect it to the negative of the external power supply. INVKG3_ST2, INVMG3_ST5A Carry out the same process on power supply G02. INVKG3_ST2, INVMG3_ST5A



G01 without external power supply



G01 with external power supply



G02 without external power supply



G02 with external power supply

- 4.4. Proceed to change one component after another in the power stage until the fault is solved. To change a component, it is necessary to remove the 24Vdc power supply beforehand so as not to damage the power stage. [INVKG3_ST2](#), [INVMG3_ST5A](#)
- 4.5. The order to follow must be as follows: [INVKG3_ST2](#), [INVMG3_ST5A](#)
- Flat cables.
 - E01 distribution board.
 - GATE DRIVE board (IGBT01-09).
 - Module control board E02.

EN

5.35. Fault code F119-F124: Desaturation Inners

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting.	
	Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction”.	
Descripción	A desaturation has been triggered in one of the IGBT (inners).	
Aplicable a	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Fallos o warnings asociados	N/A.	

Analysis procedure

The following table lists the fault codes with the components to be checked during the following analysis procedure. If during the procedure it is determined that an IGBT needs to be replaced. (See document INVG3IOP88076).

Fault code	IGBT to be checked
F119	R2(P)
F120	R2(N)
F121	S2(P)
F122	S2(N)
F123	T2(P)
F124	T2(N)



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Misconfiguration. The modules may give sporadic desaturations at operating points with a low DC bus voltage, which is lower than the mains rectified voltage ($V_{AC} \cdot 1.41$) + 50V. If this occurs:

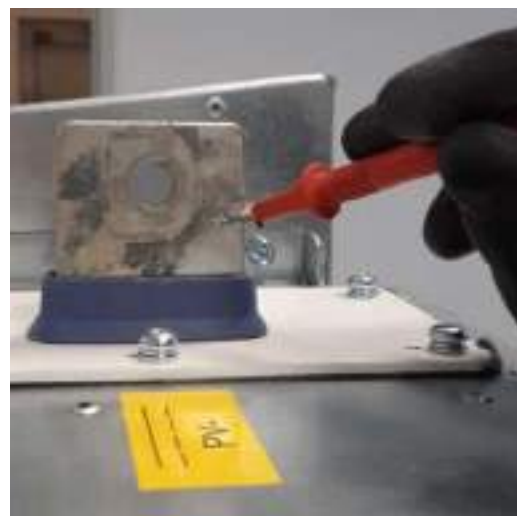
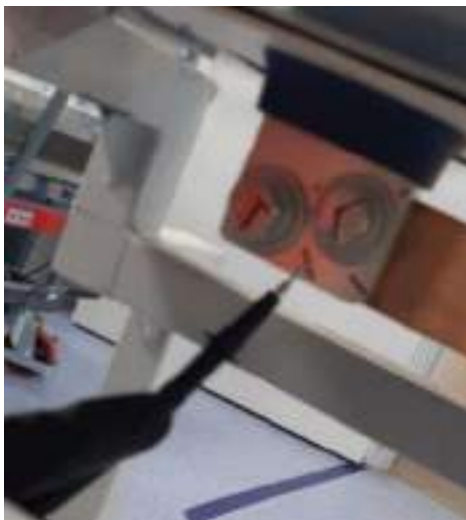
1. Adjust the minimum Vdc voltage and the minimum MPPT voltage (parameter ([Vdc mín] within group [G2.2-Stop]) y [Vdc min] within group [G4.1-MPPT]) must be set correctly according to the available DC power in order not to sink the DC bus voltage of the inverter. By PowerCommsV **INVKG3_ST3A**, **INVMG3_ST3A**

One or more IGBT of the power stage are damaged.

2. Technically check with 1500V multimeter the IGBT and replace any damaged IGBTs. **INVKG3_ST2**, **INVMG3_ST5A**. (See document *INVG3IOP88076*).

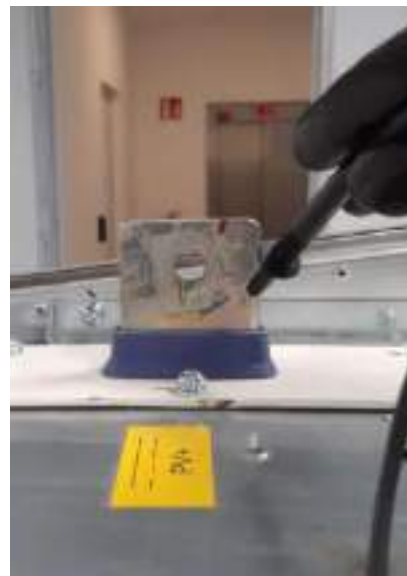


EN



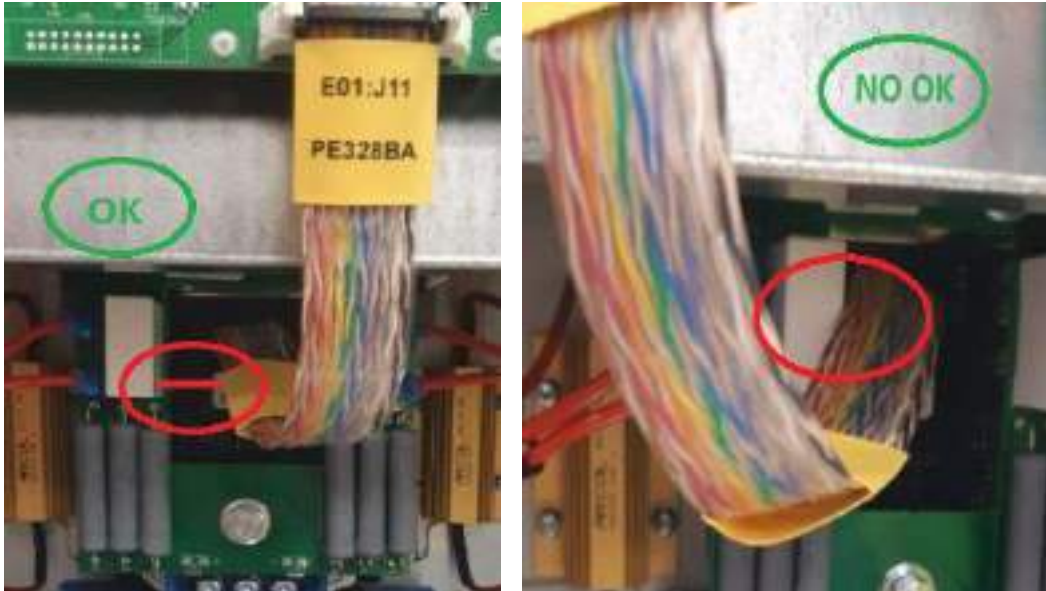


EN



Bad ribbon cables or touching Snubbers capacitors (see document HEMkPCSMkG3ECOID77720).
INVKG3_ST2, INVMG3_ST5A.

3. Check that all ribbon cables are in good condition and properly connected. If they are not in good condition or are touching any of the snubber capacitors, replace/repair the lead(s). INVKG3_ST2, INVMG3_ST5A.



EN

4. If the fault persists and is not due to any of the causes described above, contact Power Electronics to replace the power stage (see document INVG3/OP85369) INVKG3_ST2, INVMG3_ST5A. If replacement is not possible, the following procedure must be carried out:
 - 4.1. Disassemble the power stage, for this purpose refer to document INVG3IOP85369. INVKG3_ST2, INVMG3_ST5A
 - 4.2. Connect the power supply and the fibres of the power stage to the power supply and the fibres of the inverter. INVKG3_ST2, INVMG3_ST5A
 - 4.3. Supply 24Vdc to the output of sources G01 and G02 of the inverter in order to have all the inverter cards powered. To do this, disconnect the cable connected to the (+) terminal of power supply G01 and connect it to the positive of the external power supply, and disconnect the cable from the (-) terminal and connect it to the negative of the external power supply. Carry out the same process on power supply G02. INVKG3_ST2, INVMG3_ST5A



G01 without external power supply



G01 with external power supply



G02 without external power supply



G02 with external power supply

- 4.4. Proceed to change one component after another in the power stage until the fault is solved. To change a component, it is necessary to remove the 24Vdc power supply beforehand so as not to damage the power stage. [INVKG3_ST2](#), [INVMG3_ST5A](#)
- 4.5. The order to follow must be as follows: [INVKG3_ST2](#), [INVMG3_ST5A](#)
- Flat cables.
 - E01 distribution board.
 - GATE DRIVE board (IGBT01-09).
 - Module control board E02.

5.36. Fault code F5-F8: Grid failures

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting.	
	Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	A desaturation has been triggered in one of the IGBT (inners).	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK

Associated faults or warnings	May be related to: F11: Active Anti-island F28: Passive Anti-island
--------------------------------------	---

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

The inverter's nominal frequency or nominal voltage setting is not correct. If the device triggers this fault often without explanation, it is possible that the protection is too close to the instantaneous grid voltage and frequency values due to a wrong setting of its nominal values in the Drive Select (SG10 group). If this occurs:

1. Check that the nominal frequency (SG10.1.7-Nominal Frequency) and nominal voltage (SG10.1.6-Nominal Vac) match the inverter frame and country regulations. **INVG3_ST1B.**

The thresholds and times of the protections are not very permissive. If the thresholds are very close to the nominal voltage and frequency and the times are low, this will lead to grid faults with a certain frequency. In this case:

2. Check by PowerCommsV that the configuration of group G5: Protections (subgroups [Low Vac], [High V ac], [Low F ac] and [High F ac]) is in accordance with the default country standard. **INVG3_ST1B.**

The mains voltage or frequency measurement is not correct. In case the Vac measurement or frequency calculation by the main control is not correct, frequent or sporadic mains failures may occur. If this happens:

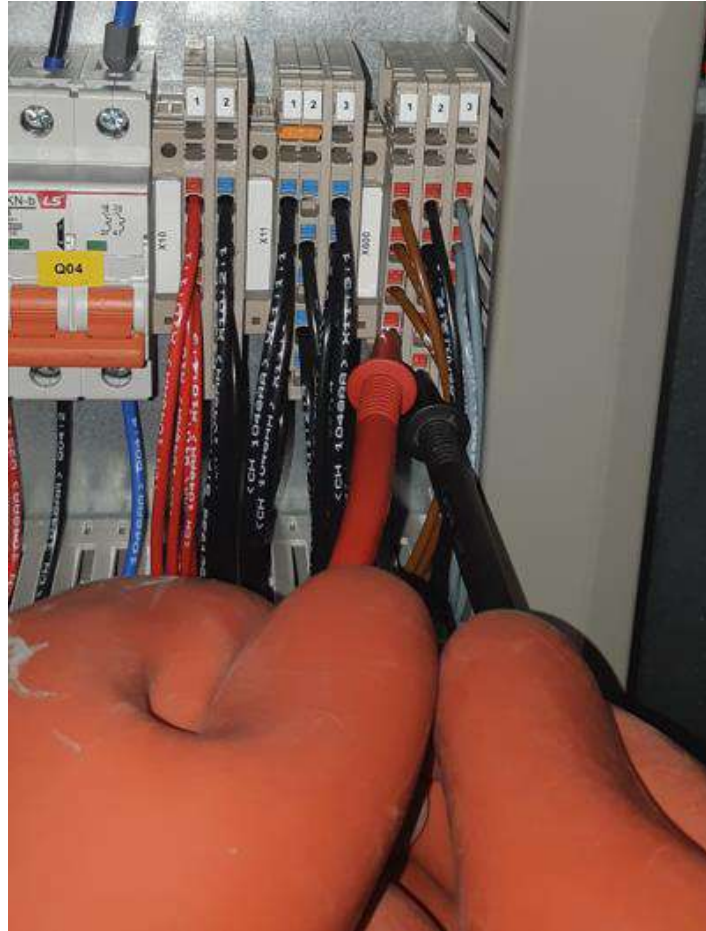
3. Analyse what the fault is. If they are sporadic frequency faults with inconsistent values, reload the DSP software and check that the software versions of the microphone and DSP match. **INVKG3_ST3A; INVMG3_ST3A.** If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88849*.

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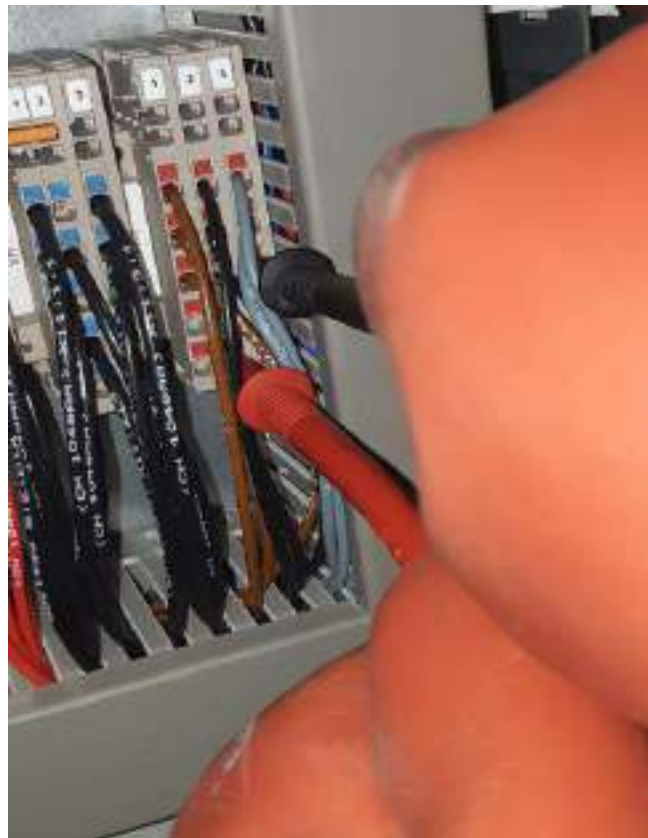
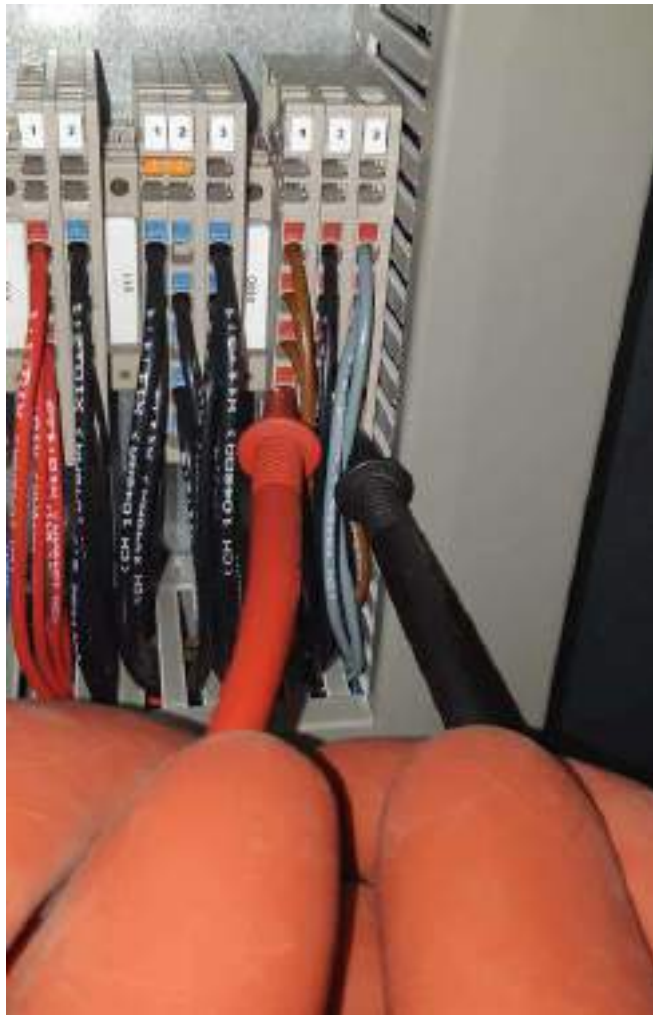
The frequency or mains voltage measurement is permanently incorrect.

4. Check that the voltage between phases (Vrs, Vst, Vtr) on the Vac voltage measurement terminal block on the control board is balanced. Check that its frequency is very close to the nominal frequency of the mains. To do this, use a 1500V multimeter set to AC.
- 4.1. Starting from the **INVMG3_ST6**, **INVKG3_ST6** status and using the electrical risk PPE, carry out the measurement on the terminal block:
 - 4.1.1. First insert one of the prongs and once the first one is inserted in the terminal block, proceed to insert the second one, taking the measurement at that moment.



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- 4.1.2. For removal, proceed in the same way: first remove one prong and once it is out, remove the second.



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5. If one phase is very unbalanced or the frequency has values very different from the mains frequency, check the wiring from the AC busbar to the control board. **INVKG3_ST2, INVMG3_ST5A.**





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6. If the fault persists, check that the low voltage Vac voltage is within the permitted range (normally $\pm 10\%$ [V] above the nominal VAC) and that the tapping of the transformer is correctly configured according to the project. **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B.**

Opening of the main disconnect in the high voltage part of the plant. In case of opening of the main disconnect of the plant, faults F5, F6, F7, F8 and to a lesser extent anti-islanding faults (F11, F28, if these protections are activated) will be observed. If this occurs:

7. Talk to the customer or the plant manager to find out the causes of the opening.

Opening of AC circuit breaker (only HEMK / PCSK). In this case:

8. Follow the instructions given in F107 for HEMK / PCSK inverters.

Fault in the IGBT or GATE DRIVE boards (IGBT01-09). If the grid faults have the following associated faults: F66, F106, F65, F112, F108, it is likely that some FRU stage is damaged. If this occurs:

9. Via PowercomsV extract the SD and Blackbox logs from the power stages. **INVG3_ST1B**. If abnormal currents are seen in any module after analysis, inspect the stage for damage, broken IGBTs, short circuits or defects *in the GATE DRIVE boards*. **INVKG3_ST3B, INVMG3_ST3B**. Refer to document INVG3IOP85369 for power stage removal.

There has been a plant disconnection from the grid, but the inverter has not triggered a grid fault and has remained in operation. This situation is dangerous because an island has been generated and the inverter has not recognised it or has blocked the protections that prevent it from failing in this situation. In this case:

10. Check the configuration of the protections (subgroups [Low Vac], [High V ac], [Low F ac] and [High F ac] of group G5: Protections), check that they are activated and with the appropriate thresholds. **INVG3_ST1B**.
11. Check that [group SG4: Test mode] is configured with the default parameters. That is to say, that there is no parameter that bypasses the mains protections. **INVG3_ST1B**.

5.37. Fault code: F14: Drive-Select DSP and F44: Drive-Select MCU

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Wrong Drive Select group configuration. The mainboard does not have the Drive Select programmed properly. Drive select parameters are out of range.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

The DSP misconfiguration fault occurs when, during the process of checking the Drive Select configuration by the inverter software, it is not consistent. Follow the steps below to identify the possible cause and resolve it.

The installed software version is not correct. If this happens:

4. Check that the software versions of each device (DSP, uP/MCU) correspond to those of the software package installed in PowerCommV. To do this, see the "versions" section in the platform itself. **INVG3_ST1B**. In case it does not correspond to the software version:

- 1.1. Install the correct version. **INVG3_ST1B**.

In case it does not correspond to the software version:

2. Check in PowerCommsV that the parameters configured in Drive Select (group [SG10:Drive select]) are within the set range. **INVG3_ST1B**.

Parameter initialisation has not been performed correctly. If this occurs, perform the following steps by PowerCommsV. **INVG3_ST1B**.

3. Access the Drive Select (group [SG10:Drive select]). Enter the password (parameter [Password], from group [SG10:Drive select]) and force default (parameter [Force defaults] from group [SG10:Drive select]).
3. Access the group [SG1: Service] and initialise the service group (parameter [Init service group]).
4. Access the general configuration group (G1: General) and initialise parameters (parameter [Init parameters]).

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5.38. Fault code F66: Module VDC Unbalance

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	There is a voltage unbalance between the DC bus half buses.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is reported when an unbalance appears between the module's half-buses, higher than the software configurable value. Follow the steps below to identify the possible cause and resolve the problem.

Incorrect unbalance protection setting

1. Verify from PowerCommsV that the inverter fault corresponds to F66 (group [SV7.4: Module events]). **INV3_ST1B**. If so, proceed to the next step.
2. Check via PowercommsV the configuration of the unbalance protection (parameters [Deseq. Semibus Mod] of subgroup [G5.2-Quality] and [K Vector Eq] of subgroup [FPGA Service]). In case of detecting modified values without a justification for that particular case, modify the parameters to their default values (consult previously with the plant manager to know if this modification is justified). **INV3_ST1B**.

The minimum voltage Vdc minimum is less than the rectified voltage ($V_{ac} \cdot \sqrt{2}$). It is possible that a greater bus unbalance may appear if the inverter is working with an insufficient DC bus voltage (lower than the rectified one). In this case, the control will not be able to balance the buses. If the equipment is operating under these conditions:

3. Make sure via PowercommsV that the Vdc is sufficient for the mains conditions. **INV3_ST1B**. For this purpose:
 - 3.1. Check that the Vdc voltage at which the equipment is working is at least the rectified voltage ($V_{ac} \cdot \sqrt{2}$). In other words, that the values shown in parameters [External voltage 1], [External voltage 2], [External voltage 3], [External voltage 4] of group [SV2: DC source] are at least the values shown in parameters [Grid voltage RS LV], [Grid voltage ST LV], [Grid voltage TR LV] of subgroup [SV3.2: Grid LV] multiplied by $\sqrt{2}$. **INV3_ST1B**. **Note:** The measurements shown in the parameters of group [SV2: DC source] will depend on the number of busbars in the inverter. For more information, refer to the *Software and Programming Manual* of the inverters.

Faulty balancing resistors. Before taking measurements on the module, ensure that there is no residual energy in the module.

4. Measure the module voltage with multimeter 1500V between +Vbus and -Vbus (BBIP.20 and BBIN.20) with the 1500V multimeter in DC measurement mode. **INV3_ST2**; **INV3_ST5A**.

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5. Measure on the balancing resistors of the module with 1500V multimeter that is failing due to unbalance (board E07). The value should be 100k Ω . **INVKG3_ST2; INVMG3_ST5A.**



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Deteriorated capacitors

6. Measure with 1500V multimeter the capacitance of the module's DC bus with the multimeter. The multimeter must be in capacitance mode. Measure between +Vbus and -Vbus. The measured value must be around 3080uF $\pm$ 10% [2772uF ...3388uF]. [INVKG3_ST2](#); [INVMG3_ST5A](#).



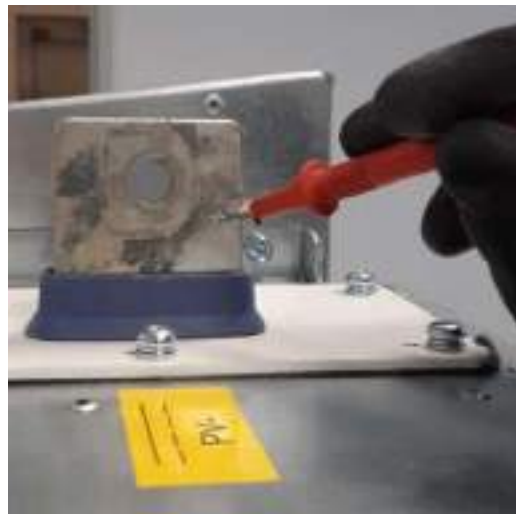
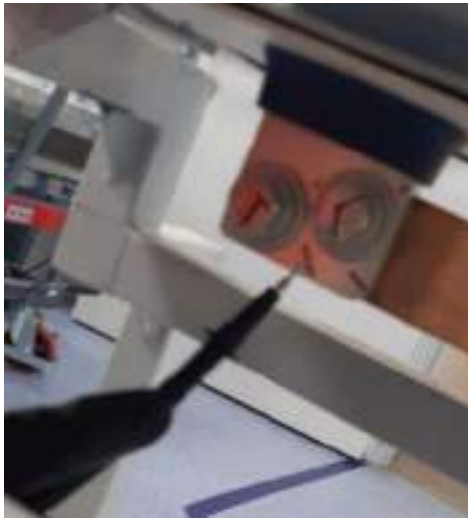
BUS_P - BUS_N

Power stage status

7. If the problem is not detected, check with 1500V multimeter the status of the power stage IGBTs. Refer to document [INVG3/OP88076](#). [INVKG3_ST2](#); [INVMG3_ST5A](#).

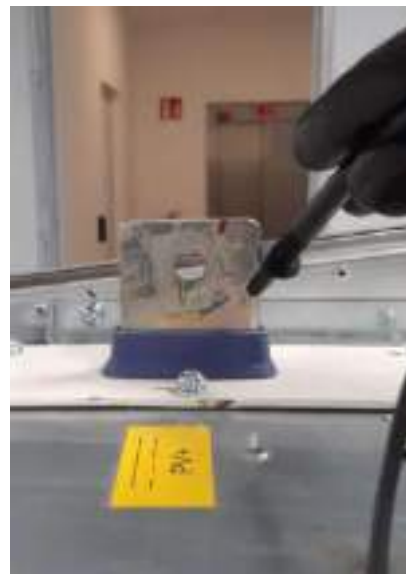


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5.39. Fault code F69 – F70: Overvoltage DC+/-

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.			
Description	An overvoltage appears on the positive side of the DC bus (F69) or on the negative side (F70).			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

Analysis procedure

The module reports F69 or F70, when it detects a voltage on one of the semibuses above 850V. Follow the steps below to identify the possible cause and resolve the problem.

Incorrect configuration of the overvoltage protection on the semibus

1. From PowerCommsV, verify that the inverter fault corresponds to F69/70. To do this, check in [SV7.1: Module events]. If so, proceed to the next step. **INV G3_ST1B**.
2. From PowerCommsV, check the configuration of the DC half bus overvoltage protection (parameter [V max bus medium] of subgroup [G5.10-Internal Prot]). In case of detecting modified values without a justification for that particular case, modify the parameters to their default values (previously consult with the plant manager to know if this modification is justified). **INV G3_ST1B**.

Incorrect configuration of the unbalance protection in the semi-buses.

3. Via PowercommsV, if the unbalance protections have been modified, the module will allow a greater unbalance between the buses, which will cause a greater overvoltage in the buses. **INV G3_ST1B**. To check this, verify that the following parameters are set to their default values:
 - Parameter [Deseq. Semibus Mod] of subgroup [G5.2-Quality].
 - Parameter [K Vector Eq] of subgroup [SG1.3-FPGA Service].
 - Parameter [Bus Desired Threshold] of subgroup [G5.10-Internal protections].

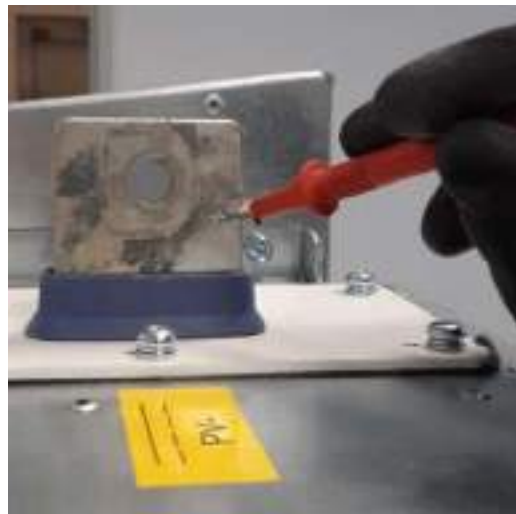
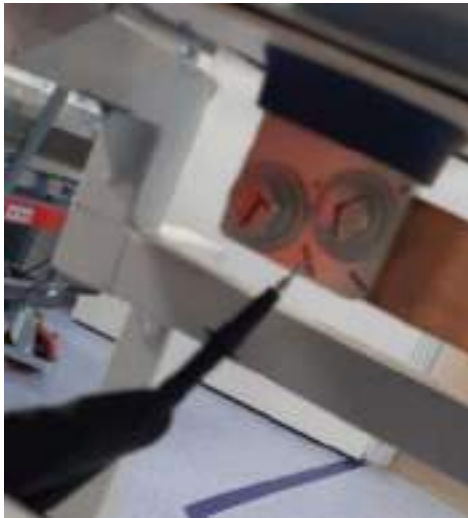
IGBT short-circuited.

4. In the event that one of the IGBT of a branch is short-circuited to the midpoint, the other half bus will see the full Vbus voltage. Perform the necessary measurements with a 1500V multimeter and check the status of the IGBT. **INVKG3_ST2, INVMG3_ST5A**. See document *INV G3/OP88076*.

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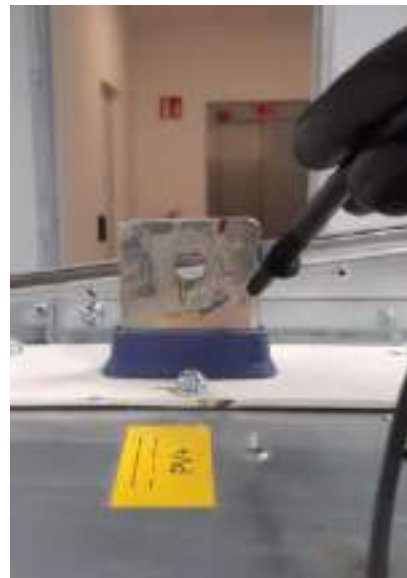


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5. If there is continuity, check it with a 1500V multimeter, between any of the half buses and the midpoint, the shorted IGBT needs to be replaced. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3IOP88076*.



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Start-up of the inverter with very high reactive power (Q).

6. If the inverter reports the fault at start-up, it is possible that it has a high Q reference (parameter [Cos phi reference] of subgroup [G8.1-References]) with very fast start-up gradient (parameter [Start] of subgroup [G4.2-Gradients]). **INV63_ST1B**. If this is the case, it will lead to bus unbalance and higher overvoltage on the DC bus. In this case contact Power Electronics to discuss the specific case.

5.40. Fault code F85: Timeout soft charge

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".			
Description	This fault shows if there is a module doing soft charging (module status: soft charging) and the voltage has not reached the set value within the defined time.			
Applicable to	☒ HEM	☒ PCSM/MPCSM		
	☒ HEMK	☒ PCSK/MPCSK		
Associated faults or warnings	Related to: F3: SoftCharge. F32: Error Mod selfdiagnosis.			

Analysis procedure

The timeout fault during soft charging of the module occurs when the module is not charged within a set time. The established failure times are identified as: not reaching a voltage of 50V in 5 seconds or not reaching the busbar voltage (parameters [Dc Voltage external 1], [Dc Voltage external 2], [Dc Voltage external 3], [Dc Voltage external 4] within the group [SV2: DC source]) in a time configured by the DSP depending on the type of inverter, these times being: PCS = 120s, Solar = 240s and BusPlus = 360s. In this case:

1. Analyse the SD logs to check what was the reason for the failure by PowerCommsV. **INV3_ST1B**. To do this:
 - 1.1. Analyse via PowercommsV in Log_2 the Vdc voltage on the modules at the moment when the soft load starts (MOD_STATUS=2). **INV3_ST1B**.
 - 1.2. If it is observed via PowercommsV that the Vdc voltage on the modules does not increase **INV3_ST1B**, checks must be made on the hardware of the inverter. In this case, continue with the following steps.

The verifications are carried out according to the component codes of module 1. If the fault is found in a different module, the codes vary as shown in the following table: **INVMG3_ST5A**, **INVKG3_ST2**.

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
Soft charge board	E60	E61	E62	E63
RJ45 cable	W013	W014	W015	W016
Contactors	KM10-KM11	KM12-KM13	KM14-KM14	KM16-KM17
Terminals	X25	X26	X27	X28
Soft-load resistors	R30-R31	R32-R33	R34-R35	R36-R37
Soft-load fuses	F101	F102	F103	F104
Soft-load positive terminals	BBIP.07	BBIP.05	BBIP.03	BBIP.01

Note that the checks to be performed on the soft-load circuit vary depending on the type of inverter.

In the case of solar inverters (HEM / K). The steps to be followed are as follows:

2. To carry out the checks at hardware level:

2.1. Check the wiring between the soft load board and the module PCB board (E02). **INVKG3_ST2, INVMG3_ST5A**. To do so:

2.1.1. Disconnect the RJ45 cable connected to J2 of the soft-load board (E60). Check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W013) connecting J2 of the soft load input/output board (E60) to J11 of the module PCB (E02). **INVKG3_ST2, INVMG3_ST5A**.



2.1.2. In case there is no continuity on any of the pins, replace the RJ45 cable. **INVKG3_ST2, INVMG3_ST5A**.

2.2. Check the soft-charging board (E60). **INVKG3_ST2, INVMG3_ST5A**. Check power supply to the card. **INVKG3_ST2, INVMG3_ST5A**. To do so:

2.2.1. Check continuity with 1500V multimeter on the positive from pin 1 of connector J1 of the soft charge board (E60) to terminal block X100. Use a multimeter at the following points:

- From pin 1 of connector J1 to X125:4:2.
- From X125:4:2 to X100:3:2.

2.2.2. Check continuity with 1500V multimeter on the negative from pin 3 of connector J1 on the soft charger board (E60) to terminal block X100. Use a multimeter at the following points:

- From pin 3 of connector J1 to X125:4:4.
- From X125:4:4 to X100:3:4.





Positive pole
From J1 pin 1 (E60) to X125:4:2



Negative pole
From J1 pin 3 (E60) to X125:4:4



Positive pole
From X125:4:2 up to X100:3:2



Negative pole
From X125:4:4 up to X100:3:4

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2.3. Check continuity with 1500V multimeter on bus+ and bus- of the soft charge board (E60). **INVKG3_ST2, INVMG3_ST5A**. To do so:

- 2.3.1. Use a multimeter in continuity mode to measure from pin J3 of the soft load board (E60) to the BBIP.07 board.
- 2.3.2. Use a multimeter in continuity mode to measure from pin J4 of the soft charge board (E60) to the BBIN.07 board.



Positive pole
From J3 pin (E60) to BBIP.07 plate





Negative pole
From J4 pin (E60) to BBIN.07 plate

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In battery inverters (PCSK / PCSM), the steps to follow are as follows:

3. To carry out the checks at hardware level, the power supply to the modules must be disconnected. **INVKG3_ST2, INVMG3_ST5A**. Once this step has been carried out:
 - 3.1. Proceed to perform the measurements on the soft load board (E60) by externally powering the soft load circuit. The wires connected to the (+) and (-) terminals must be disconnected from the power supply output (G01) and feed these wires with the external 24Vdc power supply. After this is done:



4. G01 without external power supply



G01 with external power supply



3.1.1. Check the wiring between the soft-load board and the module PCB. **INVKG3_ST2, INVMG3_ST5A**. To do this:

- Disconnect the RJ45 cable connected to J2 of the E60 soft load input/output board and check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W013) connecting J2 of the soft load input/output board to J11 of the module PCB (E02).



- If there is no continuity on any of the pins, replace the RJ45 cable. **INVKG3_ST2, INVMG3_ST5A**.

3.2. Check the soft load board (E60). To do so:

3.2.1. Jumper pins 1 and 2 of connector J7 of the soft charge board (E60). **INVKG3_ST2, INVMG3_ST5A**. Check that contactors KM10 and KM11 close.

3.2.2. Measure with a 1500V multimeter in continuity mode between pins A1 and A2 of both contactors. **INVKG3_ST2, INVMG3_ST5A**. If there is no continuity, replace the contactor. **INVKG3_ST2, INVMG3_ST5A**.



3.2.3. Check that the coils of contactors KM10 and KM11 are energised:

- Measure with a 1500V multimeter in Vdc mode between terminals X25:1:2 and X25:1:1 on terminal block X25. The measurement should be 24 Vdc. **INVKG3_ST3A, INVMG3_ST3A.**



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- If the measurement is not 24 Vdc, the wiring to the power supply (G01) must be checked. **INVKG3_ST2, INVMG3_ST5A.** In this case:
 - o Using a 1500V multimeter in continuity mode, measure between terminal X25:5:1 and the wire connected to pin (+) of power supply G01. **INVKG3_ST2, INVMG3_ST5A.**



- o Using a 1500V multimeter in continuity mode, measure between terminal X25:1:1 and the wire connected to pin (-) of power supply G01. **INVKG3_ST2, INVMG3_ST5A.**



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3.2.4. Having checked that the contactors are correct, relay R3 on the E60 soft-load board must be checked. To do this

- Disconnect the wires from connector J7 of the E60 soft-load board, and by jumpering pins 1 and 2 of connector J3 of the E60 soft-load board, measure using a 1500V multimeter in continuity mode between pins 1 and 2 of connector J7. **INVKG3_ST2, INVMG3_ST5A.**



- If there is no continuity, replace the E60 soft charge board. **INVKG3_ST2, INVMG3_ST5A.**

3.3. Check the soft load resistors (R30-R31).

- 3.3.1. Measure the value of resistor R30 and R31. **INVKG3_ST2, INVMG3_ST5A.** Use a 1500V multimeter in resistance mode between pin A2 of the KM10 contactor and the BBIP.07 board **INVKG3_ST2, INVMG3_ST5A.** This value should be $1200\Omega \pm 5\%$.



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3.4. Check fuse F101.

3.4.1. Remove the fuse and measure with a 1500V multimeter in continuity mode. This value should be 0.2 ohms. **INVKG3_ST2, INVMG3_ST5A**. If the value is different, replace the fuse.



5.41. Fault code F106: Mod. Vdc unbalanced

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Critical bus unbalance. This fault trips when the voltage unbalance exceeds the 350Vdc threshold for 64us (fixed thresholds).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F2: HW Vbus F66: Module VDC Unbalance F69: Overvoltage DC+ F70: Overvoltage DC-	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Bus balancing configuration.

1. Via PowercommsV check the configuration of the bus balancing algorithm control parameters (parameter [K Eq-Vectorial Eq] of subgroup [SG1.3: FPGA service]). If not default:
 - 1.1. Consult with the plant manager if there is a reason why this parameter is not at its default value. If there is no reason, return via PowercommsV to default value. **INVG3_ST1B.**

Working point with high reactive power.

2. Via PowercommsV check in the SD card (Log_1) the working point where the inverter is located. If it is a working point with a lot of reactive power (more than 60% of the equipment) and high gradients, reset the fault if the overvoltage is not maintained and inform Power Electronics. **INVG3_ST1B.**

Capacitor in bad condition.

3. Dismantle the power stage and check the capacity of the half buses. **INVKG3_ST3B, INVMG3_ST3B.** To do this, it is necessary to have a 1500V multimeter capable of measuring at least 7000uF.
 - 3.1. Check capacitance with the 1500V multimeter between points BBIP.20 and BBIM.20 and between points BBIM.20 and BBIN.20. In both cases the measured capacitance must be between 6776uF and 5544uF and there must not be more than a difference of 200uF between the two. If any of the checks are not correct, replace the capacitor bus (refer to document **INVG3/OP88076. INVKG3_ST3B, INVMG3_ST3B.**

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BUS/2 - BUS_P



BUS/2 - BUS_N

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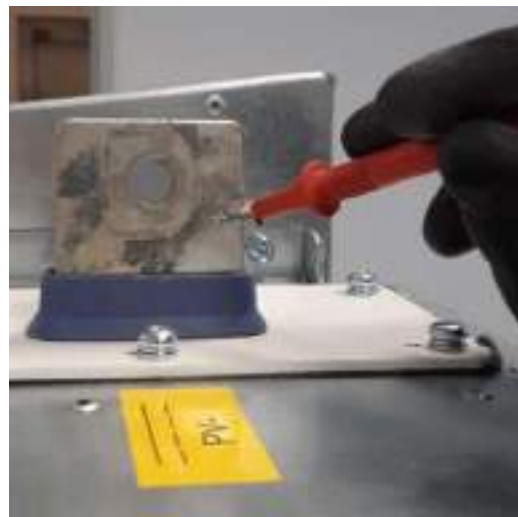
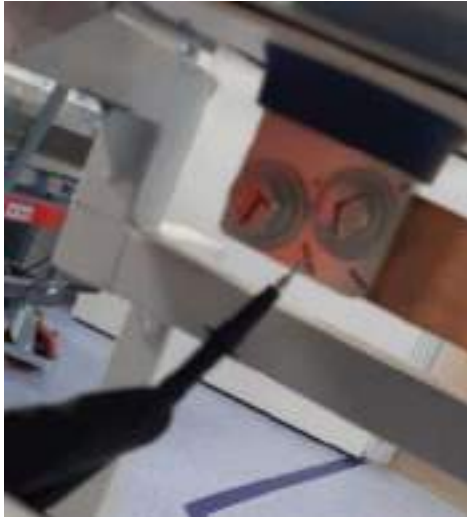


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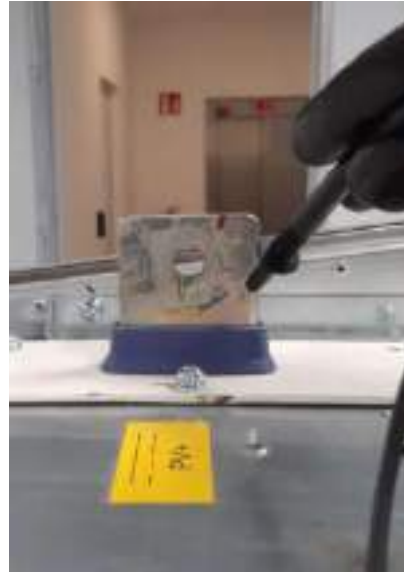
Damaged IGBT.

4. Inspect with 1500V multimeter the IGBTs and replace if any damaged IGBT are found. See document [INVKG3_ST3B](#), [INVMG3_ST3B](#).





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Balancing PCB board (E07).

5. Inspect the balancing PCB board (E07). Measure with 1500V multimeter the resistances of the board and check that they are 100k Ω . If not, replace the balancing board (E07) (see document *INVG3IOP88076*). [INVKG3_ST2](#), [INVMG3_ST5A](#).





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If the cause of the fault cannot be found, contact Power Electronics.

5.42. Fault code F108: Mod. High Vdc

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Critical overvoltage of a semibus. This fault is triggered if the module detects that one of the buses has exceeded 875Vdc.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F2: HW Vbus F66: Module VDC Unbalance F69: Overvoltage DC+ F70: Overvoltage DC- F106: : Mod. Vdc unbalanced	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Bus balancing configuration.

1. Check via PowercommsV the configuration of the bus balancing algorithm control parameters (parameter [K Eq-Vectorial Eq] of subgroup [SG1.3: FPGA service]). If not default:
 - 1.1. Consult with the plant manager if there is a reason why this parameter is not at its default value. If there is no reason, return to default value. **INVG3_ST1B.**

Vdc control parameters are not at their default value.

2. Check via PowercommsV that the parameters in group [G8.2: Advanced Settings] are at their default value. If this is not the case, check with the plant manager if there is a reason for this change. If not, return to default value. **INVG3_ST1B.**

Working point with high reactive power.

3. Check via PowercommsV in the SD card (Log_1) the working point where the invertir is located. If it is a working point with high reactive power (more than 60% of the inverter) and high gradients, reset the fault if the overvoltage is not maintained and inform Power Electronics. **INVG3_ST1B.**

Improper battery or DCDC operation.

4. Check via PowercommsV SD card (Log_4 and Log_1) and check the following items: **INVG3_ST1B.**
 - 4.1. If the fault occurred at shutdown and the equipment is BUSPLUS, check via PowercommsV if the DCDC was running when the inverter tried to shutdown ([Current Status] parameter of group [SV1.1: Status], see *DCDC Software and Programming Manual*). **INVG3_ST1B.** If so, inform the customer that this is an improper manoeuvre and that they must always stop the DCDC before stopping the inverter.
 - 4.2. If the failure has occurred while charging batteries (negative power), check via PowercommsV if the batteries have been opened while the equipment was running (customer SCADA). **INVG3_ST1B.** If so, talk to the customer about the reason for opening the battery and inform that it is an improper manoeuvre.

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Capacitor in bad condition.

5. Dismantle the power stage and check the capacity of the half-buses (see document *INVG3IOP85369*). **INVKG3_ST3B**, **INVMG3_ST3B**. For this purpose, it is necessary to have a 1500V multimeter capable of measuring at least 7000uF.
- 5.1. Check capacitance with the 1500V multimeter between points BBIP.20 and BBIM.20, and between points BBIM.20 and BBIN.20. In both cases the measured capacitance must be between 6776uF and 5544uF, and there must not be more than a difference of 200uF between the two. If any of the checks are not correct, replace the capacitor bus (see document *INVG3IOP88076*). **INVKG3_ST3B**, **INVMG3_ST3B**.



BUS/2 - BUS_P



BUS/2 - BUS_N

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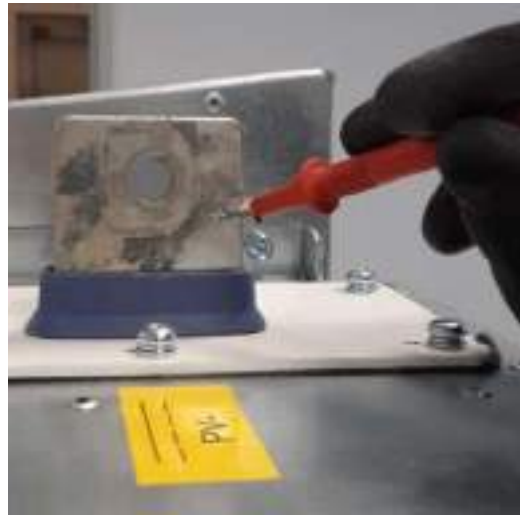
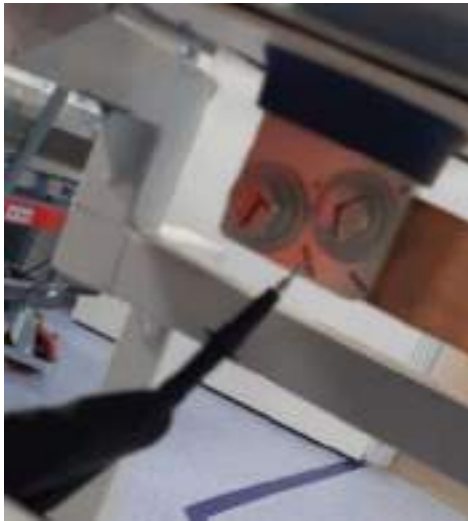


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Damaged IGBT.

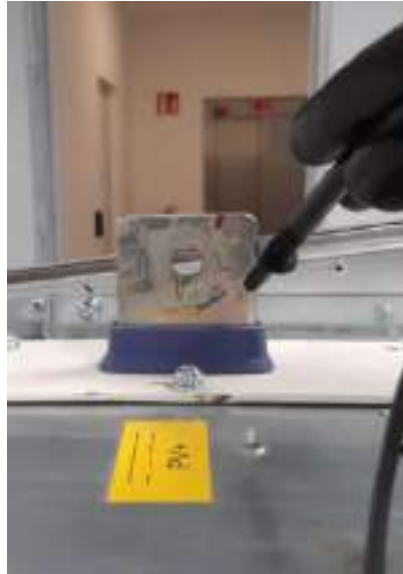
6. Inspect the IGBT with 1500V multimeter and replace if any damaged IGBT are found. See document *INVG3IOP88076*. *INVKG3_ST3B*, *INVMG3_ST3B*.





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Balancing PCB board (E07).

7. Inspect the balancing PCB (E07). Measure with 1500V multimeter the resistors on the board and check that they are 100k Ω . If not, replace the balancing PCB (E07) (see document *INVKG3/OP88076*). [INVKG3_ST2](#), [INVMG3_ST5A](#).





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If the cause of the fault is not found, contact Power Electronics.

5.43. Fault code F109: Mod. Low Vdc

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module has detected a very low DC voltage or the voltage has remained between low thresholds for 500us.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

The configuration is not correct.

1. Check the configuration of the module's DC undervoltage fault. To do this, check via PowerCommsV that the parameters [Low module Vdc], [Low mod Vdc High] and [ow mod Vdc Low] of the subgroup [G5.10: Internal Prot] have their default values. **INVG3_ST1B**. If they are not default:
 - 1.1. Preguntar al responsable de planta si hay algún motivo por el que los valores estén cambiados. Si no hay motivo, devolverlos a su valor por defecto. **INVG3_ST1B**.

Short-circuit of the DC bus of the power module.

2. Extract the Blackbox file of the failed module **INVKG3_ST3B**, **INVMG3_ST3B** and check the DC voltage behaviour. If either or both half-buses are drastically reduced, it is likely that there is a hardware problem or even a short circuit on the bus. If this occurs:
 - 2.1. Remove the power stage and check the IGBT. If any IGBT is short-circuited, replace it (see document **INVG3/OP88076**). **INVKG3_ST3B**, **INVMG3_ST3B**.

Loss of AC (night mode only).

3. Extract the Blackbox file of the failed module **INVKG3_ST3B**, **INVMG3_ST3B** and check if the module has lost AC voltage while running. To do this, check against the AC measurement recorded in Log_1 of the SD. **INVKG3_ST1B**; **INVMG3_ST1B**. If the module has lost AC voltage while running and it is not due to a voltage dip (the difference is that in a voltage dip the AC voltage recovers), then:
 - 3.1. Check if the AC boards of the modules are in good condition. Check that they are not connected incorrectly, with a change of colour or without the corresponding torque. **INVKG3_ST2**, **INVMG3_ST5A**.



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The table below shows the relationship between plates and each module with its corresponding phase.

Module/phase	Inductance side plate	Intermediate plate	Module side plate
Module 1 - Phase L1	M217C01873AA	M217C01386BA	M217C01302AA
Module 1 - Phase L2	M217C01874AA	M217C01386BA	M217C01303AA
Module 1 - Phase L3	M217C01875AA	M217C01387DA	M217C01303AA
Module 2 - Phase L1	M217C01876AA	M217C01386BA	M217C01302AA
Module 2 - Phase L2	M217C01877AA	M217C01386BA	M217C01303AA
Module 2 - Phase L3	M217C01878AA	M217C01386BA	M217C01303AA
Module 3 - Phase L1	M217C01879AA	M217C01386BA	M217C01302AA
Module 3 - Phase L2	M217C01880AA	M217C01386BA	M217C01303AA
Module 3 - Phase L3	M217C01881AA	M217C01386BA	M217C01303AA
Module 4 - Phase L1	M217C01882AA	M217C01388DA	M217C01302AA
Module 4 - Phase L2	M217C01883AA	M217C01388DA	M217C01303AA
Module 4 - Phase L3	M217C01884AA	M217C01613AAA	M217C01303AA

Short circuit in the DU.

- With the panel strings disconnected, and the modules **INVKG3_ST5B**, **INVMG3_ST5B** disconnected, check with a 1500V multimeter for a short circuit between positive and negative and between positive and negative to ground. **Important:** this check cannot be carried out with the octopus that short-circuits the busbars.





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5. Check with a 1500V multimeter for each circuit breaker that there is no continuity between external positive and external negative, between external positive and earth, between external negative and earth, between internal positive and internal negative, between internal positive and earth and between internal negative and ground. **INVKG3_ST5B, INVMG3_ST5B.**

Note: GFDI must be open if present.

Switch disconnecter according to module	Module 4	Module 3	Module 2	Module 1
External positive busbar	BBEP.75	BBEP.77	BBEP.79	BBEP.81
External negative busbar	BBEN.60	BBEN.62	BBEN.64	BBEN.66
Internal positive busbar	BBIP.01	BBIP.03	BBIP.05	BBIP.07
Internal negative busbar	BBIN.01	BBIN.03	BBIN.05	BBIN.07



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7. If there is continuity at any of the points, find the point where they are connected and repair.
INVKG3_ST2, INVMG3_ST5A.

5.44. Fault code F114-F116: Ir, Is, It module current sensing.

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	The module's FPGA has read an unexpected current in one of its Hall sensors.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A	

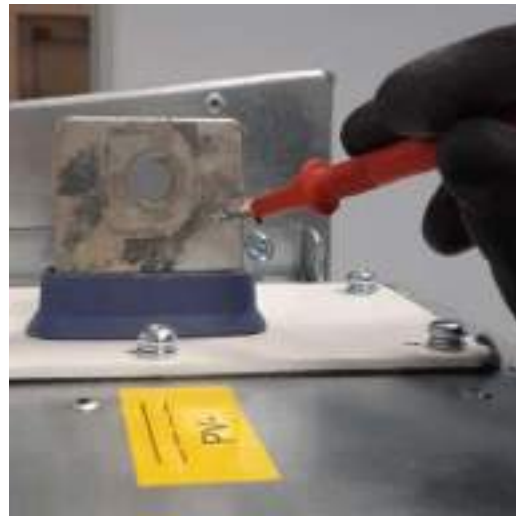
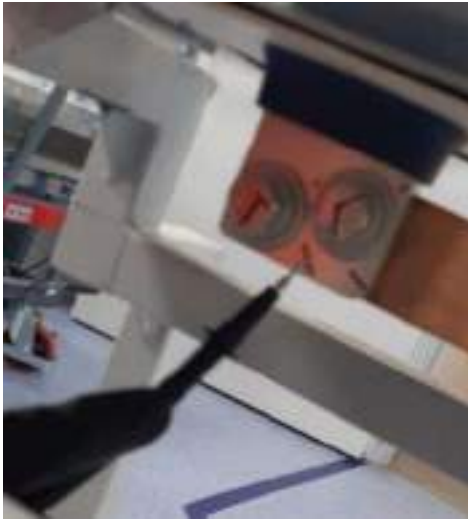
Analysis procedure

This fault is triggered when the module's FPGA reads more than $\pm 87\text{A}$ in DC current and the module is in the "stopped" or "ready" state.

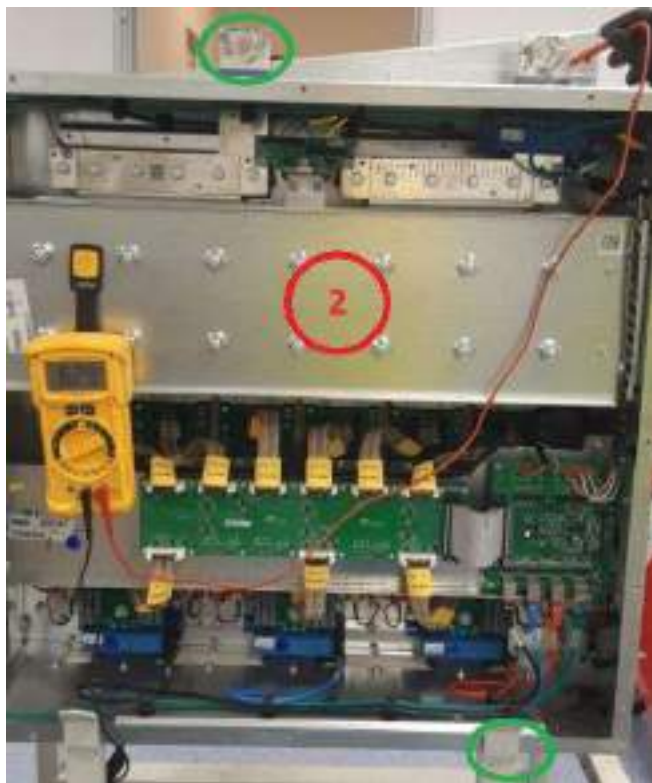
1. Disassemble the power stage and check with 1500V multimeter the IGBTs. [INVKG3_ST3B](#), [INVMG3_ST3B](#). (See document [INVG3IOP85369](#)).

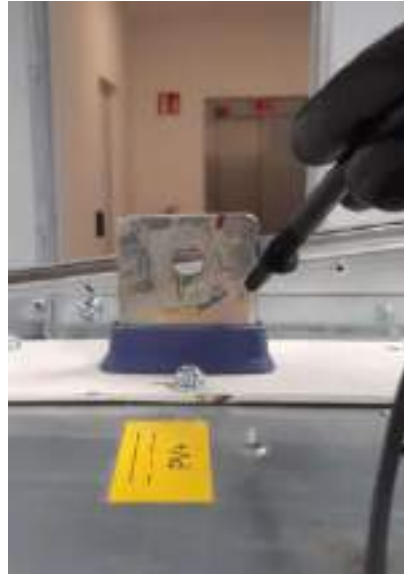


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2. The RJ45 cable is not properly connected. Disassemble the power stage and check all RJ45 connectors are properly connected. If any connector is not properly connected or is damaged and can not be connected properly, replace it. **INVKG3_ST3B**, **INVMG3_ST3B**.

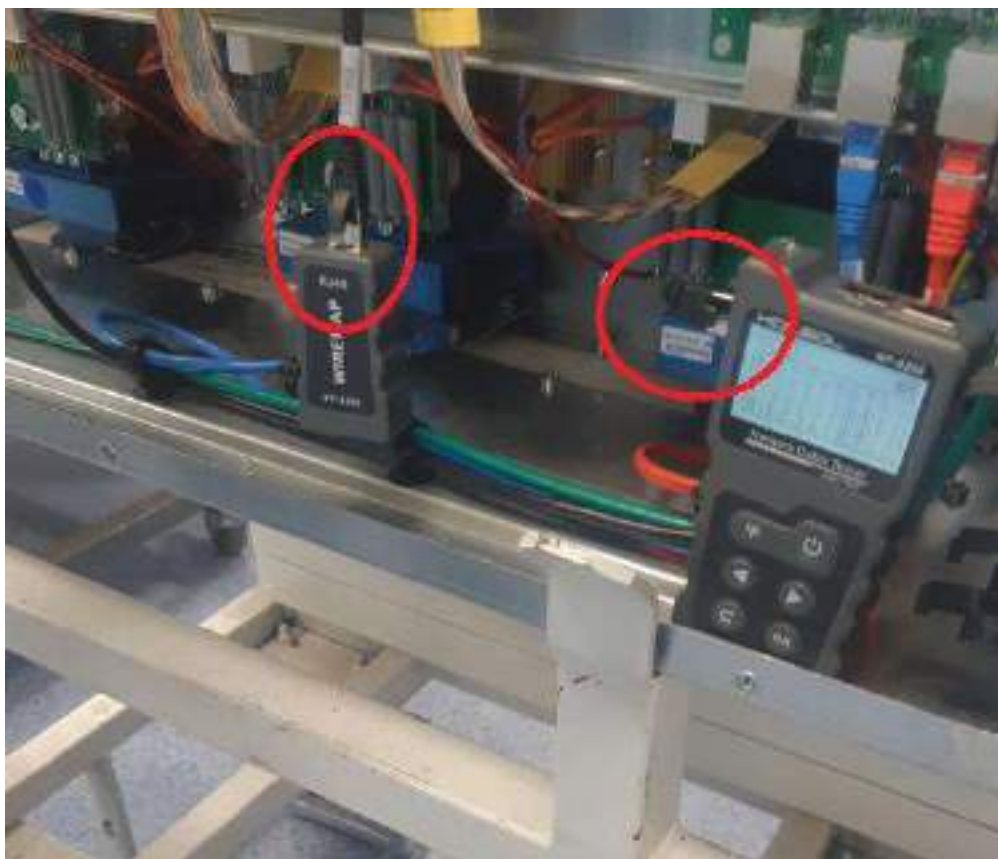


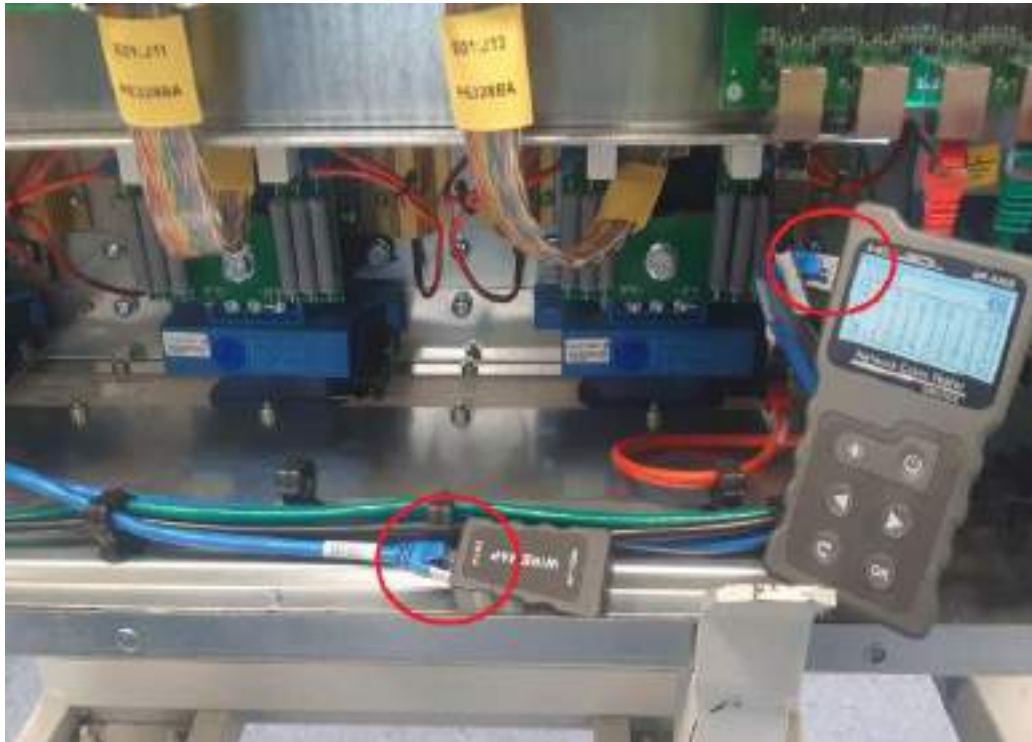


Following table indicates on which board is located the RJ45 cable and the RJ45 connector depending on the fault:

Fault	Board + Hall sensor	RJ45 cable	Conector (E02 board)
F114	E04+HT02	W030	U52
F115	E05+HT03	W031	U53
F116	E06+HT04	W032	U54

3. Check the continuity of the RJ45 cables with a network cable tester (RJ45 Tester).





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If there is no continuity in any of the RJ45 cables, replace the damaged RJ45 cable.

If the RJ45 cables are in good condition and well connected, replace the board and the corresponding Hall sensor according to the fault. Refer to the table above. **INVKG3_ST2, INVMG3_ST5A.**

4. If the fault still occurs after replacing the board and the Hall sensor, replace the E02 board. **INVKG3_ST2, INVMG3_ST5A.**

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5.45. Fault code F152: DGPT2 Temp failure

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The oil temperature of the MV transformer has exceeded 100°C. This fault is reported by the cubicle protection relay (DGPT2).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

1. Check the status of the digital input of the MV transformer overtemperature fault via PowerCommsV (parameter [DSP Status RJ16], within the subgroup [SV1.5: Digital I/O status] bit 2). **INVKG3_ST1A**.
2. If the input is set (bit 2 = X), it means that at the moment the fault is being analysed, the oil temperature is below 110°C. If this occurs:
 - 2.1. Check in Drive Select (group [SG10-Drive select]) that the equipment is configured according to the specifications of the MV transformer. **INVKG3_ST1A**.
 - 2.2. Check that there are no elements obstructing the ventilation of the MV transformer fins. **INVKG3_ST2, SKID_ST2**.
 - 2.3. Start up the equipment and monitor if the fault recurs.
3. If it is not activated (bit 2 = 0) it may be because the transformer oil temperature is higher than 110°C or there is a problem in the digital input circuit that reports this fault. In this case, start inspection on the MV Skid:
 - 3.1. Check the temperature indicated on the temperature sensor incorporated in the MV transformer. **INVKG3_ST2, SKID_ST2**.





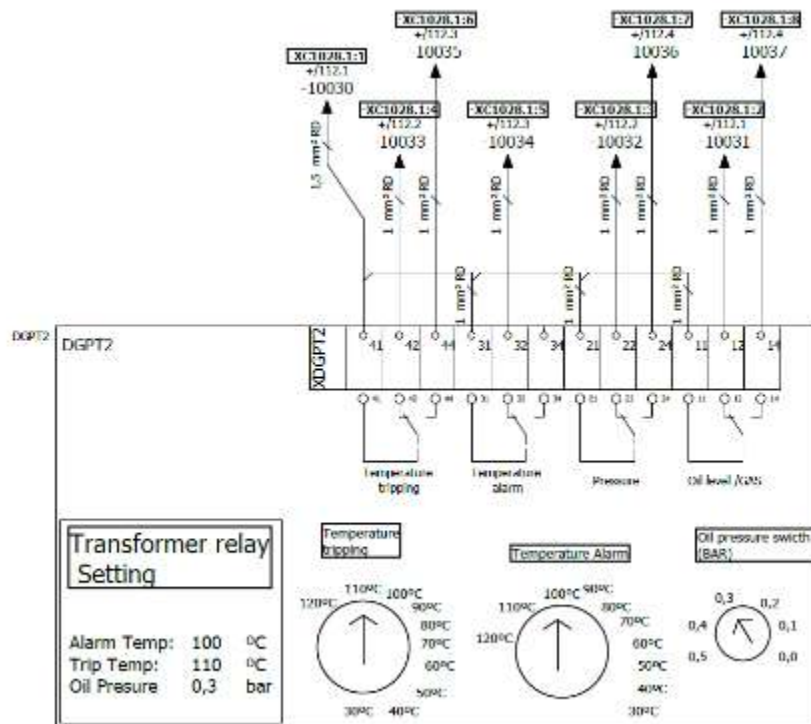
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3.2. If the temperature is higher than 110°C:

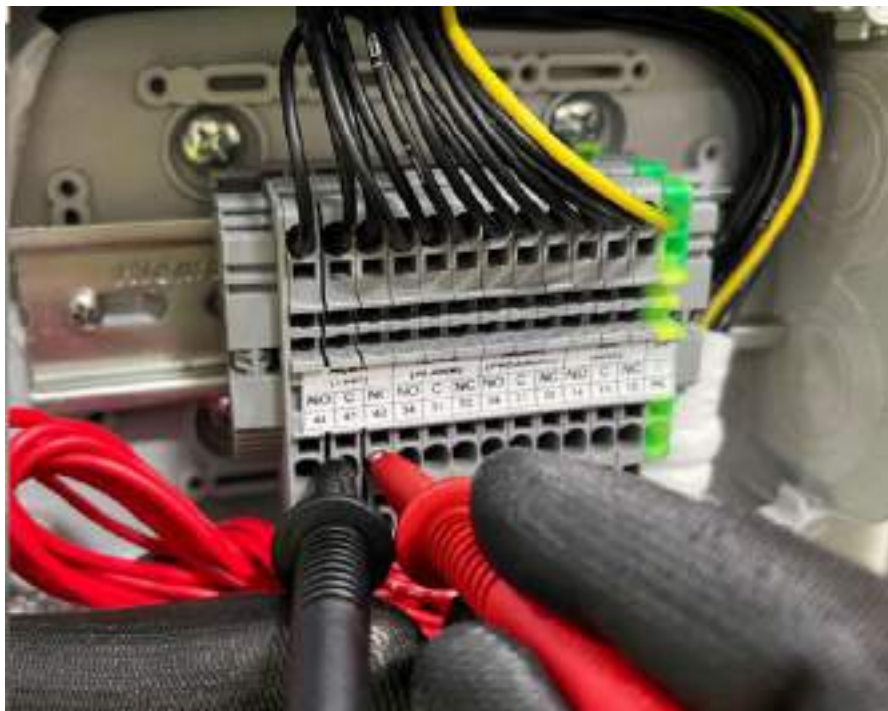
- 3.2.1. Check in Drive Select (group [SG10-Drive select]) that the inverter is configured according to the specifications of the MV transformer. **INVKG3_ST1A.**
- 3.2.2. Check that there are no elements obstructing the ventilation of the MV transformer fins. **INVKG3_ST2, SKID_ST2.**



- 3.3. If the temperature is below 110°C it means that it is not a real thermal problem and it is necessary to check the electrical trip circuit of the fault. **INVKG3_ST2, SKID_ST2.**
- 3.3.1. Disconnect pins 41, 42 of the DGPT2 relay on the MV Skid and measure with a 1500V multimeter in resistance mode, continuity between pins 41 and 42.

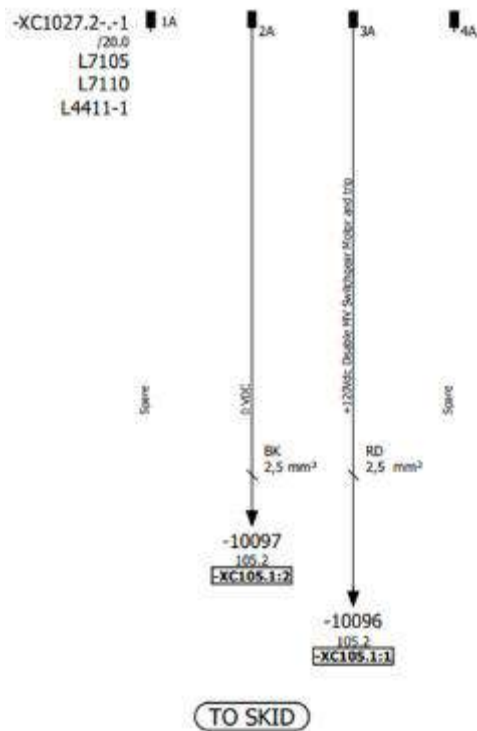


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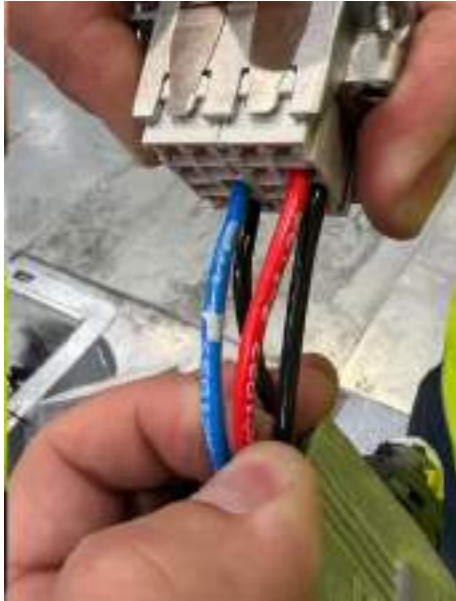


- 3.3.2. If the resistance measured with 1500V multimeter between the pins is greater than 100kOhm, this indicates that the DGPT2 relay is faulty. In that case, **replace the DGPT2.** See document *INVKG3IOP87922*.
- 3.3.3. If the resistance measured with 1500V multimeter between the pins is approximately 0Ω, this indicates that the problem is not in relay DGPT2. Proceed to the next point.

- 3.3.4. Reconnect pins 41 and 42 of relay DGPT2 and disconnect the haring connecting the MV Skid to the inverter (XC1027.2) and measure continuity with 1500V multimeter between pins XC1027.2 1A and XC1027.2 2B. **INVKG3_ST2, SKID_ST2.**



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- If the resistance measured with 1500V multimeter between the pins is greater than 100kΩ, this indicates a problem in the MV Skid wiring. Check that the wiring from the DGPT2 relay (pins 41 and 42) and the inverter hatching is wired according to the wiring diagrams. **INVKG3_ST2, SKID_ST2.**
- If the resistance measured with 1500V multimeter between the pins is approximately 0Ω, this indicates that the problem is not in that circuit. Proceed to the next point. **INVKG3_ST2, SKID_ST2.**

3.3.5. Disconnect the XC1026 hatching and check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W368) connecting XC1026.2 1D to connector J1 on the E103 board of the MV Skid. **INVKG3_ST2, SKID_ST2.**



- If any of the pins do not give continuity, replace the cable (W368). **INVKG3_ST2, SKID_ST2.**
- If all pins of the W368 cable show continuity, measure continuity with a 1500V multimeter, between pin 2 of connector J5 on the E103 board and pin 11 of the KA14 relay. **INVKG3_ST2, SKID_ST2.**



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- If there is no continuity, (resistance greater than 100kOhm) check cable and replace. **INVKG3_ST2, SKID_ST2.**
 - o If there is continuity (Resistance = 0 Ohm, measure continuity between pins 11 and 12 of the KA14 relay. **INVKG3_ST2, SKID_ST2.**



- If there is no continuity (resistance greater than 100 kΩ) replace relay. **INVKG3_ST2, SKID_ST2.**
 - If there is continuity, replace the E103 board. **INVKG3_ST2, SKID_ST2.**
4. If the problem persists, inspect the inverter. **INVKG3_ST2.** To do this:
- 4.1. Disconnect the harting XC1026.2 and check continuity between all pins of the RJ45 cable (W063) using an RJ45 cable tester. **INVKG3_ST2, SKID_ST2.**



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4.1.1. If there is no continuity between the pins (resistance greater than 100 kOhm), replace the cable. **INVKG3_ST2, SKID_ST2.**

If all the above steps have not resolved the problem, contact Power Electronics.

5.46. Fault code F153: DGPT2 Pressure failure

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction” .			
Description	The MV Skid protection cell has reported a high oil pressure fault. This fault is reported by the cell protection relay (DGPT2).			
Applicable to	☐	HEM	☐	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	N/A.			

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

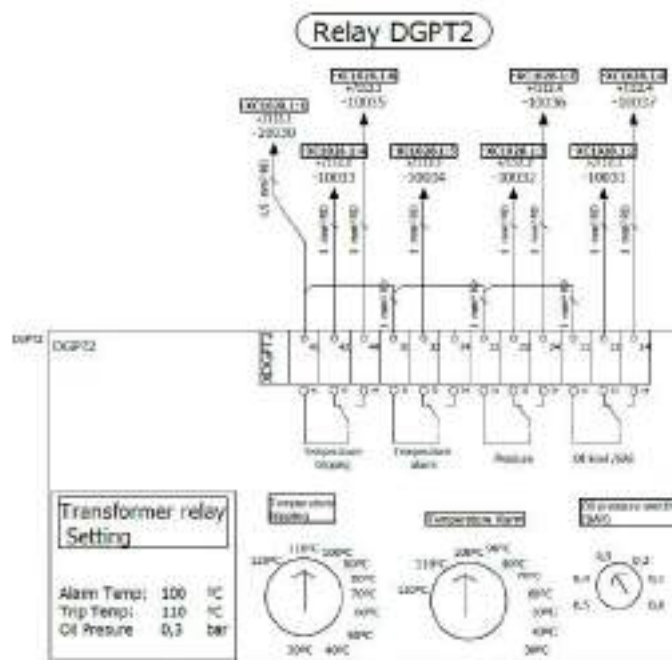
1. Check the status of the digital input of the MV transformer oil pressure failure via PowerCommsV (parameter [DSP Status RJ16], in subgroup [SV1.5: Digital I/O status], bit 3). **INVKG3_ST1A**.
2. If the input is set (bit 3 = X), it means that at the moment the fault is being analysed, the oil pressure in the protection cell of the MV Skid is high. If this occurs:

2.1. Check in Drive Select (group [SG10-Drive select]) that the inverter is configured according to the specifications of the MV transformer. **INVKG3_ST1A.**

2.2. Check that there are no elements obstructing the ventilation of the MV transformer fins.
INVKG3_ST2, SKID_ST2.

2.3. Start up the equipment and monitor if the fault recurs.

3. If it is not enabled (bit 3 = 0) it may be because the transformer pressure is lower than 0.3bar (than the set limit) or there is a problem in the digital input circuit that reports this fault. In this case, start inspection on the MV Skid:
 - 3.1. Due to the fact that it is not possible to measure the pressure externally and compare it with the pressure measured by DGPT2, check the electrical trip circuit of the fault **INVKG3_ST2**, **SKID_ST2**. To do this, the following steps must be followed.

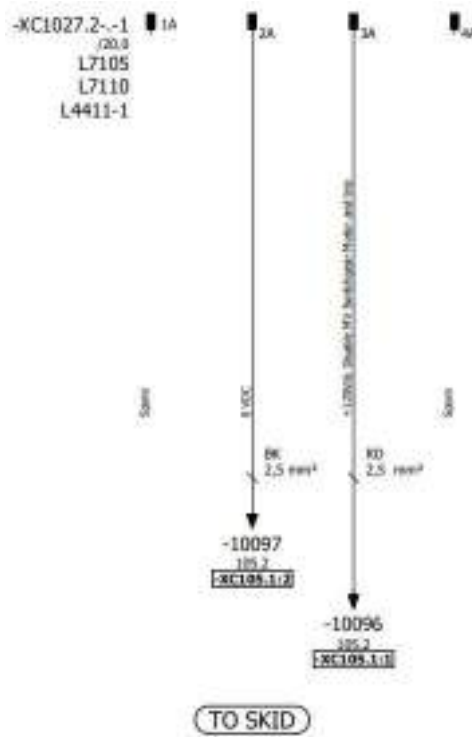


- 3.1.1. Disconnect pins 22, 24 of the DGPT2 relay in the SKID and measure with a 1500V multimeter in resistance mode, continuity between pins 21 and 22. **INVKG3_ST2, SKID_ST2.**



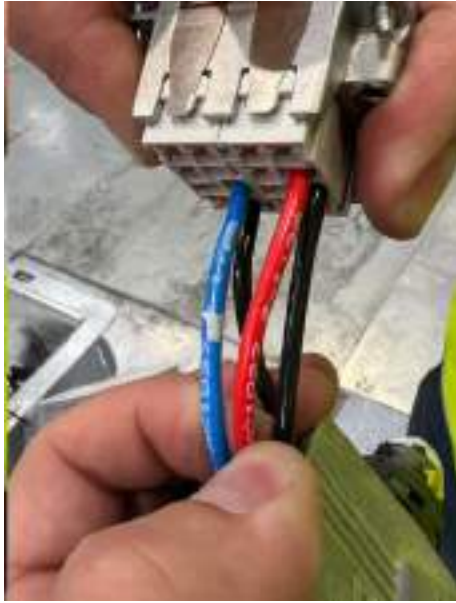
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- 3.1.2. If the resistance measured with 1500V multimeter between the pins is greater than 100kOhm, this indicates that the DGPT2 relay is faulty. In that case, replace the DGPT2. See document *INVKG3/OP87922*. **INVKG3_ST2, SKID_ST2.**
- 3.1.3. If the resistance measured with 1500V multimeter between the pins is approximately 0Ω, this indicates that the problem is not in relay DGPT2. Proceed to the next point.
- 3.1.4. Reconnect pins 21 and 22 of relay DGPT2 and disconnect the harting connecting the MV Skid to the inverter (XC1027.2) and measure continuity with 1500V multimeter between pins XC1027.2 1A and XC1027.2 2B. **INVKG3_ST2, SKID_ST2.**



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- If the resistance measured with 1500V multimeter between the pins is greater than 100kΩ, this indicates a problem in the MV Skid wiring. Check that the wiring from the DGPT2 relay (pins 21 and 22) and the inverter hating is wired according to the wiring diagrams. **INVKG3_ST2, SKID_ST2.**
- If the resistance measured with multimeter 1500V between the pins is approximately 0Ω, this indicates that the problem is not in that circuit. Proceed to the next point. **INVKG3_ST2, SKID_ST2.**

3.1.5. Disconnect the XC1026 hating and check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W368) connecting XC1026.2 1D to connector J1 on the E103 board of the MV Skid. **INVKG3_ST2, SKID_ST2.**



- If any of the pins do not give continuity, replace the cable (W368). **INVKG3_ST2, SKID_ST2.**
- If all pins of the W368 cable show continuity, measure continuity with a 1500V multimeter, between pin 2 of connector J4 on the E103 board and pin 11 of the KA13 relay. **INVKG3_ST2, SKID_ST2.**



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- If there is no continuity, (resistance greater than 100kΩ) check cable and replace. **INVKG3_ST2, SKID_ST2.**
 - o If there is continuity (Resistance = 0 Ohm, measure continuity with 1500V multimeter between pins 11 and 12 of the KA13 relay. **INVKG3_ST2, SKID_ST2.**



- If there is no continuity (resistance greater than 100kΩ) replace relay. **INVKG3_ST2, SKID_ST2.**
- If there is continuity, replace the E103 board. **INVKG3_ST2, SKID_ST2.**

4. If the problem persists, inspect the inverter. **INVKG3_ST2**. To do this:
 - 4.1. Disconnect the harting XC1026.2 and check continuity between all pins of the RJ45 cable (W063) using an RJ45 tester. **INVKG3_ST2, SKID_ST2**.



- 4.1.1. If there is no continuity between the pins (resistance greater than 100k Ω), replace the cable. **INVKG3_ST2, SKID_ST2**.
5. If all the above steps have not resolved the problem, contact Power Electronics.

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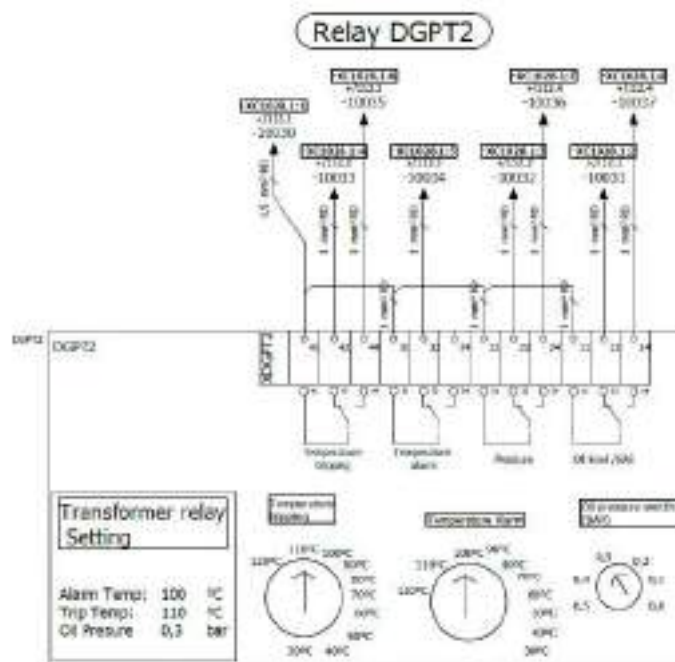
5.47. Fault code F155: DGPT2 oil or gas level

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction”.	
Description	The MV Skid protection cell has reported a low oil or gas level fault. This fault is reported by the cell protection relay (DGPT2).	
Applicable to	☐ HEM ☒ HEMK	☐ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

- Check the status of the digital input of the MV transformer oil pressure failure via PowerCommsV (parameter [DSP Status RJ16], in subgroup [SV1.5: Digital I/O status], bit 4). **INVKG3_ST1A**.
- If the input is set (bit 4 = X), it means that at the moment the fault is being analysed, the oil pressure in the protection cell of the MV Skid is high. If this occurs:
 - Check in Drive Select (group [SG10-Drive select]) that the inverter is configured according to the specifications of the MV transformer. **INVKG3_ST1A**.
 - Check that there are no elements obstructing the ventilation of the MV transformer fins. **INVKG3_ST2, SKID_ST2**.
 - Start up the equipment and monitor if the fault recurs.
- If it is not enabled (bit 4 = 0) may be due to the transformer pressure being below the set level or there is a problem in the digital input circuit that reports this fault. In that case, start inspection on the MV Skid:
 - Due to the fact that it is not possible to measure the pressure externally and compare it with the pressure measured by DGPT2, check the electrical trip circuit of the fault **INVKG3_ST2, SKID_ST2**. To do this, the following steps must be followed.

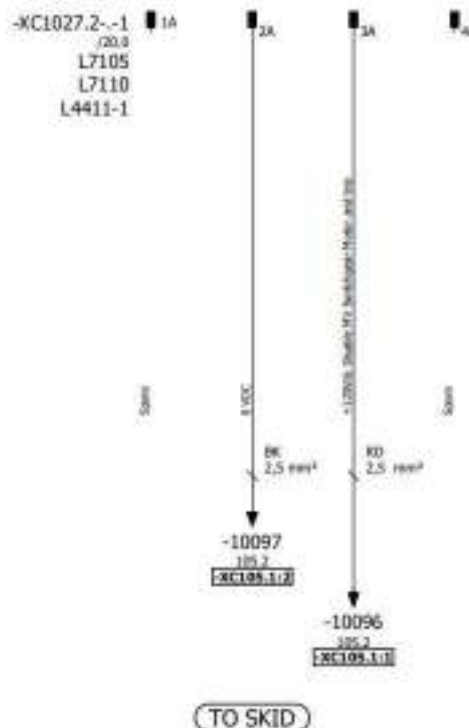


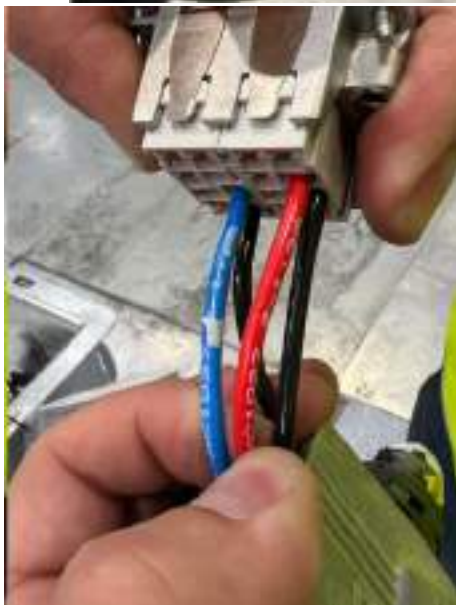
- 3.1.1. Disconnect pins 11, 12 of the DGPT2 relay in the SKID and measure with a 1500V multimeter in resistance mode, continuity between pins 11 and 12. **INVKG3_ST2, SKID_ST2.**



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- 3.1.2. If the resistance measured with multimeter 1500V between the pins is greater than 100k Ω , this indicates that the DGPT2 relay is faulty. In that case, replace the DGPT2. See document *INVKG3/OP87922*. **INVKG3_ST2, SKID_ST2.**
- 3.1.3. If the resistance measured with 1500V multimeter between the pins is approximately 0 Ω , this indicates that the problem is not in relay DGPT2. Proceed to the next point.
- 3.1.4. Reconnect pins 11 and 12 of relay DGPT2 and disconnect the harting connecting the MV Skid to the inverter (XC1027.2) and measure continuity with 1500V multimeter between pins XC1027.2 1A and XC1027.2 2B. **INVKG3_ST2, SKID_ST2.**





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- If the resistance measured with 1500V multimeter between the pins is greater than 100k Ω , this indicates a problem in the MV Skid wiring. Check that the wiring from the DGPT2 relay (pins 11 and 12) and the inverter harging is wired according to the wiring diagrams. **INVKG3_ST2, SKID_ST2.**
- If the resistance measured with 1500V multimeter between the pins is approximately 0 Ω , this indicates that the problem is not in that circuit. Proceed to the next point. **INVKG3_ST2, SKID_ST2.**

3.1.5. Disconnect the XC1026 harging and check continuity with an RJ45 pin tester on all pins of the RJ45 cable (W368) connecting XC1026.2 1D to connector J1 on the E103 board of the MV Skid. **INVKG3_ST2, SKID_ST2.**



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- If any of the pins do not give continuity, replace the cable (W368). **INVKG3_ST2, SKID_ST2.**
- If all pins of the W368 cable show continuity, measure continuity with a 1500V multimeter, between pin 3 of connector J3 on the E103 board and pin 11 of the KA12 relay. **INVKG3_ST2, SKID_ST2.**



- If there is no continuity, (resistance greater than 100kΩ) check cable and replace. **INVKG3_ST2, SKID_ST2.**
- If there is continuity (Resistance = 0 Ohm, measure continuity between pins 11 and 12 of the KA12 relay. **INVKG3_ST2, SKID_ST2.**



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- If there is no continuity (resistance greater than 100kΩ) replace relay. **INVKG3_ST2, SKID_ST2.**
 - If there is continuity, replace the E103 board. **INVKG3_ST2, SKID_ST2.**
4. If the problem persists, inspect the inverter. **INVKG3_ST2.** To do this:
- 4.1. Disconnect the harting XC1026.2 and check continuity between all pins of the RJ45 cable (W063) using an RJ45 tester. **INVKG3_ST2, SKID_ST2.**



- 4.1.1. If there is no continuity between the pins (resistance greater than 100kΩ), replace the cable. **INVKG3_ST2, SKID_ST2.**
5. If all the above steps have not resolved the problem, contact Power Electronics.

5.48. Fault code F165: Improper Modulation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".			
Description	The inverter has been working longer than allowed with undue modulation.			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

Analysis procedure

This fault occurs when the inverter is in double signal modulation for more than 2 minutes. The type of modulation in which the inverter is operating, the modules working in this modulation and the number of times it has been switched to double signal, is displayed by PowerCommsV within the group [\[SV4.17: Modulation\]](#). **INVG3_ST1B.**

The following are possible reasons why the inverter may be working in double signal:

inverter has been forced by configuration to operate in double signal.

1. Check via PowercommsV whether the value of the [\[Modulation type\]](#) parameter in the [\[SG1.3-FPGA service\]](#) group is set to "Double". If so, change to "Single". **INVG3_ST1B.**

The inverter is supporting an LVRT/HVRT event during this time.

2. Via PowercommsV analyse the SD logs and check if the inverter has been in an HVRT event for more than 2 minutes. In case the HVRT/LVRT event has passed, reset and check if the fault disappears. **INVG3_ST1B.**
3. Check via PowercommsV in the Blackbox file that during the HVRT/LVRT event the inverter was operating in double signal. In case the HVRT/LVRT event has passed, reset and check if the fault disappears. **INVG3_ST1B.**
4. If the inverter repeatedly enters HVRT/LVRT, the high-voltage protections must be checked via PowercommsV. **INVG3_ST1B.** For this purpose:
 - 4.1. Check that the high voltage protection (parameter [\[Fast protection\]](#) of subgroup [\[G5.1-Voltage-frequency\]](#) is not above the HVRT trip (parameter [\[Threshold\]](#) of subgroup [\[G4.4: OVRT\]](#)). **INVG3_ST1B.**
 - 4.2. Analyse via PowercommsV the reason for the switch to double signal. To do so:
 - 4.2.1. Add the addresses of the parameters [\[Active Mod.\]](#), [\[Modules in DS\]](#), [\[N° DS Activations\]](#) of the visualization group [\[SV4.17: Modulation\]](#) to the custom log. Set the configuration of the F165 as "abrupt stop", with manual reset and user **PE Staff**, so that in case of a retriggering of the F165 fault, the Blackbox file can be downloaded at the time of the fault. **INVG3_ST1B.**

. If the fault continues to occur, contact Power Electronics.

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5.49. Fault code F170: Thermal L Feedback

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	the inductance core of a module overheats, this module will trigger this fault and all modules will stop.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault occurs when any inductance overheats. Each inductance has four thermal sensors (NC contacts) connected in series and connected to the filter panel board (PFC) (E19). These thermal sensors are factory set to 150°C. When one of the thermal sensor exceeds 150°C, it opens and causes the fault.

1. Check in Powercomms the visualization group [SV1.5 Digital I/O status] the DSP status RJ18 - bit 5, if set to 1 it indicates that the inductance feedback has been activated.. **INVKG3_ST1B; INVMG3_ST1B.**
2. If the fault has been caused by the activation of one of the thermal sensor of the inductance, the sensor contact will close automatically when the temperature returns to below 150°C.
3. Check that the temperature of the inductances is less than 150°C, visualising in Powercomms in the group [SV1.6-environment] in the PT100 measurements of the inductances. If the temperature is lower than 150°C and the value of the digital signal of the RJ18 remains at 1, carry out the following checks:
 - 3.1. Check continuity with 1500V multimeter between pins J4:1 - J4:2 of the filter digital inputs and outputs board (E19) of filter panel C (PFC).



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If there is no continuity, the wiring is correct and the inductance has not exceeded 150°C. In this case, the problem is between the input/output board (E19) and the central control board (E35).
INVKG3_ST2; INVMG3_ST5A.

- 3.2. Check with an RJ45 pin tester that there is continuity in the cable connected to J2 of the E19 board and J18 of the central control board (E35). If there is no continuity, replace the cable.
INVKG3_ST2; INVMG3_ST5A



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If there is continuity, replace the I/O board (E19). See document *INVKG3/OP88849*.

If there is no continuity, the wiring or the inductance is damaged.

- Check continuity with 1500V multimeter between pins XTS:C - XTS:NC of the inductance.
INVKG3_ST2; INVMG3_ST5A



- If there is continuity, the wiring is faulty. Check wiring between inductance and I/O board (E19).

Measure continuity with 1500V multimeter between XTS:C – E19:J4:1



Measure continuity with 1500V multimeter between XTS:NC – E19:J4:2



Replace the damaged cable and its terminals and carry out the checks again.

INVKG3_ST2; INVMG3_ST5A

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- If there is no continuity between pins XTS:C - XTS:NC of the inductance and the temperature of the PTC100 probes is less than 150°C, replace the inductance. See document *INVKG3/OP85375*. **INVKG3_ST2; INVMG3_ST5A**
4. Once the above steps have been carried out and if it is observed that high temperatures are reached repeatedly, check the ventilation affecting the inductance.

4.1. Check LC filter ventilation check.

- Check via PowerCommsV that the LC filter fans are working properly. Set in fixed reference mode to 100% speed reference (parameter [Mode], in group [G11.6: LC filter config]) I and check that they are rotating at the configured rpm. If they do not, check the fan display [SV11 Ventilation], check the fault/warning given by the fan, and act according to the fan manual. **INVKG3_ST1B; INVMG3_ST1B.**
- Check with PowerCommsV that there is no fan warning during the fault. **INVKG3_ST1B; INVMG3_ST1B.** If there is a warning, check the fans for damage, check that they do not rub against any part of the equipment and that they are securely anchored. **INVKG3_ST2; INVMG3_ST5A**
- Check the fans, check that the rotor blades are not broken or worn, check the power and communication cables that they do not rub against the moving parts of the fan (check that the cables are connected correctly and without forcing to avoid connectivity faults, bent pins, etc.). If they are not OK, replace the fan or cables as appropriate. **INVKG3_ST2; INVMG3_ST5A**

4.2. Check module ventilation

- Check via PowerCommsV that the module fans are working correctly. Set in fixed reference mode to 100% speed reference (parameter [Mode], in group [G11.5: Module Config]) and check that they are rotating at the configured speed. If they do not, check the fan display [SV11 Ventilation], check the fault/warning given by the fan, and act according to the fan manual. **INVKG3_ST1B; INVMG3_ST1B.**
- Check with PowerCommsV that there is no fan warning during the fault. **INVKG3_ST1B; INVMG3_ST1B.** If there is a warning, check the fans for damage, check that they do not rub against any part of the equipment and that they are securely anchored. **INVKG3_ST2; INVMG3_ST5A**
- Check the fans, check that the rotor blades are not broken or worn, check the power and communication cables that they do not rub against the moving parts of the fan (check that the cables are connected correctly and without forcing to avoid connectivity faults, bent pins, etc.). If they are not OK, replace the fan or cables as appropriate. **INVKG3_ST2; INVMG3_ST5A.**

5.50. Fault code F197: HW Overvoltage

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	An overvoltage has been reached on the AC low side, higher than the hardware overvoltage limit of the inverter.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is reported by the inverter when an overvoltage appears on the low voltage Vac, higher than the hardware overvoltage profile. The hardware limit for continuous operation is 115% of the nominal voltage.

- Check the voltage reached at the time of the fault. It is necessary to ensure in the SD the voltage Vac reached by the inverter at the time of the fault. To do this:
 - Check via PowercommsV in Log_1 the values of Vrs, Vst and Vtr. The voltages must be higher than 115% of the nominal voltage (parameter [Nominal Vac] of subgroup [SG10.1-General]). This is a network event for which the inverter has been correctly protected. **INVG3_ST1B**.

Incorrect Drive select configuration.

- If a nominal voltage value lower than the operating voltage of the inverter (parameter [Nominal Vac] of subgroup [SG10.1-General]) has been entered in the Drive Select, the inverter will report the fault at an admissible overvoltage. To do so:
 - Ensure via PowerCommsV that the nominal voltage entered (parameter [Nominal Vac] of subgroup [SG10.1-General]) coincides with the operating voltage of the inverter. If the value differs, enter the correct nominal voltage. **INVG3_ST1B**.

Incorrect transformer tapping.

- If the MV tapping of the transformer has been set to a lower voltage level than the one existing in the network, the undervoltage of the inverter will be higher than the nominal voltage and therefore it will be easier to reach the hardware overvoltage. If this happens:
 - Check that the MV tapping of the transformer corresponds to the MV voltage of the plant. This requires access to the transformer drawer and a visual inspection. If an incorrect tapping is detected, correct it. **INVKG3_ST3B**, **INVMG3_ST1B3B**, **SKIDG3_ST3B**. If the correct tapping is not known, consult Power Electronics.



Overvoltages due to POI opening.

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4. If overvoltages occur sporadically, ask the plant manager if they are opening the POI, and check if they coincide with the overvoltages of the inverter.

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5.51. Fault code F198: HW Undervoltage

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	An undervoltage has been reached on the AC low side, lower than the hardware undervoltage limit of the inverter.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is reported by the equipment when an undervoltage appears on the LV Vac, lower than the HW undervoltage profile. The HW limit for continuous operation is 76.5% of nominal voltage.

- Check the voltage reached at the time of the fault. It is necessary to ensure in the SD the Vac voltage reached by the inverter at the time of the fault. To do this:
 - Check via PowercommsV in Log_1 the values of Vrs, Vst and Vtr. The voltages must be lower than 76.5% of the rated voltage (parameter [Nominal Vac] of subgroup [SG10.1-General]). This is a gap in the network against which the inverter has been correctly protected. **INVG3_ST1B**.

Incorrect Drive select configuration.

- If a nominal voltage value lower than the operating voltage of the inverter has been entered in the Drive Select (parameter [Nominal Vac] of subgroup [SG10.1-General]), the inverter will report the fault at an admissible undervoltage. To do so:
 - Ensure by PowerCommsV that the nominal voltage entered (parameter [Nominal Vac] of subgroup [SG10.1-General]) coincides with the operating voltage of the inverter. If the value differs, enter the correct nominal voltage. **INVG3_ST1B**.

Incorrect transformer tapping.

- If the MV tapping of the transformer has been set at a lower voltage level than the one existing in the network, the undervoltage of the inverter will be higher than the nominal voltage and therefore it will be easier to reach the hardware undervoltage. If this happens:
 - Check that the MV tapping of the transformer corresponds to the MV voltage of the plant. This requires access to the transformer drawer and a visual inspection. If an erroneous tapping is detected, correct it. **INVKG3_ST3B**, **INVMG3_ST1B3B**, **SKIDG3_ST3B**. If the correct tapping is not known, consult Power Electronics.



One of the Vac voltage measurements does not reach the control board. It may be the case that the unit reports F198, because it has lost one of the AC voltage measurements. If this happens, after tripping F198, the inverter will lose voltage start conditions as it does not detect LV mains. Carry out the following checks to detect the source of the problem:

4. With the AC inverter closed (AC circuit breaker closed in the case of HEMK / PCSK inverter or switchgear closed in the case of HEM / PCSM inverter), check the LV AC voltage measurements from PowerCommsV (parameters [Grid voltage RS LV], [Grid voltage ST LV], [Grid voltage TR LV] of subgroup [SV3.2: Grid LV]). **INVKG3_ST1B**.
5. If a null or erroneous measurement is detected in some of the line voltages, this is the cause of the fault. To solve it, it is necessary to detect whether the problem lies in the wiring or in the central control board itself.

5.1. Checking connectors J3, J4 and J5 central control board (E35), for this purpose:

- 5.1.1. With the inverter isolated from DC and AC, check the correct connection of faston connectors J3, J4 and J5. **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B**.



- 5.1.2. With the AC inverter shut down, check by PowerCommsV if the voltage measurements measure correctly again. **INVKG3_ST1B, SKIDG3_ST1**.

- 5.1.3. If the measurements are still not correct, go to the next wiring check.

5.2. Checking F100 Fuse.

- 5.2.1. With the inverter isolated from DC and AC, check that the F100 fuses are in the 'ON' position. Measure the continuity between both parts of the F100 fuse by carrying out a check on the 3 phases. If any of the fuses do not give continuity, replace it. **INVKG3_ST2, INVMG3_ST5A**.

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- 5.2.2. If one of the fuses has blown, physically check the status of the SPD P01 protection to see if it has tripped. If the cartridge has blown, the marker will be red and it will be necessary to replace it. Otherwise, the marker will be green, which indicates good condition.
INVKG3_ST2, INVMG3_ST5A.



SPD OK

- 5.3. Checking wiring. Carry out the ringing in the AC voltage measurement wiring. To do this, check continuity with 1500V multimeter at the following points: **INVKG3_ST2, INVMG3_ST5A.**
- Measure continuity between point F100.1 and faston connector J3 on the central control board (E35).



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- Measure continuity between point F100.3 and faston connector J4 on the central control board (E35).



- Measure continuity between point F100.5 and faston connector J5 on the central control board (E35).



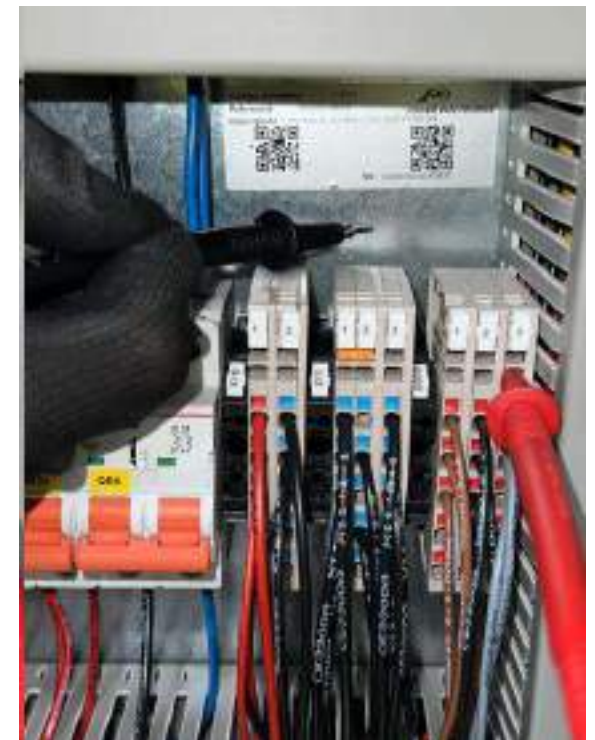
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- If all 3 phases give a correct continuity measurement, move on to the next possible cause, otherwise proceed to the next step.

5.3.1. Once detected which is the phase that does not give continuity, perform the ringing by parts to detect where the measurement is cut off. **INVKG3_ST2, INVMG3_ST2.**

- Check continuity with 1500V multimeter between point F100.X (X will be 1,3 or 5 depending on the phase detected in the previous point) and terminal block X600.X (X will be 1,2,3). If there is no continuity, replace the wiring. **INVKG3_ST2; INVMG3_ST2.**





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Replacement of the E35 central control board. The failure in the Vac voltage measurement may be due to the breakage of the acquisition circuit of this measurement. If this occurs:

6. Replace the central control board (see document *INV3/OP88849*). **INVKG3_ST2**, **INVMG3_ST5A**. Check if the equipment recovers voltage measurements by PowerCommsV (parameters [LV RS mains voltage], [LV ST mains voltage], [LV TR mains voltage] of subgroup [SV3.2-LV mains]). **INV3_ST1B**.

Faulty boards. It is possible that one of the boards involved in the measurement process is faulty.

7. Check the following by PowerCommsV. **INVKG3_ST1B**.
 - 7.1. Check if there are no measurements on the central board (parameters [Voltage extern] of group [SV2: DC source]).
 - 7.2. Check if there are measurements on the module board (parameters [DC voltage P], [DC voltage N] of group [SV12: Module information]).
 - 7.3. Check if the DC disconnecter is closed (parameters [DC disconnecter status] of group [SV1: Status]).
 - 7.4. In this case, replace central board E35 (see document *INVKG3/OP88849*). **INVKG3_ST2**, **INVMG3_ST5A**.
8. If, on the other hand, this is the case via PowercommsV: **INVKG3_ST1B**.
 - 8.1. Verify the measurements on the SV2 DC Source central board in the External Voltage measurement corresponding to the module in fault. (parameters [Voltage extern] within the group [SV2 DC source]).
 - 8.2. There are no measurements on the module board (parámetros [DC Voltage P], [DC voltage N] within the group [SV12: Module information]).
 - 8.3. The DC disconnecter is closed (parameters [DC disconnecter status] within the group [SV1: Status]).
 - 8.4. In this case, replace the E02 board of the module that is causing the fault (see document *INVKG3/OP88849*). **INVKG3_ST2**, **INVMG3_ST5A**
 - 8.5. If none of the cases corresponds, check all the above points. If the fault continues:
 - 8.6. Action. Contact with Power Electronics.

5.52. Fault code F208: MV cri overtemperature

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The critical temperature threshold of the MV transformer has been exceeded.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	F203	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

MV transformer overtemperature problem (T01).

1. Check in the invertir SD card (Log_3) if the temperature increase is real or if it was a noise problem in the temperature measurement INVG3_ST1B.
 - If the temperature has risen gradually and without ripple, it is a problem of overtemperature of the MV transformer (T01). Go to step 3, 4, 5 and 6.
 - If there is excessive ripple in the PT100 temperature measurement, it may be a wiring problem. Go to step 7.

Misconfiguration of temperature thresholds.

2. Check the fault configuration, to do so, consult Powercomms ([Transformer] in group [G5.4-Temperature]), if the critical temperature of the MV transformer (T01) is below its default value. In this case, consult Power Electronics in case it is a special configuration of this plant. INVG3_ST1B.

Possible obstruction of the cyclone filters.

3. Check that the air inlet is not obstructed by anything. If any of the filters are damaged or clogged in any way, replace them. INVKG3_ST2, INVMG3_ST5A. (See document INVG3/OP88813).



Possible MT transformer fault (T01).

4. Check the condition of the MV transformer (T01), so that there are no breaks in the resin or hot spots in the transformer. INVKG3_ST3B, INVMG3_ST5A, SKIDG3_ST3B.
In this case, consult Power Electronics about the possibility of replacing the MV transformer (T01). See document INVG3/OP85379.

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5. Check that there are no hot spots or bad tightness on the low voltage PCBs and high voltage cables of the MV transformer (T01).
In this case, repair and if not possible replace it. [INVKG3_ST3B](#), [INVMG3_ST5A](#), [SKIDG3_ST3B](#). (See document [INVG3IOP85379](#)).



6. Check the ventilation ducts of the MV transformer (T01), so that they are not obstructed or the windings are displaced and obstruct the ventilation.
If they are obstructed, clean them and remove the obstruction. [INVKG3_ST3B](#), [INVMG3_ST5A](#), [SKIDG3_ST3B](#).
If they are dislocated, contact Power Electronics to see what action can be taken to correct the problem.



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PT100 wiring problem.

7. Check continuity between earth and ground of all PT100 cables (W043, W044, W045) of the MV transformer (T01).



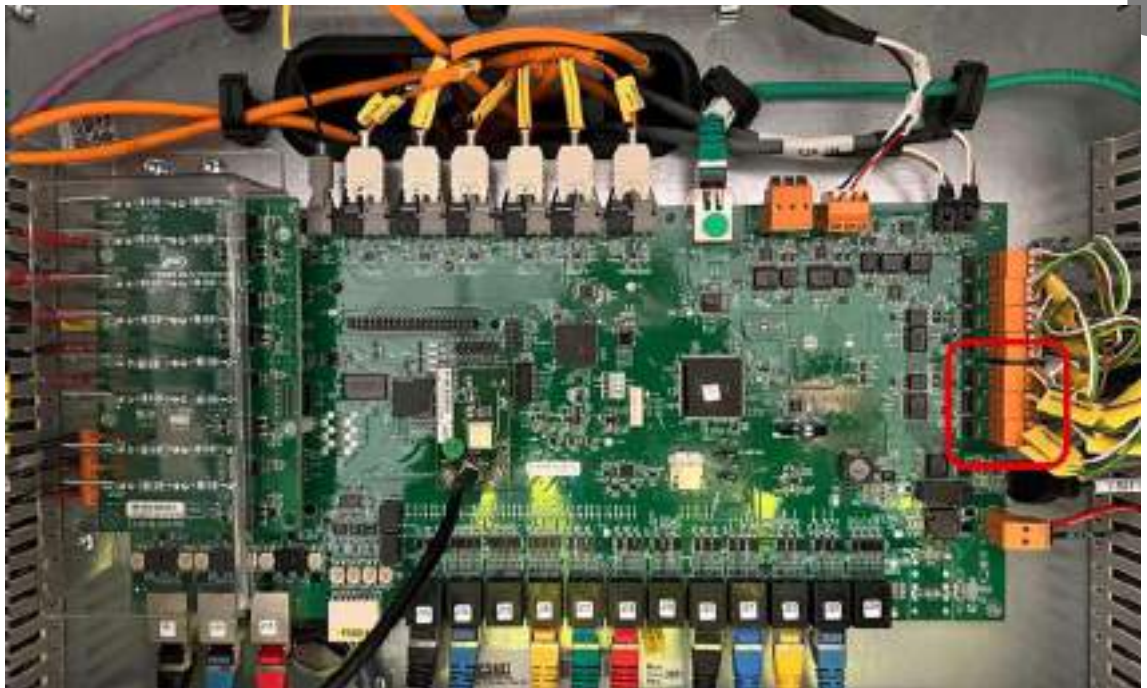
If there is no continuity, replace the corresponding cable. **INVKG2_ST3B, INVMG3_ST5A, SKIDG3_ST3B.**

8. Check for loose or incorrectly tightened wires at the terminals of the MV transformer (T01) and the central control board (E35). **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B.**
 - Check on the MV transformer (T01) that the cables W405, W406 and W407 are properly connected to the terminals XPT1, XPT2 and XPT3 of the MV transformer (T01). **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B.**



- check on the control board (E35) that the connectors J29, J28 and J27 are properly connected. **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B.**

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In the case of loose or incorrectly tightened cables, retighten and clean them. **INVKG3_ST2, INVMG3_ST5A, SKIDG3_ST3B.**

Erroneous temperature measurement.

The temperature of the transformer is correct (checked in point 1), but an erroneous temperature measurement has occurred and F208 has appeared without explanation. In this case, follow the steps below:

9. Check in SD card (Log_3), the temperatures at the MV transformer (T01) at the time of fault ([MV PT100 (1)], [MV PT100 (2)], [MV PT100 (3)]. **INVG3_ST1B.**

If the temperature has a very high value (≈ 65535), the cause may be due to a loose wiring problem (the value will appear for several seconds in Log_3) or the software has not read the temperature at that time.

If the problem occurs frequently, check the wiring connections and grounding grid if present.

If the wiring has been checked and secured, contact Power Electronics.

5.53. Fault code F217: Critical temperature filter LC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .		
Description	The critical temperature of the filter LC is over the configured limit during a configured time		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	W172		

Analysis procedure

This fault is activated when the temperature read by the resistance NTC of the filter is above the configured limit during a configured time.

When the fault releases, a ramp stop is performed, and after coming back to these conditions, the fault is not autoreset.

There are different causes that could lead into this fail. Follow the next steps in order to identify the possible cause and solve it.

1. Download via PowercommsV remotely all the data from the inverter of the SD and check at the moment of fault if the temperature of the filter LC was in the correct value of fault. In case an elevated temperature is reached or this elevated temperature continues, go to the following point. **INVG3_ST1B.**
2. Check in PowerCommsV ([LCL filter Temp] in the group [SV1.6-Ambient]) if the LC filter temperature is above the configured limits [LC filter] in the group [G5.4-Temperature]). Check the value of the configured critical temperature threshold. **INVG3_ST1B.**

This value should be default, unless otherwise justified.

If the temperature is above the limits and it is set correctly, proceed to the next step.

LC filter NTC probe in bad condition.

3. Check the resistance between the NTC and the central control board (E35), for this purpose disconnect connector J34 from the central control board (E35) and measure with a 1500V multimeter between connector pins XC307.1:1 and XC307.1:2. **INVKG3_ST2, INVMG3_ST5A.**

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The resistance varies as a function of the actual temperature as shown in the table below.

Temp °C	Resistance	Temp °C	Resistance	Temp °C	Resistance	Temp °C	Resistance
-35°C	144.30	0°C	27.27	30°C	6.70	71°C	2.18
-34°C	135.82	1°C	26.12	31°C	6.47	72°C	2.12
-33°C	129.96	2°C	25.03	32°C	6.25	73°C	2.04
-32°C	125.34	3°C	23.99	33°C	6.04	74°C	1.98
-31°C	117.15	4°C	23.06	34°C	5.83	75°C	1.93
-30°C	111.34	5°C	22.36	35°C	5.63	76°C	1.87
-29°C	106.91	6°C	21.18	36°C	5.44	77°C	1.82
-28°C	103.08	7°C	20.38	37°C	5.25	78°C	1.77
-27°C	98.54	8°C	19.49	38°C	5.08	79°C	1.72
-26°C	95.07	9°C	18.70	39°C	4.92	80°C	1.67
-25°C	90.55	10°C	17.95	40°C	4.75	81°C	1.62
-24°C	87.37	11°C	17.24	41°C	4.60	82°C	1.58
-23°C	83.42	12°C	16.58	42°C	4.45	83°C	1.53
-22°C	79.68	13°C	15.91	43°C	4.30	84°C	1.49
-21°C	75.13	14°C	15.29	44°C	4.15	85°C	1.45
-20°C	67.77	15°C	14.59	45°C	4.03	86°C	1.41
-19°C	64.59	16°C	14.12	46°C	3.90	87°C	1.37
-18°C	61.58	17°C	13.58	47°C	3.77	88°C	1.34
-17°C	58.72	18°C	13.06	48°C	3.68	89°C	1.30
-16°C	55.01	19°C	12.58	49°C	3.54	90°C	1.27
-15°C	51.44	20°C	12.09	50°C	3.43	91°C	1.23
-14°C	47.01	21°C	11.63	51°C	3.32	92°C	1.20
-13°C	42.89	22°C	11.20	52°C	3.22	93°C	1.17
-12°C	40.50	23°C	10.79	53°C	3.12	94°C	1.14
-11°C	41.41	24°C	10.38	54°C	3.02	95°C	1.11
-10°C	42.40	25°C	10.00	55°C	2.93	96°C	1.08
-9°C	40.55	26°C	9.65	56°C	2.84	97°C	1.05
-8°C	38.75	27°C	9.32	57°C	2.75	98°C	1.02
-7°C	37.06	28°C	8.95	58°C	2.67	99°C	1.00
-6°C	35.44	29°C	8.62	59°C	2.51	100°C	0.97
-5°C	33.91	30°C	8.31	60°C	2.44	101°C	0.95
-4°C	32.44	31°C	8.02	61°C	2.37	102°C	0.92
-3°C	31.05	32°C	7.73	62°C	2.30	103°C	0.90
-2°C	29.72	33°C	7.46	63°C	2.23	104°C	0.88
-1°C	28.47	34°C	7.20	64°C	2.16	105°C	0.86
		35°C	6.95				

- 3.1. Refer to the table for the equivalent in °C of the resistance value measured above. If it is below the configured limit (checked in point 2), reconnect the J34 connector to the central control board (E35) and check in PowerCommsV ([LCL filter Temp] in the group [SV1.6-Ambient]) if a temperature is higher than the fault temperature. **INVKG3_ST1B.**

If this is the case, substitute the main control board (E35). **INVKG3_ST2, INVMG3_ST5A.** (Check document *INVG3/OP88849*).

- 3.2. If the measured resistance values show a large difference with respect to the ambient temperature, replace the NTC probe. **INVKG3_ST2, INVMG3_ST5A.** Check document *INVG3/OP88849*. After that, check another time in PowerCommsV ([LCL filter Temp] in the group [SV1.6-Ambient]) that the measured resistance is the same that the one that is being read. **INVG3_ST1B.**

If they are the same value, the NTC is found correctly. Go to the next checkpoint.

LC filter fans failure.

4. Configure PowerCommsV ([Mode] in the group [G11.6-Config. LC Filter]) the speed of the fans to 100% and check that they rotate at the configured revolutions in the group [SV11.1-Fans].

If not, check PowerCommsV in the group ([SV11-Fans]) which failure or warning appears and proceed following fan manual instructions. **INVG3_ST1B.**

If any ventilation warning appears during the fault, continue to the next checkpoint.

5. Check that no fan rubs against any part of the equipment and that they are well anchored. **INVKG3_ST2, INVMG3_ST5A.**
6. Check that the fan rotor Blades are not broken or worn.
7. Check that the power and communications cables do not rub against the moving parts of the fan. The cables must be well connected and guided to avoid connectivity failures and pin bending, ...

If there is any cable in bad conditions, you can proceed to change the fan or the cables as appropriate. **INVKG3_ST2, INVMG3_ST5A.** Check document *INVG3/OP85370*.

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5.54. Fault code F220: PT100 Inductance no connected

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	One or more of the PT100 of the inductance is not connected to one of its terminals.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

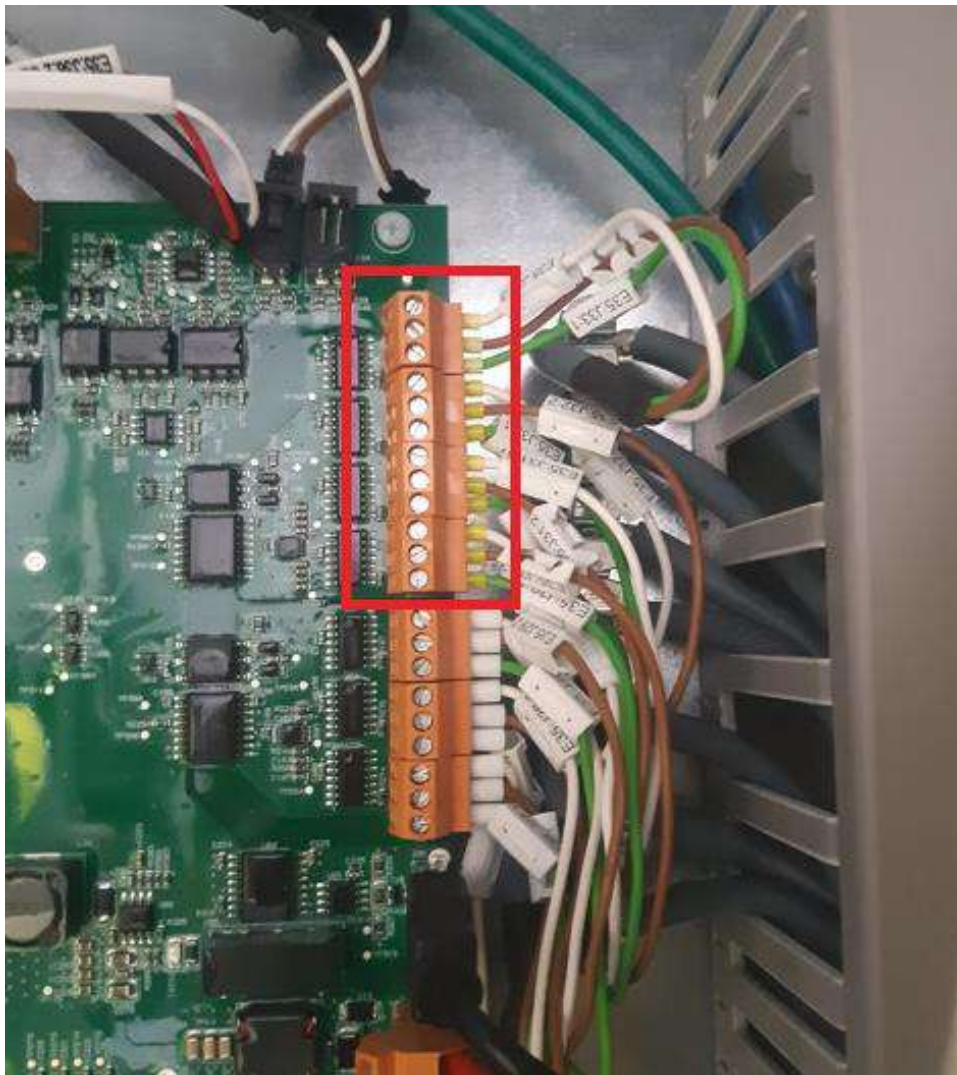
Analysis procedure

This fault is triggered when any of the PT100s located on the LC filter measures a temperature of -40°C or lower for more than 5 seconds. If for 60 seconds a temperature of 5°C higher than the activation temperature (-40°C) is measured, the fault disappears.

1. Check in PowerCommsV (group [SV1.6: Ambient]) the temperature measurements of each PT100 to know which one is triggering the fault according to the instance. **INVKG3_ST1B, INVMG3_ST1B.**
2. The following table shows the connectors and pins corresponding to each PT100 to perform the next checks:

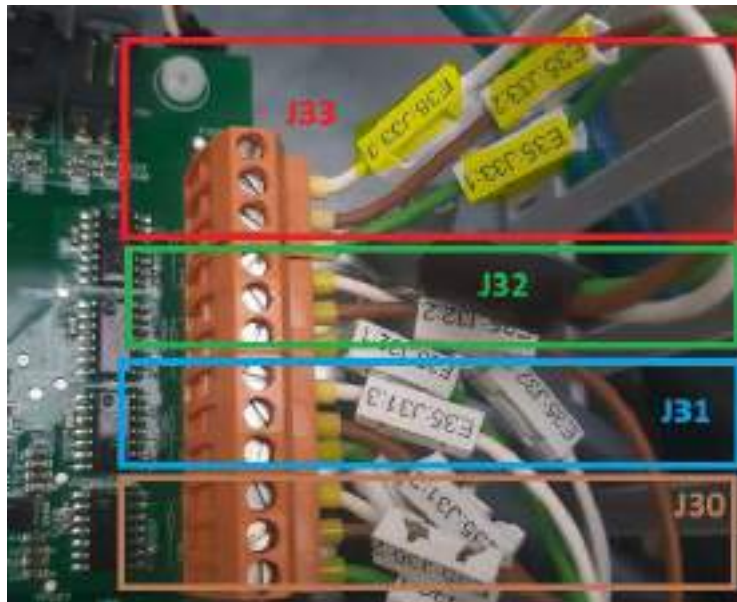
Fault instance	Connector at the central control board (E35)	Terminal block at the main inductance (L01)	Pins of the terminal block at the main inductance (L01)	Connection cable
1	J30	XPT1	A B1 B2	W052
2	J31	XPT2	A' B1' B2'	W053
3	J32	XPT3	A'' B1'' B2''	W054
4	J33	XPT4	A''' B1''' B2'''	W055

- 2.1. Check that there is continuity with a 1500V multimeter between the central control board connector (E35) and the terminal block pins on inductance (L01) according to the PT100 that is triggering the fault. **INVKG3_ST2, INVMG3_ST5A.**
 - Measure between pin 1 of the board connector and pin B2 of the terminal block.
 - Measure between pin 2 of the board connector and pin B1 of the terminal block.
 - Measure between pin 3 of the board connector and pin A of the terminal block.

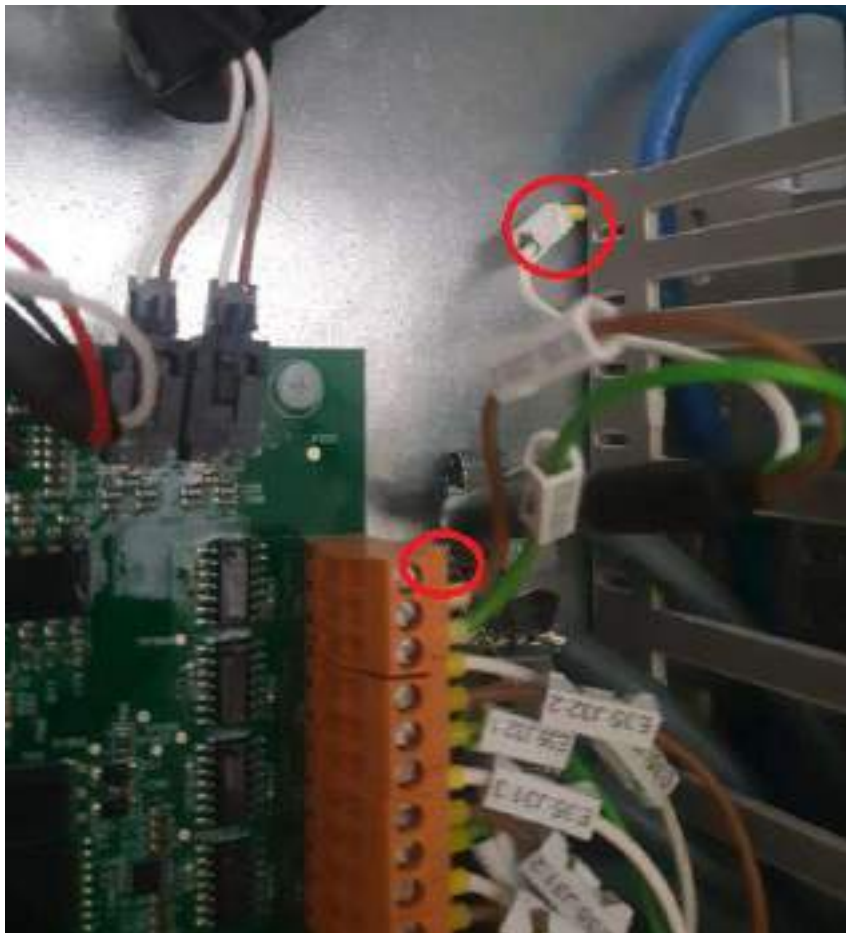


Connectors at the central control board (E35)

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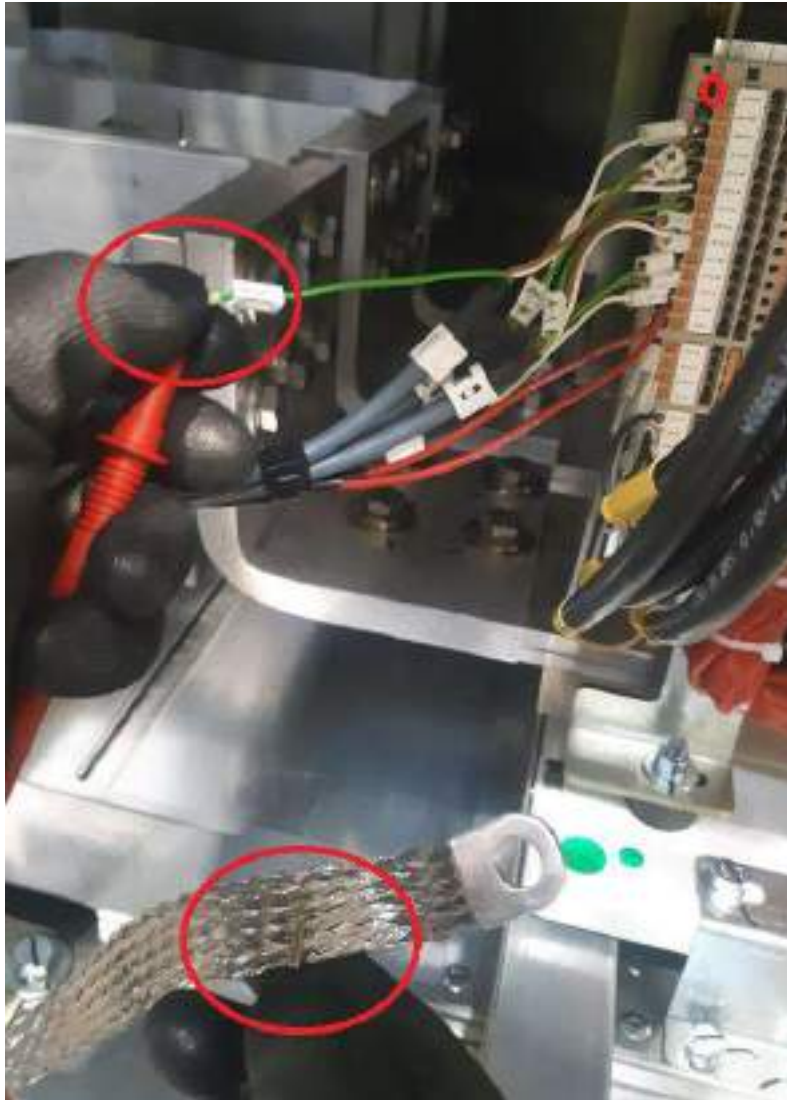


Each connector and pin of the central control board (E35)



Pin 1 of J33 to the ground

EN



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XPT2 terminal block at the main inductance (L01) of the pin B2 in order to measure continuity with the pin of the connector J33 of the central control board (E35)

Do the same for each connector and pin of the central control board (E35) with each corresponding terminal block of the main inductance.

- 2.2. If there is no continuity at any of the points measured above, replace the corresponding cable. **INVKG3_ST2, INVMG3_ST5A.**
- 2.3. Check the resistance between the PT100 which is triggering the fault and the central control board (E35) by disconnecting the connector of the central control board (E35) and measuring with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**
 - Between pins 1-2 of the connector, measured value $\approx 0\Omega$.
 - Between pins 2-3 of the connector, the measured value varies depending on the inductance temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).



Connector J33 of the central control board (E35) between pins 1-2



Connector J33 of the central control board (E35) between pins 2-3

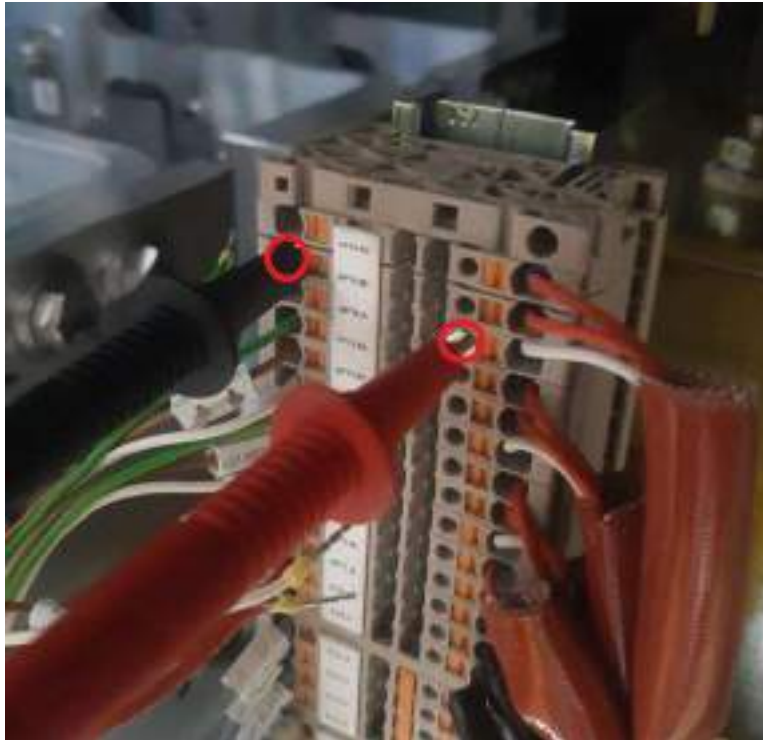
Repeat it for all the pins of each connector of the central control board (E35)

If the measured values are correct, replace the central control board (E35). **INVKG3_ST2**, **INVMG3_ST5A**. (Refer to document *INVG3/OP88849*).

Otherwise, proceed to the next check point.

2.4. Check the resistance at the pins of the inductance (L01) terminal block according to the PT100 which is triggering the fault. To do so, measure with a 1500V multimeter: **INVKG3_ST2**, **INVMG3_ST5A**.

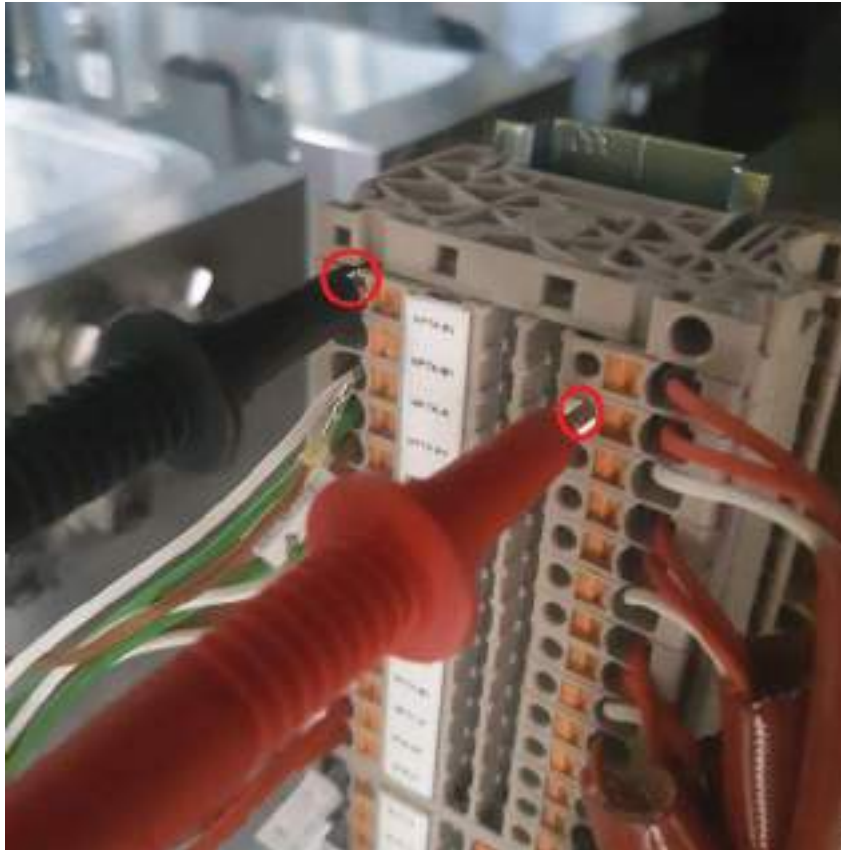
- Between pins A-B1 of the terminal block, the measured value varies depending on the inductance temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).
- Between pins B1-B2 of the terminal block, measured value $\approx 0\Omega$.



Pins A-B1 of the terminal block XPT4 at the main inductance (L01)



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Pins B1-B2 of the terminal block XPT4 at the main inductance (L01)



Repeat it for the rest of the pins if the terminal blocks at the main inductance (L01)

If the measured values are not correct, replace the inductance (L01). **INVKG3_ST2, INVMG3_ST5A.**
(Refer to document *INVG3/OP85375*).

5.55. Fault code F221: PT100 MV Transformer no connected.

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	Una o varias de las PT100 del transformador no está conectada en alguno de sus terminales.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

Analysis procedure

This fault is triggered when any of the PT100s located on the MV transformer measures a temperature of -40°C or lower for more than 5 seconds. If for 60 seconds a temperature of 5°C higher than the activation temperature (-40°C) is measured, the fault disappears.

1. Check in PowerCommsV (group [SV1.6: Ambient]) the temperature measurements of each PT100 to know which one is triggering the fault. **INVMG3_ST1B**.
2. If the inverter is an HEM / PCSM:

The following table shows the connectors and pins corresponding to each PT100 to perform the next checks:

Fault instance	Connector at the central control board (E35)	Terminal block at the MV transformer (T01)	Connection cable*
1	J27	XPT1	W043, W405
2	J28	XPT2	W044, W406
3	J29	XPT3	W045, W407

* Cable reference at both sides of the interconnection hating.

- 2.1. Check that there is continuity with 1500V multimeter between the central control board connector (E35) and the terminal block pins on MV transformer (T01) according to the PT100 that is triggering the fault. **INVMG3_ST5A**.
 - Measure between pin 1 of the board connector and pin B2 of the terminal block.
 - Measure between pin 2 of the board connector and pin B1 of the terminal block.
 - Measure between pin 3 of the board connector and pin A of the terminal block.

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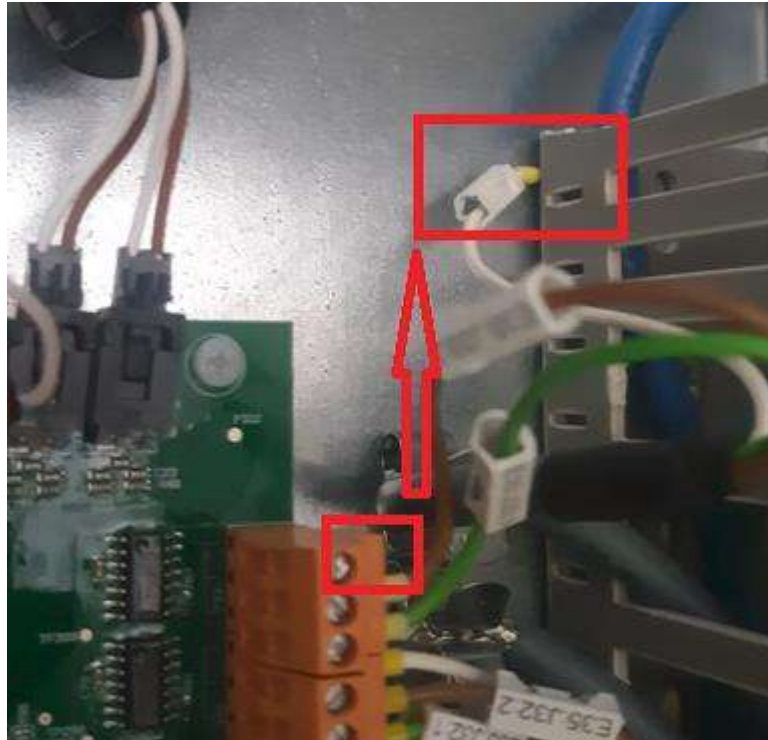


Each connector and pin of the central control board(E35)

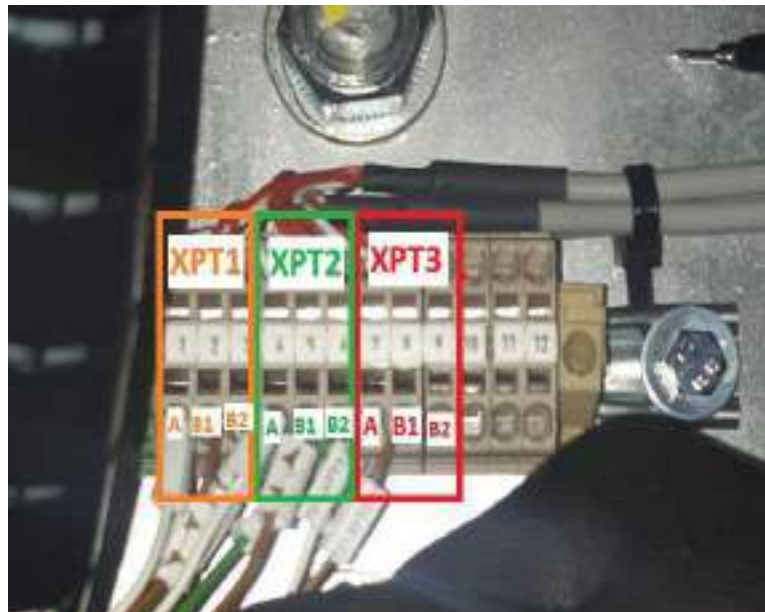


Connect separately to the ground each pin of each connector for each measure. Example below:

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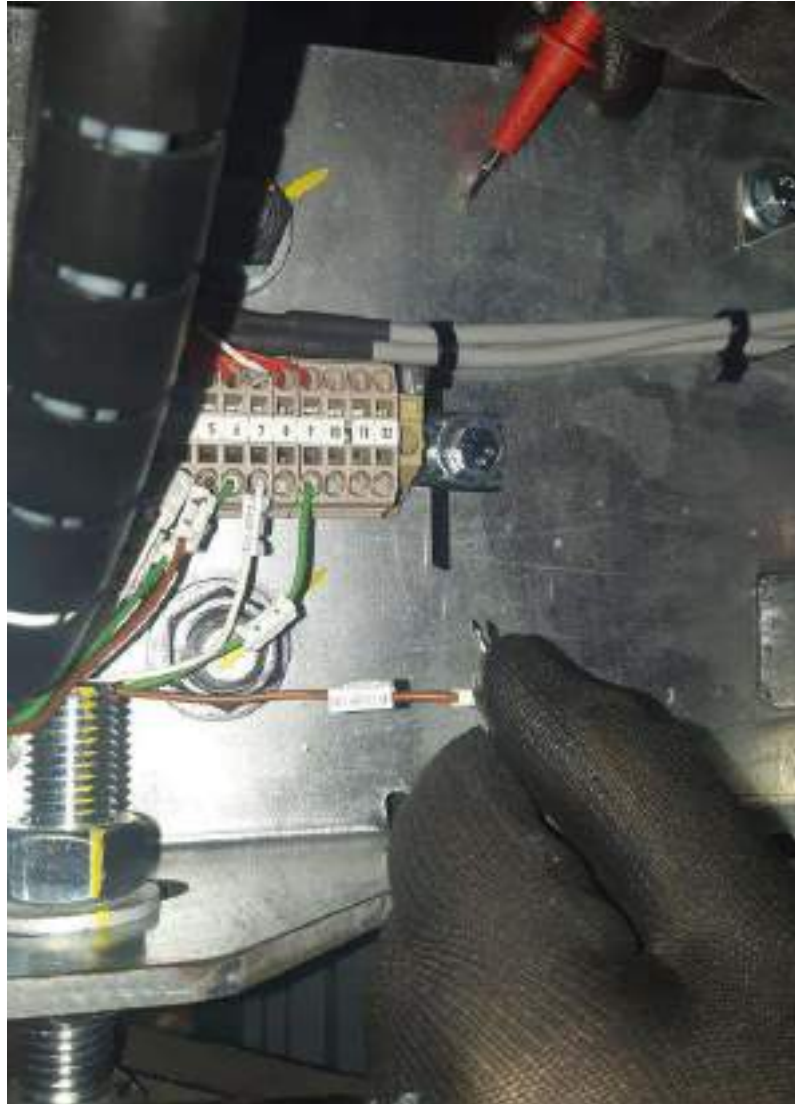


Example of wire connected to the ground



Terminal block of MV transformer

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Example of checking the continuity from MV transformer

- 2.2. If there is no continuity at any of the points measured above, replace the corresponding cable. **INVMG3_ST5A.**
- 2.3. Check the resistance between the PT100 which is triggering the fault and the central control board (E35) by disconnecting the connector of the central control board (E35) and measuring with a 1500V multimeter: **INVMG3_ST5A.**
 - Between pins 1-2 of the connector, measured value $\approx 0\Omega$.
 - Between pins 2-3 of the connector, the measured value varies depending on the inductance temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).

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Example checking resistance of pins 1-2. Realice it for all the connectors of the central control board J29, J28 and J27.



Example checking resistance of pins 2-3. Realice it for all the connectors of the central control board J29, J28 and J27.

If the measured values are correct, replace the central control board (E35). **INVMG3_ST5A**. (Refer to document *INVG3IOP88849*).

Otherwise, proceed to the next check point.

2.4. Check the resistance at the pins of the MV transformer (T01) terminal block according to the PT100 which is triggering the fault. To do so, measure with a 1500V multimeter: **INVMG3_ST5A**.

- Between pins A-B1 of the terminal block, the measured value varies depending on the MV transformer temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).
- Between pins B1-B2 of the terminal block, measured value $\approx 0\Omega$.



Example checking resistance of pins A-B1 of the terminal block XPT3. Realice if for the rest of the terminal blocks too.



Example checking resistance of pins B1-B2 of the terminal block XPT3. Realice ir for the rest of the terminal blocks too.

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If the measured values are not correct, replace the MV transformer (T01). **INVMG3_ST5A**. (Refer to document *INVG3IOP85379*).

3. If the inverter is an HEMK / PCSK:

The following table shows the connectors and pins corresponding to each PT100 to perform the next checks:

Fault instance	Connector at the central control board (E35)	Terminal block at the MV transformer (T01)	Connection cable*
1	J27	XPT1	W043, W480

* Cable reference at both sides of the interconnection harting. HEMK: cable W043, SKID: cable W480.

- 3.1. Check that there is continuity between the central control board connector (E35) and the terminal block pins on MV transformer (T01) according to the PT100 that is triggering the fault. **INVKG3_ST2, SKIDG3_ST2**.
- Measure between pin 1 of the board connector and pin B2 of the terminal block.
 - Measure between pin 2 of the board connector and pin B1 of the terminal block.
 - Measure between pin 3 of the board connector and pin A of the terminal block.

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Connector J27 and pin of the central control board (E35). In HEMK/PCSK J28 and J29 do not exist J28 and J29.

- 3.2. If there is no continuity at any of the points measured above, replace the corresponding cable.
INVKG3_ST2, SKIDG3_ST2.
- 3.3. Check the resistance between the PT100 which is triggering the fault and the central control board (E35). To do so, disconnect the J27 connector of the central control board (E35) and measure with a 1500V multimeter: **INVKG3_ST2, SKIDG3_ST2.**
- Between pins 1-2 of the connector, measured value $\approx 0\Omega$.
 - Between pins 2-3 of the connector, the measured value varies depending on the MV transformer temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).



Example of checking resistance of pins 1-2.

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Example of checking resistance of pins 2-3.

If the measured values are correct, replace the central control board (E35). **INVKG3_ST2**, **SKIDG3_ST2**. (Refer to document *INVG3/OP88849*).

Otherwise, proceed to the next check point.

3.4. Check the resistance at the pins of the MV transformer (T01) terminal block. To do so, measure with a 1500V multimeter: **INVKG3_ST2**, **SKIDG3_ST2**

- Between pins A-B1 of the terminal block, the measured value varies depending on the MV transformer temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).
- Between pins B1-B2 of the terminal block, measured value $\approx 0\Omega$.



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Example of checking resistance of pins A-B1.



Example of checking resistance of pins B1-B2

If the measured values are not correct, replace the MV transformer (T01). **INVKG3_ST2, SKIDG3_ST2** (Refer to document *INVG3/OP85379*).

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5.56. Fault code F222: NTC admission temperature no connected.

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Admission temperature NTC resistor is not connected.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is triggered when the NTC resistor located at the air inlet of the AC cabinet measures an admission temperature of -28°C or lower for more than 5 seconds. If for 60 seconds a temperature of 5°C higher than the activation temperature (-28°C) is measured, the fault can be reset manually, as it is not self-resettable. [INVKG3_ST3A](#), [INVMG3_ST3A](#).

1. Check in PowerCommsV (group [SV1.6: Ambient]) the value of the admission temperature measurement. [INVKG3_ST1B](#), [INVMG3_ST1B](#).
2. Check the resistance between the NTC and the central control board (E35) by disconnecting the J35 connector of the central control board (E35) and measuring with a 1500V multimeter between pins of XC308.1:1 and XC308.1:2 connectors. [INVKG3_ST2](#), [INVMG3_ST5A](#).



The resistance varies as a function of temperature; therefore, the measured values should be $\approx 10\text{K}\Omega$ for a temperature of 25°C.

Otherwise:

- If the measured resistance value is between $95,64\Omega$ (-27°C) and $0,86\text{K}\Omega$ (105°C) with a tolerance of $\pm 1\%$, replace the central control board (E35). [INVKG3_ST2](#), [INVMG3_ST5A](#). See document INVKG3IOP88849.

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- If the measured value is $> 100.58\Omega$ (-28°C) o $< 0.86\text{K}\Omega$ (105°C) with a tolerance of $\pm 1\%$, reconnect J35 connector on the central control board (E35), **INVKG3_ST2**, **INVMG3_ST5A**, and check in PowerCommsV (grupo [SV1.6: Ambiente]) the value of the admission temperature measurement, **INVKG3_ST1B**, **INVMG3_ST1B**. If this is a temperature $> -28^{\circ}\text{C}$, replace the NTC temperature sensor. **INVKG3_ST2**, **INVMG3_ST5A**. See document INVG3IOP88494.

5.57. Fault code F223: NTC LC filter no connected.

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	NTC LC filter is not connected.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is triggered when the NTC resistor located on the LC filter measures an admission temperature of -28°C or lower for more than 5 seconds. If for 60 seconds a temperature of 5°C higher than the activation temperature (-28°C) is measured, the fault can be reset manually, as it is not self-resettable. [INVKG3_ST3A](#), [INVMG3_ST3A](#).

1. Check in PowerCommsV (group [SV1.6: Ambient]) the value of the LC filter temperature measurement. [INVKG3_ST1B](#), [INVMG3_ST1B](#).
2. Check the resistance between the NTC and the central control board (E35) by disconnecting the J34 connector of the central control board (E35) and measuring with a 1500V multimeter between pins of XC307.1:1 and XC307.1:2 connectors. [INVKG3_ST2](#), [INVMG3_ST5A](#).



The resistance varies as a function of temperature; therefore, the measured values should be $\approx 10\text{K}\Omega$ for a temperature of 25°C.

Otherwise:

- If the measured resistance value is between $95,64\Omega$ (-27°C) and $0,86\text{K}\Omega$ (105°C) with a tolerance of $\pm 1\%$, replace the central control board (E35). [INVKG3_ST2](#), [INVMG3_ST5A](#). See document [INVKG3IOP88849](#).
- If the measured value is $> 100.58\Omega$ (-28°C) or $< 0.86\text{K}\Omega$ (105°C) with a tolerance of $\pm 1\%$, reconnect J34 connector on the central control board (E35), [INVKG3_ST2](#), [INVMG3_ST5A](#), and check in

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PowerCommsV (grupo [SV1.6: Ambiente]) the value of the LC filter measurement, **INVKG3_ST1B**, **INVMG3_ST1B**. If this is a temperature $>-28^{\circ}\text{C}$, replace the NTC temperature sensor. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3/OP88493*.

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5.58. Fault code F90: T^a IGBTs HF

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The temperature measurement of an IGBT has exceeded the maximum threshold defined by hardware (this threshold is not configurable).	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	F103	

Analysis procedure

This fault has the same causes as fault F103 *Temperature IGBTs* and both may have been reported in the same fault case, only in this case the threshold is fixed and always higher. Therefore, the diagnosis and resolution method is the same:

To solve this fault, proceed to check if the IGBT temperature readings have reached the threshold of 127°C.

The module ventilation is not working properly:

1. This can be caused by a problem in the fans, or in the intake of the inverter. If this occurs, the temperature of all IGBT in the module will be abnormally high with respect to adjacent modules. For this, check the state of the module ventilation:

- 1.1. Check the fan settings for PowerCommsV. It should be at its default value unless otherwise justified (group [G11.5: Config Modules]). **INVKG3_ST1A**.
- 1.2. Check that all fans are rotating at the configured setpoint and without unjustified speed variations (groups [SV11.3: Fan 3], [SV11.4: Fan 4], [SV11.5: Fan 5] and [SV11.6: Fan 6]). If this is not the case, check the fan warnings and correct according to the relevant warning. **INVKG3_ST1A**.
- 1.3. Check that there are no strange noises in the fan and that they rotate correctly. Don't open the equipment. **INVKG3_ST1A**.

If there are noises, check that the fan support is fixed and that the nozzle is in good condition without rubbing against the fan, that there is no object obstructing the air inlet and outlet of the inverter, check the fan bearing and also the air inlet filters. **INVKG3_ST3B, INVKG3_ST3B**.

If there is anything obstructing the air inlet, remove it. **INVKG3_ST3B, INVKG3_ST3B**.

If the problem persists:

- 1.3.1. Check that there is no dirt or foreign matter on the top of the module heatsink. To access the air duct where the heatsink is located, the stage must be removed. **INVKG3_ST3B, INVKG3_ST3B**. (See document *INVKG3/OP85369*).

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- 1.3.2. Check the AC and DC power connections of the module for the presence of hot spots. **INVKG3_ST3B, INVMG3_ST3B.**

Excess current for that phase/module. When this happens, one more phase or module will normally be displayed with a higher temperature than another (parameters [Current R], [Current S] and [Current T], from group [SV12: Modules Information]). **INVG3_ST1A.** In this case:

2. Activate the reverse sequence of the device (parameter [Inverse sequence], in the [Advanced config] group) to balance the AC currents per phase. If the problem persists, contact Power Electronics. **INVG3_ST1B.**

There is a connection problem in the ribbon cables (W033 and W034) connecting the E01 board to the E02 module FPGA board. If there is a bad connection, the temperature of the IGBTs displayed will be approximately 203°C (group [SV12.x.3: Temperature Measurements]). **INVG3_ST1B.** If this occurs:

3. Check that the cable is well connected and not damaged. If it is in good condition but still fails, replace the flat cables **INVKG3_ST3B, INVMG3_ST3B.** If it still fails:



- 3.1. Replace (E01) IGBT board. **INVKG3_ST3B, INVMG3_ST3B.** (See document *INVG3/OP88076*).
- 3.2. If the problem persists, replace module control board (E02). **INVKG3_ST3B, INVMG3_ST3B.** (See document *INVG3/OP88076*).

There is a problem connecting the flat cables (W020 to W028) from the E01 distribution board to the GATE DRIVE board of the IGBT (IGBT01-09). **INVKG3_ST3B, INVMG3_ST3B**. If this occurs, an IGBT at a temperature of approximately 201°C will be displayed. In this case:

4. Check that the IGBT cable is well connected and not damaged. If it is in good condition but still fails, replace the IGBT GATE DRIVE board (IGBT01-09) and ribbon cable. **INVKG3_ST3B, INVMG3_ST3B**. (See document *INVG3IOP88076*).
- 4.1. If the problem persists, replace the IGBT board (E01) board. **INVKG3_ST3B, INVMG3_ST3B**. (See document *INVG3IOP88076*).
- 4.2. If the problem persists, replace the module control board (E02). **INVKG3_ST3B, INVMG3_ST3B**. (See document *INVG3IOP88076*).

There is an IGBT in bad condition due to a bad tightening of its electrical connections, internal defect, overheated points or thermal interface (graphite) in bad condition. It is detected because the thermal behaviour of the damaged IGBT is different from the rest, rising much faster than the others in the same position and reaching higher temperature than the rest for the same current. Likewise, the damaged IGBT will drop in temperature faster than the rest. In steady state with the same current it will remain about $\geq 5-10^{\circ}\text{C}$ higher than other IGBTs in same position. If this happens:

5. Visually check IGBT connections for: hot spots, discoloured or degraded platens or screws, connections susceptible to retightening, or degraded adjacent parts **INVKG3_ST3B, INVMG3_ST3B**. If this is the case:



- 5.1. Re-tighten connections and repair what has been detected. Check all power module hardware. **INVKG3_ST3B, INVMG3_ST3B**.





- 5.2. Check temperature of the IGBT in operation and see its behaviour by PowerCommsV (group [SV12: Modules Information]). **INVG3_ST1A**. In case the IGBT follows the same thermal behaviour, disassemble the power module and replace the problematic IGBT. **INVKG3_ST3B**, **INVMG3_ST3B**. (See document *INVG3/OP88076*).
- 5.3. If the problem persists, then replace the power module. **INVKG3_ST3B**, **INVMG3_ST3B**. (See document *INVG3/OP85369*).
- 5.4. If the problem persists, please contact Power Electronics.

EN

5.59. Fault code F224: Ind overtemperature

Obligatorio	El usuario debe consultar las Instrucciones de Seguridad para la Operación, Resolución de Problemas y Mantenimiento antes de empezar. Cada acción descrita en la secuencia tiene un estado definido. No existen estados generales. El usuario debe leer y comprender los estados descritos en el documento mencionado en la sección “Introducción” .			
Descripción	La temperatura de alguna de las inductancias de módulo es superior a la configurada.			
Aplicable a	☒ HEM	☒ PCSM/MPCSM		
	☒ HEMK	☒ PCSK/MPCSK		
Fallos o warnings asociados	F170, F225			
Procedimiento de análisis				

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it. Problema de sobrettemperatura en la inductancia (L01).

- Review via PowercommsV on SD card (Log_3) of the inverter if the temperature increase is real or if it was a noise problem in the temperature measurement. **INV G3_ST1B**.
 - If the temperature has risen progressively and without ripple, it is a problem of overtemperature in the inductance. Perform steps 5-12.
 - If there is excessive ripple in the PT100 temperature measurement, it may be a wiring problem. Go to step 3.

Poor configuration of the temperature thresholds.

- Check the configuration of the fault, for this purpose consult via PowercommsV ([Inductance] within the group [G5.4-Temperature]), if the value of the inductance failure threshold temperature (L01) and hysteresis are below their default values. In this case, consult Power Electronics in case it is a special configuration of this plant. **INV G3_ST1B**.

If the configuration is correct, continue with the following:

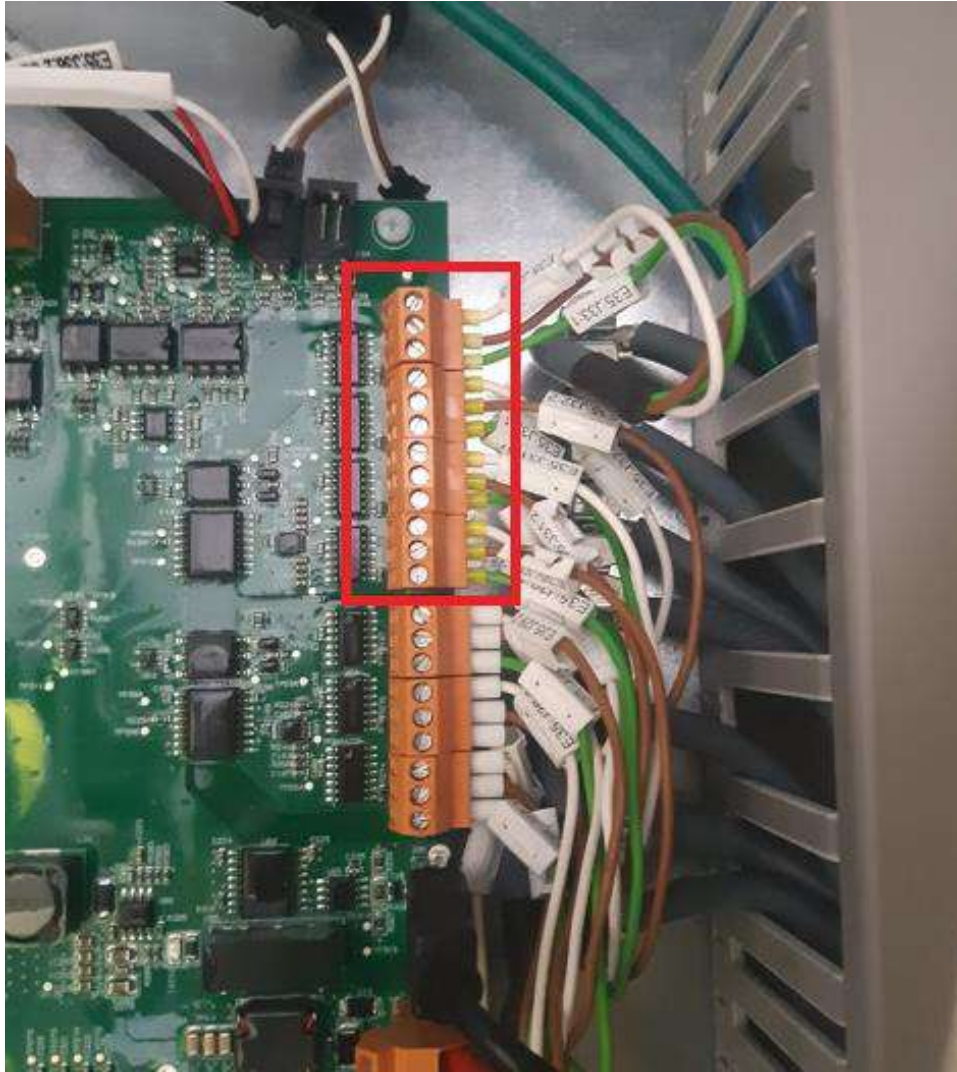
PT100 wiring problem.

- The following table lists the connectors and pins corresponding to each PT100 to perform the following checks::

Fault instance	Connector at the central control board (E35)	Terminal block at the main inductance (L01)	Pins of the terminal block at the main inductance (L01)	Connection cable
1	J30	XPT1	A B1 B2	W052
2	J31	XPT2	A' B1' B2'	W053
3	J32	XPT3	A'' B1'' B2''	W054
4	J33	XPT4	A''' B1''' B2'''	W055

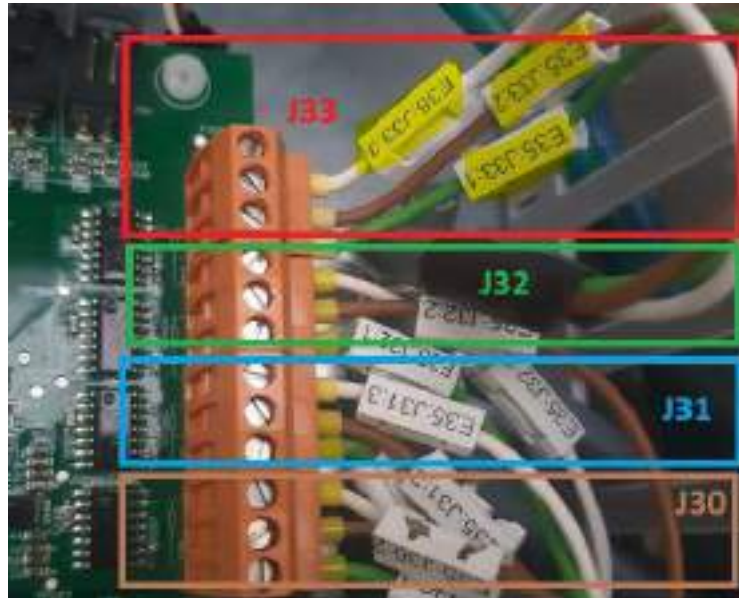
- Check with 1500V multimeter that there is continuity between the central control board connector (E35) and the terminal block pins on the inductance (L01) according to the PT100 that is failing. **INV K G3_ST2, INV M G3_ST5A**.

- medir entre el pin 1 del conector de la tarjeta y B2 del bornero
- medir entre el pin 2 del conector de la tarjeta y B1 del bornero
- medir entre el pin 3 del conector de la tarjeta y A del bornero.



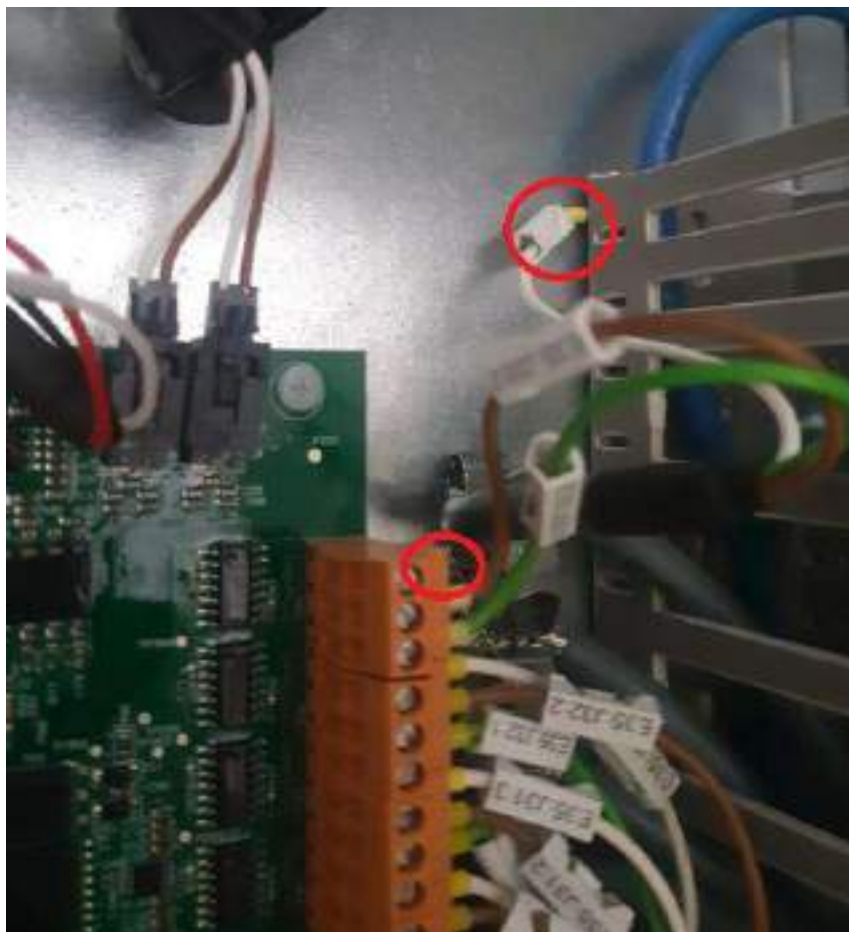
Connectors at the central control board (E35)

EN

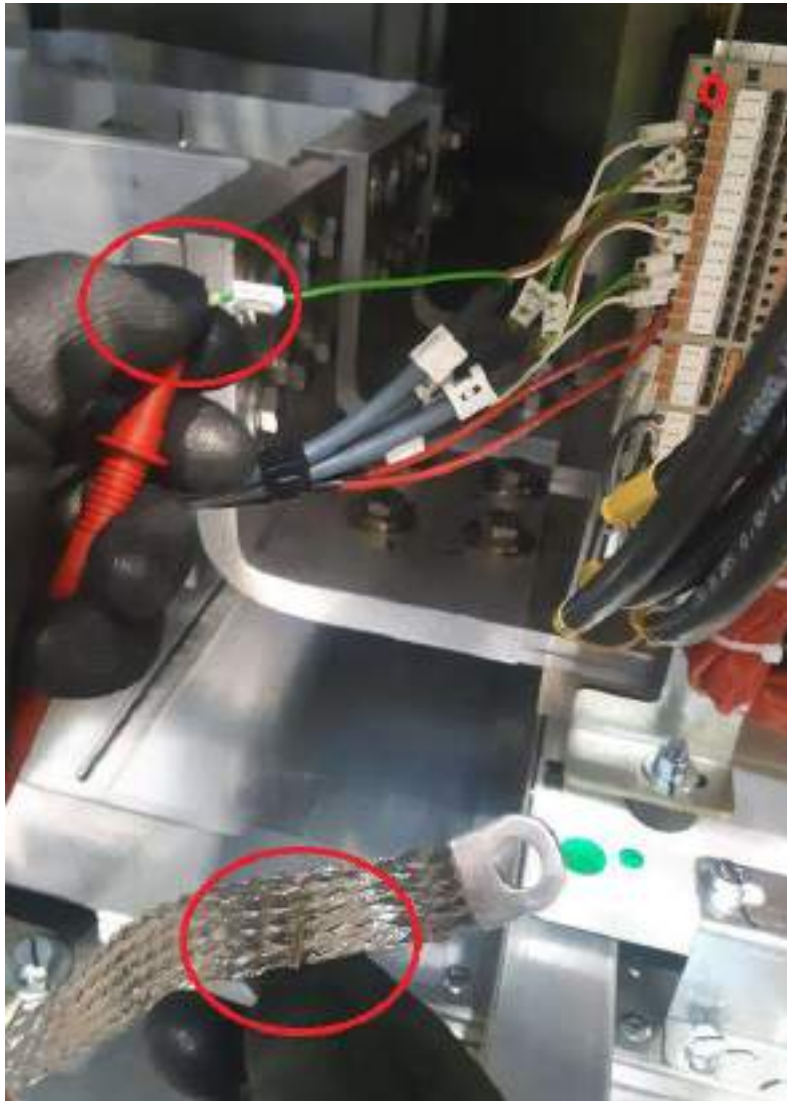


Each connector and pin of the central control board (E35)

EN



Pin 1 of J33 to the ground



EN

XPT2 terminal block at the main inductance (L01) of the pin B2 in order to measure continuity with the pin of the connector J33 of the central control board (E35)

Do the same for each connector and pin of the central control board (E35) with each corresponding terminal block of the main inductance.

3.2. If there is no continuity at any of the points measured above, replace the corresponding cable. **INVKG3_ST2, INVMG3_ST5A.**

3.3. Check the resistance between the PT100 which is triggering the fault and the central control board (E35) by disconnecting the connector of the central control board (E35) and measuring with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

- Between pins 1-2 of the connector, measured value $\approx 0\Omega$.
- Between pins 2-3 of the connector, the measured value varies depending on the inductance temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).



Connector J33 of the central control board (E35) between pins 1-2



Connector J33 of the central control board (E35) between pins 2-3

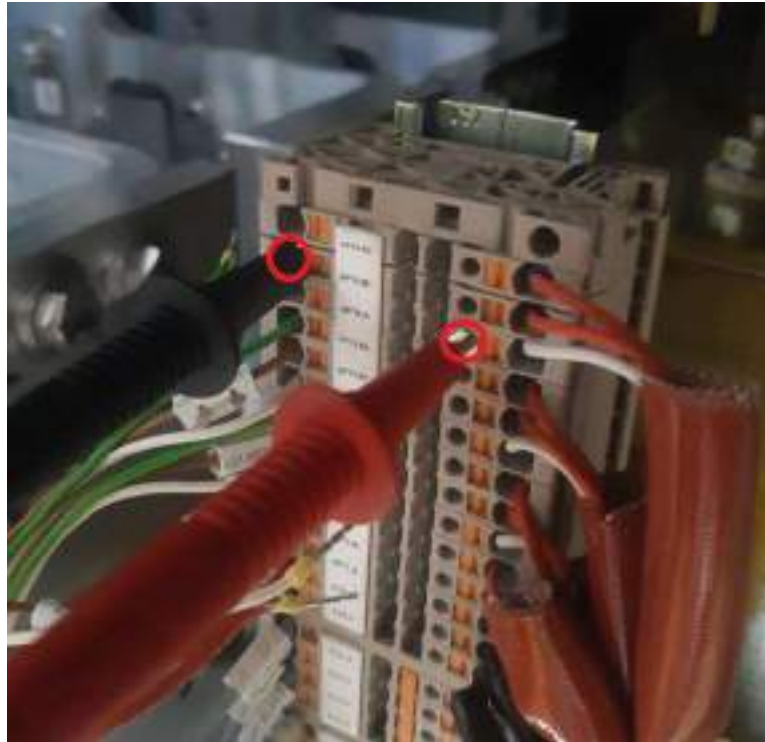
Repeat it for all the pins of each connector of the central control board (E35)

If the measured values are correct, replace the central control board (E35). **INVKG3_ST2**, **INVMG3_ST5A**. (Refer to document *INVG3/OP88849*).

If not, proceed to the next check point.

3.4. Check the resistance on the pins of the inductance terminal block (L01) according to the failed PT100 by measuring with a 1500V multimeter: **INVKG3_ST2**, **INVMG3_ST5A**.

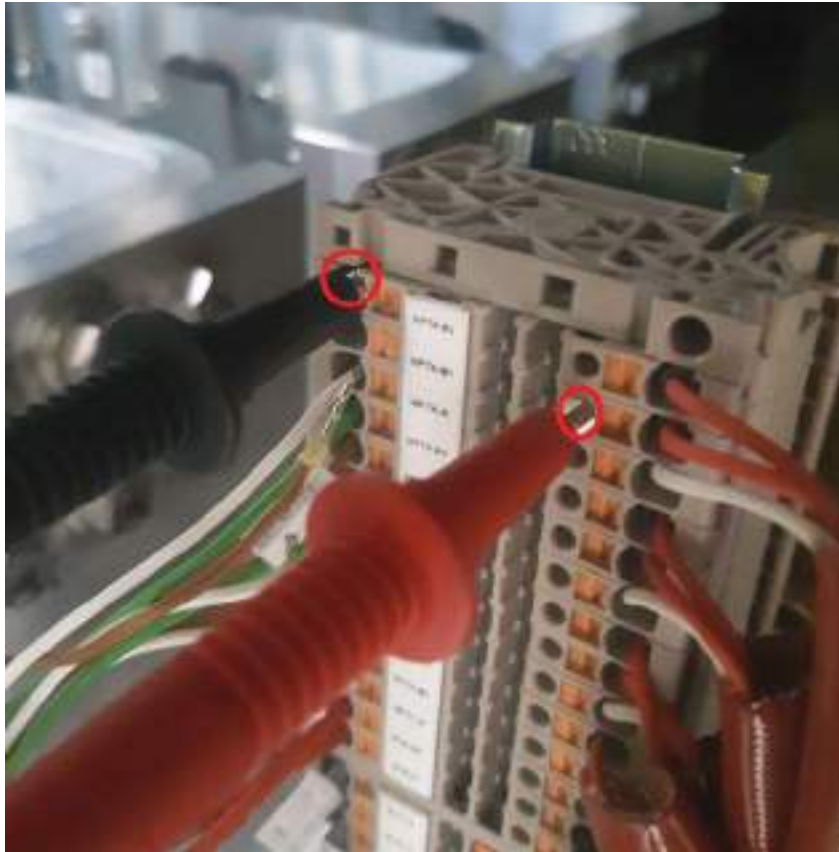
- between pins A-B1 of the terminal block, the value varies according to the temperature of the inductance. $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).
- between the pins B1-B2 of the terminal block, the measured value $\approx 0\Omega$.



Pins A-B1 of the terminal block XPT4 at the main inductance (L01)



EN



Pins B1-B2 of the terminal block XPT4 at the main inductance (L01)



Repeat it for the rest of the pins if the terminal blocks at the main inductance (L01)

If the measured values are not correct, replace the inductance (L01). INVKG3_ST2, INVMG3_ST5A.
(Refer to document *INVG3IOP85375*)

Wrong temperature measurement.

The temperature of the inductance is adequate (checked in point 1), but an erroneous temperature point measurement has occurred and the F225 has appeared without explanation. In this case, follow the steps below:

4. Check via PowercommsV of the SD board (Log_3), temperatures at the choke (L01) at the time of failure ([IND PT100 (1)], [IND PT100 (2)], [IND PT100 (3)], [IND PT100 (4)]). **INVK3_ST1B.**

If the temperatura has a very high value (≈ 65535), the cause may be due to a loose wiring problem (the value will appear for several seconds in the Log_3) or the software has not read the temperature at that time.

If the problem occurs frequently, check the wiring and grounding screen connections if present. **INVK3_ST1B, INVMG3_ST5A.**

If the wiring has been checked and secured, but high inductance temperatures are still measured, proceed to the next check point.

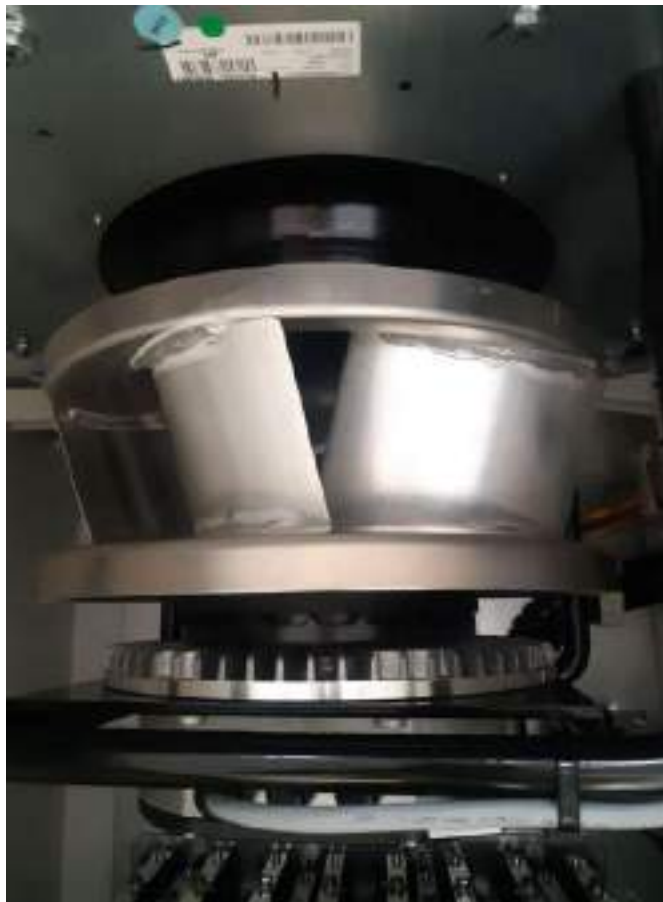
LC filter fan fault.

5. Configure via PowercommsV ([Mode] within the group [G11.6-Config. LC Filter]) the fan speed to 100% and check that the fans are rotating at the set speed within the group [SV11.1-Fan]). **INVG3_ST1B.**

If not, check via PowerCommsV in group ([SV11-Fan]) waht fault or warning appears and act according the fan manual. **INVG3_ST1B.**

If a ventilation warning appears during the fault, continue to the next step.

6. Check that no fans are rubbing against any part of the equipment and that they are properly anchored. **INVKG3_ST2, INVMG3_ST5A.**

**EN**

7. Check that the fan rotor blades are not broken or worn.
8. Check that the power and communication cables do not rub against the moving parts of the fan. Cables must be well connected and routed to avoid connectivity failures, bent pins, ... [INVKG3_ST2](#), [INVMG3_ST5A](#).
If any cables are in poor condition, replace the fan or cables as appropriate. [INVKG3_ST2](#), [INVMG3_ST5A](#). See document [INVG3/OP85370](#).

Module fans failure.

9. Configure via PowerCommsV ([Mode] within the group [G11.5-Config. Modules]) the fan speed to 100% and check that the fans are rotating at the set speed en el grupo [SV11.1-Fan]). [INVG3_ST1B](#).

If not, check via PowerCommsV within group ([SV11-Fan]) which fault or warning appears and act according to the fan's manual. [INVG3_ST1B](#).

If a ventilation warning appears during the failure, continue to the next test step.

10. Check that no fans are rubbing against any part of the equipment and that they are securely anchored. [INVKG3_ST2](#), [INVMG3_ST5A](#).



11. Check that the fan rotor blades are not broken or worn. [INVKG3_ST2](#), [INVMG3_ST5A](#).

EN

12. Comprobar que los cables de alimentación y comunicaciones no rocen con las partes móviles del ventilador. Cables must be well connected and routed to avoid connectivity failures, bent pins, ... **INVKG3_ST2, INVMG3_ST5A.**
If any cables are in poor condition, replace the fan or cables as appropriate. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP85370*.

5.60. Fault code F225: Ind cri overtemperature

Obligatorio	El usuario debe consultar las Instrucciones de Seguridad para la Operación, Resolución de Problemas y Mantenimiento antes de empezar. Cada acción descrita en la secuencia tiene un estado definido. No existen estados generales. El usuario debe leer y comprender los estados descritos en el documento mencionado en la sección “ Introducción ”.			
Descripción	La temperatura de alguna de las inductancias de módulo es superior a la configurada.			
Aplicable a	☒ HEM	☒ PCSM/MPCSM		
	☒ HEMK	☒ PCSK/MPCSK		
Fallos o warnings asociados	F170, F225			
Procedimiento de análisis				

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it. Problema de sobretemperatura en la inductancia (L01).

- Review via PowercommsV on SD card (Log_3) of the inverter if the temperature increase is real or if it was a noise problem in the temperature measurement. **INV3_ST1B**.
 - If the temperature has risen progressively and without ripple, it is a problem of overtemperature in the inductance. Perform steps 5-12.
 - If there is excessive ripple in the PT100 temperature measurement, it may be a wiring problem. Go to step 3.

Poor configuration of the temperature thresholds.

- Check the configuration of the fault, for this purpose consult via PowercommsV ([Inductance] within the group [G5.4-Temperature]), if the value of the inductance failure threshold temperature (L01) and hysteresis are below their default values. In this case, consult Power Electronics in case it is a special configuration of this plant. **INV3_ST1B**.

If the configuration is correct, continue with the following:

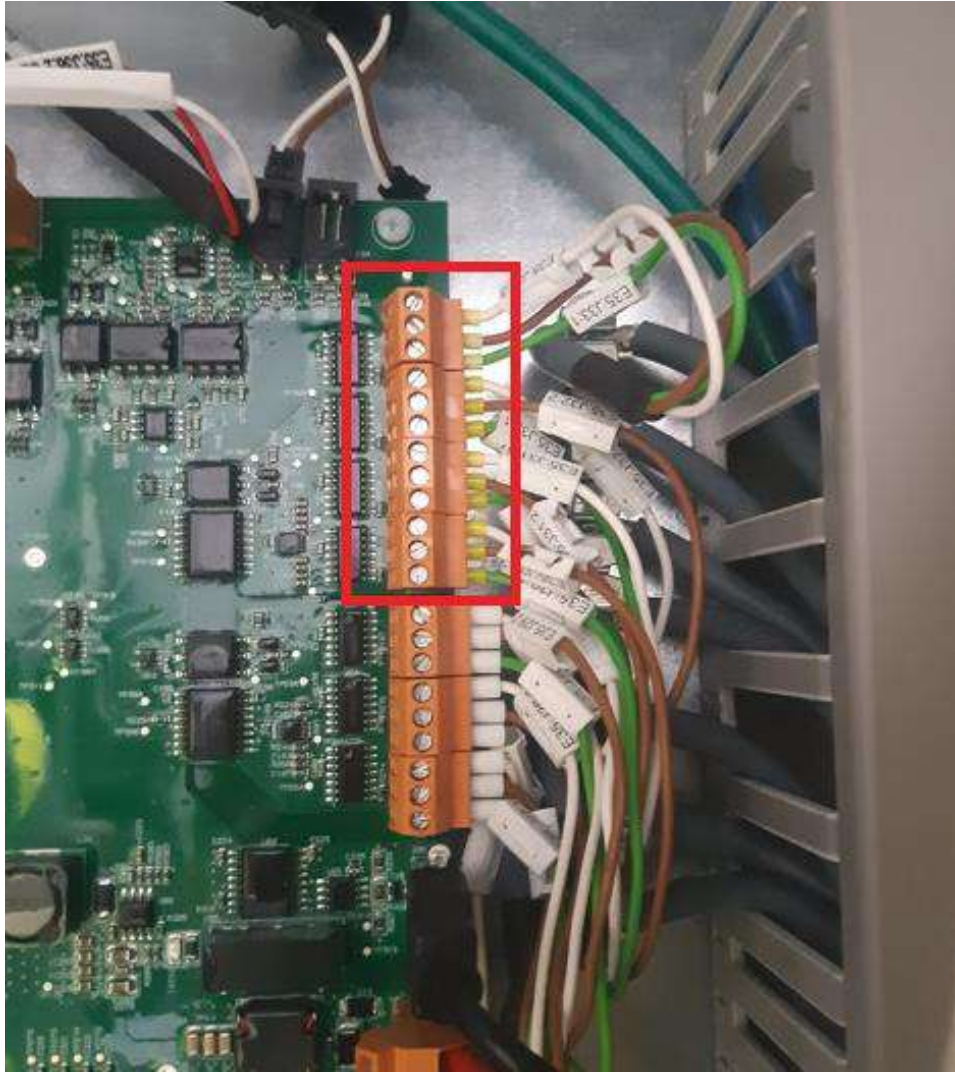
PT100 wiring problem.

- The following table lists the connectors and pins corresponding to each PT100 to perform the following checks::

Fault instance	Connector at the central control board (E35)	Terminal block at the main inductance (L01)	Pins of the terminal block at the main inductance (L01)	Connection cable
1	J30	XPT1	A B1 B2	W052
2	J31	XPT2	A' B1' B2'	W053
3	J32	XPT3	A'' B1'' B2''	W054
4	J33	XPT4	A''' B1''' B2'''	W055

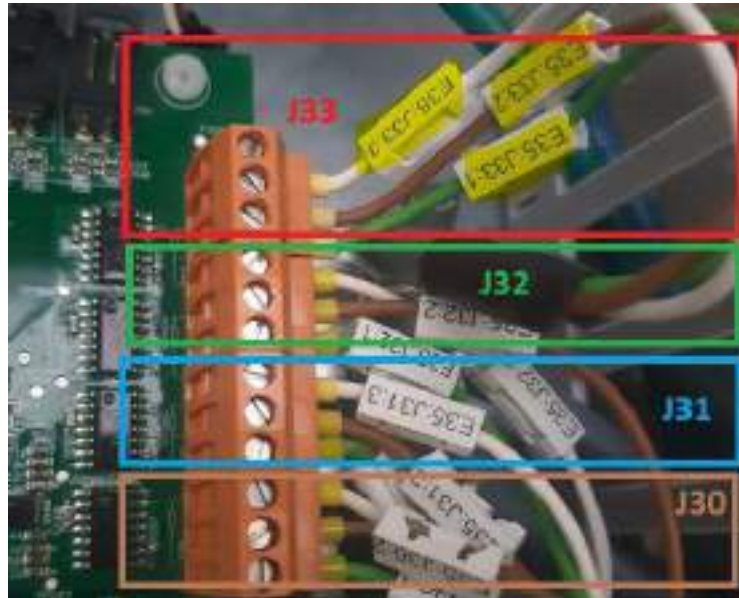
- Check with 1500V multimeter that there is continuity between the central control board connector (E35) and the terminal block pins on the inductance (L01) according to the PT100 that is failing. **INV3_ST2, INV3_ST5A**.

- medir entre el pin 1 del conector de la tarjeta y B2 del bornero
- medir entre el pin 2 del conector de la tarjeta y B1 del bornero
- medir entre el pin 3 del conector de la tarjeta y A del bornero.



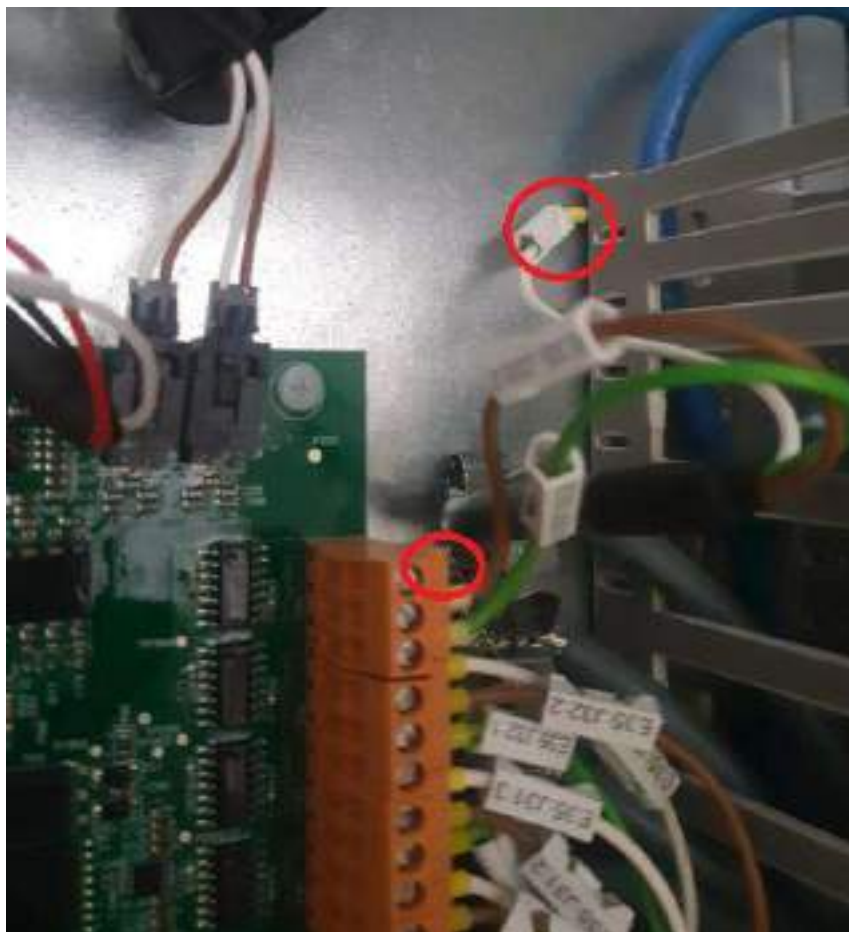
Connectors at the central control board (E35)

EN

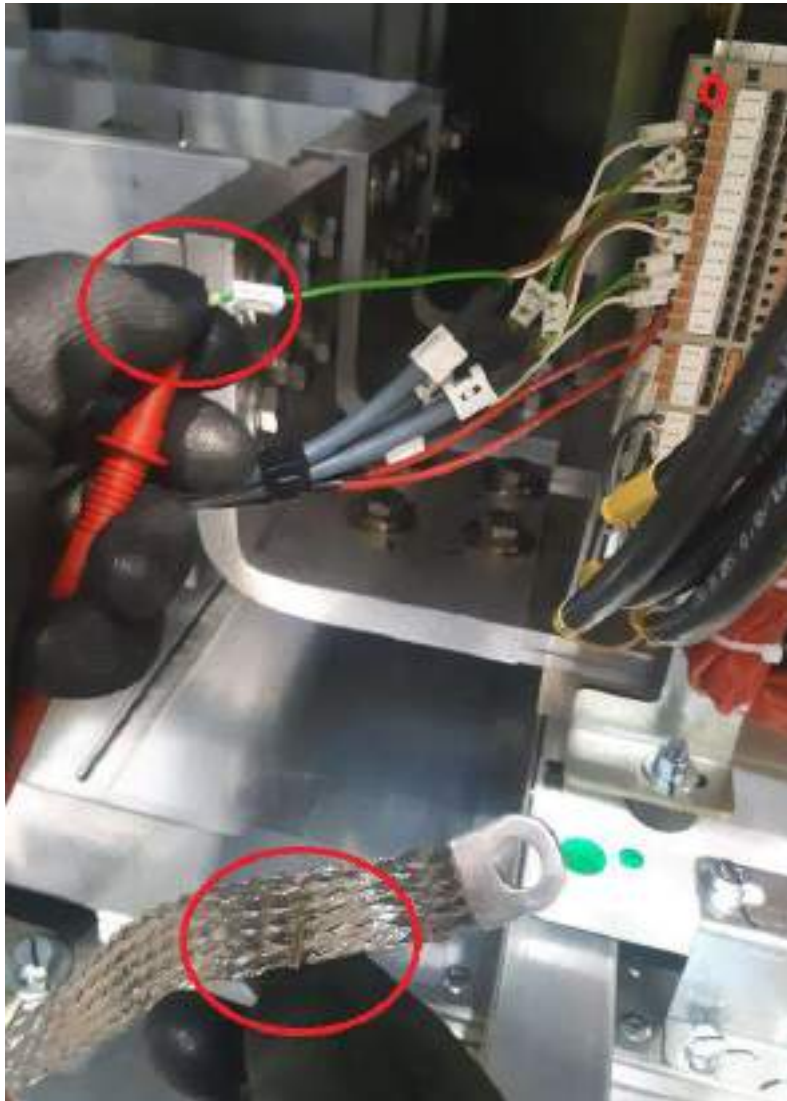


Each connector and pin of the central control board (E35)

EN



Pin 1 of J33 to the ground



EN

XPT2 terminal block at the main inductance (L01) of the pin B2 in order to measure continuity with the pin of the connector J33 of the central control board (E35)

Do the same for each connector and pin of the central control board (E35) with each corresponding terminal block of the main inductance.

3.2. If there is no continuity at any of the points measured above, replace the corresponding cable.
INVKG3_ST2, INVMG3_ST5A.

3.3. Check the resistance between the PT100 which is triggering the fault and the central control board (E35) by disconnecting the connector of the central control board (E35) and measuring with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

- Between pins 1-2 of the connector, measured value $\approx 0\Omega$.
- Between pins 2-3 of the connector, the measured value varies depending on the inductance temperature $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).



Connector J33 of the central control board (E35) between pins 1-2



Connector J33 of the central control board (E35) between pins 2-3

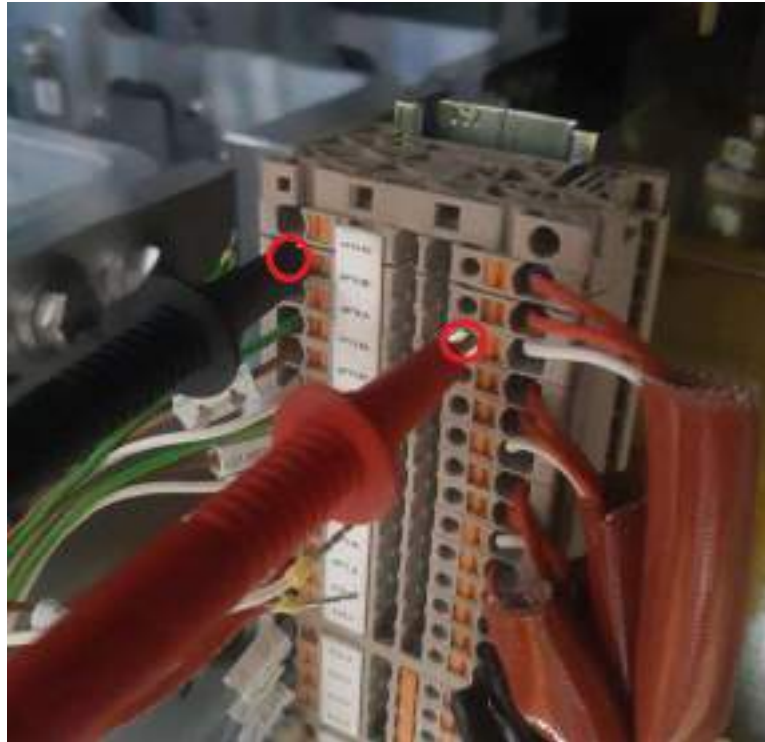
Repeat it for all the pins of each connector of the central control board (E35)

If the measured values are correct, replace the central control board (E35). **INVKG3_ST2**, **INVMG3_ST5A**. (Refer to document *INVG3/OP88849*).

If not, proceed to the next check point.

3.4. Check the resistance on the pins of the inductance terminal block (L01) according to the failed PT100 by measuring with a 1500V multimeter: **INVKG3_ST2**, **INVMG3_ST5A**.

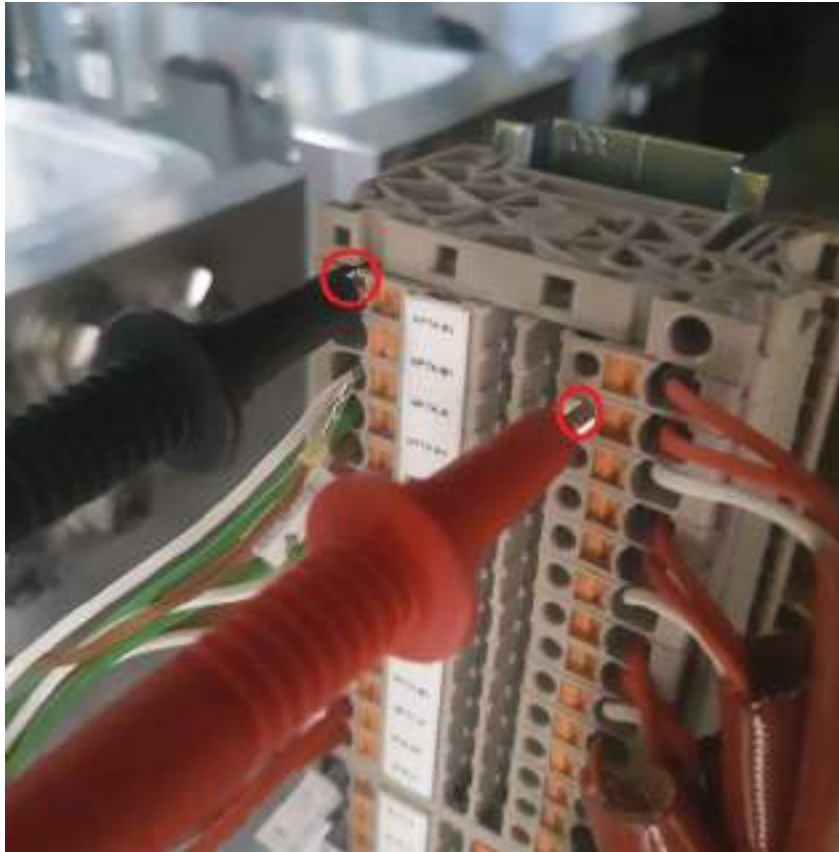
- between pins A-B1 of the terminal block, the value varies according to the temperature of the inductance. $\approx 100\Omega$ (0°C) or $\approx 150\Omega$ (130°C).
- between the pins B1-B2 of the terminal block, the measured value $\approx 0\Omega$.



Pins A-B1 of the terminal block XPT4 at the main inductance (L01)



EN



Pins B1-B2 of the terminal block XPT4 at the main inductance (L01)



Repeat it for the rest of the pins if the terminal blocks at the main inductance (L01)

If the measured values are not correct, replace the inductance (L01). INVKG3_ST2, INVMG3_ST5A.
(Refer to document *INVG3IOP85375*)

Wrong temperature measurement.

The temperature of the inductance is adequate (checked in point 1), but an erroneous temperature point measurement has occurred and the F225 has appeared without explanation. In this case, follow the steps below:

4. Check via PowercommsV of the SD board (Log_3), temperatures at the choke (L01) at the time of failure ([IND PT100 (1)], [IND PT100 (2)], [IND PT100 (3)], [IND PT100 (4)]). INVKG3_ST1B.

If the temperatura has a very high value (≈ 65535), the cause may be due to a loose wiring problem (the value will appear for several seconds in the Log_3) or the software has not read the temperature at that time.

If the problem occurs frequently, check the wiring and grounding screen connections if present. INVKG3_ST1B, INVMG3_ST5A.

If the wiring has been checked and secured, but high inductance temperatures are still measured, proceed to the next check point.

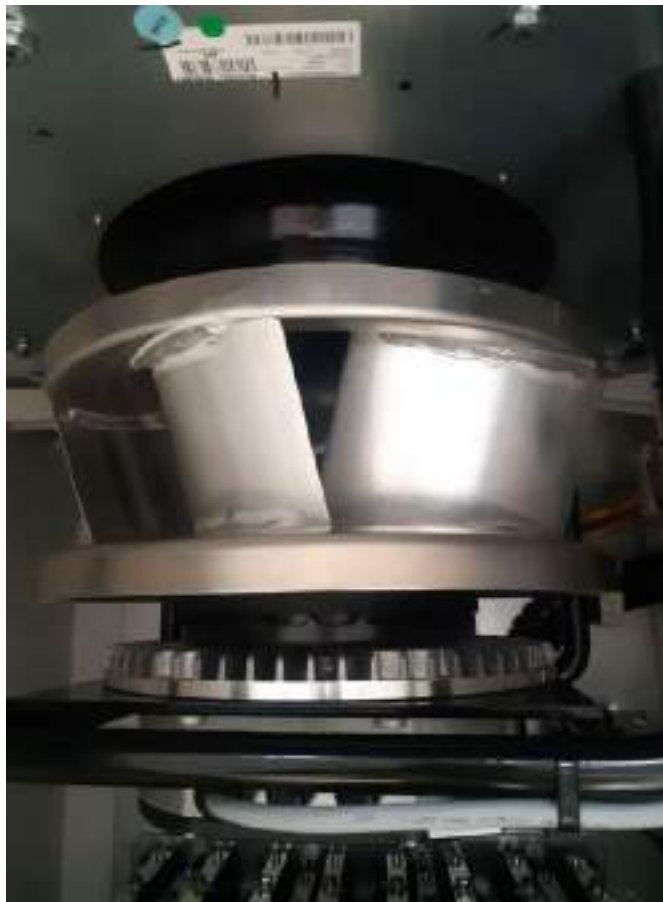
LC filter fan fault.

5. Configure via PowercommsV ([Mode] within the group [G11.6-Config. LC Filter]) the fan speed to 100% and check that the fans are rotating at the set speed within the group [SV11.1-Fan]. INVG3_ST1B.

If not, check via PowerCommsV in group ([SV11-Fan]) waht fault or warning appears and act according the fan manual. INVG3_ST1B.

If a ventilation warning appears during the fault, continue to the next step.

6. Check that no fans are rubbing against any part of the equipment and that they are properly anchored. INVKG3_ST2, INVMG3_ST5A.



7. Check that the fan rotor blades are not broken or worn.
8. Check that the power and communication cables do not rub against the moving parts of the fan. Cables must be well connected and routed to avoid connectivity failures, bent pins, ... [INVKG3_ST2](#), [INVMG3_ST5A](#).
If any cables are in poor condition, replace the fan or cables as appropriate. [INVKG3_ST2](#), [INVMG3_ST5A](#). See document [INVG3/OP85370](#).

Module fans failure.

9. Configure via PowerCommsV ([Mode] within the group [G11.5-Config. Modules]) the fan speed to 100% and check that the fans are rotating at the set speed en el grupo [SV11.1-Fan]). [INVG3_ST1B](#).

If not, check via PowerCommsV within group ([SV11-Fan]) which fault or warning appears and act according to the fan's manual. [INVG3_ST1B](#).

If a ventilation warning appears during the failure, continue to the next test step.

10. Check that no fans are rubbing against any part of the equipment and that they are securely anchored. [INVKG3_ST2](#), [INVMG3_ST5A](#).

**EN**

11. Check that the fan rotor blades are not broken or worn. [INVKG3_ST2](#), [INVMG3_ST5A](#).
12. Comprobar que los cables de alimentación y comunicaciones no rocen con las partes móviles del ventilador. Cables must be well connected and routed to avoid connectivity failures, bent pins, ...
[INVKG3_ST2](#), [INVMG3_ST5A](#).
If any cables are in poor condition, replace the fan or cables as appropriate. [INVKG3_ST2](#), [INVMG3_ST5A](#). See document *INVG3/OP85370*.

EN

5.61. Fault code F234: Thermal Bus Plus

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	Appears when the digital input connected to the Bus Plus cabinet thermal is activated.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	W182: Thermal Bus Plus		

Analysis procedure

This protection can be configured as Fault (F234) or Warning (W182) from PowercommsV in group [G5.5 I/O Protections] Thermal Bus Plus.

When the fault is tripped, a ramp stop with DU opening occurs, and must be reset manually with Power Electronics Staff role.

There is a bimetal sensor placed on the BBIN.117 plate. This sensor integrates a normally closed (NC) contact in series and wired up to the power supply panel (PSA) E31 input/output card. This bimetal is factory set at 110-120°C depending on the manufacturer:

PE Reference	Manufacturer	Model	Temperature
L9865	SAFTTY ELECTRONIC TE	BW-BCP 110C	110°C
L9824	THERMIK	S06 - 110.05 0100/0100	110°C
L9889	THERMIK	S06.120.05.0100.0100	120°C

1. Visually verify that the equipment has the Bus Plus Advanced Backpack cabinet connected to the equipment, both the power boards and its interconnection that integrates the 248Vac power supply and the bimetallic sensor feedback to connector J4 of the input and output board E31 of the power supply panel (PSA). [INVKG3_ST2](#), [INVMG3_ST5A](#).

In case you do not see this interconnection, check if it is missing on the inverter side or on the Bus Plus side. Once identified, the necessary wiring should be included:

- 1.1. Inverter: wiring from the power supply panel (PSA) and control panel (PCL) to connector XC802.2. [INVKG3_ST2](#), [INVMG3_ST5A](#).
- 1.2. Bus Plus cabinet: wiring from connector XC802.1 to the M12 fan and to connector XC803 integrating the bimetal sensor. [INVKG3_ST2](#), [INVMG3_ST5A](#).

There are different causes that can trigger this fault. Follow the steps below to identify the possible cause and solve it.

2. Check via PowercommsV in group [SV1.5 Digital I/O Status] the LC contactor feedback (bit 8: Retro Aux 2). [INVKG3_ST1B](#).

This signal may be active due to:

- the temperature continues to be higher than the setpoint
- there is a wiring problem
- the measurement sensor

Wiring failure.

EN

3. With the inverter stopped and de-energized, check continuity with a 1500V multimeter between pins J4:5 and J4:6 on the E31 input/output board of the power supply panel (PSA). **INVKG3_ST2, INVMG3_ST5A.**

- 3.1. If there is continuity, the wiring is correct and the bimetallic sensor has not exceeded the factory setting.

In this case, the problem is between the input/output board E31 and the central control board E35.

- 3.1.1. Check the continuity of the RJ45 cable with the network cable continuity tester between connector J2 on the E31 input/output board and connector J23 on the E35 central control board.

- 3.1.2. If there is no continuity, replace the cable. **INVKG3_ST2, INVMG3_ST5A.**

- 3.1.3. If the RJ45 cable does not fail, replace the input/output board E31. See document *INVKG3IOP88849*. **INVKG3_ST2, INVMG3_ST5A.**

- 3.1.4. If the fault persists, replace the central control board E35. See document *INVKG3IOP88849*. **INVKG3_ST2, INVMG3_ST5A.**

- 3.2. If there is no continuity, the wiring or the sensor is defective. Disconnect connector XC802 connecting inverter and Bus Plus cabinet. **INVKG3_ST2, INVMG3_ST5A.**

- 3.2.1. Measure continuity in the inverter at the following points: **INVKG3_ST2, INVMG3_ST5A.**

- between connector J4:5 of the input/output board E31 and connector XC802.2:3.
- between connector J4:6 on the E31 I/O board and connector XC802.2:4.

If there is no continuity at any of the measured points, replace the wiring and perform the above checks again.

- 3.2.2. To perform the tests in the Bus Plus cabinet, disconnect connector XC803 and measure continuity at the following points: **INVKG3_ST2; INVMG3_ST5A.**

- between connector XC802.1:3 and connector XC803.2:1.
- between connector XC802.1:4 and connector XC803.2:2.

If there is continuity in both checks, replace the sensor.

If not, replace the wiring and check again.

Bus Plus cabinet fan fault.

4. If the fault occurs frequently, check if the M12 fan in the Bus Plus cabinet is working properly when the temperature is higher than 60°C. **INVKG3_ST2, INVMG3_ST5A.**
5. Check fan rotor blades for broken or worn fan blades.
6. Check that the power and communication cables do not rub against the fan's moving parts. Cables should be well connected and routed to avoid connectivity failures, bent pins, etc. **INVKG3_ST2, INVMG3_ST5A.**
7. Check continuity using a 1500V multimeter between pins 1-2 and 3-4 of the circuit breaker Q81 (closed). **INVKG3_ST2, INVMG3_ST5A.**
8. Measure with a 1500V multimeter the capacitance of capacitor C66 between pins 1 and 2. The measured value should be $\approx 3.5\mu\text{F}$.

If not, replace the capacitor. **INVKG3_ST2, INVMG3_ST5A.**

Check thermostat ST01, to do this lower the activation threshold a few degrees below ambient temperature and check that the thermostat contact closes by measuring continuity between pins 1 and 2. **INVKG3_ST2, INVMG3_ST5A.**

- 8.1. With this activation threshold set, connect the power supply and check that the fan turns on. **INVKG3_ST1B.**

Note: This check will be performed with all equipment closed. From the outside you can verify that it is running by the noise.

If the fan does not turn on, replace the fan. INVKG3_ST2, INVMG3_ST5A.

Note: After performing this check re-set the thermostat activation level to 60°C. INVKG3_ST2, INVMG3_ST5A.

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5.62. Fault code F132: High voltage DC module

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module has detected an overvoltage on the entire module bus. The fault threshold is set to 1550V total bus voltage.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F2: HW Vbus	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect night mode regulation.

1. If the fault occurred while the unit was in night mode control, check the value of the parameters [Kp Vdc night] and [Ki Vdc night] in group [G8.2: Advanced config] by PowerCommsV. **INV3_ST1B**. If they are not set to their default value:

1.1. Contact the plant manager to determine if it is correct. If not, set to default value. **INV3_ST1B**.

Solar field oversizing.

2. If the PV field voltage exceeds 1500V and the hardware of the inverter (wiring and E35 board), such as the calibration, is correct, in this case talk to the customer and contact Power Electronics. **Note:** in case of checking the hardware of the inverter, the statuses are **INVKG3_ST2**, **INVMG3_ST5A**.

Desconexión de las baterías de forma repentina estando en carga (aplica solo equipos PCS).

3. Check with the customer whether the order to open the battery has been given. If the customer has opened the battery voluntarily and intentionally, inform the customer that this is an improper action and may damage the inverter. If it was an unintentional operation (battery protection), ask the customer if it could have been caused by the inverter and contact PowerElectronics.

Sudden DCDC shutdown-stop with inverter and DCDC in battery charging mode (only for inverters containing Bus Plus). A sudden DC shutdown with the DCDC and inverter in battery charging mode will naturally cause the bus voltage of the inverter to rise. It is necessary to ensure that the bus voltage does not remain close to 1600 V which could compromise the hardware. In this case:

4. Check via PowerCommsV that the configuration of the subgroup [Control Loops] of group [G8: Control] is in accordance with the project specifications. **INV3_ST1B**.
5. Download the configuration parameters of the inverter, DCDC and their respective logs and contact Power Electronics. **INV3_ST1B**.

Failure in the module board measurement (E02)

If the module measures voltage above the 1550Vdc threshold and is not consistent with respect to the field voltage, the rest of the modules and/or the voltage presence LEDs and, replace the module control board (E02) and check that when reassembling it does not trigger the fault. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INV3IOP88076*

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5.63. Fault code F101: Mod. measure DC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction” .			
Description	The DC voltage measurement of the module is below the average of the central control board. The difference between the two must be greater than 200V for the fault to be triggered.			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	N/A.			

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Failure in the metering system.

1. It is possible that the difference caused by the central metering system and modules is caused by a disconnection or failure of the metering wiring. To check this, it is necessary to check the module where the fault has been triggered. **INVKG3_ST2, INVMG3_ST5A.**
 - 1.1. Check the positive semibus. Check continuity with 1500V multimeter between E02 terminal J4 and BBIP.20 and continuity between E02 terminal J5 and BBIM.20.
 - 1.2. Check negative half bus. Check continuity with 1500V multimeter between E02 terminal J7 and BBIN.20 and the continuity between E02 terminal J6 and the BBIM.20.



Example measure

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Measurement points (BBIN.20 / BBIP.20 / BBIM.20)



Measurement points (E02:J4/J5/J6/J7)

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2. It is also possible that the problem is in the Bus DSP measurement:

2.1. In the case of PCS inverters:

2.1.1. Check continuity with 1500V multimeter between the host card and the corresponding bus by measuring the cases against the table: **INVKG3_ST2**, **INVMG3_ST5A**.

1 BUSBAR	2 BUSBARS	3 BUSBARS	4 BUSBARS
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- a. Positive 1: Check continuity from E35 terminal J1 to BBEP.82, from BBEP.82 to BBEP.81 and from BBIP.07 to BBIP.20.
- b. Negative 1: Check continuity from E35 terminal J2 to BBEN.67, from BBEN.67 to BBEN.66 and from BBIN.07 to BBIN.20.



E35:J1



BBEP.82



E35:J2



BBEN.67

- c. Positive 2: Check continuity from E36 terminal J1 to BBEP.80, from BBEP.80 to BBEP.79 and from BBIP.05 to BBIP.20.
- d. Negative 2: Check continuity from E36 terminal J2 to BBEN.65, from BBEN.65 to BBEN.64 and from BBIN.05 to BBIN.20.

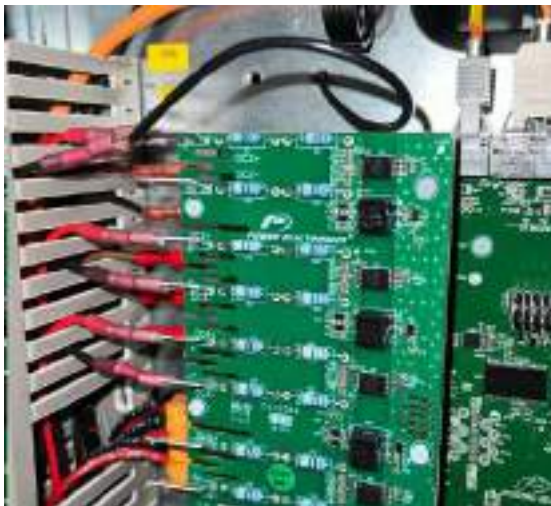
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E36:J1



BBEP.80



E36:J2



BBEN.65

- e. Positive 3: Check continuity from E36 terminal J3 to BBEP.78, from BBEP.78 to BBEP.77 and from BBIP.03 to BBIP.20.
- f. Negative 3: Check continuity from E36 terminal J4 to BBEN.63, from BBEN.63 to BBEN.62 and from BBIN.03 to BBIN.20.



E36:J3



BBEP.78



E36:J4



BBEN.63

- g. Positive 4: Check continuity from E36 terminal J5 to BBEP.76, from BBEP.76 to BBEP.75 and from BBIP.01 to BBIP.20.
- h. Negative 4: Check continuity from E36 terminal J6 to BBEN.61, from BBEN.61 to BBEN.60, and from BBIN.01 to BBIN.20.



E36:J5



BBEP.76



E36:J6



BBEN.61

2.2. HEM / HEMK inverters:

2.2.1. Check continuity with 1500V multimeter between the central board and the bus.
INVKG3_ST2, INVMG3_ST5A.

- a. Positive 1: Check continuity from E35 terminal J1 to BBEP.82, from BBEP.82 to BBEP.81 and from BBIP.07 to BBIP.20.
- b. Negative 1: check continuity from E35 terminal J2 to BBEN.67, from BBEN.67 to BBEN.66 and from BBIN.07 to BBIN.20.



E35:J1



BBEP.82

EN



E35:J2



BBEN.67

2.3. Replace the affected cable section, check plates and/or replace them.**INVKG3_ST2, INVMG3_ST5A.**

Incorrect tightening torque

3. It is possible that the tightening torque between PCBs or the power cable connections is incorrect and this is affecting the measurements of both the module and the main board:

3.1. In case of PCS inverters:

3.1.1. Check tightening torques between boards, measuring the cases with respect to the table **INVKG3_ST2, INVMG3_ST5A.**

1 BUSBAR	2 BUSBARS	3 BUSBARS	4 BUSBARS
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- a. Positive 1: Check tightening torque of BBEP.82 to BBEP.81 and BBIP.07 to BBIP.20.
- b. Negative 1: Check tightening torque from BBEN.67 to BBEN.66 and BBIN.07 to BBIN.20.
- c. Positive 2: Check tightening torque from BBEP.80 to BBEP.79 and BBIP.05 to BBIP.20.
- d. Negative 2: Check tightening torque from BBEN.65 to BBEN.64 and from BBIN.05 to BBIN.20.
- e. Positive 3: Check tightening torque from BBEP.78 to BBEP.77 and BBIP.03 to BBIP.20.
- f. Negative 3: Check tightening torque from BBEN.63 to BBEN.62 and from BBIN.03 to BBIN.20.
- g. Positive 4: Torque check BBEP.76 to BBEP.75 and BBIP.01 to BBIP.20.
- h. Negative 4 Check tightening torque of BBEN.61 to BBEN.60 and BBIN.01 to BBIN.20.

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Check of all screws of the 4 DC switch-disconnectors and DC busbars

HEM / HEMK inverters:

3.1.2. Check continuity between the central board and the bus. **INVKG3_ST2, INVMG3_ST5A.**

- a. Positive 1: Check tightening torque of BBEP.82 to BBEP.81 and BBIP.07 to BBIP.20.
- b. Negative 1: Check tightening torque from BBEN.67 to BBEN.66 and BBIN.07 to BBIN.20.



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Check of all screws of the 4 DC switch-disconnectors and DC busbars

- 3.2. Bring the corresponding plates to the proper torque and that the power wiring is correctly connected **INVKG3_ST2, INVMG3_ST5A**.

Damaged DC switch disconnector. It is possible that the DC disconnector is faulty and does not operate properly. If this occurs:

4. On the module that has triggered the fault, carry out a visual inspection of the disconnector. **INVKG3_ST2, INVMG3_ST5A**.
5. If blackening is detected on the connections to the plates or on the component itself, then replace the disconnector of the affected module (see document *INVG3IOP85377*). **INVKG3_ST2, INVMG3_ST5A**.

Damaged boards. It is possible that one of the boards involved in the measurement process is damaged.

6. Check by PowerCommsV if the following cases are present. **INVG3_ST1B**.
 - 6.1. There are no measurements on the central board (parameters [Voltage extern] of group [SV2: DC source]).
 - 6.2. There are measurements on the module board (parameters [DC voltage P], [DC voltage N] of group [SV12: Module information]).
 - 6.3. The DC disconnector is closed (parameters [DC disconnector status] of group [SV1: Status]).
 - 6.4. In this case, replace central board E35 (see document *INVG3IOP88849*). **INVKG3_ST2, INVMG3_ST5A**.
7. If, on the other hand, this is the case via PowercommsV: **INVG3_ST1B**.
 - 7.1. Measurement exists on the central card (parameters [Voltage extern] within the group [SV2 DC source]).
 - 7.1. There are no measurements on the central board (parámetros [DC Voltage P], [DC voltage N] within the group [SV12: Module information]).
 - 7.2. The DC disconnector is closed (parameters [DC disconnector status] within the group [SV1: Status])
 - 7.3. In this case, replace the E02 board of the module that is causing the fault (see document *INVG3IOP88849*). **INVKG3_ST2, INVMG3_ST5A**

8. If none of these cases apply, check all of the above points. If the fault persists, contact Power Electronics.

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5.64. Fault code F2: HW Vbus

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The voltage measured on the DC channels has exceeded the threshold (1600Vdc).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F132: High voltage DC module	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Sudden disconnection of the batteries with negative current references (applies to PCS inverters)

1. If the connection to the batteries is lost during the charging process, the internal bus may reach this protection threshold. There are two possibilities:
 - 1.1. The inverter has suddenly opened the DC disconnectors and triggered another fault. In this case:
 - 1.1.1. Check in PowerCommsV the fault history (group [SV7: Faults and warnings]) or in the SYS of the SD and resolve the fault related to the sudden opening with what is indicated in this document. **INVG3_ST1B**.
 - 1.2. Another possibility could be that the customer has suddenly disconnected the batteries during normal plant operation. If this is the case:
 - 1.2.1. Contact the plant manager to determine the cause of opening and customer information.

Sudden disconnection-stop of DCDC with inverter and DCDC in battery charging mode (inverters with optional Bus Plus).

2. A sudden DC shutdown with the DCDC and inverter in battery charging mode can cause the bus voltage of the inverter to rise naturally. It must be ensured that the bus voltage does not remain close to 1600V which could compromise the hardware. If this happens:
 - 2.1. Check the cause of the sudden stop in the DCDC. If the shutdown has been caused by the customer, contact the plant manager and indicate that this handling is inappropriate. **INVG3_ST1B**.

Incorrect regulation in night mode

3. If the night mode regulation loops are incorrect, it is possible that overvoltage may occur when regulating in this control mode. In this case:
 - 3.1. Check in PowerCommsV that the parameters of group [G8: Control] are set to their default values. If they are not, check with the plant manager to see if they are appropriate. If they are not, set in PowerCommsV the default value. **INVG3_ST1B**.

There is a problem in the Vdc measurement.

4. It is possible that either the central board (E35) or the expansion board is faulty. In this case:
 - 4.1. For the case of inverters:
 - 4.1.1. Replace the central board (E35). **INVKG3_ST2**, **INVMG3_ST5A**. See document **INVG3/OP88849**.
 - 4.1.2. If the fault persists, contact Power Electronics.

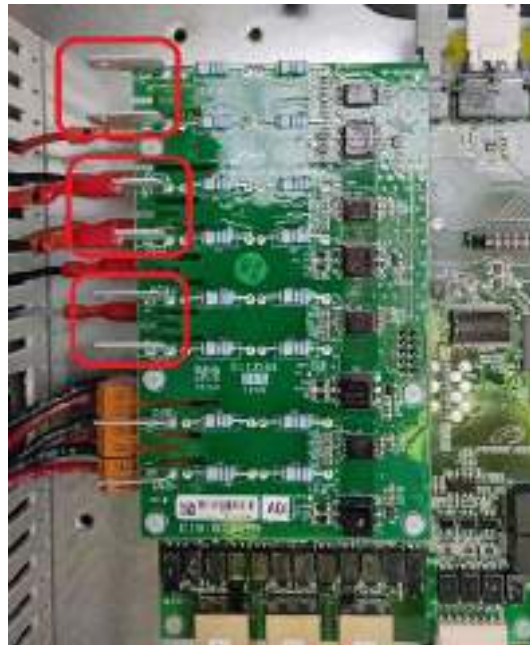
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4.2. For inverters with optional Bus Plus or PCS inverters:

- 4.2.1. Exchange the connectors of the channel that is giving the fault with another one and check if the fault remains in the new channel that has been connected. The measurement channels to be exchanged are two by two (positive and negative) and are E35 J1 and J2, E36 J1 and J2, J3 and J4, J5 and J6. **INVKG3_ST2; INVMG3_ST5A.**



J1 and J2 of board E35



J1 and J2, J3 and J4, J5 and J6 of board E36

- If changing the channels does not fail, replace the corresponding board (E35 or E36 depending on the channel giving the fault). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3IOP88849*.
- If when changing channels the new channel fails, it is possible that there is a real PV or BESS overvoltage. If this occurs, check with the plant manager. **INVKG3_ST2, INVMG3_ST5A.**

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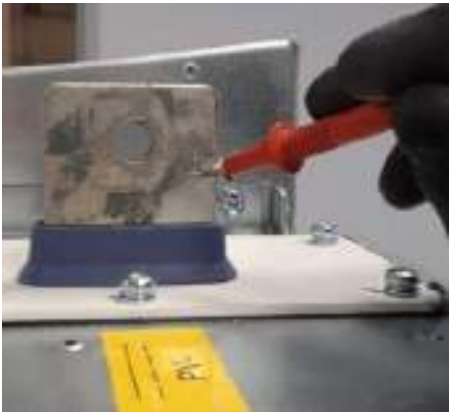
5.65. Fault code F4: Discharge

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	It has taken more than 60 seconds to go below 10 Vdc since it starts discharging the module buses.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A		

Analysis procedure

There may be several causes for this failure:

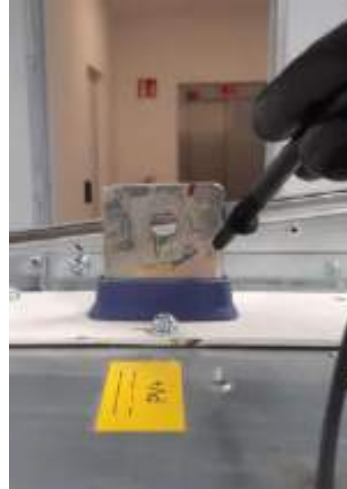
- 1. The IGBT are in bad condition. Check the condition of the IGBTs of the power stage.
 - 1.1. Technically check with 1500V multimeter the IGBT and replace any damaged IGBTs. **INVKG3_ST2, INVMG3_ST5A.** (See document *INVG3/OP88076*).



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2. The modules/busbars are disabled. Check in the SD card (SYSLOG) and check if at the moment of unloading buses the modules, there was a disabled or locked module.
 - 2.1. If the module has been interlocked/disabled due to a module fault, check the equipment history log (subgroup [SV7.3: Events]) and the event history (subgroup [Module x events] within group [SV7.4: Module events]) and check the module fault.
 - 2.2. If the module has been disabled/locked by the customer at the time of module download, talk to the customer and tell him that this is an improper procedure in the inverter.
3. If the source of the fault cannot be found, contact Power Electronics.

5.66. Fault code F11: Active Anti-island

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	The active island detection algorithm has detected a possible island condition in the AC part of the network.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	Related to: F28: Passive Anti-island. F5: High Vac. F6: Low Vac. F7: High frequency. F8: Low frequency.	

Analysis procedure

The Anti-island fault occurs when the inverter detects a series of anomalous grid conditions and trips the inverter, stopping it and avoiding the option of it becoming an island. In other words, injecting power into the grid but without the grid being present (this indicates that the plant's head-end disconnecter or part of the electrical distribution grid has been opened).

This algorithm trips when it detects a frequency difference between the voltage grid waveform and the voltage waveform generated by the inverter itself. Under normal conditions there is a small difference in degrees between the two voltage waves. If the difference between the two is 0° then the inverter and grid loads are balanced, and the inverter is in islanded condition so it will fail and protect the equipment and the users. This is a fault that can occur just by the opening of the main plant disconnecter that feeds the medium voltage part of the inverter. The typical trip time for this fault is between 0.5 - 1s. This fault is related to faults due to grid conditions, i.e. low or high frequency (F7 and F8) and high or low grid voltage (F5 and F6).

This fault is disabled by default. If the inverter were to fail due to these faults, then:

1. Check in PowercommsV that the inverter parameterisation is set to its default value (parameter [Mode] of subgroup [G5.3:Anti islanding]). In case the plant requires a special configuration for this algorithm, check that it is according to the requirements. **INVG3_ST1B**.
2. In case the Anti-island protection is activated, its parameterisation is as required and a sporadic fault (F11 or F28) has occurred due to a network event, it should not be problematic as it is the function of the algorithm itself. **INVG3_ST1B**.
3. If the Anti-island protection is activated, its parameterisation is as required, there is a sudden disconnection of the system from the mains and a device is left running, this indicates a dangerous and undesired situation. If this occurs:
 - 3.1. Check in PowercommsV that the configuration of the parameters corresponding to the "test mode" are in their default values (group SG4: Test mode). **INVG3_ST1B**.
4. If the Anti-Island protection is activated, its parameterisation is as required and there are frequent Anti-Island faults (F11 or F28) without apparent cause, **INVG3_ST1B** this is the case:
 - 4.1. Contact Power Electronics.

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5.67. Fault code F23: Unbalanced Vac

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	There is an unbalance between phases greater than the protection of the inverter.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This fault is reported by the equipment when an unbalance appears between the AC voltages. To detect it, the error between the phase with the highest RMS value and the average of the three phases is calculated:

$$(\text{Vrms_max} - \text{Vrms_avg}) / \text{Vrms_avg} * 100$$

If this error is higher than the one set, it will trigger this fault. Check in PowerCommsV the parameter [Unbalance V] of the subgroup [G5.2: Quality]). **INV3_ST1B.**

Follow the steps below to identify the possible cause and resolve it.

Unbalance in the grid voltages.

1. If the network presents an unbalanced three-phase system, where the three phases are not equal in amplitude, it will cause the fault in the inverter. In this case, to check that it is a problem in the network and not in the inverter:
 - 1.1. Check in the SD card the Vac voltages read at the time of the fault. To do this, check in Log_1, the values of Vrs, Vst and Vtr. It should be seen that the RMS values shown for the three phases differ in some of them. If this is the case, the inverter has been correctly protected from the unbalance caused. Try to reset the fault in case it has been a transient unbalance due to some event in the mains and if it has already been reset. **INV3_ST1B.**
2. If the fault persists, consult with the plant manager to check if the POI shows the same voltage unbalance. If so, it is a network fault and therefore the fault does not come from the inverter.

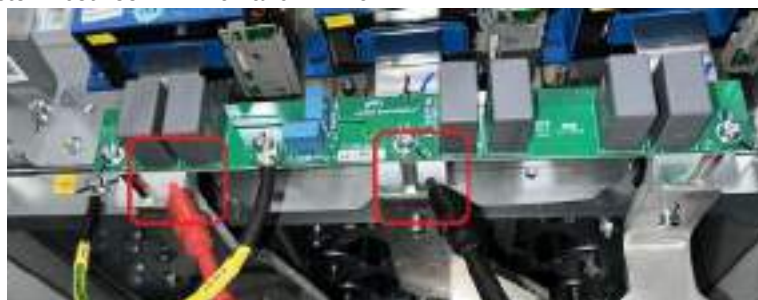
Incorrect setting of the fault threshold.

3. Check if the fault threshold has been configured to a minimum unbalance, being the protection very strict and causing the fault to be triggered by the usual unbalances of the network itself. To do this:
 - 3.1. Check from PowerCommsV the configured fault threshold (parameters [Unbalance V] and [Delay unbalance V] of subgroup [G5.2-Quality]). In the event of detecting modified values without justification, modify the parameters to their default values. **INV3_ST1B.**

Note: consult the plant manager beforehand to find out if this modification is justified.

Voltage measurement error Vac.

4. The inverter may report F23 due to an error in the AC voltage measurements. In this case:
 - 4.1. Ensure that the inverter is isolated from DC and AC. **INVKG3_ST2, INVMG3_ST5A.**
 - 4.1.1. Connect a 1500V multimeter to each of the AC terminals to obtain the line measurements:
 - Multimeter 1 between BBI1.02 and BBI2.02.



- Multimeter 2 between BBI2.02 and BBI3.02.



- Multimeter 3 between BBI3.02 and BBI1.02.



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- 4.1.2. Use alligator clips as you cannot be holding the multimeter when measuring. **INVKG3_ST2, INVMG3_ST5A.**

If there is only one 1500V multimeter, check each of the measurements separately, bearing in mind that between measurements it will be necessary to re-isolate the inverter by changing the clamps and re-measure as will be seen later.

Note: close all the inverter, go to INVKG3_ST1B status to perform the measurement. When the measurement is finished, return to the INVKG3_ST2, INVMG3_ST2 state to remove the measuring inverter.

- With the AC inverter closed (AC circuit breaker closed in the case of HEMK / PCSK inverter or switchgear closed in the case of HEM / PCSM inverter), check the measurement displayed by the 1500V multimeter and compare it with the measurement displayed in PowerCommsV (subgroup [SV3.2- Grid LV]). **INVKG3_ST1B. After the measurement has been completed, return to INVKG3_ST2, INVKG3_ST2 status to remove the measuring device.**

	Inverters HEM / PCSM	Inverters HEMK / PCSK
Multimeter 1 - BBI1.02 and BBI2.02	[Grid voltage RS LV]	[Grid voltage TR LV]
Multimeter 2 - BBI2.02 and BBI3.02	[Grid voltage ST LV]	[Grid voltage ST LV]
Multimeter 3 - BBI3.02 and BBI1.02	[Grid voltage TR LV]	[Grid voltage RS LV]

- If it is detected that any measurement differs from the one taken with the 1500V multimeter, there is a problem in the measurement of the voltages of the inverter. To solve it, it is necessary to detect whether the problem lies in the wiring or in the central control board itself.
- Check connectors J3, J4 and J5 on card E35 (central control card).
 - With the inverter isolated from DC and AC, check the correct connection of the faston connectors J3, J4 and J5 of the E35 board. **INVKG3_ST2, INVMG3_ST5A.**



- 7.2. With the inverter isolated from AC, check if the voltage measurements are correct again. **INVKG3_ST1B.**
- 7.3. the measurements are still not correct, go on to the next point.
8. Check the wiring.
- 8.1. Check that the AC voltage measurement wiring is in correct condition and that there is no slack in the connections that could cause voltage drops (BB1.02:L1, BB1.02:L2, BB1.02:L3 and F100:2, F100:4, F100:6). **INVKG3_ST2, INVMG3_ST5A.**
- 8.2. Perform a visual inspection to ensure that the voltage measurements are connected to the correct phase (BB1.02:L1 connected to F100:2; BB1.02:L2 connected to F100:4; BB1.02:L3 connected to F100:6). **INVKG3_ST2, INVMG3_ST5A.**
- 8.3. Before starting the continuity test, disconnect the voltage measurement cables from the F100 fuses (F100:1, F100:3 and F100:5). This is necessary to disconnect the wires of the auxiliary transformer circuit and to be able to carry out the continuity test properly. **INVKG3_ST2, INVMG3_ST5A.**
- 8.4. Carry out the following AC voltage measurements. Measure continuity with a 1500V multimeter between:
- 8.4.1. The previously disconnected cable of F100.1 and the faston connector J3 of the central control board E35. **INVKG3_ST2, INVMG3_ST5A.**

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8.4.2. The previously disconnected cable from F100.3 and the faston connector J4 on the E35 central control board. **INVKG3_ST2, INVMG3_ST5A.**



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8.4.3. The previously disconnected cable from F100.5 and the faston connector J5 of the central control board E35.4. **INVKG3_ST2, INVMG3_ST5A.**



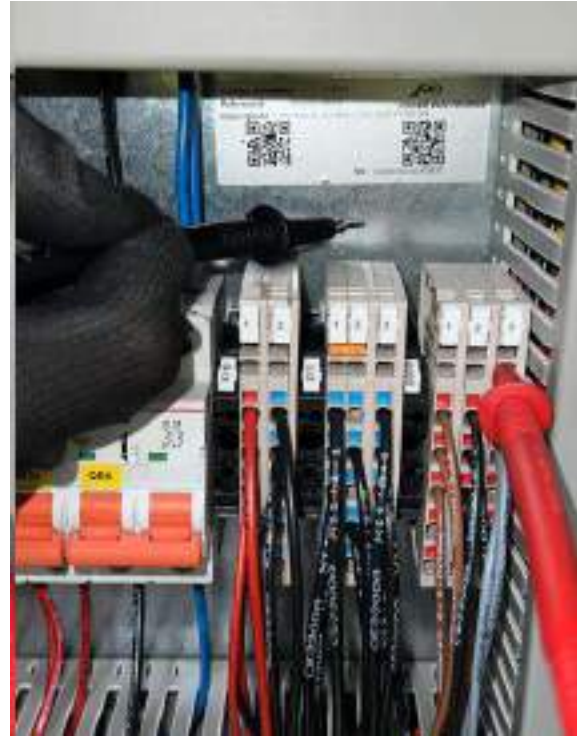
If all three phases give a correct measure of continuity, move on to the next possible cause, otherwise proceed to the next step.

8.5. Once it has been detected which phase does not give continuity, measure by parts to detect where the measurement is cut off. Check at each measurement point that the wiring is not defective.

- 8.5.1. Check the continuity between the cable previously disconnected from F100.X (X will be 1,3 or 5 depending on the phase detected in the previous point) and the terminal block X600.X (X will be 1,2,3). If there is no continuity, replace the wiring. **INVKG3_ST2, INVMG3_ST5A.**



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Replacement of the E35 central control board.

9. If no fault has been detected in the wiring check, the failure in the Vac voltage measurement may be due to a break in the acquisition circuit for this measurement.

Replace the central control board (E35) **INVKG3_ST2**, **INVMG3_ST5A** and check if the inverter recovers the voltage measurements in [Grid voltage RS LV], [Grid voltage ST LV] and [Grid voltage TR LV]. **INVG3_ST1B**. See document *INVG3/OP88849*.

False contacts on the AC boards.

10. If there are connection points on the PCBs with a bad tightening torque, voltage drops can be generated in the phase in question. It is therefore necessary to check at the following points that the tightening torque is correct, and that the entire AC power circuit is correctly connected. **INVKG3_ST2**, **INVMG3_ST5A**.
 - Between BBI1.10 and BBI1.02; BBI2.10 and BBI2.02; BBI3.10 and BBI03.02.
 - Between BBI1.02 and BBI1.03; BBI2.02 and BBI2.03; BBI3.02 and BBI3.03.
 - Between BBI1.03 and L01:1; BBI2.03 and L01:2; BBI3.03 and L01:3.
 - Between L01 to MODX R; L01 to MODX S; L01 to MOD1 T (being X module 1,2,3 and 4).

Loss of insulation on one phase.

11. If one of the phases reduces its impedance to earth, an unbalance in the voltages could be generated. Under normal conditions, this loss of insulation would be detected by the IMD, but in the case of a slight loss of insulation that does not cause a fault, it is possible to observe unbalance in the voltages due to the recirculation of the zero-sequence component. To detect this, carry out the following steps:
 - 11.1. Ensure that the inverter is isolated from DC and AC. **INVKG3_ST2**, **INVMG3_ST5A**.
 - 11.2. Carry out different insulation measurements on the AC phases in both the low and high parts of the transformer. For this purpose, use an insulation meter set to a voltage of 1000Vdc with a measurement time of 1 minute.
 - 11.3. Perform insulation measurement separately from points BB1.10 BBI2.10 and BBI3.10 to earth. In case of detecting a phase with lower impedance to earth than the other two, check the connections of that phase to see if the distance to chassis has been reduced or if there is any defect in the LV circuit. **INVKG3_ST2**, **INVMG3_ST5A**.



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- 11.4. Repeat the previous point but from the transformer high zone. Carry out the insulation measurement separately from the transformer connection of each phase to earth (H1, H2 and H3). If any defect is detected in the medium voltage wiring, repair. **INVKG3_ST2, SKIDG3_ST2, INVMG3_ST2.**



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5.68. Fault code F43: Emergency stop

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	Indicates that the emergency button has been pressed.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	F56: Remote emergency stop. F166: E-STOP.	

Analysis procedure

1. Visually check whether the emergency button (S02) located on the control door has been pressed. If it has been pressed, identify the reason. If it is part of the LOTOTO procedure or an unexpected event has occurred, the activation of this fault is normal. [INVG3_ST1B](#).
2. If this fault has not been pressed, there may be several reasons for this fault to be triggered:
 - 2.1. Check via PowerCommsV in group [SV1.5 Digital I/O status] the status of [parameter \[DSP RJ22\]](#). If it is set to X, fault F43 is triggered. [INVG3_ST1B](#).
 - 2.2. If the bit is set even though the button is not pressed, continuity must be checked with a 1500V multimeter between the following pins: [INVKG3_ST2](#), [INVMG3_ST5A](#)
 - 2.2.1. E41:J4:3 - E41:J4:4 of the door panel input/output (PPU) board. Corresponds to channel 1 (normally closed contact) of the emergency stop button (S02).



- 2.2.2. E41:J4:3 - E42:J4:4 of the door panel input/output boards (PPU). Corresponds to channel 2 (normally closed contact) of the emergency stop button (S02).

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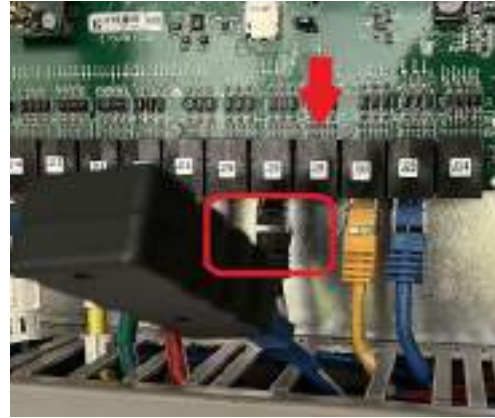
- 2.3. If there is no continuity, use a 1500V multimeter to measure continuity between pins 11 and 12 of emergency stop button (S02), and between pins 21 and 22 of the same pushbutton with the emergency stop button disconnected. **INVKG3_ST2, INVMG3_ST5A.**



- 2.3.1. If there is no continuity, replace emergency stop button (S02). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP89283*.
- 2.4. If there is continuity, the wiring is correct. In this case, the problem is between the input/output boards (E41 / E42) and the central control board (E35). In that case:
- 2.4.1. Check the continuity of the two RJ45 cables of the input/output boards, from connectors J2 of the boards E41 (W124) and E42 (W125) to connectors J21 and J22 of the central control board E35 respectively with the Ethernet cable tester. If any cable fails, replace it. **INVKG3_ST2, INVMG3_ST5A.**



E41:J2:1



E35:J21:1



E42:J2:1



E35:J22:1

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- 2.4.2. If the RJ45 cables do not fail, replace the input and output boards (E41 / E42). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP88849*.
- 2.4.3. If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP88849*.

5.69. Fault code F54: DU Overcurrent

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction” .			
Description	Overcurrent in one of the DU channels. The overcurrent is detected from the average of the last 100 samples, discarding those that are 25% above the average.			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	N/A.			

Analysis procedure

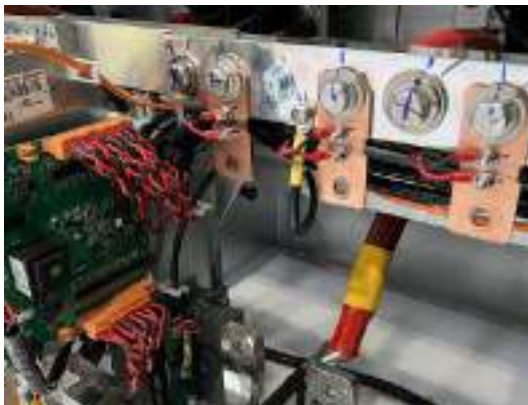
There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect parameterisation.

1. The fault setting is incorrect and fires when it is not due. If this occurs:
 - 1.1. Check by PowerCommsV the default value of the protection (**parameter [DU overcurrent]** of group [G5: Protections]). If it is not the default value, check with the plant manager for the appropriate value. **INVG3_ST1B**.

Incorrectly configured channels.

2. If the channels of the inverters are not configured according to the corresponding hardware, it is possible that the fault is triggered by a value outside the full scale. If this occurs:
 - 2.1. Check that the input strings of the inverter that are connected to BBEP.63 or BBEP.64 and have the shunt connected to the E66 or E65 board. **INVKG3_ST2, INVMG3_ST5A**.



- 2.2. Configure by PowerCommsV the type of DU of the inverter (**parameter [DU Fuses]** of subgroup [SG10.6: DU]). **INVG3_ST1B**. It is possible to differentiate it visually as the option "Both Poles" of parameter [DU Fuses] requires that there are fuses in the positive busbar strings BBEP.63 or BBEP.64 as well as in the negative BBEN.03. In the case of positive pole configuration, the positive strings are connected to BBEP.63 or BBEP.64. **INVKG3_ST2, INVMG3_ST5A**.
- 2.3. Depending on the number of channels connected, this must be configured by PowerCommsV in the parameter **[DU Number of Channels]** of the subgroup [SG10.6: DU]. **INVG3_ST1B**.
- 2.4. If the configuration of the **[DU Fuses]** parameter does not match any of the options, in this case use the "Customisable" option and disable in the group [G10: DU] the channels that are not physically connected. **INVG3_ST1B**.

Note: Configure the channels of the inverter correctly to avoid false measurement failures.

Damaged wiring or shunt.

3. If the shunt resistors or measurement wiring is damaged it is possible to trigger the fault. If this occurs:

- 3.1. Check that the SHX correspond to the current measurement shunt resistors. INVKG3_ST2, INVMG3_ST5A.
- 3.2. Check that these resistors give continuity with a 1500V multimeter. INVKG3_ST2, INVMG3_ST5A.



- 3.3. Check that the cables that go from the SHX measurement resistors to the E66 or E65 boards are correctly wired, paying special attention to the connections on the resistor itself. INVKG3_ST2, INVMG3_ST5A.

String configuration.

4. If one of the inverter's input strings is poorly distributed, the amount of current that will enter through that measurement channel will be significantly higher and may cause a fault. This may be due to a bad distribution of the inputs by the wiring of the project. If this happens:
 - 4.1. Check that the DU strings are evenly distributed at the input of the inverter. If this is not the case, contact the plant manager so that he/she can check this wiring with the customer. INVKG3_ST2, INVMG3_ST5A.

Internal recirculation in the measurement board.

5. If none of the above cases apply, it is possible that the overcurrent is caused by a recirculation problem in the DU metering board. In this case:
 - 5.1. Check if the DU measurement boards are model E21801204BA or later. If this is not the case, replace these boards with this model INVKG3_ST2, INVMG3_ST5A and check the operation of the inverter by monitoring so that the fault is not repeated.



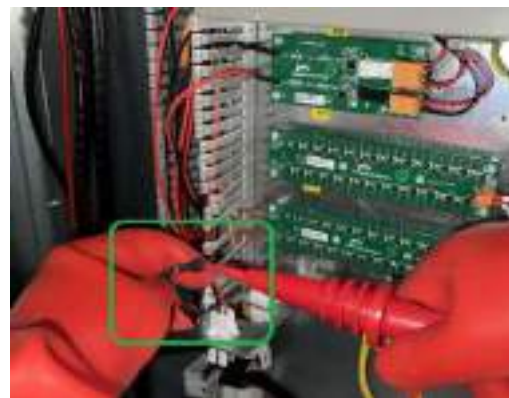
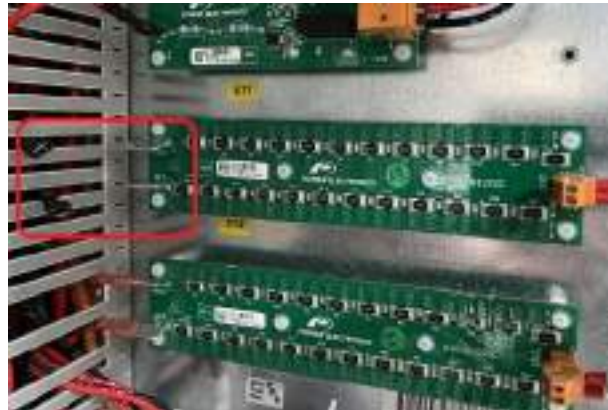
If it is repeated, check other causes. See document INVKG3/OP88849.

Short-circuit in the channel.

6. It is possible that the overcurrent is due to a short circuit or shunt on the corresponding channel. If this is the case:
 - 6.1. Visually inspect the inside of the DU for damaged cables, blackened or bad busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU. INVKG3_ST2, INVMG3_ST5A.

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- 6.2. With the strings / batteries disconnected, connect an isolation meter between positive and negative of each input of the inverter and make a measurement with a maximum of 1000Vdc. Note that previously it is necessary to disconnect the RFI boards and the isolation measurement boards.
INVKG3_ST5B, INVMG3_ST5B.



- 6.2.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables from each of the busbars and repeat the measurement.
- 6.2.2. If the measurement exceeds $1\text{M}\Omega$ when any of the input cables to the inverter are disconnected, this means that this cable has a shunt between positive and negative. If this occurs, inform the customer so that this channel can be repaired.
- 6.2.3. If the measurement does not improve after disconnecting all external cables, consult Power Electronics.

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5.70. Fault code F56: Remote emergency stop

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	The inverter has received an emergency stop from the customer via communications (Modbus or PowerCommsV).	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

The remote emergency stop is activated when the inverter receives the command via Modbus or if it has been sent directly from PowerCommsV. The activation of this emergency stop can only be carried out by a user with a 'Qualified Client' role or higher. In the same way, to reset the F56, it is necessary to do it manually with a 'Qualified Customer' role or higher.

There are different causes that can trigger this fault. Follow the steps below to identify the possible cause and resolve it.

Report recurring fault.

1. If the F56 trips on a recurring basis, check with the plant manager what actions are being taken to cause this to happen. A possible cause could be a Modbus address being written to by the customer. If this occurs in PowercommsV: **INVG3_ST1B**.
 - 1.1. Check with the customer if he has a remote inverter control (SCADA type) that could be commanding the writing in the Modbus addresses that activate the F56 (Password Address = 3037; Fault activation address = 3038).
 - 1.2. Ask the customer about the equipment shutdown procedure. Ensure that they are not shutting down the equipment by means of remote emergency shutdown.

Impossibility to reset the fault.

2. F56 has tripped and it is not possible to reset the fault manually via the PowerCommsV reset button. If this occurs:
 - 2.1. Check with PowercommsV user role (parameter [User] of group [G1: General]). It must be 'Qualified Customer' or higher to be allowed to reset it. In case of a lower role, ask Power Electronics for an account with a higher role. **INVG3_ST1B**.
 - 2.2. If the role is 'Qualified Customer' and the fault is not reset by pressing the reset button, perform the following check: **INVG3_ST1B**.
 - 2.2.1. Access PowerCommsV with role 'PE staff'.
 - 2.2.2. Select F56 in the parameter [Fault selected] of the subgroup [G5.7-Fault/warning configuration].
 - 2.2.3. Check in the parameter [Manual Reset User] of the subgroup [G5.7-Fault/Alert Configuration] that it is configured as 'Qualified Client'. In case it is in a higher role, change it to 'Qualified Customer'.
 - 2.2.4. If the account has role 'Qualified Customer' and in addition to F56 has another more restrictive fault active, i.e. requiring a higher role than 'Qualified Customer' to be reset, then it will not be possible to reset F56. In this case it will be necessary to reset the F56 with the user role that allows the more restrictive fault to be reset.

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5.71. Fault code F65: Overcurrent AC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction”.		
Description	One of the AC currents of a module has exceeded the protection threshold.		
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK	
Associated faults or warnings	Related to: F66: Module VDC Unbalance F69: Overvoltage DC+ F70: Overvoltage DC- F83: Multiple desaturation F112: OCAC emergency F106: Mod. Vdc unbalanced F108: Mod. High Vdc		

Analysis procedure

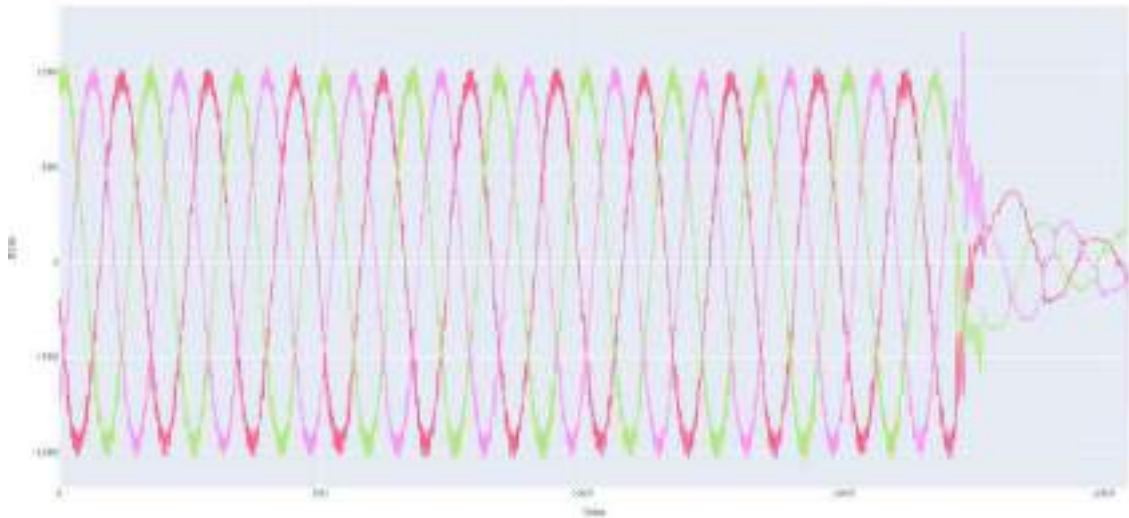
There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Threshold misconfiguration.

- Check by PowerCommsV if the configured value of the current threshold (parameter [Overcurrent threshold] of the subgroup [SG10.2: Protections]) is at its default value. **INVG3_ST1B.**
 - If it is not at its default value, contact the plant manager to find out if there is a reason. If not, set to default value. **INVG3_ST1B.**

Network event. If the inverter has suffered a mains event (a rapid voltage or frequency variation), it is normal for it to fail by F65.

- Download the Blackbox file from the SD card after the failure. Check the Blackbox file of the module and check if at the moment of the fault there is a sudden variation of the AC voltage or its frequency, for this it is necessary to distinguish between a gap and a short-circuit voltage drop (see attached pictures as an example). If so, it is a normal fault. Reset the fault and restart. **INVG3_ST1B.**



Example of AC short-circuit voltage variation

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Example of AC gap voltage variation

Measurement of crossed voltages.

3. If the AC voltage measurements of the central control board (E35) are crossed, the inverter will be constantly tripping F65 when starting up the modules. If this occurs, perform the following checks:

- 3.1. Measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

Note: To carry out the continuity checks, it is necessary to disconnect the measurement cables (W071) from the F100 so that there is no false continuity with the auxiliary transformer.

- Between F100:1 and J3 of E35 (J5 for HEMK / PCSK).



- Between F100:3 and J4 of E35.

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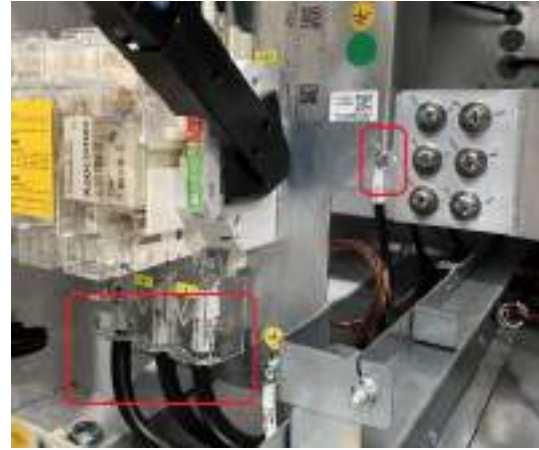
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- Between F100:5 and J5 of E35 (J3 for HEMK / PCSK).



3.2. Visually check that:

- F100:2 is connected to busbar L1.
- F100:4 is connected to busbar L2.
- F100:6 is connected to busbar L3.

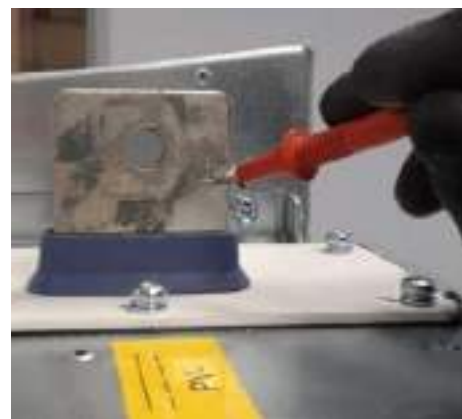


3.3. If any of the points do not give continuity or are not visually connected where they should be, correct the wiring or replace the wiring. **INVKG3_ST2, INVMG3_ST5A.**

IGBT damaged.

4. One of the power stage IGBTs is damaged. Check the IGBT of the power stage.

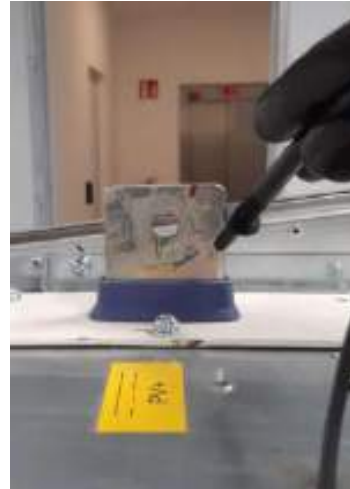
4.1. Technically check with 1500V multimeter the IGBT and replace any damaged IGBTs. **INVKG3_ST2, INVMG3_ST5A.** (See document *INVG3IOP88076*).



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Short-circuit in the AC part of the inverter.

5. If the F65 fault is accompanied by an AC circuit breaker overcurrent trip (Q01 for PCSK / HEMK) or a medium voltage switchgear overcurrent fault (Q02), it is possible that the cause of the F65 fault is an internal short-circuit in the inverter. If this occurs, carry out the following checks:
 - 5.1. Visually check that the main choke (L01) has no blackened points, also check that the busbars that connect to the modules are in good condition and no points where the insulation between phases is lost (i.e. joined or very close busbars of different phase) can be seen. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.2. Visually check the output busbars for any foreign object that could be shorting between phases. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.3. Visually check that the insulation of the notch choke (L07, L08 and L09) is in perfect condition. If not, replace the notch choke. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP85367*.
 - 5.4. Check all capacitors of the capacitor filter (C25-C48). To do, measure with a 1500V multimeter the capacitance between: **INVKG3_ST2, INVMG3_ST5A.**

5.4.1. Between L1 and L2 of the E20, E21, E22 and E23 discharge cards.

5.4.2. Between L2 and L3 of the E20, E21, E22 and E23 discharge cards.

5.4.3. Between L3 and L1 of the E20, E21, E22 and E23 discharge cards.

The measured value must be $\approx 184\mu\text{F}$ (170-200 μF).

NOTE: The difference in the measured capacitance value between the points should not exceed 10 μF .



L1-L2



L2-L3



L1-L3

If any measured value is not correct, replace the corresponding capacitor. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3/OP85367*.

- 5.5. Disconnect the AC boards of all modules **INVKG3_ST2**, **INVMG3_ST5A** and measure with an inter-phase resistance meter at the outputs of each of the modules. The resistance measured at that point should be greater than approximately 2 m Ω . If the resistance is less than measured, check the resistance of the following components with the resistance meter to see if they are correct.



L1-L2



L2-L3



L1-L3

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Note: These checks shall be made with each of the components disconnected at all their interconnection points. **INVKG3_ST2, INVMG3_ST5A.**

- Resistance between phase and neutral of the MV transformer (T01). It must be greater than (approximately) $0.8\text{m}\Omega$.

NOTE: To measure between neutral and phase L2 it is necessary to remove the fiber plate.



Neutral-L1



Neutral-L3

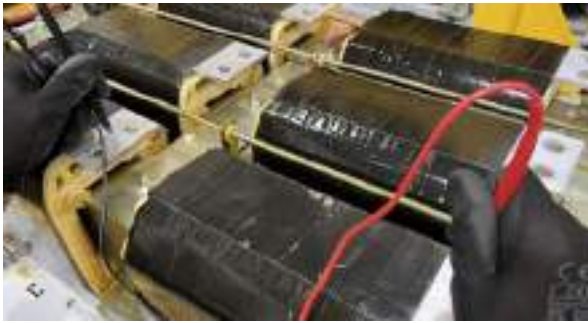


Without fiber plate



Neutral-L2

- Phase resistance of each of the module inductances (L01). Must be greater than (approximate) $0.9\text{m}\Omega$.



Fase R de L01



Fase S de L01



Fase T de L01

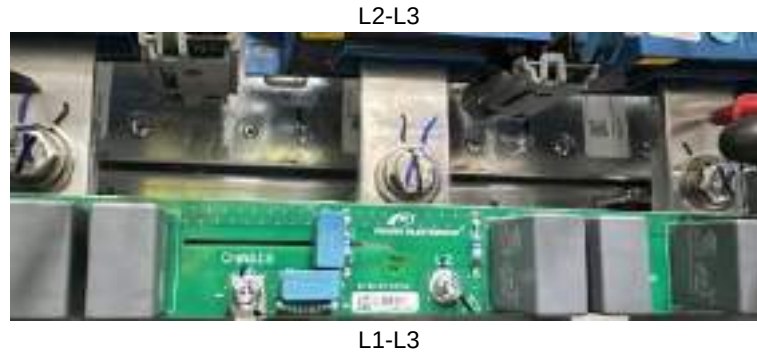
- Phase-to-phase resistance of filter C greater than $10\text{M}\Omega$. Measure between BBI1.12-BBI2.12, BBI2.12-BBI3.12 and BBI3.12-BBI1.12.



L1-L2



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- If any of the points have a lower resistance than indicated, replace the element that does not meet its resistance.

Main inductance damaged.

6. If all the checks have gone well, and it is only one module that is failing due to F65, replace this power stage with another one that works well. **INVKG3_ST2, INVMG3_ST5A.**
7. If the fault remains in module "x" even after replacing it with a working power stage, it means that the main choke is damaged. Consult Power Electronics for the possibility of replacing the main choke (L01). See document *INVG3/OP85375*.

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5.72. Fault code F86: Feedback breaker

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.			
Description	The DC switch disconnecter is not closing/opening when required. This fault appears when there is a discrepancy between the command and the feedback from the DC switch-disconnector.			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings				

Analysis procedure

This protection can be configured as Fault or Warning in PowerCommsV (parameter [Feedback breaker DC] of group [G5.5 I/O Protections]).

1. Check in PowerCommsV the status of the DC disconnectors to locate the failed disconnector (parameters [DC x switch status] of the [SV1.4-Equipment] subgroup). **INVG3_ST1B.**
2. Check in PowerCommsV on which module it has been reported via the digital input and output signals (parameters [Mod x Digital In], [Mod x Digital Out] of subgroup [SV1.5 Digital I/O Status]). **INVG3_ST1B.**
3. The following checks must be carried out on the above signals:
 - 3.1. To carry out the opening and closing of the DC switch disconnectors, perform the following steps in PowerCommsV: **INVG3_ST1B.**
 - 3.1.1. Set the configuration of the DC virtual selector in communications (parameter [DC Virtual Sel conf] of subgroup [G2.6: Selectors]) and activate and deactivate the DC virtual selector depending on whether you want to check the opening or closing (parameter [DC Virtual Selector] of subgroup [G2.6: Selectors]). **INVG3_ST1B.** This will start the manoeuvre.
 - 3.1.2. To check the opening or closing command of the DC disconnectors, check that bit 3 (DC breaker close command) or bit 4 (DC breaker open command) of the module digital outputs are set (X) (parameters [Mod x Digital Out] of subgroup [SV1.5 Digital I/O Status]).
 - Open command. Verify that bit 4 (feedback DC breaker open) is set (X) in the module digital inputs (parameters [Mod x Dig Inputs], status [dis_open], of subgroup [SV1.5 Digital I/O Status]). **INVG3_ST1B.**
 - Close command. Verify in the module digital inputs that bit 5 (feedback DC breaker closed) is set (X) (parameters [Mod x Digital Out], status [dis_close] of subgroup [SV1.5 Digital I/O Status]). **INVG3_ST1B.**
4. If the results of these tests are not as expected, carry out the following steps:
 - 4.1. Check that the switch disconnector opens and closes correctly. To do this, command an open and close command via PowerCommsV and check that it opens and closes correctly (it should be heard to close). **INVG3_ST1B, with the doors closed.**
 - 4.2. If it physically opens and closes correctly, but the fault occurs, check the status of the feedback by means of the following hardware checks: **INVKG3_ST2, INVMG3_ST5A.**

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4.2.1. The checks are carried out according to the component codes of module 1. If the fault is in a different module, the codes vary as shown in the table below:

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
DC switch disconnecter	SI01	SI02	SI03	SI04
Coil motor connector	XC470.1	XC475.1	XC480.1	XC485.1
Feedback connector	XC471.1	XC476.1	XC481.1	XC486.1
Input and output board	E55	E56	E57	E58
RJ45 cable	W009	W010	W011	W012

4.2.2. Visually check that the wiring is connected at the points indicated in the diagram with origin / destination: **INVKG3_ST2, INVMG3_ST5A**

- E55:J4:3 – XC471.1:11.
- E55:J4:4 – XC471.1:12.
- E55:J4:5 – XC471.1:21.
- E55:J4:6 – XC471.1:24.

4.2.3. If the wiring is connected to the points indicated above, check continuity by means of a 1500V multimeter in resistivity mode on the disconnecter feedback (open/closed). **INVKG3_ST2, INVMG3_ST5A**. With the DC disconnecter (SI01) in the open position:

- There must be no continuity between E55:J4:5 - E55:J4:6. This is the closed disconnecter feedback whose internal contact is normally open. **INVKG3_ST2, INVMG3_ST5A**.



- o If there is continuity between E55:J4:5 - E55:J4:6, release connector XC471.1 from the disconnecter and measure continuity on pins 21 - 24 of disconnecter SI01. **INVKG3_ST2, INVMG3_ST5A**. If there is continuity, replace the disconnecter SI01 as it is due to an internal fault and this contact must be open. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP85377*.
- o If there is no continuity between E55:J4:5 - E55:J4:6, the isolating switch is working correctly. Check the wiring from the disconnecter to the input/output board (E55). **INVKG3_ST2, INVMG3_ST5A**.
- There must be continuity between E55:J4:3 - E55:J4:4. This is the open disconnecter feedback whose internal contact is normally closed.



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- o If there is no continuity between E55:J4:3 - E55:J4:4, identify whether it is the wiring or the disconnecter that is faulty. **INVKG3_ST2, INVMG3_ST5A**. To do this:
 - Release connector XC471.1 from the disconnecter and measure continuity on pins 11 - 12 of disconnecter SI01. **INVKG3_ST2, INVMG3_ST5A**.
 - If there is continuity, the disconnecter is functioning correctly. The wiring from the isolator to the input/output board (E55) should be replaced. **INVKG3_ST2, INVMG3_ST5A**.
 - Otherwise, the SI01 disconnecter must be replaced as it is due to some internal fault. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP85377*.

4.2.4. If the above checks are satisfactory, the disconnecter and its wiring are correct. In this case, the problem is between the input/output board (E55) and the module control board (E02).

- Disconnect the RJ45 cable (W009) connecting the input/output board E55:J2 to the module control board E02:J13. Check the continuity of the RJ45 cable with the network cable continuity tester. Replace the faulty cable. **INVKG3_ST2, INVMG3_ST5A**.



- If the RJ45 cable does not fail, replace the input/output board (E55). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88849*.

- If fault persists, replace module control board (E02). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88076*.

4.2.5. If the above test does not open and close correctly, the switch disconnector coils are most likely not supplied with power or are damaged. Carry out the following steps.

Note: These checks must be carried out in the **INVKG3_ST2, INVMG3_ST5A** state with external 110Vdc power supply.

- Disconnect the wires connected to pins X125:1:2, X125:2:2.
- Connect the positive of the external 110Vdc power supply to pin X125:1:2 (use electrical hazard EPIS 1).
- Disconnect the wires connected to pins X125:1:1, X125:2:1.
- Connect the negative of the external 110Vdc power supply to the X125:1:1 pin.
- Check that the circuit breaker has 110V power. Proceed to next step. **Note:** Required status **INVKG3_ST2, INVMG3_ST5A** with external 110Vdc power supply.

4.2.6. To check that the switch disconnector has 110V power supply, it is necessary to check that 110Vdc power is supplied to the main terminal block of the disconnecter cabinet.

Note: These checks must be carried out in the **INVKG3_ST2, INVMG3_ST5A** state with external 110Vdc power supply.

- With a 1500V multimeter set to measure DC voltage, measure between X125:1:2 (Positive 110V) and X125:1:1 (Negative 110V) (use electrical hazard EPIS 1).



- If 110V is present, check step by step that voltage is present at each of the following points:
 - Between positive pole of terminal X125:1:2 of the SI01 disconnecter board and the negative of the closing coil (XC470.1 A2), opening coil (XC470.1 C2), minimum coil (XC470.1 D2) and spring motor (XC470.1 U2). (use of electrical hazard EPIS 1).



X125:1:2



XC470.1 D2



XC470.1 U2 – X125:1:2

- If there is no 110Vdc at all these points, the negative lead from the main terminal block (X125:1:1) to the point where we do not measure 110Vdc voltage is broken, check and replace. Remove the external 110Vdc power supply. **INVKG3_ST2, INVMG3_ST5A.**
 - Measure DC voltage with the 1500V multimeter between the negative pole of the disconnecter board terminal block (X125:1:1) and the positive of the activation relays of the input/output board (E55, J5:1, J6:1, J7:1), and the positive of the UVT undervoltage coil (terminal D1 of connector XC470 of the switch disconnecter). These checks must be carried out in the **INVKG3_ST2, INVMG3_ST5A state with external 110Vdc power supply.**

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E55:J5:X



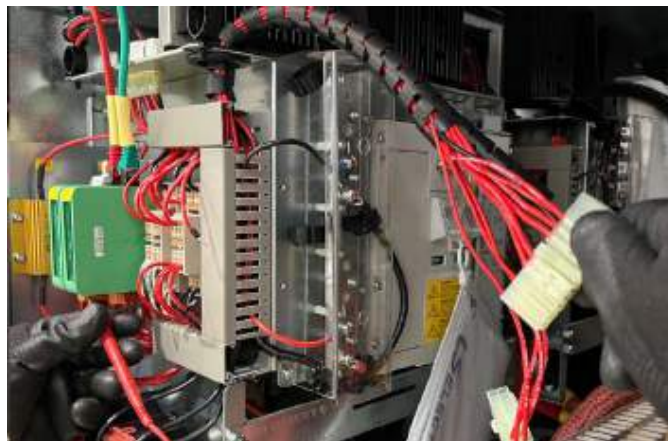
X125:1:1

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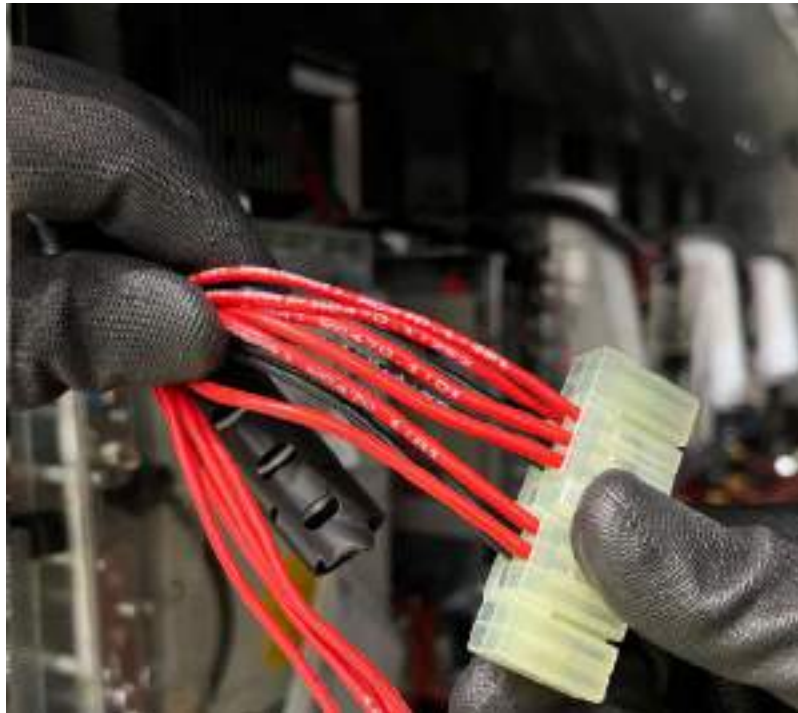


- If 110Vdc is not present at any of the points, check the wiring and replace. **Remove external 110Vdc power supply. INVKG3_ST2, INVMG3_ST5A.**

4.2.7. If 110V is present in all circuits verified above, check continuity between the relay outputs of the input/output board E55, J5:2, J6:2, J7:2, and the connection point with the DC isolator XC470.1 A1, XC470.1 C1, XC470.1 U1. **Remove the external 110Vdc power supply. INVKG3_ST2; INVMG3_ST5A.**



- 4.2.8. If there is no continuity at any of the points, check and replace the wiring. **INVKG3_ST2; INVMG3_ST5A.**
- 4.2.9. If all checks are correct, verify that the motor resistance is approximately 30Ω XC470.1:U1 and XC470.1:U2 (with the spring unloaded). **INVKG3_ST2; INVMG3_ST5A.**



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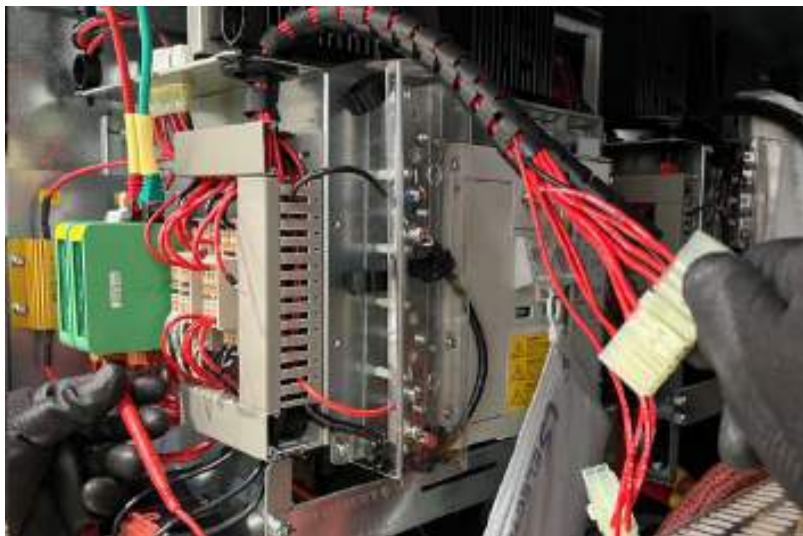
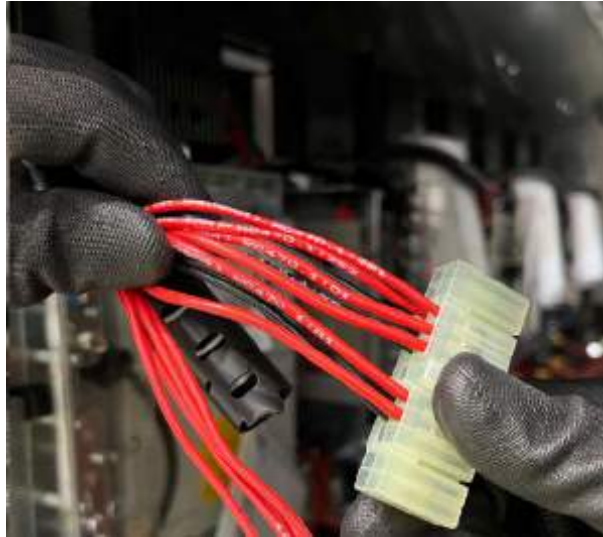
- 4.2.10. To check the opening and closing coils, the following actions must be performed:

Note: these checks, it is necessary to perform them in the **INVKG3_ST2, INVMG3_ST5A state with external 110Vdc power supply.**

- Externally power supplies G01 and G02 of the central control panel.
- Supply the external 110Vdc power supply to sources G01 and G02 as follows:
 - Disconnect the wires connected to the L(+) and N(-) terminals of both sources.
 - Connect the positive of the external power supply to the L(+) terminal and the negative to the N(-) of the G01 and G02 power supplies.
 - At the output of power supply G01, check the switch disconnectors Q13 (Mod 1), Q14 (Mod 2), Q14 (Mod 3), Q16 (Mod 4) that cut off the power supply to the module control boards.
 - Only the switch disconnector of the module on which the fault has been reported should be left closed.
- Tests on the opening and closing coils:

Note: These checks, it is necessary to perform them in the **INVKG3_ST2, INVMG3_ST5A state with external 110Vdc power supply.**

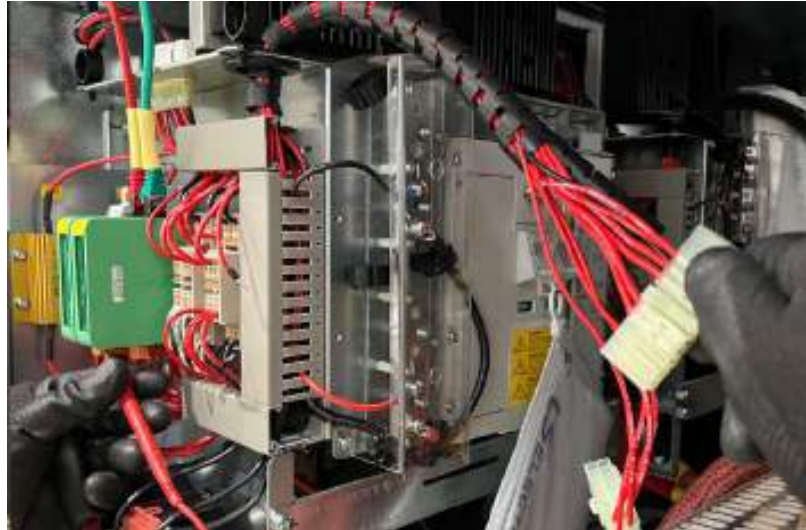
- Opening coil. Command opening and measure using a 1500V multimeter in Vdc mode between terminals XC470.1:C1 and XC470.1:C2. If 110Vdc is not present, check wiring up to E55:J6:2, and if correct, replace coil. Remove external 110Vdc power supply. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP85377*.



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- Closing coil. Command opening and measure by 1500V multimeter in Vdc mode between terminals XC470.1:A1 and XC470.1:A2. If 110Vdc is not present, check wiring up to E55:J5:2, and if correct, replace coil. Remove external 110Vdc power supply. [INVKG3_ST2](#); [INVMG3_ST5A](#). See document [INVG3IOP85377](#).





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5.73. Fault code F98: Idc derivation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module has measured an average current greater in absolute value than configured in the DC current sensor with the module stopped for 250ms.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Threshold misconfiguration.

1. Check by PowerCommsV if the configured value of the current threshold (parameter [Deriv. Idc Threshold] of subgroup [G5.10: Internal Prot]) is at its default value. **INVG3_ST1B**.
 - 1.1. If it is not in its default value, contact the plant manager to find out if there is any reason. If not, set to default value. **INVG3_ST1B**.

Hall sensor in bad condition. The hall sensor in the module is in bad condition. If all checks have gone well, it is possible that the module's sensor is in poor condition. If so, the following checks are necessary:

2. The RJ45 cable is not properly connected. If this occurs:
 - 2.1. Dismantle the power stage and check that all RJ45 connectors are properly connected (W029). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88076*.

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- 2.2. If one of them is not properly connected or is damaged and cannot be connected properly, replace it. [INVKG3_ST2](#), [INVMG3_ST5A](#).
3. The RJ45 cable has no continuity. If this occurs:
 - 3.1. Check using an RJ45 tester, that the RJ45 cable (W029) has continuity on all its pins. If this is not the case, replace the damaged cable. [INVKG3_ST2](#), [INVMG3_ST5A](#).



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Cables to check continuity and an example of the screen of the RJ45 tester

- 3.2. If the RJ45 cable is in good condition and well connected, replace the board and the measurement sensor (E03+HT01). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88849*.
- 3.3. If the fault still occurs after replacing the sensor and module board (E02). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88076*.

Excessive current consumption in the DU. There is some consumption point that is consuming energy from the DU in an uncontrolled manner.

4. If the inverter has the optional Bus Plus, check in the SD logs (DCDC Log) that the DCDCs are stopped at the time of the fault. If this is not the case, inform the customer that this is an incorrect way of manoeuvring the inverter. **INVG3_ST1B**.
5. In the case of PCS inverters, check that the battery voltage is above the rectified mains voltage of the inverter. In other words, that the values shown in parameters [External voltage 1], [External voltage 2],

[External voltage 3], [External voltage 4] of group [SV2: DC source] are at least the values shown in parameters [Grid voltage RS LV], [Grid voltage ST LV], [Grid voltage TR LV] of subgroup [SV3.2: Grid LV] multiplied by $\sqrt{2}$. **INV3_ST1B**. **Note:** The measurements shown in the parameters of group [SV2: DC source] will depend on the number of busbars in the inverter. For more information, refer to the *Software and Programming Manual* of the inverters.

- 5.1. If this is not the case, inform the customer that the batteries are not in a voltage range necessary for the correct operation of the inverter. **INV3_ST1B**.
6. With the DC cables disconnected from the module and without their ends touching any metallic point, connect an insulation meter between positive and negative of the module cables (in case of Bus Plus, it is necessary to disconnect all the modules and DCDC). Make a measurement with a maximum of 1000Vdc. If the measurement is less than 10Mohms. **INVKG3_ST5B, INVMG3_ST5B**.



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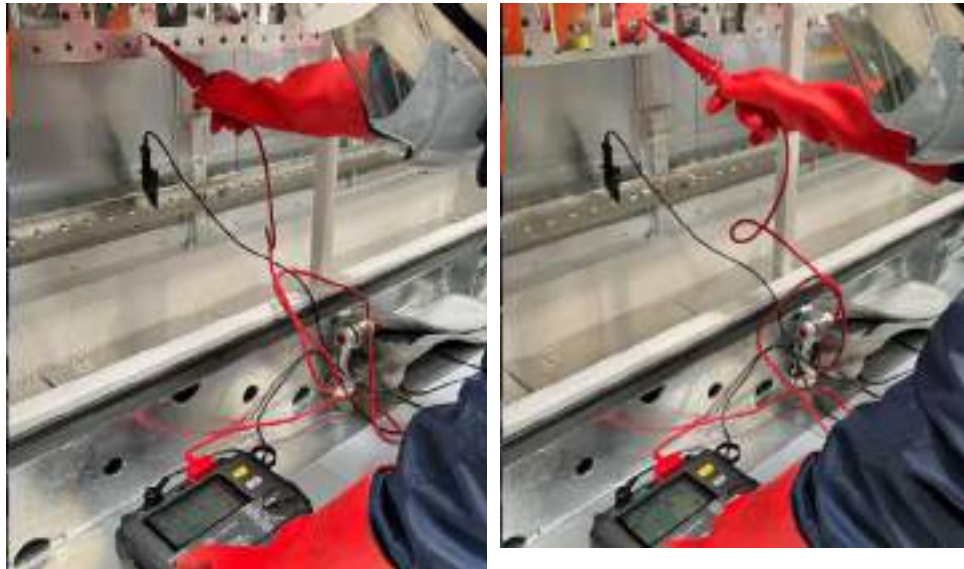




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- 6.1. Replace the cables that go from the DC switch disconnecter to the module. In the case of Bus Plus replace all wires from all modules. **INVKG3_ST5A, INVMG3_ST5A.**
- 6.2. Visually inspect inside the DU for damaged cables, blackened or damaged busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU. **INVKG3_ST5A, INVMG3_ST5A.**
- 6.3. If any of the above problems are found, repair or replace the damaged elements. **INVKG3_ST5A, INVMG3_ST5A.**
7. With the strings / batteries disconnected, connect an insulation meter between positive and negative of each one of the inputs of the inverter (DC switch disconnecter boards by means of extenders and banana cables). Subsequently, make a resistance measurement with a maximum of 1000Vdc. **INVKG3_ST5B INVMG3_ST5B.**





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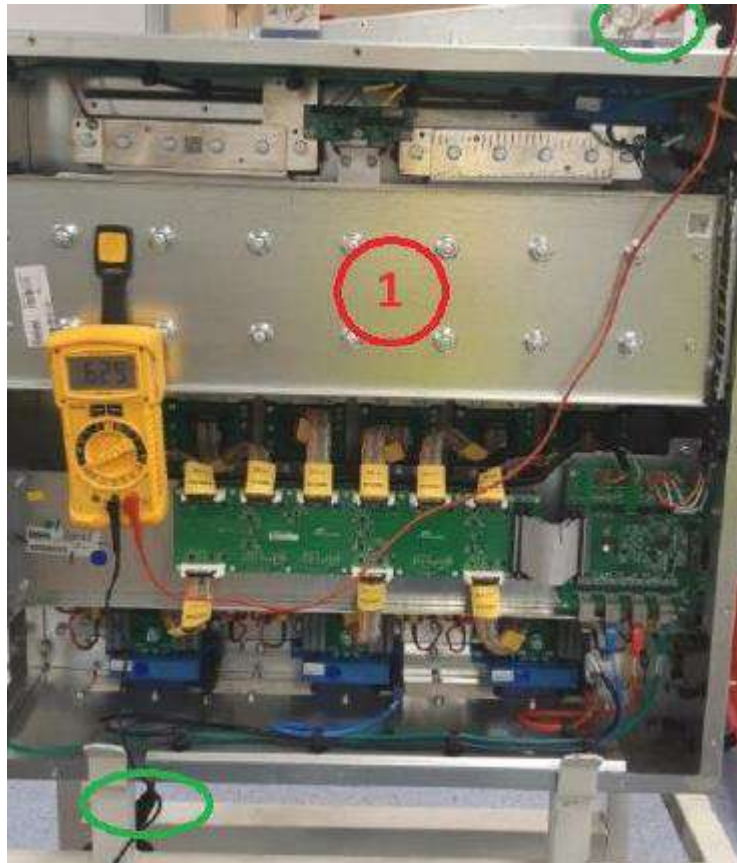
Note: Please note that the RFI boards from ground (E65 and E68 for solar inverter, E50 and E51 for PCS with one busbar and E50, E51, E52 and E53 for PCS with more than one busbar), and ISOPRO measurement boards (E71 for solar inverter, E71, E72, E73 and E74 for each of the busbars for PCS) must be disconnected beforehand. **INVKG3_ST5B INVMG3_ST5B**

- 7.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables of each busbar and repeat the measurement.
- 7.2. If the measurement exceeds 1Ω when any of the input cables to the inverter are disconnected, this indicates that the cable has a shunt between positive and negative. If this occurs, inform the customer of the situation so that the channel can be repaired. If the measurement does not improve after disconnecting all external wires, consult Power Electronics.

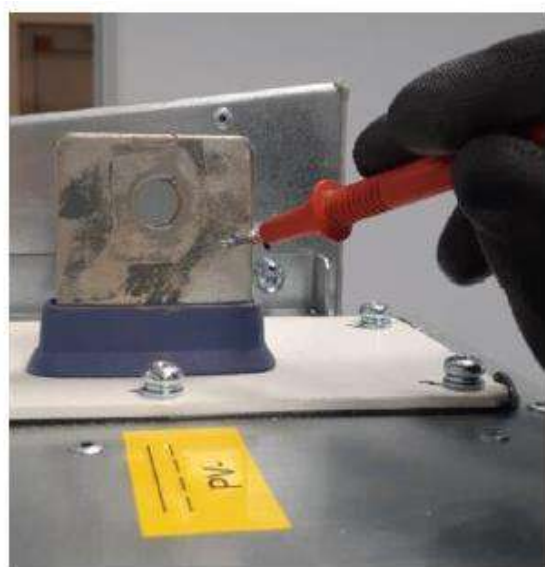
Busbar	Positive plate	Negative plate
1	BBEP.81	BBEN.66
2	BBEP.79	BBEN.64
3	BBEP.77	BBEN.62
4	BBEP.75	BBEN.60

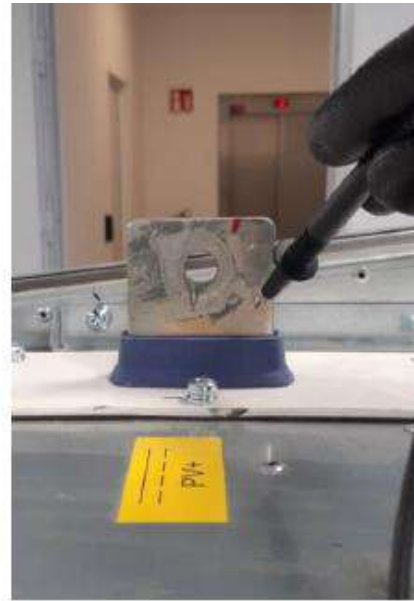
Internal derivation of the module.

8. Check with the 1500V multimeter the module IGBT. See document INVKG3IOP88076.



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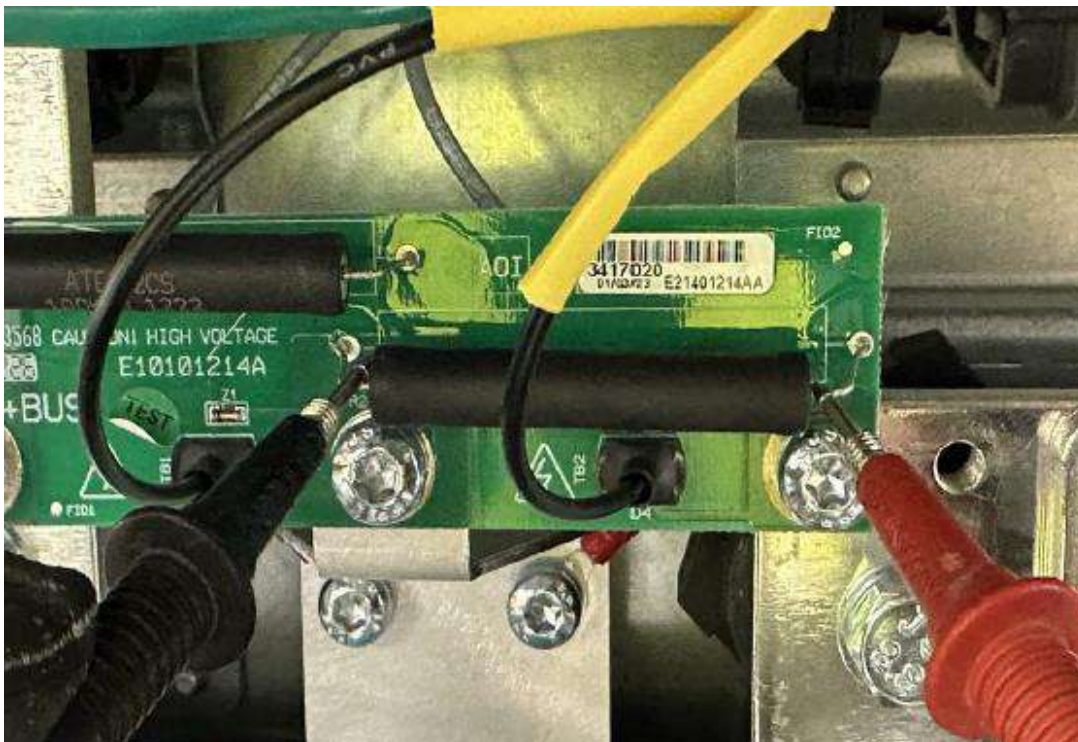


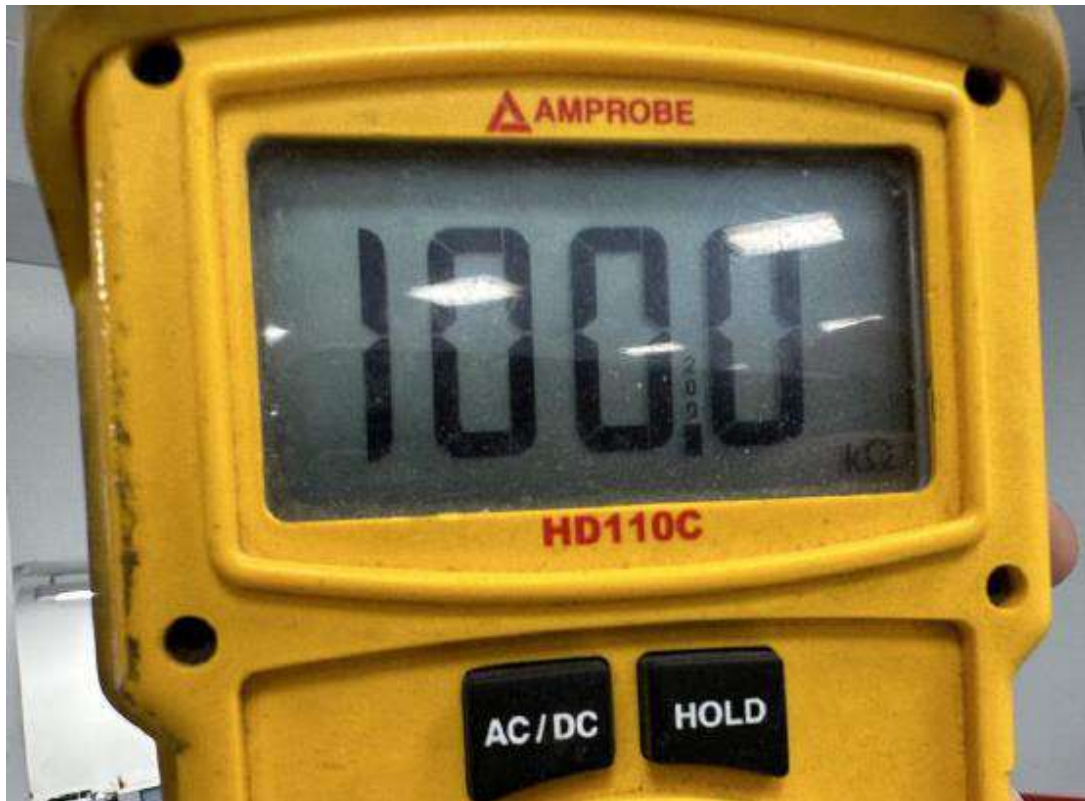


9. Check the resistance between positive (BBIP.20) and negative (BBIN.20) of the module input. In all cases it must be greater than 100kΩ. The measurement should be taken after connecting the 1500V multimeter and waiting at least 10s and with the DC cables disconnected from the module.
[INVKG3_ST2](#), [INVMG3_ST5A](#).



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- 9.1. If the measurement is lower than this value, consult Power Electronics for replacement of the power stage. See document *INVG3IOP88076*.

5.74. Fault code F99: Mod. Iac derivation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module has measured a higher RMS current than configured in one of the AC sensors with the module stopped for 250ms.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Threshold misconfiguration.

1. Check by PowerCommsV if the configured value of the current threshold (parameter [Deriv. Idc Threshold] of subgroup [G5.10: Internal Prot]) is at its default value. **INVG3_ST1B**.
 - 1.1. If it is not in its default value, contact the plant manager to find out if there is any reason. If not, set to default value. **INVG3_ST1B**.

Hall sensor in bad condition. The hall sensor in the module is in bad condition. If all checks have gone well, it is possible that the module's sensor is in poor condition. If so, the following checks are necessary:

Board + sensor hall	RJ45 cable	Board connector E02
E04+HT02	W030	U52
E05+HT03	W031	U53
E06+HT04	W032	U54

2. The RJ45 cable is not properly connected. If this occurs:
 - 2.1. Dismantle the power stage and check that all RJ45 connectors are properly connected (according to table). **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3/OP88076*.

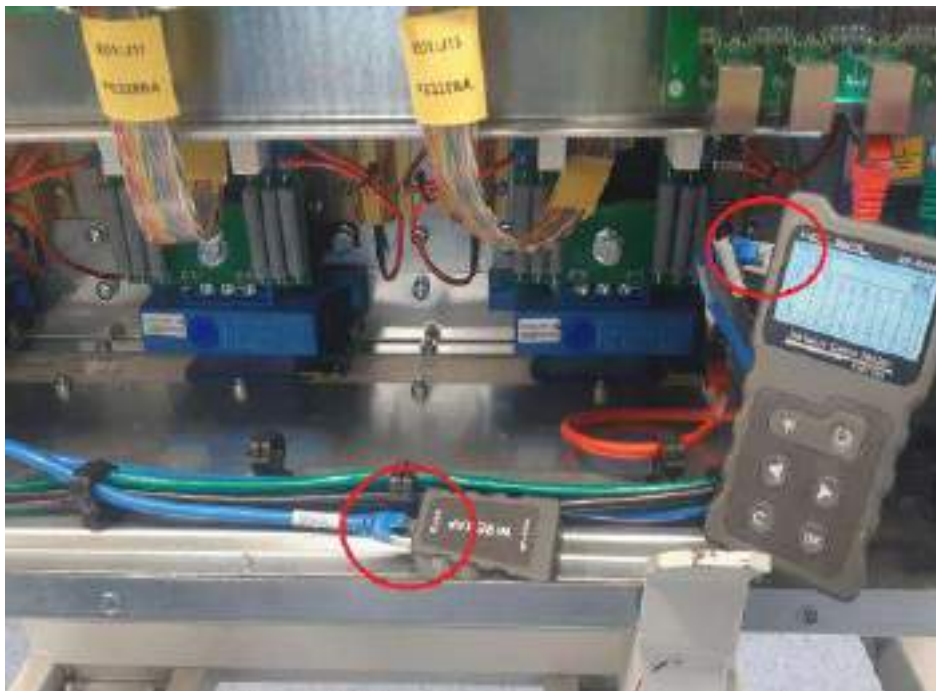
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- 2.2. If one of them is not properly connected or is damaged and cannot be connected properly, replace it. **INVKG3_ST2, INVMG3_ST5A.**
3. The RJ45 cable has no continuity. If this occurs:
 - 3.1. Check using an RJ45 tester, that the RJ45 cable (according to table) has continuity on all its pins. If this is not the case, replace the damaged cable. **INVKG3_ST2, INVMG3_ST5A.**



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- 3.2. If the RJ45 cable is in good condition and well connected, replace the board and the measurement sensor (according to table). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88849*.
- 3.3. If the fault still occurs after replacing the sensor and module board (E02). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88076*.

Excessive current consumption in the DU. There is some consumption point that is consuming energy from the DU in an uncontrolled manner.

4. If the inverter has the optional Bus Plus, check in the SD logs (DCDC Log) that the DCDCs are stopped at the time of the fault. If this is not the case, inform the customer that this is an incorrect way of manoeuvring the inverter. **INVG3_ST1B**.
5. In the case of PCS inverters, check (by PowerComms V or SCADA client) that the battery voltage is above the rectified mains voltage of the inverter. In other words, that the values shown in parameters [External voltage 1], [External voltage 2], [External voltage 3], [External voltage 4] of group [SV2: DC source] are at least the values shown in parameters [Grid voltage RS LV], [Grid voltage ST LV], [Grid voltage TR LV] of subgroup [SV3.2: Grid LV] multiplied by $\sqrt{2}$. **INVG3_ST1B**. **Note:** The measurements shown in the parameters of group [SV2: DC source] will depend on the number of busbars in the inverter. For more information, refer to the *Software and Programming Manual* of the inverters.
 - 5.1. If this is not the case, inform the customer that the batteries are not in a voltage range necessary for the correct operation of the inverter. **INVG3_ST1B**.
6. With the DC cables disconnected from the module and without their ends touching any metallic point, connect an insulation meter between positive and negative of the module cables (in case of Bus Plus, it is necessary to disconnect all the modules and DCDC). Make a measurement with a maximum of 1000Vdc. If the measurement is less than 10Mohms. **INVKG3_ST5B, INVMG3_ST5B**.



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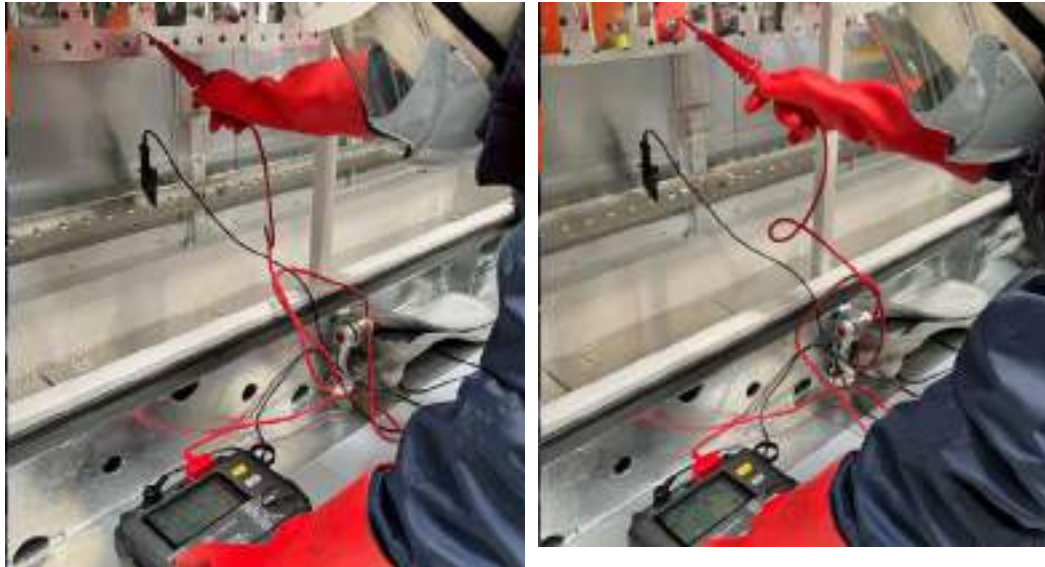




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- 6.1. Replace the cables that go from the DC switch disconnector to the module. In the case of Bus Plus replace all wires from all modules. [INVKG3_ST5A](#), [INVMG3_ST5A](#).
- 6.2. Visually inspect inside the DU for damaged cables, blackened or damaged busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU. [INVKG3_ST5A](#), [INVMG3_ST5A](#).
7. With the strings / batteries disconnected, connect an insulation meter between positive and negative of each one of the inputs of the inverter (DC switch disconnector boards by means of extenders and banana cables). Subsequently, make a resistance measurement with a maximum of 1000Vdc. [INVKG3_ST5B](#) [INVMG3_ST5B](#).





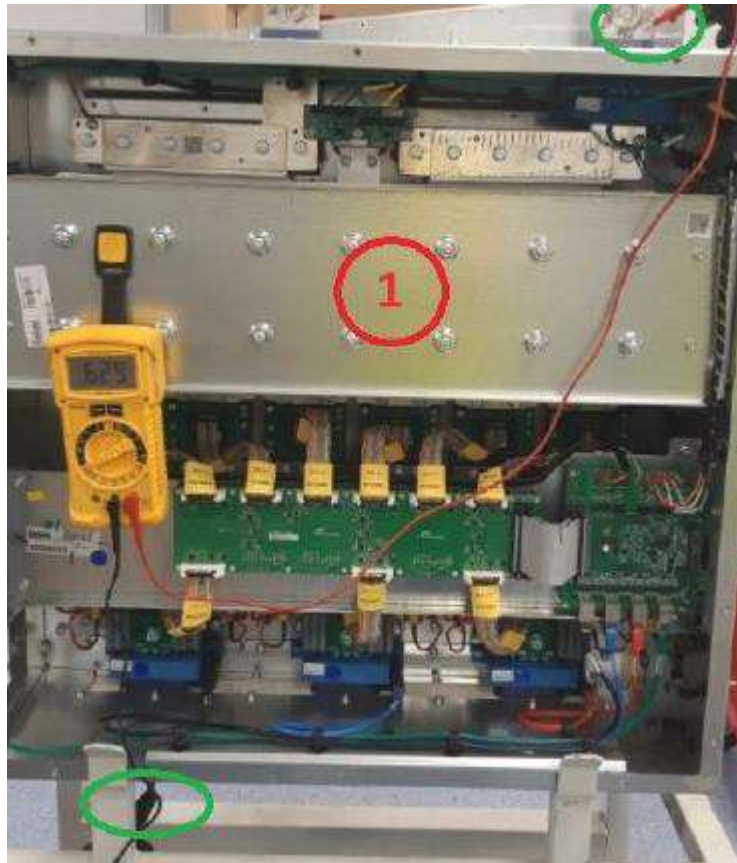
Note: Please note that the RFI boards from ground (E65 and E68 for solar inverter, E50 and E51 for PCS with one busbar and E50, E51, E52 and E53 for PCS with more than one busbar), earth and ISOPRO measurement boards (E71 for solar inverter, E71, E72, E73 and E74 for each of the busbars for PCS) must be disconnected beforehand. [INVKG3_ST5B](#) [INVMG3_ST5B](#)

- 7.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables of each busbar and repeat the measurement.
- 7.2. If the measurement exceeds 1Ω when any of the input cables to the inverter are disconnected, this indicates that the cable has a shunt between positive and negative. If this occurs, inform the customer of the situation so that the channel can be repaired. If the measurement does not improve after disconnecting all external wires, consult Power Electronics.

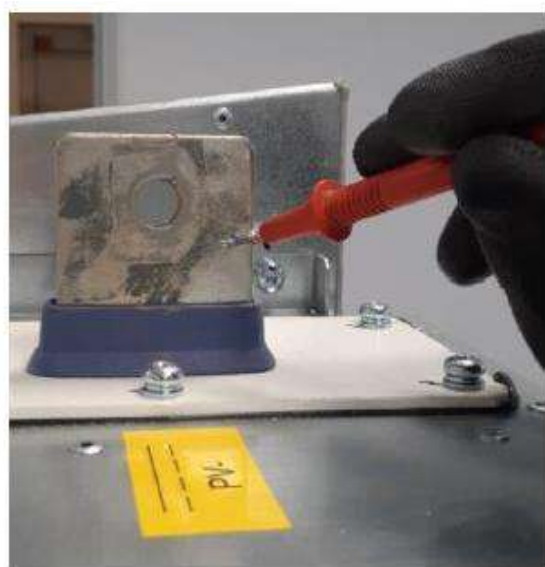
Busbar	Positive plate	Negative plate
1	BBEP.81	BBEN.66
2	BBEP.79	BBEN.64
3	BBEP.77	BBEN.62
4	BBEP.75	BBEN.60

Internal derivation of the module.

8. Check with the 1500V multimeter the module IGBT. See document INVKG3IOP88076.

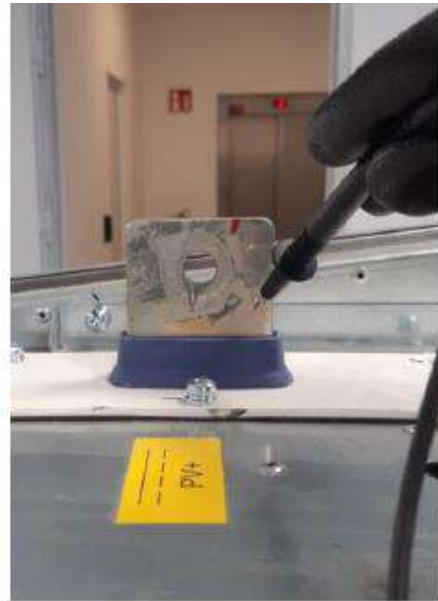


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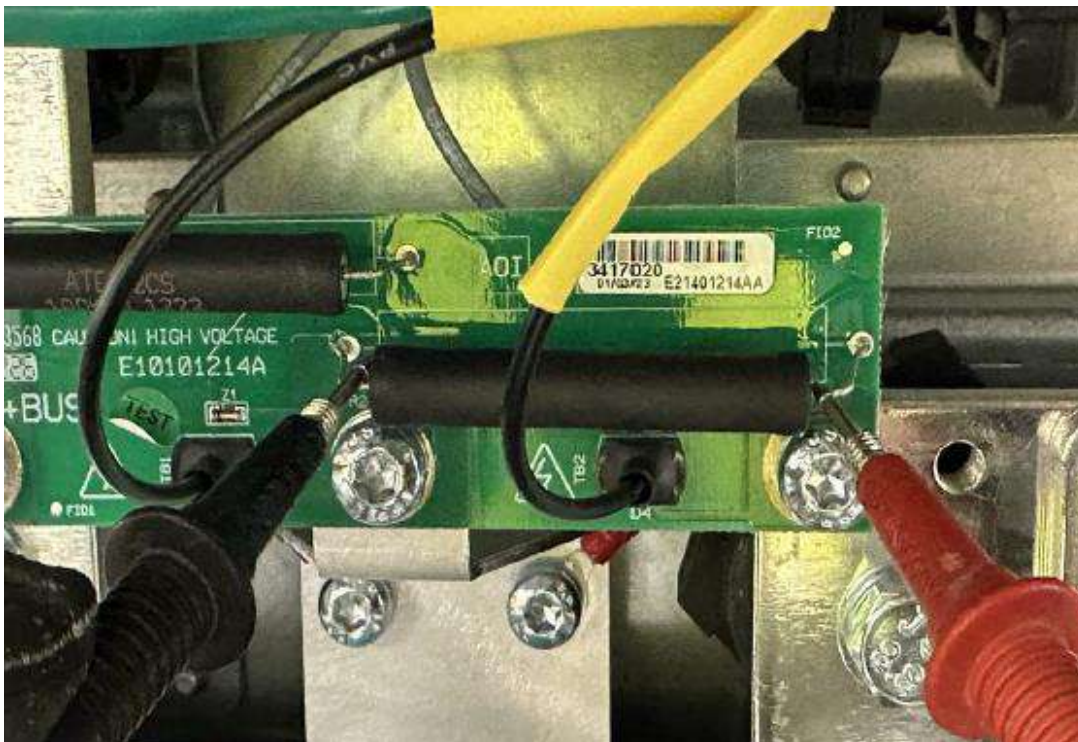




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9. Check the resistance between positive (BBIP.20) and negative (BBIN.20) of the module input. In all cases it must be greater than 100k Ω . The measurement should be taken after connecting the 1500V multimeter and waiting at least 10s and with the DC cables disconnected from the module.
INVKG3_ST2, INVMG3_ST5A.



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- 9.1. If the measurement is lower than this value, consult Power Electronics for replacement of the power stage. See document *INVG3IOP88076*.

5.75. Fault code F107: Iac not reached

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module FPGA shall trigger this fault if the output current does not exceed at least 50 % of the reference current, the latter having exceeded the defined threshold.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	Related to: F157: Feedback AC breaker F164: Excessive Maneuvers F45: General Isolation For HEMK / PCSK inverters and if there is a circuit breaker opening: F5: High Vac F6: Low Vac F7: High frequency F8: Low frequency	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Bus voltage low during operation.

1. The fault has been triggered by operating with the bus voltage too low or at values very close to or below the parameter **[Min Vdc]** (subgroup **[G4.1-MPPT]**).
 - 1.1. Increase by PowerCommsV the parameters **[Min Vdc]** by +25V and **[Vac Hyst]** from 2% to 5%. Test the inverter for a few days and observe that the fault does not recur. If so, reset the subgroup **[G4.1: Mppt]**. **INVG3_ST1A**.

Incorrect active power control mode.

2. If the inverter operates during the day in P reference control mode (parameter **[P control mode]** of subgroup **[G8.1: Reference]**) then this fault may occur under certain conditions:
 - With low power on the DC side (low irradiation on PV modules).
 - With high reference power (parameter **[Reference P]** of subgroup **[G8.1: Reference]**).
 - 2.1. If this occurs, lower the power reference to values equal to or lower than real DC power or switch the control mode to "MPPT". Set in PowerCommsV (parameter **[Control mode P]** of subgroup **[G8.1: Reference]**). **INVG3_ST1A**.

The AC circuit breaker has opened for no apparent reason (applies to HEMK / PCSK inverters).

3. May occur when the inverter is operating above a certain power. It could have associated faults such as F5, F6, F7, F8. There are several reasons that can cause it:

3.1. Misconfiguration of the AC circuit breaker protections.

- 3.1.1. Check that the AC circuit breaker settings are in accordance with the size of the inverter and its power. **INVKG3_ST3A, INVMG3_ST3A.**



Protección	Parámetros
Iu	$1 \cdot I_n = 4000A$
Ir	$0.95 \cdot I_u = 3800A$
Ir	0.5
Iud	$2 \cdot I_r = 7600A$
tud	0.2s (off)
Ii (kA)	$3 \cdot I_n = 12000A$
Ig	$0.2 \cdot I_n = 800A$
Ig	0.05 (off)

Protección	Parámetros
Iu	$0.8 \cdot I_n = 3200A$
Ir	$0.9 \cdot I_u = 2880A$
Ir	0.5
Iud	$2 \cdot I_r = 5760A$
tud	0.2s (off)
Ii (kA)	$2 \cdot I_n = 8000A$
Ig	$0.2 \cdot I_n = 800A$
Ig	0.05 (off)

Protección	Parámetros
Iu	$0.5 \cdot I_n = 2000A$
Ir	$0.95 \cdot I_u = 1900A$
Ir	0.5
Iud	$2 \cdot I_r = 3800A$
tud	0.2s (off)
Ii (kA)	$2 \cdot I_n = 8000A$
Ig	$0.2 \cdot I_n = 800A$
Ig	0.05 (off)

- 3.2. Overheating of the input or output plates of the AC circuit breaker which could indicate that internal parts may be damaged or affected. If so:

- 3.2.1. Visually inspect for correct connections looking for hot spots, discoloured screws (they are usually dull grey in colour), blast or flashes, adjacent parts or elements (especially plastics, wiring, trunking) deteriorated by heat.



If so, repair elements that are damaged **INVKG3_ST2, INVMG3_ST5A** and test the equipment with incremental power until trying to bring it to failure again. **INVG3_ST1A**. If it fails again and the AC circuit breaker opens, there may be internal damage.

- 3.3. Internal toroid of the AC circuit breaker does not measure correctly for some reason and trips above a certain power/current.

- 3.3.1. Check that the circuit breaker really trips from a certain power, starting the equipment and increasing power in steps of 5% until the fault is reached. **INVG3_ST1A**. This can also be verified by PowerCommsV from the analysis of recent logs. **INVG3_ST1A**. If this happens repair internal toroid or replace the whole AC circuit breaker assembly. **INVKG3_ST2, INVMG3_ST5A**. See document **INVG3IOP85374**.

Ingress of water and moisture into the equipment.

4. Technically inspect the inside of the device and check whether there are any traces of moisture or water ingress near or on conductive parts. This can lead to shunts from the busbar and AC busbars to the chassis or to other live parts.

In this case it is likely that the insulation resistance measured by the insulation monitor IMD is punctually or permanently low. Typical values considered correct are 1M-600K. Fault values considered low would be 150K or lower. **INVKG3_ST2, INVMG3_ST5A.**

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5.76. Fault code F128:Spring loaded

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".		
Description	The spring of the DC switch disconnecter is not charged when it should be.		
Applicable to	☒ HEM	☒ PCSM/MPCSM	
	☒ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A		

Analysis procedure

1. Check in PowerCommsV the status of the DC disconnectors to locate the failed disconnector (parameters [DC x switch status] of the [SV1.4-Equipment] subgroup). **INV3_ST1B.**
2. Check via PowerCommsV the fault history (subgroup SV7: Faults and warnings) if the emergency stop (F43) or LOTO DC (F162) has been pressed on the F128. In that case, F128 is expected to trip. **INV3_ST1B.**
3. The verifications are carried out according to the component codes of module 1. If the fault is in a different module, the codes vary as shown in the table below:

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
DC disconnector	SI01	SI02	SI03	SI04
Motor connector and feedback	XC470.1	XC475.1	XC480.1	XC485.1
Terminal block	X125	X126	X127	X128
Input and output board	E55	E56	E57	E58
RJ45 cable	W009	W010	W011	W012

4. The order to load the spring is carried out when the following conditions are met:
 - Module enabled. Check if the module is enabled by PowerCommsV (group [SG1.5: Module configuration]). **INV3_ST1B.**
 - Module is not faulted. Check the status of each module in PowerCommsV (group [SV12: Module information]). **INV3_ST1B.**
 - Spring unloaded. Check the status of the spring in PowerCommsV (parameters [DC x spring status] of subgroup [SV1.4: Equipment]). **INV3_ST1B.**
 - DC switch disconnecter open. Check the status of the spring in PowerCommsV (parameters [DC x switch status] of subgroup [SV1.4: Equipment]). **INV3_ST1B.**
 - No other spring is loaded. Check the status of the spring in PowerCommsV (parameters [DC x spring status] of subgroup [SV1.4: Equipment]). **INV3_ST1B.**
 - The emergency stop button is not activated. Check the status of the DSP RJ22 (parameter [DSP RJ22 status] of group [SV1.5 Digital I/O status]) and bit 10 of the uC digital inputs (parameter [uC Digital inputs] of group [SV1.5 Digital I/O status]) via PowerCommsV. **INV3_ST1B.**
 - The AC and/or DC LOTO selectors are not activated. Check the status of bits 0 and 1 of the uC digital inputs via PowerCommsV (parameter [uC Digital inputs digit] of group [SV1.5 Digital I/O status]). **INV3_ST1B.**
 - E-STOP is not activated. Check the status of bit 3 of the DSP RJ20 via PowerCommsV (parameter [DSP RJ20 status] in group [SV1.5 Digital I/O status]).

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5. If all the above conditions are met and fault F128 occurs, check the 120V power supply to the disconnecter and the rest of the wiring. To do this: **INVKG3_ST2, INVMG3_ST5A.**

- 5.1. Check continuity with the 1500V multimeter between X100:1:2--X125:1:1:2 and X100:1:1--X125:1:1:1.



X100:1:2



X125:1:2



X100:1:1



X125:1:1

- 5.2. Check with a 1500V multimeter in continuity mode between the following points: **INVKG3_ST2, INVMG3_ST5A.**

- Motor load (M). **INVKG3_ST2, INVMG3_ST5A.**
 - o X125:1:3 – E55:J7:1



o E55:J7:2 – XC470.1:U1



o XC470.1:U2 – X125:1:4



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- Minimum coil (UVT). **INVKG3_ST2, INVMG3_ST5A.**
 - o X125:3:2 – XC470.1:D1

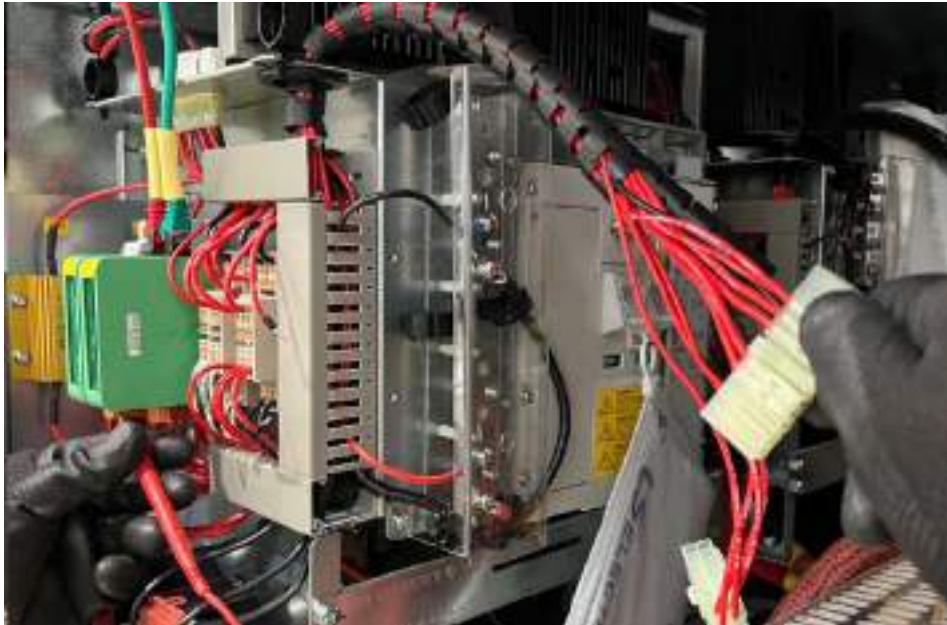


- o X125:2:4 – XC470.1:D2



- RCS (ready to close). **INVKG3_ST2, INVMG3_ST5A.**
 - o E55:J4:1 – XC470.1:251
 - o E55:J4:2 – XC470.1:252

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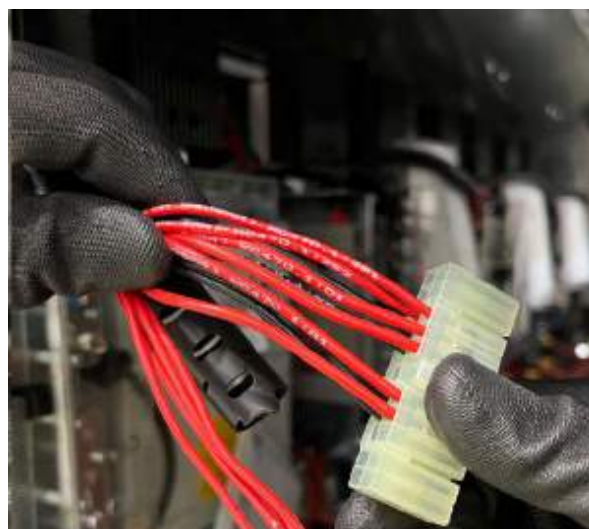
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E55:J4:1



E55:J4:2



XC470.1:252 || XC470.1:251

- 5.3. If at any point there is no continuity, replace wiring. **INVKG3_ST2, INVMG3_ST5A.**
6. Visually check the state of the DC disconnecter (SI01). It must be open (OFF position) and the spring must be unloaded. If it has been left with the spring loaded:
- 6.1. Close and open the disconnecter to unload the spring, as indicated below.
7. To diagnose the reason for the fault, it is necessary to have 110Vdc power supply on the DC disconnecters. To do this, the following steps should be carried out:
- Note:** These verifications must be carried out in the **INVKG3_ST2, INVMG3_ST5A state with external 110Vdc power supply. Always wear electrical hazard PPE.**
- 7.1. Disconnect the cables connected to pins X125:1:2, X125:2:2.
- 7.2. Connect the positive of the external 110Vdc power supply to pin X125:1:2.
- 7.3. Disconnect the wires connected to pins X125:1:1, X125:2:1.
- 7.4. Connect the negative of the external 110Vdc power supply to pin X125:1:1.
8. To carry out the opening and closing of the spring, perform the following steps in PowerCommsV: **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.**
- 8.1. Set the configuration of the DC virtual selector in communications (parameter [Conf DC Sel Virtual] of subgroup [G2.6: Selectors]) and activate or deactivate the DC virtual selector (parameter [DC Virtual Selector] of subgroup [G2.6: Selectors]). **INVG3_ST1B.** This will start the manoeuvre.
- 8.2. To check the opening or closing command of the spring, verify in the module digital outputs that bit 3 (DC isolating switch closing command) or bit 4 (DC switch disconnecter opening command) are activated (X) (parameters [Mod x Digital Out] of subgroup [SV1.5 Digital I/O Status]).
9. Measure voltage with the 1500V multimeter in Vdc mode of approximately 110 Vdc between the pins: **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.**
- 9.1. Measure voltage with the 1500V multimeter in Vdc mode of approximately 110 Vdc between the pins:
- X125:3:2 - X125:2:4 (120Vdc distribution terminal block).



- 9.2. In case of having the correct voltage (110 Vdc), proceed to the next step spring charging. The minimum coil must be powered in order to be able to carry out any operation on the SI01 disconnecter.
10. Check that the conditions for spring loading are met as indicated above. Verify audibly that the spring charging is performed and completed by changing the status on the disconnecter. It is possible to check the spring status by PowerCommsV (parameters [DC x spring status] of subgroup [SV1.4: Equipment]). **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.**
- 10.1. In case of successful dock loading, proceed to the step of checking the status of the RCS (ready to close) confirmation dock loaded.
- 10.2. Check that the relay RL3 of the inputs and outputs board E55 is activated when the spring is loaded. **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.** To do this:
- 10.2.1. Place the 1500V multimeter to measure voltage on terminals E55:J7:2 - X125:1:4. **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.**



E55:J7:2



X125:1:4

- 10.2.2. Check that the conditions are fulfilled for the spring charge activation to be performed again.
- 10.2.3. Verify that 110Vdc voltage arrives with the activation. **INVKG3_ST2, INVMG3_ST5A with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.**
- 10.2.4. If this is verified, we are at the previous point and the fault is internal to the disconnecter.
- 10.2.5. Otherwise, the problem is between the inputs and outputs board E55 and the module control board (E02).
- Disconnect the RJ45 cable (W009) connecting the input/output board E55:J2 to the module control board E02:J13. **INVKG3_ST2, INVMG3_ST5A.**



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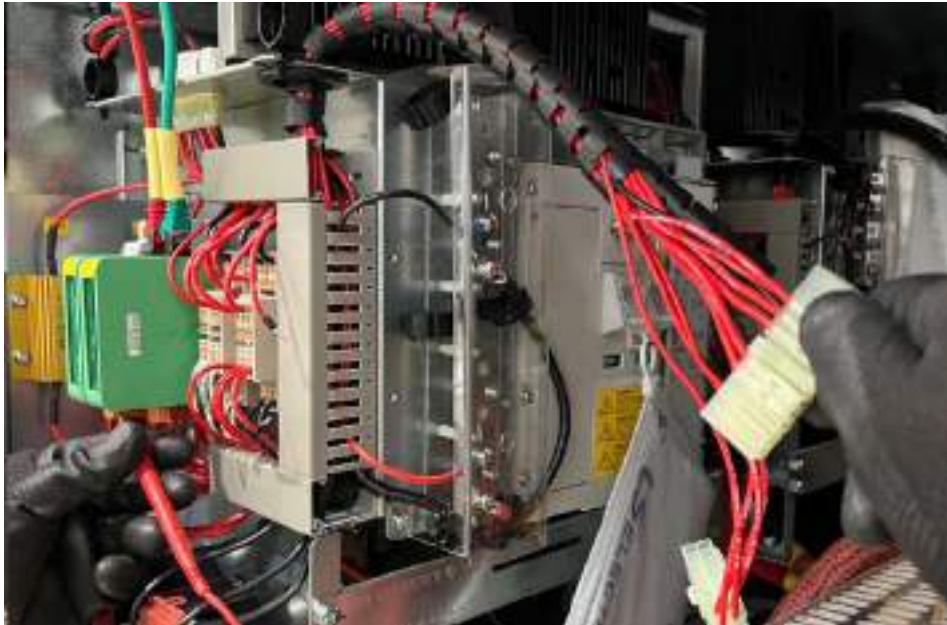
- Check the continuity of the RJ45 cable with the RJ45 tester. Replace the faulty cable. *INVKG3_ST2, INVMG3_ST5A.*
 - If the RJ45 cable does not fail, replace the input/output board E55. Refer to document *INVG3/OP88849.* *INVKG3_ST2, INVMG3_ST5A.*
 - If the fault persists, replace the module control board E02. See document *INVG3/OP88076.* *INVKG3_ST2, INVMG3_ST5A.*
11. With the SI01 disconnecter and the spring loaded, check that the RCS (ready to close) feedback is received correctly. *INVKG3_ST2, INVMG3_ST5A.* To do this:
- 11.1. There must be continuity, check with the 1500V multimeter, between E55:J4:1 - E55:J4:2. This is the RCS feedback whose internal contact is normally open. *INVKG3_ST2, INVMG3_ST5A.*



E55:J4:1

E55:J4:2

- 11.1.1. If there is no continuity between E55:J4:1 - E55:J4:2, identify whether it is the wiring or the disconnecter that is faulty. *INVKG3_ST2, INVMG3_ST5A.*
- Measure with a 1500V multimeter in continuity mode between E55:J4:1 - XC470.1:251 and E55:J4:2 - XC470.1:252. *INVKG3_ST2, INVMG3_ST5A.*



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E55:J4:1



E55:J4:2



XC470.1:252 || XC470.1:251

- If there is continuity, the wiring is correct and the SI01 disconnecter must be replaced as it is due to an internal fault. See document *INVG3/OP85377*. [INVKG3_ST2](#), [INVMG3_ST5A](#).
- If there is no continuity, the wiring from the disconnecter to the inputs and outputs board E55 should be replaced. See document *INVG3/OP88849*. [INVKG3_ST2](#), [INVMG3_ST5A](#).

11.1.2. If the above check is satisfactory, the disconnecter and its wiring are correct.

11.2. If there is continuity between E55:J4:1 - E55:J4:2 and this status is not displayed in PowerCommsV, check bit 3 of the digital inputs of each module (subgroup [FPGA module interf] of group [SV1.5 Digital I/O status]). If bit 3 is set to 0, the problem is between the input/output board (E55) and the module control board (E02). If so, the above points must be repeated. [INVKG3_ST2](#), [INVMG3_ST5A](#) with external 110Vdc power supply with use of electrical hazard PPE at all times. Power the electronics at 24V to perform the marked tasks.

11.2.1. Disconnect the RJ45 cable (W009) connecting the I/O board E55:J2 to the module control board E02:J13. Check the continuity of the RJ45 cable with the RJ45 tester. [INVKG3_ST2](#), [INVMG3_ST5A](#). Replace failed cable. [INVKG3_ST2](#). [INVMG3_ST2](#).



11.2.2. If the RJ45 cable does not fail, replace the E55 input/output board. See document *INVG3/OP88849*. [INVKG3_ST2](#), [INVMG3_ST5A](#).

11.2.3. If the fault persists, replace the module control board E02. See document *INVG3/OP88076*. [INVKG3_ST2](#), [INVMG3_ST5A](#).

5.77. Fault code F161: LOTO AC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The AC LOTOTO is activated, and the inverter must be disconnected from AC and DC.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F162: LOTO DC.	

Analysis procedure

The AC selector (S04) which is part of the inverter's LOTOTO procedure has been activated or interlocked.

1. Visually check the position of the AC selector (S04) located on the control door. **INV G3_ST1B**. This selector has three positions: "ON", "OFF", Interlock (lockout with padlock). Its internal contacts are normally closed (NC) in "ON" position. This fault is activated in case of having the AC selector (S04) in the "OFF" position or in the interlocked position.



ON position



OFF position



Interlock position

2. Check in PowerCommsV the status of bit 1 corresponding to LOTO AC (parameter **[uC Digital inputs]** of group [SV1.5 Digital I/O Status]). **INV G3_ST1B**. If it is in X with the AC selector (S04) in "ON" position and the fault is still active, check the following points:
 - Wiring.
 - Input and output boards (E41 / E42).
 - Central control board (E35).
3. For wiring check, check continuity using a 1500V multimeter in resistivity mode between pins E41:J4:3 - E42:J4:6 of the input and output boards (E41 / E42) of the door panel (PPU). **INV K G3_ST2**, **INV M G3_ST5A**.

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4. If there is no continuity between pins E41:J4:3 - E42:J4:6, the wiring is faulty and continuity must be measured on each wire separately: **INVKG3_ST2**, **INVMG3_ST5A**.
- Between E41:J4:3 – S02:11.



- Between S02:11 – S02:21.



- Between S02:21 – S04:1.



- Between S04:2 - E42:J4:6.



- If any of these wiring has no continuity, replace it. **INVKG3_ST2, INVMG3_ST5A.**
5. In case of continuity, check that the AC selector (S04) is working correctly. To do so, measure continuity in the following points. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.1. Between S04:1 - S04:2. If the AC selector (S04) is in the "ON" position, it should give continuity.

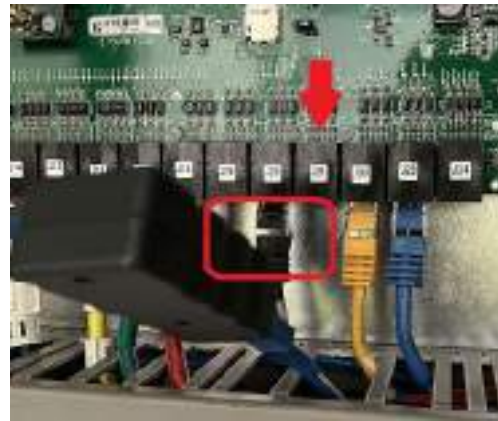


If not, there is an internal fault, and it must be replaced. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP89283*.

- 5.2. If there is continuity between pins E41:J4:3 - E42:J4:6, the wiring and the AC selector (S04) are working correctly. In this case, the problem is between the input/output boards (E41 / E42) and the central control board (E35). If this is the case:

- 5.2.1. Check the continuity of the two RJ45 cables of the input and output cards E41 (W124) and E42 (W125) with the Ethernet cable tester. **INVKG3_ST2, INVMG3_ST5A**. To do this:

- Release the hose W124 from the input/output board E41:J2:1 and from the central control board E35:J21:1.



- Also detach the hose W125 from the input/output board E42:J2:1 and from the central control board E35:J22:1.



- Replace failed cable. **INVKG3_ST2, INVMG3_ST5A**.
- If the RJ45 cables do not fail, replace the input and output boards (E41 / E42). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP88849*.
- If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP88849*.

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5.78. Fault code F162: LOTO DC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The DC LOTOTO is activated, and the inverter must be disconnected from AC and DC.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F161: LOTO AC	

Analysis procedure

The DC selector (S03) which is part of the inverter's LOTOTO procedure has been activated or interlocked.

1. Visually check the position of the DC selector (S03) located on the control door. **INVG3_ST1B**. This selector has three positions: "ON", "OFF", Interlock (lockout with padlock). Its internal contacts are normally closed (NC) in "ON" position. This fault is activated in case of having the DC selector (S03) in the "OFF" position or in the interlocked position.



ON position



OFF position



Interlock position

2. Check in PowerCommsV the status of bit 0 corresponding to LOTO DC (parameter **[uC Digital inputs]** of group [SV1.5 Digital I/O Status]). **INVG3_ST1B**. If it is in X with the DC selector (S03) in "ON" position and the fault is still active, check the following points:
 - Wiring.
 - Input and output boards (E41).
 - Central control board (E35).
3. For wiring check, check continuity using a 1500V multimeter in resistivity mode between pins E41:J4:3 – E41:J4:2 of the input and output boards (E41) of the door panel (PPU). **INVKG3_ST2, INVMG3_ST5A**.

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3.1. If there is no continuity between pins E41:J4:3 – E41:J4:2, the wiring is faulty and continuity must be measured on each wire separately: **INVKG3_ST2**, **INVMG3_ST5A**.

- Between E41:J4:3 – S02:11.



- Between S02:11 – S02:21.

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- Between S02:21 – S04:1.



- Between S04:1 – S03:1.



- Between S03:2 - E41:J4:2.



- If any of those wiring has no continuity, replace it. **INVKG3_ST2, INVMG3_ST5A.**

3.1.1. In case of continuity, check that the DC selector (S03) is working correctly. To do so, measure continuity in the following points. **INVKG3_ST2, INVMG3_ST5A.**

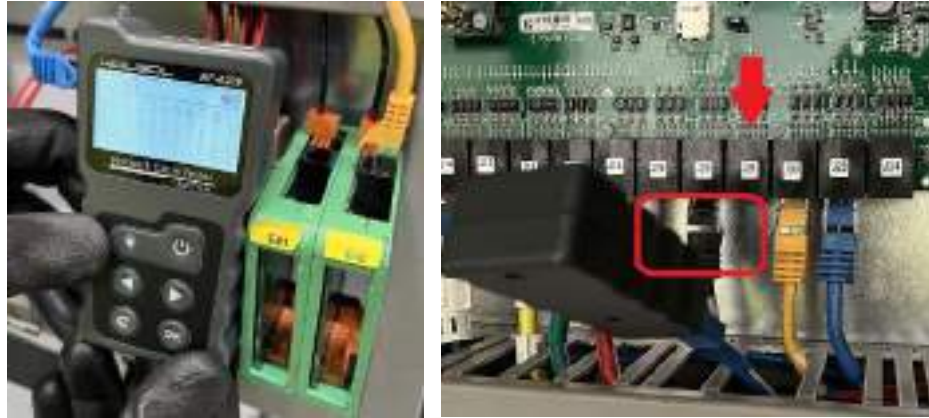
- Between S03:1 – S03:2. If the DC selector (S03) is in the "ON" position, it should give continuity. If not, there is an internal fault, and it must be replaced. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3IOP89283*.



3.2. If there is continuity between pins E41:J4:3 – E41:J4:2, the wiring and the DC selector (S03) are working correctly. In this case, the problem is between the input/output board (E41) and the central control board (E35). If this is the case:

3.2.1. Check the continuity of the two RJ45 cables of the input and output board E41 (W124) with the Ethernet cable tester. **INVKG3_ST2, INVMG3_ST5A.** To do this:

- Release the hose W124 from the input/output board E41:J2:1 and from the central control board E35:J21:1.



- Replace failed cable. **INVKG3_ST2, INVMG3_ST5A.**
- If the RJ45 cables do not fail, replace the input and output boards (E41). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88849*.
- If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88849*.

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5.79. Fault code F163: WD MV Cell

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	The DSP has detected a WD failure of the MV switchgear.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

- In HEMK / PCSK inverter without V Skid, check that the interconnection hirting is correctly covered with the protective cap. The hirting to check in HEMK / PCSK inverter is XC1026.1, and in MV Skid inverter XC1026.2-1 and XC1026.2-2. The backflushes coming from the switchgear are normally open, so it must be ensured that there is no connection of the inlets due to moisture or dirt.
- Identify the manufacturer of the switchgear (ORMAZABAL / SIEMENS) as the watchdog (WD) backfeed operates inversely in each of these switchgears.
- In HEMK and PCSK inverters with MV Skid, confirm by Power Comms V that the configured switchgear coincides with the real switchgear of the inverter (group [SG10 Drive Select]). **INVG3_ST1B**.
- The MV watchdog status can be displayed as follows:
 - X = Watchdog activated = Watchdog closed = It is an electrical "1" = 24 Vdc
 - 0 = Watchdog not activated = Watchdog open = It is an electrical "0" = 0 Vdc
- Check the status of bit 6 of DSP RJ14 (parameter [DSP RJ14 Status] of group [SV1.5 Digital I/O Status]) via PowerCommsV. **INVG3_ST1B**.
- watchdog logic in Siemens is the reverse of ORMAZÁBAL:
 - ORMAZABAL logic:
 - ORMAZABAL to DI Watchdog relay= X → Fault.
 - ORMAZABAL to DI Watchdog relay = 0 → Ok
 - SIEMENS logic:
 - SIEMENS to DI Watchdog relay = X → Ok.
 - SIEMENS to DI Watchdog relay = 0 → Fault
- Check that the switchgear relay is powered. To do this:
 - Open the switchgear door and look at the relay status. If the display is not lit, there is a fault in the 120 V power supply. **Note: special case in INVG3_ST1B status, and with use of electrical hazard PPE. Only open the switchgear door and check if the display is on. Then close the switchgear door.**
- In HEM / PCSM inverter, use a 1500V multimeter in Vdc mode to measure if there is 120 V between pins X31:3:2 and X31:4:3. If 120 V is present but no power is supplied, open the cover of the cubicle, and check with a multimeter in continuity mode between X31:3:2 -Q02:XG:1 and X31:4:3-Q02:XG:2. If there is no continuity, replace the wiring.

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X31:3:2 – chasis



Q02:YG:1

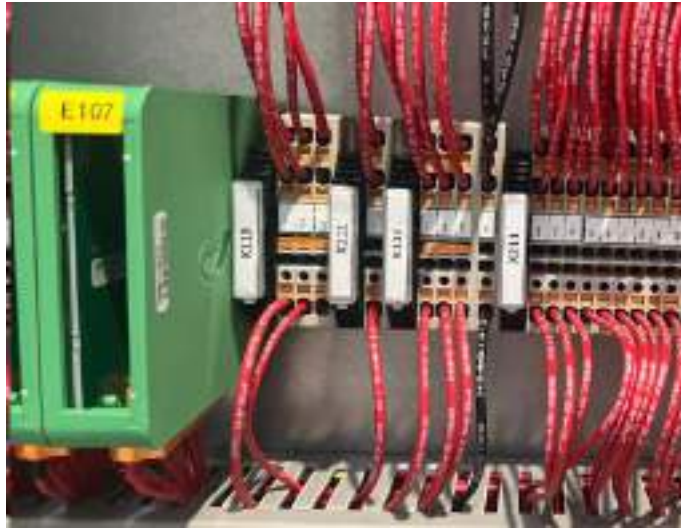


X31:4:3



Q02:YG:2

9. In HEMK / PCSK with MV Skid switchgear ORMAZABAL, use a 1500V multimeter in Vdc mode to measure if there is 120 V between pins X112:1:1 and X112:4:1. **SKIDG3_ST3.**



- 9.1. If 120V is present but no power is supplied to the relay, open the switchgear cover and check with a 1500V multimeter in continuity mode between X112:1:1-Q02:XG:1 and X112:4:1 Q02:XG:2. If there is no continuity, replace the wiring. **SKIDG3_ST2.**



Q02:XG:1



Q02:XG:2

10. In HEMK / PCSK with MV Skid switchgear SIEMENS, use a 1500V multimeter in Vdc mode to measure if there is 24V between pins X011:1:1 and X011:4:1. **SKIDG3_ST3.**



- 10.1. If there is 24 V but no power supply to the relay, open the cover of the switchgear and check with a 1500V multimeter in continuity mode between X011:1:1-HZ02:F71:12 and X011:11:1-HZ02:F71:14. **SKIDG3_ST2**.

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HZ02:F71:12



X011:1:1



X011:1:1



HZ02:F71:14

11. Checking without voltage on HEM (ORMAZABAL switchgears only).

- 11.1. To check the wiring with the relay OFF, check continuity using a 1500V multimeter in resistivity mode between pins E82:J2:1 - E82:J2:2 of the E82 input board of the medium voltage panel (PMC). **INVMG3_ST4.**



EN

- 11.2. If there is no continuity between pins E82:J2:1 - E82:J2:2, the wiring is faulty, or the relay is not functioning correctly. Continue with the voltage checks in the next step.

12. Checking with auxiliary voltage on HEM (ORMAZABAL switchgears only).

- 12.1. Connect to the switchgear relay via the micro USB port and extract its fault file in txt format. This will check if the watchdog fault has been activated. **INVMG3_ST3A.**

- 12.2. To check if the wiring is correct, open the switchgear cover. Before having to open the switchgear cover, proceed to supply 120Vdc voltage to the relay and check if the fault is coming from the relay. **INVMG3_ST4 with external 120Vdc power supply.**

- 12.2.1. Supply 120Vdc power to the switchgear relay to check if the fault comes from the relay. Release the terminal block wires X31:3:1 and X31:4:1 and connect the 120Vdc positive and negative respectively of the external power supply. **INVMG3_ST4 with external 120Vdc power supply.**

- 12.2.2. Supply 24Vdc to the wires of power supply G02 of the central control panel in order to have central control and thus receive the statuses of the input board E82 of the medium voltage panel. **INVMG3_ST4 with external 120Vdc power supply.**

- 12.2.3. This external power supply can be done in two ways:

- Disconnect the wires connected to terminals (+) and (-) of G02 and connect these to the positive and negative, respectively, of the external 24 Vdc power supply.
- Disconnect the wires connected to terminals L(+) and N(-) of G02 and connect the positive and negative of the external 120 Vdc power supply to terminals L(+) and N(-).
- In the event of having 0Vdc between terminals E82:J2:1-E82:J2:2 and visualise via PowerCommsV that its status is not correct (bit 6 parameter [DSP Status RJ14] of group [SV1.5 Digital I/O Status]). **Shut down the device and go to INVMG3_ST3A.** If no switchgear fault appears, the problem is between the input board (E82) and the central control board (E35).

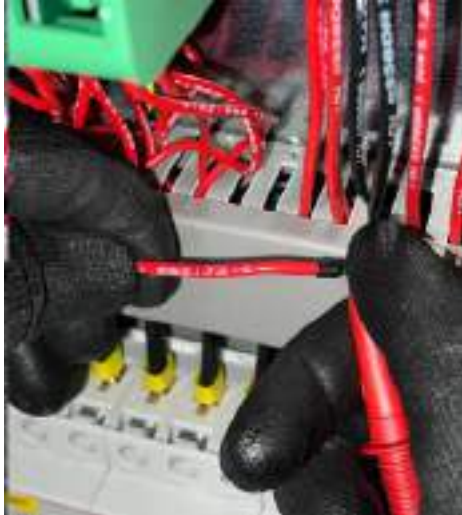
- 12.3. Disconnect the RJ45 cable (W042) connecting the E82:J1 board to the E35:J14 central control board. Check the continuity of the RJ45 cable with the network cable continuity tester. **INVMG3_ST3A.**



EN

- 12.3.1. Replace failed cable. **INVMG3_ST2.**
- 12.3.2. If the RJ45 cable does not fail, replace the digital input board E82. **INVMG3_ST2.** See document *INVG3IOP88076*.
- 12.3.3. If the fault persists, replace the central control board E35. **INVMG3_ST5A.** See document *INVG3IOP88076*.
- 12.4. With the relay powered ON, after 600ms from its power on, the watchdog contact must be zero and have no 24Vdc between pins E82:J2:1 - E82:J2:2. Check with a 1500V multimeter in Vdc mode between these terminals. **INVMG3_ST3A.**
- 12.5. If 24 Vdc is not present, go to the next point as the relay and its wiring are correct.
- 12.6. In case of 24 Vdc, if the relay has not faulted and there is 24Vdc between pins E82:J2:1 - E82:J2:2, then the wiring is faulty. The protection switchgear cover should be opened.
 - 12.6.1. Measure continuity at the following points by loosening the intermediate connectors and removing the front cover of the MV switchgear. **INVMG3_ST2.**

– E82:J2:1 – XC752.1:8	– Q02:J18 - Q02:52:27
– E82:J2:2 – XC752.1:7	– Q02:52:27 - Q02:52:25
– XC752.2:8 – Q02:J8	– Q02:52:25 - Q02:52:67
– Q02:J8 - Q02:J10	– Q02:52:67 - Q02:XG:3
– Q02:J10 - Q02:J16	– XC752.2:7 – Q02:XG:5
– Q02:J16 - Q02:J18	



E82:J2:1



XC752.1:8



E82:J2:2



XC752.1:7

EN



XC752.2:8



Q02:J8



Q02:J8 - Q02:J10



Q02:J10 - Q02:J16

EN



Q02:J16 - Q02:J18



Q02:J18 - Q02:52:27

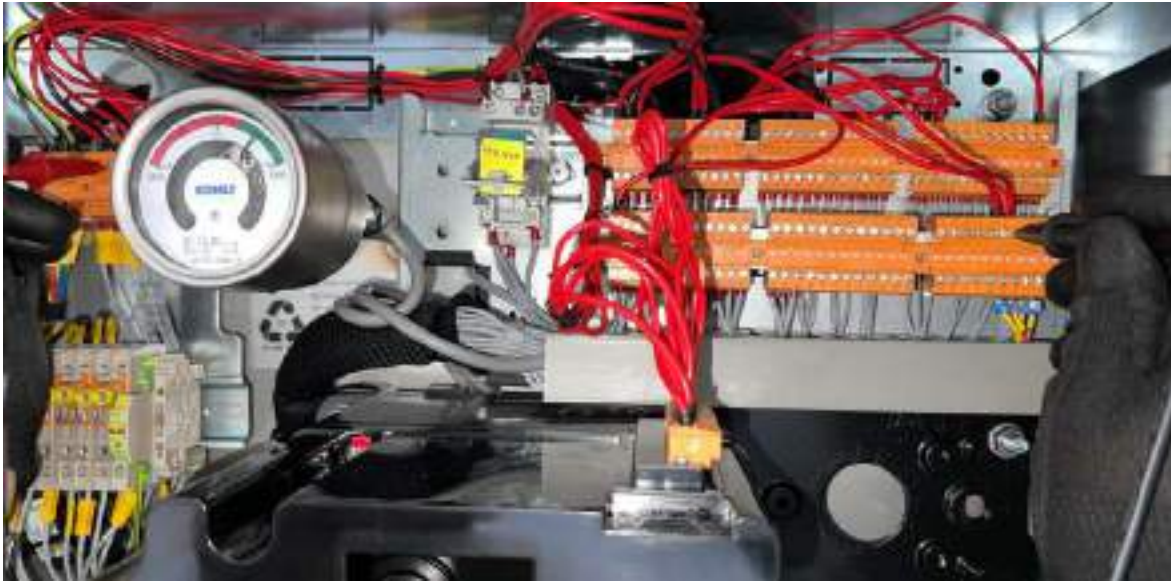


Q02:52:27 - Q02:52:25



Q02:52:25 - Q02:52:67

EN



Q02:52:67 - Q02:XG:3

EN



XC752.2:7



Q02:XG:5

12.6.2. If any of these wiring harnesses have NO continuity, it must be replaced. **INVMG3_ST2**.

12.6.3. If the wiring is correct at all points, it is due to an internal relay error and should be replaced. **INVMG3_ST2**.

13. Check without voltage on SKID/TWIN (ORMAZÁBAL/SIEMENS switchgears).

13.1. The difference between a MV Skid and Twin Skid with respect to the watchdog is that this signal is duplicated from the switchgear relay, in order to take it to the two inverters by means of the different input boards E101 and E105.

13.2. To check the wiring with the relay OFF, continuity must be checked using a 1500V multimeter in resistivity mode between the pins: **SKIDG3_ST2**

- E101:J2:1 - E101:J2:2 of the input board E101 of the MV Skid / Twin (LVE) panel.



- E105:J2:1 - E105:J2:2 of the E105 input board of the Twin Skid (LVE) panel.



- If there is no continuity between the above pins, the wiring is faulty or the relay is not working properly.

14. Checking with auxiliary voltage in MV Skid / Twin (ORMAZÁBAL / SIEMENS switchgears).

14.1. Connect to the switchgear relay via the micro USB port and extract its fault file in txt format. This will check if the watchdog fault has been activated. **SKIDG3_ST3**.

14.2. To check if the wiring is correct, open the switchgear cover. Before opening the switchgear cover, supply 120Vdc voltage to the relay and check if the fault is coming from the relay. **SKIDG3_ST2 with external 120Vdc power supply**.

14.2.1. Supply 120Vdc to the switchgear relay to check if the fault comes from the relay. Disconnect the terminal wires X112:1:1 & X112:4:1 and connect the positive and negative 120Vdc respectively (ORMAZABAL only). **SKIDG3_ST2 with external 120Vdc power supply**.

14.2.2. The switchgear relay must be supplied with 24Vdc. Disconnect the wires from the terminals X011:1:1 & X011:4:1 and connect the positive and negative of the external

24Vdc power supply respectively (SIEMENS only). SKIDG3_ST2 with external 120Vdc power supply.

- 14.2.3. The wires of the power supply G02 of the central control panel of inverter 1 (MV Skid / Twin) and inverter 2 (Twin) must be supplied with 24Vdc in order to have central control and thus receive the states of the input card E101 and E105 of the MV Skid panel. **SKIDG3_ST2** with external 120Vdc power supply.

- 14.2.4. This external power supply can be done in two ways:

- Disconnect the wires connected to the (+) and (-) terminals of G02 and connect these to the positive and negative, respectively, of the external 24 Vdc power supply. **SKIDG3_ST2** with 120Vdc external power supply.
- Disconnect the wires connected to terminals L(+) and N(-) of G02 and connect the positive and negative of the 120 Vdc external power supply to terminals L(+) and N(-). **SKIDG3_ST2** with 120Vdc external power supply.

- 14.2.5. In case of having 0 Vdc (ORMAZABAL) or 24 Vdc (SIEMENS) between terminals E101:J2:1 - E101:J2:2 of the input card E101 of the MV Skid/Twin (LVE) panel, display via PowerCommsV its status is not correct (bit 6 parameter [DSP Status RJ14] of group [SV1.5 Digital I/O Status]). If no switchgear fault appears, the problem is between the input board E101 and the central control board E35.

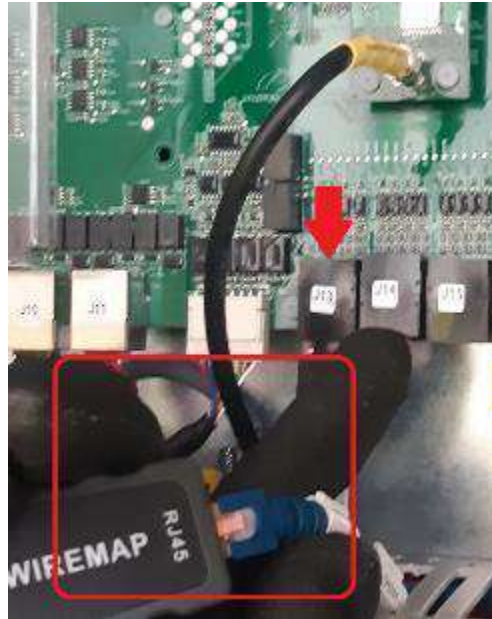
- 14.2.6. Disconnect the RJ45 cable (W367) connecting the input card E101:J1 to the central control card E35:J14 from the inverter 1. Check the continuity of the RJ45 cable with the network cable continuity tester. **INVKG3_ST3A**, **SKIDG3_ST3A**. Replace the faulty cable. **INVKG3_ST2**. **INVMG3_ST2**.



E101:J1



XC.1026.2



E35:J14

EN

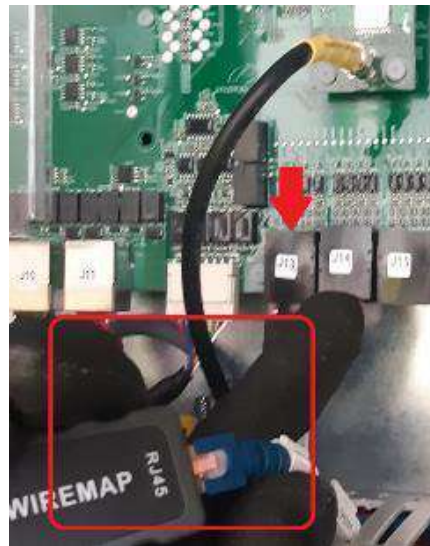
- If the RJ45 cable does not fail, replace the E101 input board. SKIDG3_ST2. See document INVG3IOP89958.
 - the fault persists, replace the central control board E35 of inverter 1. INVKG3_ST5A. See document INVG3IOP88849.
 - Repeat the same process for inverter 2.
- 14.2.7. In case of having 0 Vdc (ORMAZABAL) or 24 Vdc (SIEMENS) between terminals E105:J2:1 - E105:J2:2 of the input board E105 of the MV Skid/Twin (LVE) panel and displaying via PowerCommsV its status is not correct (bit 6 parameter [DSP Status RJ14] of group [SV1.5 Digital I/O Status]). If no cell fault appears, the problem is between the input board E105 and the central control board E35.
- 14.2.8. Disconnect the RJ45 cable (W371) connecting the input board E105:J1 to the central control board E35:J14 from the inverter 2. Check the continuity of the RJ45 cable with the network cable continuity tester. **INVKG3_ST3A, SKIDG3_ST3A**. Replace the failed cable. **INVKG3_ST2, SKIDG3_ST2**.



E105:J1



XC.1026.2



E35:J14

EN

- If the RJ45 cable does not fail, replace the E105 input board. See document INVG3IOP89958. **SKIDG3_ST2**.

14.3. If the fault persists, replace the central control card E35 of inverter 2. **INVKG3_ST5A**. Con el relé con alimentación (ON), tras 600ms de su *power on*, el contacto del *wachdog* debe estar a cero y no tener 24Vdc. Medir mediante un multímetro en modo Vdc entre los pines: **SKIDG3_ST2 con alimentación externa de 24Vdc**

- E101:J2:1 - E101:J2:2 of the E101 input board on the MV Skid/Twin (LVE) panel.



- E105:J2:1 - E105:J2:2 of the E105 input board of the Twin MV Skid (LVE) panel.



- If 24 Vdc, go to the next point as the relay and its wiring are correct.

14.4. En caso contrario, si el relé no ha entrado en fallo y tenemos 24Vdc entre los pines indicados anteriormente, el cableado es erróneo. Se debería abrir la tapa de la celda de protección.

14.4.1. Checking wiring with ORMAZABAL switchgear. **SKIDG3_ST2.**

- | | |
|--|--|
| – E101:J2:1 – X010:1:4 (MV Skid/Twin) | – XC900.2:7 – Q02:J:7 (MV Skid/Twin) |
| – E101:J2:2 – X211:5:2 (MV Skid/Twin) | – Q02:J:7 – Q02:J:9 (MV Skid/Twin) |
| – E105:J2:1 – X010:2:4 (solo Twin) | – Q02:J:9 – Q02:J:15 (MV Skid/Twin) |
| – E105:J2:2 – X211:5:3 (solo Twin) | – Q02:J:15 – Q02:J:17 (MV Skid/Twin) |
| – X211:5:1 – XC102.2:8 (MV Skid/Twin) | – Q02:J:17 - Q02:52:26 (MV Skid/Twin) |
| – XC102.1:8 – XC900.1:8 (MV Skid/Twin) | – Q02:52:26 - Q02:52:22 (MV Skid/Twin) |
| – XC900.2:8 – Q02:XG:5 (MV Skid/Twin) | – Q02:52:22 - Q02:52:68 (MV Skid/Twin) |
| – X010:3:4 – XC102.2:7 (MV Skid/Twin) | – Q02:52:68 - Q02:XG:3 (MV Skid/Twin) |
| – XC102.1:7 – XC900.1:7 (MV Skid/Twin) | |



E101:J2:1



X010:1:4



E101:J2:2 – X211:5:2

EN



X211:5:1 – XC102.2:8



XC102.1:8



XC900.1:8

EN



XC900.2:8



Q02:XG:5

EN



X010:3:4



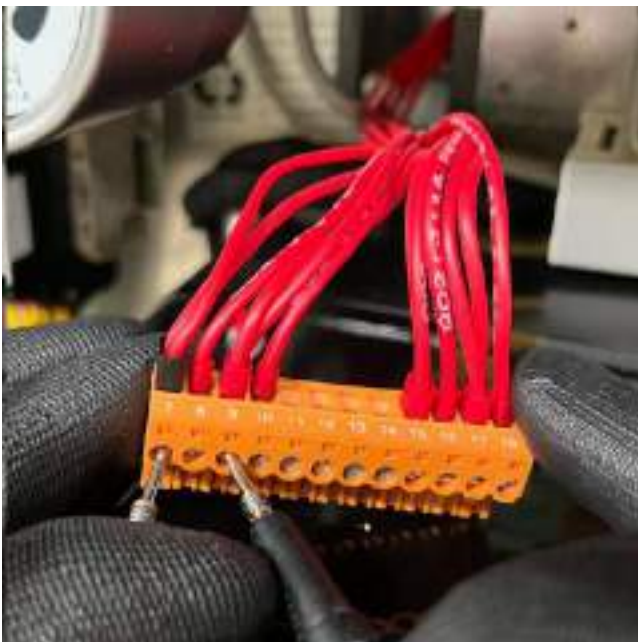
XC102.2:7



XC900.2:7



Q02:J:7



Q02:J:7 – Q02:J:9



Q02:J:9 – Q02:J:15

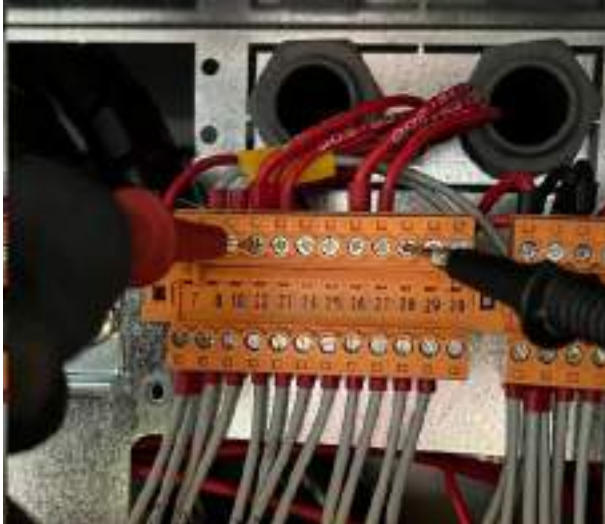


Q02:J:15 – Q02:J:17

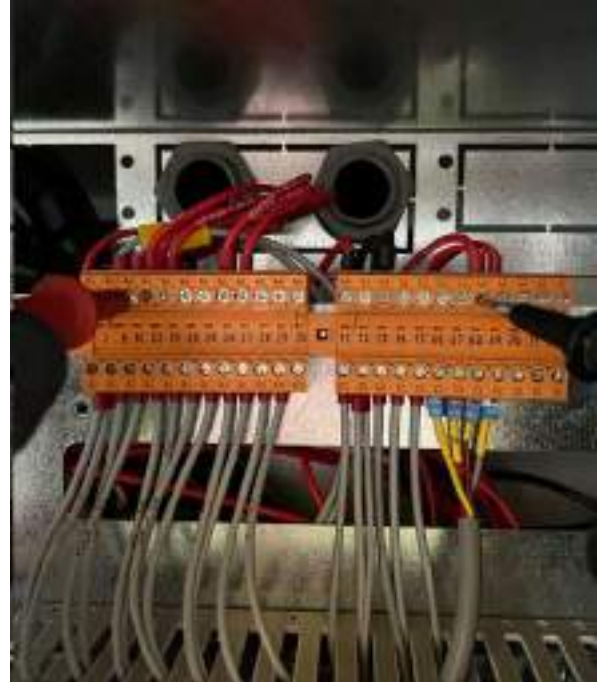


Q02:J:17 - Q02:52:26

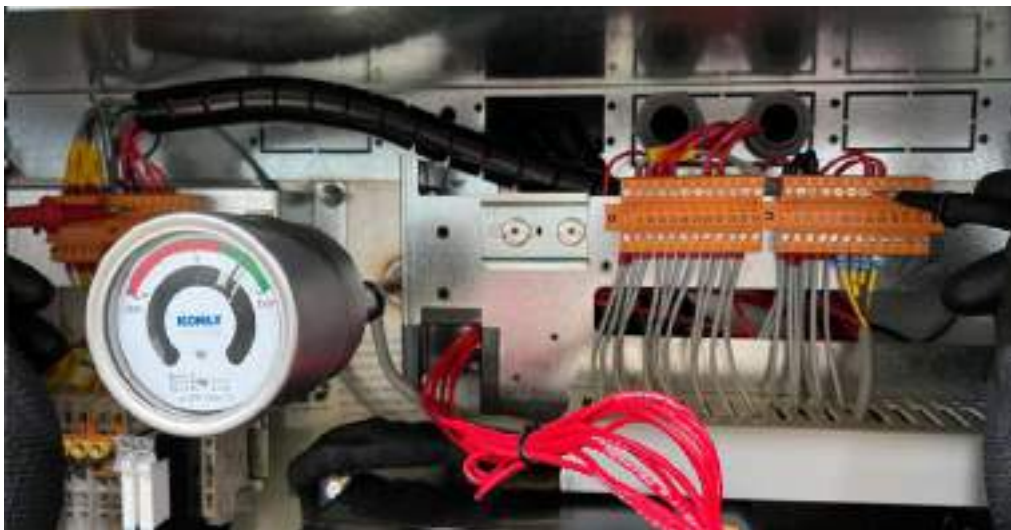
EN



Q02:52:26 - Q02:52:22



Q02:52:22 - Q02:52:68



Q02:52:68 - Q02:XG:3

1.1.1. Checking wiring with SIEMENS switchgear: **SKIDG3_ST2**.

- | | |
|---|--|
| - E101:J2:1 – X010:1:4 (MV Skid/Twin) | - XC102.1:7 – XC900.1:7 (MV Skid/Twin) |
| - E101:J2:2 – X211:5:2 (MV Skid/Twin) | - XC900.2:7 – XQ1:111 (MV Skid/Twin) |
| - E105:J2:1 – X010:2:4 (solo Twin) | - XQ1:111 – XQ1:121 (MV Skid/Twin) |
| - E105:J2:2 – X211:5:3 (solo Twin) | - XQ1:121 – XQ1:521 (MV Skid/Twin) |
| - X211:5:1 – XC102.2:8 (MV Skid/Twin) | - XQ1:521 – XQ1:511 (MV Skid/Twin) |
| - XC102.1:8 – XC900.1:8 (MV Skid/Twin) | - XQ1:511 – XQ0:31 (MV Skid/Twin) |
| - XC900.2:8 – XQ1:F71:21 (MV Skid/Twin) | - XQ0:31 - XQ0:33 (MV Skid/Twin) |
| - X010:3:4 – XC102.2:7 (MV Skid/Twin) | - XQ0:33 – HZ02:F71:18 (MV Skid/Twi) |



E101:J2:1



X010:1:4



E101:J2:2 – X211:5:2

EN



X211:5:1 – XC102.2:8



XC102.1:8



XC900.1:8

EN



XC900.2:8



HZ02:F71:21

EN



X010:3:4



XC102.2:7



XC102.1:7



XC900.1:7

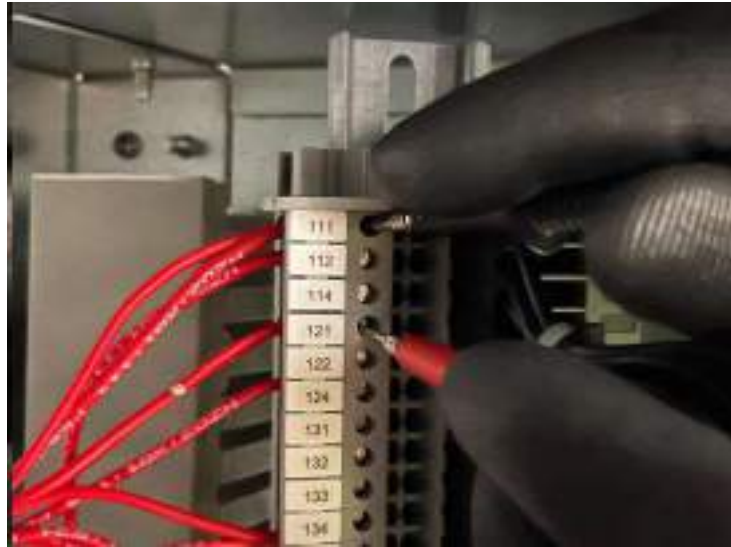
EN



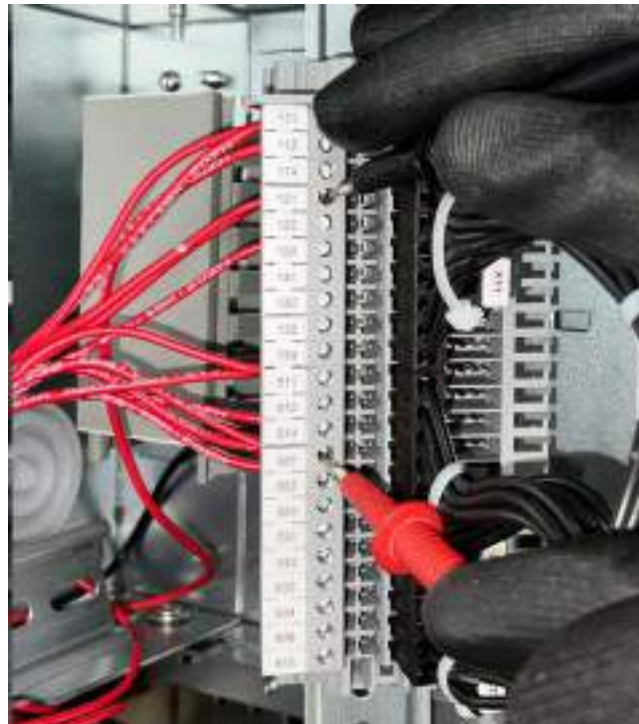
XC900.2:7



XQ1:111

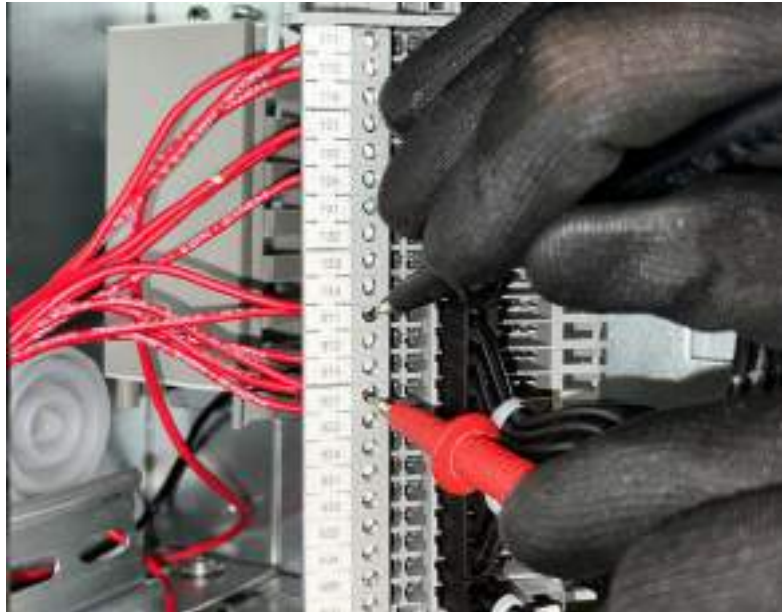


XQ1:111 – XQ1:121



XQ1:121 – XQ1:521

EN



XQ1:521 – XQ1:511



XQ1:511 – XQ0:31

EN



XQ0:31 – XQ0:33



HZ02:F71:18

Si alguno de estos cableados no tiene continuidad, sustituir. **SKIDG3_ST2**. If any of these wiring harnesses have no continuity, replace. **SKIDG3_ST2**. Si el cableado fuese correcto en todos los puntos, se debe a un error interno del relé por lo que se debería sustituir. **SKIDG3_ST2**. Consultar documento *INV3IOP87922*

EN

5.80. Fault code F166: E-STOP

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The customer has triggered an emergency stop. E-STOP	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	F56, F43	

Analysis procedure

The emergency stop of the customer's drawer has been triggered.

1. Check whether the customer has an external emergency stop button connected to the terminals X35:5:3 and X35:5:4 of the user panel (PPCU). [INVKG3_ST2](#), [INVMG3_ST2](#).



The electrical logic of this signal is "1 to External emergency / 0 to OK". The customer contact shall be potential free and normally open (NO).

2. If the customer has an external emergency mushroom connected, continuity must be checked with a 1500V multimeter between pins X35:5:3 - X35:5:4 of the user panel (PPCU). [INVKG3_ST2](#), [INVMG3_ST2](#).



This checks if the customer has activated the external emergency signal of the equipment.

2.1. If there is continuity, the customer has the external emergency ON and the display of this fault is correct. The reason for this activation should be checked and discussed with the customer. Until this signal disappears, the equipment will not be able to start up.

2.2. If there is NO continuity, the customer has deactivated the emergency. The status of bit 3 corresponding to digital input E-STOP must be checked in PowerCommsV (Group [SV1.5: Digital I/O status], (Subgroup [SV1.5.1: DSP-uC central], (Parameter [SV1.5.1.11: Status DSP RJ20])). **INVKG3_ST1B.**

If this bit is set to X, the internal wiring of the device must be checked. **INVKG3_ST2, INVMG3_ST2.**

See next point, as the checks will be the same as if there is NO external emergency mushroom connected and the fault is displayed.

3. If the customer does not have an external emergency mushroom connected, the internal wiring of the equipment must be checked. **INVKG3_ST2, INVMG3_ST2.**

Check continuity using a 1500V multimeter between pins E44:J4:5 - E44:J4:6 of the user panel (PPCU). **INVKG3_ST2, INVMG3_ST2.**



- 3.1. If there is continuity, there is a bad wire connection. Continuity should be measured between these pins: **INVKG3_ST2, INVMG3_ST2.**

NOTE: Due to the distance between connectors inside the equipment, continuity is measured with 1500V multimeter through the equipment chassis.

- X35:5:2 – E42:J4.5. This cable is the 24Vdc that runs from the door panel (PPU) E42 input/output board to the user emergency terminal block X35:5:2. Check from beginning to end, in case of no continuity check and replace the cut wire.



X35:5:2



E42:J4.5

- KA04:24 – KA02: +A1. This cable is the external customer emergency signal that activates our internal auxiliary fault relay KA02. It runs from the door panel (PPU) to the user panel (PPCU). Check from beginning to end, in case of no continuity check and replace the cable that does not give continuity.

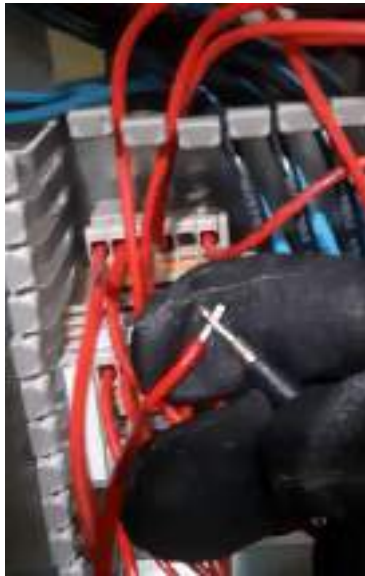


KA02: +A1



KA04:24

- X35:5:1 – KA04: +A1



X35:5:1



KA04: +A1

- KA04: -A2 – E44:J3:2



KA04: -A2

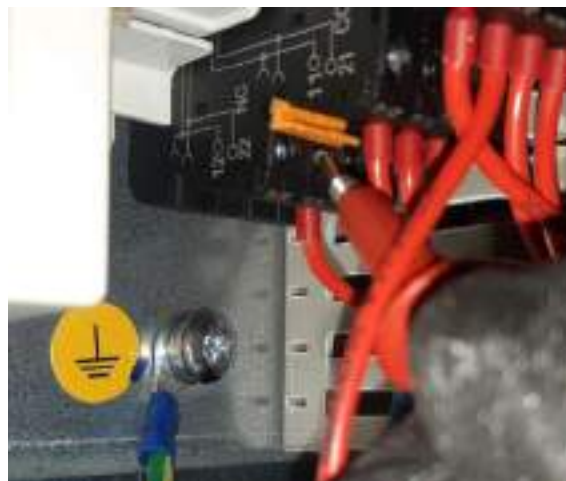


E44:J3:2

- E44:J4:5 – KA04:11



E44:J4:5



KA04:11

EN

- KA04:14 - E44:J4:6



KA04:14



E44:J4:6

Replace the wiring that has not given continuity. **INVKG3_ST2, INVMG3_ST2.**

- 3.2. If there is NO continuity, but the external emergency fault status is displayed by PowerCommsV **INVG3_ST1B**, then the problem is between the E44 input/output board and the E35 central control board.

- 3.2.1. Disconnect the RJ45 cable (W123) connecting the input board E44:J2 to the central control board E35:J20. Check the continuity of the RJ45 cable with the Ethernet cables tester. **INVKG3_ST2, INVMG3_ST2.**



Replace the failed cable. **INVKG3_ST2, INVMG3_ST2.**

- 3.2.2. If the RJ45 cable does not fail, replace the E44 input board. **INVKG3_ST2, INVMG3_ST2.** See document INVG3IOP88849.
- 3.2.3. If the fault persists, replace the central control board E35. **INVKG3_ST2, INVMG3_ST2.** See document INVG3IOP88849.

5.81. Fault code F167: Open DU

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The DC circuit breaker has been opened while the inverter is running during the day. Only applies to inverters which contain the optional Bus Plus.	
Applicable to	☒ HEM ☐ PCSM/MPCSM	☐ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect configuration.

1. If the inverter at hardware level does not match the configuration at software level, it is possible that this failure may occur. In this case:
 - 1.1. Check by PowerCommsV that the inverter is configured correctly, especially the product type. (parameter [Product type] of subgroup [SG10.1: General]). **INVG3_ST1B.**

Disabled modules.

2. The inverter with optional Bus Plus needs to have all modules enabled or interlocked in order to operate properly. If a module fault of type "Isolated DC" or "Special" occurs (parameters [Fault action] of subgroup [G5.8:Faults module config] and parameter [Fault type] of subgroup [G5.7: Fault/warning config]), in an inverter which contain Bus Plus, in addition to the fault itself, all modules will be disabled (this can be seen in subgroup [SG1.5: Module configuration] in parameters [Module x state]). In this case:
 - 2.1. Check in PowerCommsV the fault history (subgroup [SV7.1: Faults]) or the Log SYS of the SD to determine the fault that has disabled one of the modules. **INVG3_ST1B.**
 - 2.2. If the module fault is located, with PowercommsV take the actions indicated in this manual for that fault and once resolved, enable the modules (parameters [Module x state] of subgroup [SG1.5: Module configuration]). Reset the fault and check that the inverter returns to normal operation. **INVG3_ST1B.**
 - 2.3. If the fault persists, contact the plant manager to ensure with the customer that the module statuses are not being manipulated.

EN

5.82. Fault code F169: LCL Feedback

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction” .			
Description	Fault caused when the LCL capacitor contactor is activated and does not respond. Each instance indicates which capacitor bank has failed. If there is no instance, all have failed.			
Applicable to	☒ HEM	☒ PCSM/MPCSM		
	☒ HEMK	☒ PCSK/MPCSK		
Associated faults or warnings	N/A			

Analysis procedure

In PowerCommsV, (group [SV1.5 Digital I/O status]), (subgroup [SV1.5.1: DSP-uC central]), (parameter [SV1.5.1.9: Status DSP RJ17]), check the status of bits 3, 4 and 5 referring to the digital inputs of the feedbacks of contactors 1, 2 and 3 respectively, and the status of bit 3 (Parameter [SV1.5.1.10: Status DSP RJ18]) corresponding to the feedback of contactor 4, must be checked.
(INVKG3_ST1B; INVMG3_ST1B).

By viewing the fault instance and the bit that has been set, the contactor to be checked can be located.

1. De-energised checks: (INVKG3_ST2; INVMG3_ST5A).

The connections between the LCL filter contactors and the E18 / E19 input and output boards of the filter panel (PFC) shall be checked.

- 1.1. Check continuity using a 1500V multimeter in resistivity mode between the following pins:

- E18:J4:5 - E18:J4:6 - KM01 contactor feedback.
- E18:J4:5 - E18:J4:4 - KM02 contactor feedback
- E18:J4:5 - E18:J4:2 - KM03 contactor feedback
- E18:J4:5 - E19:J4:6 - KM04 contactor feedback



EXAMPLE E18:J4:5 - E18:J4:6

With the contactors open (OFF), there should be NO continuity.

We must manually force the contactor that has given failure, from the front.

Check that there is continuity on the pins mentioned above (INVKG3_ST2: INVMG3_ST5A).

1.2. If there is no continuity, check the wiring from the contactor to the input/output board using a 1500V multimeter in continuity mode between the pins: (INVKG3_ST2; INVMG3_ST5A).

- E18:J4:5 - KM01:14 = 24Vdc contactor KM01
- E18:J4:6 - KM01:13 = KM01 contactor feedback
- E18:J4:5 - KM02:14 = 24Vdc contactor KM02
- E18:J4:4 - KM02:13 = KM02 contactor feedback
- E18:J4:5 - KM03:14 = 24Vdc contactor KM03
- E18:J4:2 - KM03:13 = KM03 contactor feedback
- E18:J4:5 - KM04:14 = 24Vdc contactor KM04
- E19:J4:6 - KM04:13 = KM04 contactor feedback





PIN 13



PIN 14



E18:J4:6



E18:J4:5

EN



E18:J4:4



E18:J4:2

EN

Replace and/or modify if incorrectly connected or if there is no continuity (INVKG3_ST2; INVMG3_ST5A).

If there is still continuity (with the contactor open) or no continuity (with the contactor closed manually) at the terminals of the digital inputs of the E18 / E19 boards, the contactor (KM0X that corresponds) must be replaced as it is due to an internal fault. (INVKG3_ST2; INVMG3_ST5A).

2. Before proceeding with the voltage checks, the wiring from the contactor coils to the input and output boards shall be checked (INVKG3_ST2; INVMG3_ST5A).

Check that there IS continuity using a 1500V multimeter in resistivity mode between the following pins: (INVKG3_ST2; INVMG3_ST5A).

- X22:2:2 - E18:J5:1
- X22:3:2 - E18:J6:1
- X22:4:2 - E18:J7:1
- X22:5:2 - E19:J5:1



X22



J5:1 (E18)



J6:1 (E18)



J7:1 (E18)



J5:1 (E19)

EN

- E18:J5:2 - KM01:A1
- E18:J6:2 - KM02:A1
- E18:J7:2 - KM03:A1
- E19:J5:2 - KM04:A1



A1 (KM0X)



J5:2 (E18)



J6:2 (E18)

EN



J7:2 (E18)



J5:2 (E19)

- X22:2:1 - KM01:A2
- X22:2:1 - KM02:A2
- X22:2:1 - KM03:A2
- X22:2:1 - KM04:A2

EN



Replace and/or modify if there is NO continuity in these cables (INVKG3_ST2; INVMG3_ST5A).

If there is, continue with the voltage checks.

3. Checks WITH voltage:

An external 24 Vdc power supply must be provided to access the central control (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY).

To do this, disconnect the cables connected to the terminals (+) and (-) of the G02 source, and connect these cables to the positive and negative of the external 24 Vdc power supply.

The terminal block X22 must be supplied with 110 Vdc externally. To do this, the following steps must be carried out: (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

3.1. Disconnect the cable connected to X22:1:2 and connect the positive of the external power supply to this pin. Also disconnect the cable connected to pin X22:5:3.

3.2. Disconnect the cable connected to X22:1:1 and connect the negative of the external power supply to this pin. Also disconnect the cable connected to the X22:5:4 pin.

The 110 Vdc power supply must not be connected and disconnected at terminal block X22 in order to carry out the tests, but by switching the external power supply on and off.

As all the connections have been checked, we proceed to change the status of the contactors and check that the operation is correct (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

4. The failed contactor must be activated by following the steps below:

4.1. **Activation KM01** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

4.1.1. Only with the 24 Vdc power supply (110 Vdc power supply off), make a jumper between terminals J3:6 and J3:5 on the E18 board.



4.1.2. Once the jumper has been made, connect the 110 Vdc power supply.

4.1.3. With 110 Vdc power supply, the contactor must be closed.

4.1.4. Visually check that the contactor has entered.

4.1.5. If the contactor does not enter, go to "Point 1". If the contactor is in, go to the next step.

- 4.1.6. Measure with the 1500V multimeter in Vdc voltage mode between terminals E18:J4:5 - E18:J4:6 and you should have 24 Vdc.



EN

- 4.1.7. If you have 24 Vdc voltage, but in PowerCommsV (Consult in the groups named at the beginning of the explanation of the fault) you do not see the feedback, go to "Point 2".
- 4.2. **KM02 activation** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).
- 4.2.1. Only with the 24 Vdc power supply (110 Vdc power supply off), make a jumper between terminals J3:4 and J3:3 of the E18 board.
- 4.2.2. Once the jumper has been made, connect the 110 Vdc power supply.
- 4.2.3. With 110 Vdc power supply, the contactor must be closed.
- 4.2.4. Visually check that the contactor has entered.
- 4.2.5. If the contactor does not enter, go to "Point 1". If the contactor is in, go to the next step.
- 4.2.6. Measure with the 1500V multimeter in Vdc voltage mode between terminals E18:J4:5 - E18:J4:4 and you should have 24 Vdc



4.2.7. If you have 24Vdc voltage, but in PowerCommsV (Consult in the groups named at the beginning of the explanation of the fault) you do not see the feedback, go to "Point 2".

4.3. **KM03 activation** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

4.3.1. Only with the 24 Vdc power supply (110 Vdc power supply off), make a jumper between terminals J3:2 and J3:1 of the E18 board.



4.3.2. Once the jumper has been made, connect the 110 Vdc power supply.

4.3.3. With 110 Vdc power supply, the contactor must be closed.

4.3.4. Visually check that the contactor has entered.

4.3.5. If the contactor does not enter, move on to "Point 1". If the contactor enters, go to the next step.

4.3.6. Measure with the 1500V multimeter in Vdc voltage mode between terminals E18:J4:5 - E18:J4:2 and you should have 24 Vdc.



4.3.7. If you have 24Vdc voltage, but in PowerCommsV (Consult in the groups named at the beginning of the explanation of the fault) you do not see the feedback, go to "Point 2".

4.4. **KM04 activation** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

4.4.1. Only with the 24 Vdc power supply (110 Vdc power supply off), make a jumper between terminals J3:6 and J3:5 of the E19 board.



4.4.2. Once the jumper has been made, connect the 110 Vdc power supply.

4.4.3. With 110 Vdc power supply, the contactor must be closed.

4.4.4. Visually check that the contactor has entered.

4.4.5. If the contactor does not enter, go to "Point 1". If the contactor enters, go to the next step:

4.4.6. Measure with the 1500V multimeter in Vdc voltage mode between terminals E18:J4:5 - E19:J4:6 and we should have 24 Vdc.

4.4.7. If you have 24Vdc voltage, but in PowerCommsV (Consult in the groups named at the beginning of the explanation of the fault) you do not see the feedback, go to "Point 2".

5. **Point 1:** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

If you do not see that the contactor enters, you must measure the voltage in Vdc (approximate value of 110Vdc) at the terminals of its coils with the 110 Vdc power supply.

The points are indicated below:

- KM01:A1 - KM01:A2
- KM02:A1 - KM02:A2
- KM03:A1 - KM03:A2
- KM04:A1 - KM04:A2



As continuity has already been checked before, the fault can come from the contactor or from the input and output cards E18 / E19 or from the central control card.

If the 110Vdc voltage is measured and the contactor does not come in, the fault is internal to the contactor and must be replaced (INVKG3_ST2; INVMG3_ST5A).

6. **Point 2:** (INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY + 110Vdc EXTERNAL POWER SUPPLY).

In case of having the 110Vdc measurement, the contactor in ON position and still not receiving its feedback correctly (see status in PowerCommsV), the fault comes from the input and output cards E18 / E19 or from the central control card E35.

- 6.1. Disconnect the RJ45 cable (W059) connecting the input board E18:J2 to the central control board E35:J17. Check the continuity of the RJ45 cable with the network cable continuity tester (INVKG3_ST2; INVMG3_ST5A). Replace the failed cable (INVKG3_ST2; INVMG3_ST5A).



- 6.2. If the RJ45 cable does not fail, replace the E18 input board (INVKG3_ST2; INVMG3_ST5A).
(See document INVG3IOP88076).
- 6.3. If the fault persists, replace the central control board E35 (INVKG3_ST2; INVMG3_ST5A).
(See document INVG3IOP88076 and INVG3IOP88849).
- 6.4. Disconnect the RJ45 cable (W060) connecting the E19:J2 input board to the E35:J18 central control board. Check the continuity of the RJ45 cable with the network cable continuity tester (INVKG3_ST2, INVMG3_ST5A). Replace the failed cable (INVKG3_ST2; INVMG3_ST5A).



- 6.5. If the RJ45 cable does not fail, replace the E19 input board (INVKG3_ST2; INVMG3_ST5A).
(See document INVG3IOP88076).
- 6.6. If the fault persists, replace the central control board E35 (INVKG3_ST2; INVMG3_ST5A).
(See document INVG3IOP88076 & INVG3IOP88849).

5.83. Fault code F174: AC circuit breaker fb

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Inconsistency between expected breaker status and backfeed readings. Expected closed and read open, or expected open and read closed.	
Applicable to	☐ HEM ☒ HEMK	☐ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F157	

Analysis procedure

This fault occurs when a discrepancy is detected between the open/close command and the information received by the AC LV circuit breaker feedback.

The fault also trips to detect situations where the AC circuit breaker has opened because it has been commanded to do so by its built-in protection relay or a manual opening of the AC circuit breaker.

In cases where the circuit breaker trips a protection and opens without the inverter detecting it, it could be a problem in the circuit breaker relay, a problem with the internal current measurement in the circuit breaker or a HW problem of the equipment (short circuit, phase loss, etc.).

To rule out whether it is an AC circuit breaker problem or a backfeed wiring problem:

1. Visually check if there is any trip alarm activated on the circuit breaker relay (any led on the circuit breaker relay is on). ([INVKG3_ST6](#)).



- 1.1. If any LED is on, it indicates that the circuit breaker has tripped autonomously in operation for some reason. Go to [INVKG3_ST2](#) status and search for the cause.
- 1.2. If there are no LEDs on, it is probably a problem with the feedback wiring; go to the next step.

2. If the circuit breaker relay has tripped due to overcurrent, check that there is no short-circuit between boards in the inductance zone, measure (with a 1500V multimeter in resistance mode) continuity between phases (BBI1.03-BBI2.03, BBI103-BBI3.03, BBI2.03 - BBI3.03) with the circuit breaker open. (INVKG3_ST2).



If there is no continuity between phases (measured resistance $> 100\text{k}\Omega$) and there are no bars with a safety distance of less than 11 millimetres, check the relay and the circuit breaker toroids. (INVKG3_ST2).



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If a part is found to be damaged or in poor condition, replace the circuit breaker or relay (depending on availability). (INVKG3_ST2).

3. Check the integrity of the circuit breaker for the presence of hot spots or signs of degradation of the circuit breaker. (INVKG3_ST2).

If the circuit breaker's Ig protection is tripping, it is because the circuit breaker is sensing a very high phase-to-phase unbalance that could be closing to ground.

4. Check that all boards (BBI1.03, BBI2.03, BBI3.03) are connected and no phase is missing. (INVKG3_ST2).
5. Check that no point of the AC power circuit is connected to earth. Measure continuity with a 1500V multimeter between the AC terminals (BBI1.03, BBI2.03, BBI3.03) and earth. (INVKG3_ST2).



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If there is continuity (resistance < 1kOhm), make a visual inspection of the inverter for elements that could lead a phase to ground. (INVKG3_ST2).

6. If no fault has been detected in the previous point, bring the inverter to "READY" status and measure with the IMD, enable the manual operation of the IMD (Subgroup [G9.3: Maneuver]), (Subgroup [G9.3.0: Mnuual Maneuver]), (Parameter [G9.3.0.1: Enable]) and check the resistance of each channel in the parameter [Resistor] in the group [SV9.2: Channels]). For PowerCommsV (INVKG3_ST1B).
If everything is OK (resistance measured by IMD higher than 50kOhm), this indicates that the problem is in the relay; replace it. (INVKG3_ST2).
7. If the protection continues to trip, there is probably a problem with the internal current measurement in the circuit breaker. To rule out problems with the internal current measurement:
 - 7.1. Measure the impedance of the internal toroidals of each phase of the circuit breaker, the correct impedance value must be 83Ω. To do this, contact Power Electronics. (INVKG3_ST2).
 - 7.2. Check that the circuit breaker opens and closes correctly. To carry out the opening and closing of the AC selector, it must be done from PowerCommsV in the following way:
 - 7.2.1. Set the configuration of the AC virtual selector to communications (Group [G2.6: Selectors], (Subgroup [G2.6.1: Selectors Maneuver]), (parameter [G2.6.1.3: AC Virtual Sel conf]), and activate and deactivate the AC Virtual Selector depending on whether you want to test opening or closing (parameter [2.6.1.3a: AC Virtual Selector]). (INVKG3_ST1B)
 - 7.2.2. Switch the AC selector on and off (in "DC Ready" state). Reset the circuit breaker and check that it opens and closes correctly (you should hear it close). (INVKG3_ST1B with the equipment shut down).
 - 7.2.3. If it physically opens and closes correctly, but fails, check the status of the backfeed (next step). If it does not open and close correctly, the circuit breaker coils are most likely not supplied with power or are damaged. Go to next step.



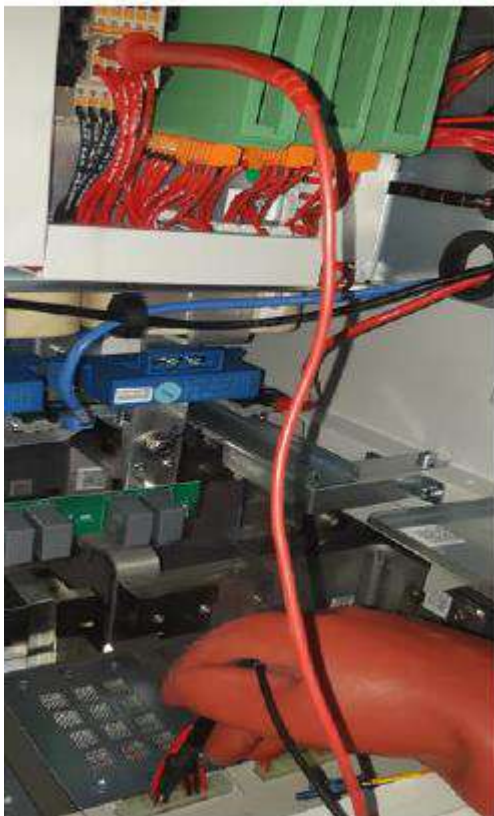
- 7.3. Check that the circuit breaker has 120V power:
Check that 120Vdc power is supplied to the main terminal block of the filter cabinet C. With a 1500V multimeter configured to measure DC voltage, measure between X22:2 (Positive 120V) and X22:1 (Negative 120V). To do this, the polycarbonate must first be removed. (INVKG3_ST2 with external 120V power supply).



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If 120V is present, check step by step with the 1500V multimeter that voltage is supplied to each of the following points:

- 7.3.1. Between positive pole of terminal X22:2 of the filter frame and the negative pole of the closing coil (XC100.1 A2), opening coil (XC100.1 C2), minimum coil (XC100.1 D2) and spring motor (XC100.1 U2). (INVKG3_ST2 with external 120V power supply).



If there is no 120Vdc at all these points, the negative cable from the main terminal block (X22:1) to the point where we do not measure 120Vdc voltage is broken, check and replace. (INVKG3_ST4).

- 7.3.2. Measure DC voltage with the 1500V multimeter between the negative pole of the filter board terminal block (X22:1) and the positive of the activation relays of the input/output board (E17, J5:1, J6:1, J7:1), and the positive of the UVT undervoltage coil (terminal D1 of connector XC100 of the circuit breaker). (INVKG3_ST2 with external 120V power supply). If 120Vdc is not available at any of the points, check the wiring and replace. (INVKG3_ST4).



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- 7.4. If 120V is present in all the circuits checked above, check continuity with 1500V multimeter between the relay outputs of the input and output boards and the connection point to the AC circuit breaker. If there is no continuity at any of the points, check and replace the wiring. Also check if the intermediate connector has a changed pin. (INVKG3_ST4).



Board E17 where continuity is checked



First point to check continuity of the E17 board



U1 point to check continuity with the first point of the E17 board



Second point to check continuity of the E17 board

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C1 point to check continuity with the second point of the E17 board



Third point to check continuity of the E17 board



A1 point to check continuity with the third point of the E17 board

- 7.5. 7.5. If all the checks have been successful, verify that the motor resistance (U1 and U2) is approximately 30Ω (with the spring unloaded). To check the opening and closing coils, the following actions should be carried out with the 1500V multimeter:



- 7.5.1. Opening coil: Command opening and measure using a 1500V multimeter in Vdc mode between terminals E17:J6:2 and XC100.1:C2. If there is no 120 Vdc, check wiring up to E517:J6:2, and if this is correct, replace the coil. (INVKG3_ST2 with an external alimentation of 120 Vdc).



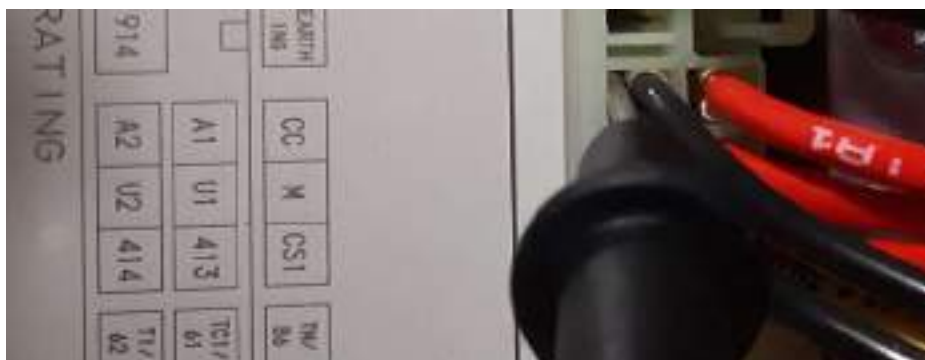
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7.5.2. Closing coil: Command opening and measure with a 1500V multimeter in Vdc mode between terminals E17:J7:2 and XC100.1:A2. If there is no 120 Vdc, check wiring up to E17:J7:2, and if this is correct, replace the coil.



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See document INVG3IOP85377. (INVKG3_ST4).

8. Problems with the wiring of the backfeed: the corresponding backfeed is not activated.

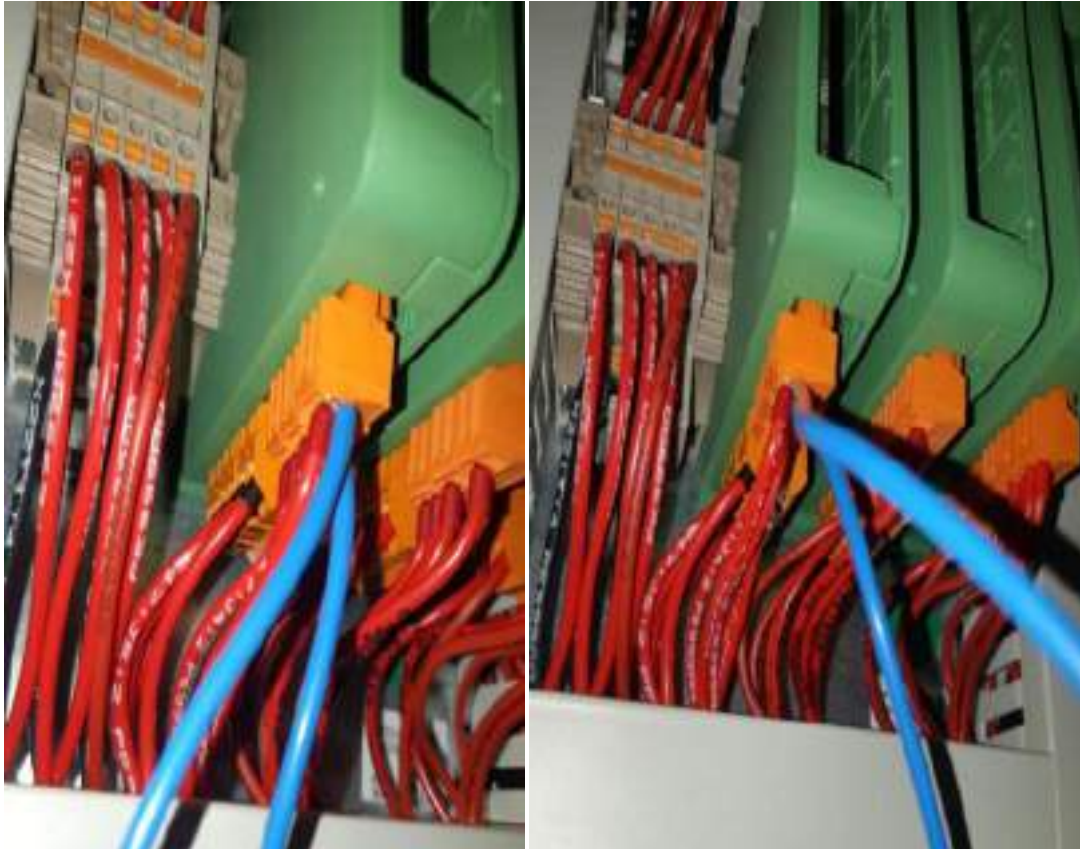
Using PowerCommsV check the status of the AC circuit breaker feedbacks (Group [SV1.5: Digital I/O status]), (Subgroup [SV1.5.1: DSP-uC central]), (Parameter [SV1.5.1.1: uC Digital inputs]) for the status of bits 4 (open) and 5 (closed) of the digital inputs of uC (INVKG3_ST1B).

If the feedbacks are correct at this time, it will be necessary to identify at what time the fault has been triggered to know the expected state at that time and to be able to determine if it is a problem with the

state of the circuit breaker. Download the SD and attempt to reproduce the fault to identify the problem. For PowerCommsV. **INVKG3_ST1B**.

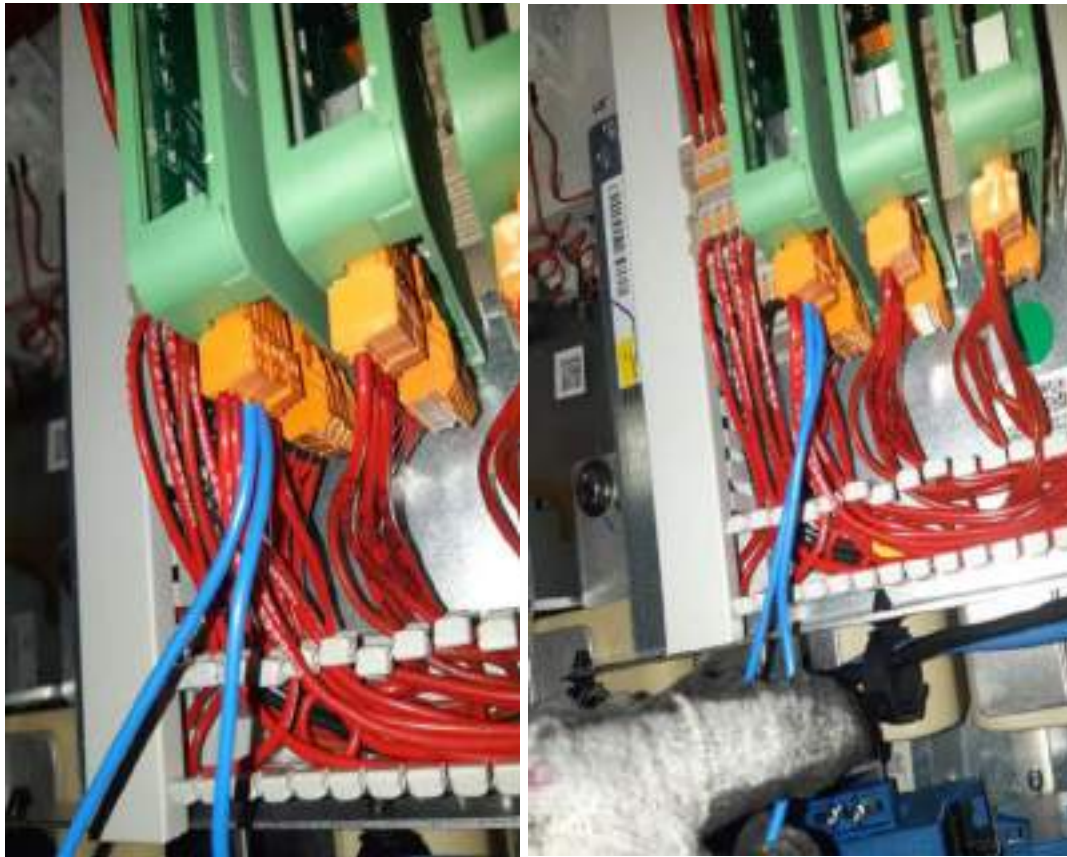
Once the faulty feedback is found, proceed as follows:

- 8.1. If the feedback appears open when it should be closed, jumper the feedback on the input/output board (E17) between pins J4.1 and J4.2 for the closing feedback and J4.3 and J4.4 for the opening feedback (**INVKG3_ST2**) and verify in PowerCommsV. Check in PowerCommsV in group [SV1.5.1.1] the status of bits 4 (open) and 5 (closed) of the uC digital inputs. (**With the device closed INVKG3_ST1B**).



Jumper between J4.1 and J4.2

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Jumper between J4.3 and J4.4

If it does not change, the problem is with the wiring to the central control or digital input/output board.

8.1.1. Check that the RJ45 cable is properly connected on both boards. (INVKG3_ST2).



- 8.1.2. Check continuity of the RJ45 cable using an RJ45 tester from E17:J2 to the central control board E35:J19. (INVKG3_ST2).



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- 8.1.3. If the RJ45 cable is OK, replace the I/O board (E17). (INVKG3_ST2). Re-energise and check if the feedback is still open. (INVKG3_ST1B with the equipment closed). If this does not work, replace the central control board (E35) (See document INVKG3/OP88849). (INVKG3_ST2).

If none of these points work, contact Power Electronics.

5.84. Fault code F213: Negative DU current

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	When the DU current is below the negative current threshold value, this fault is triggered.	
Applicable to	☒ HEM ☐ PCSM/MPCSM	☐ PCSK/MPCSK
Associated faults or warnings	F54: DU Overcurrent	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect parameterisation.

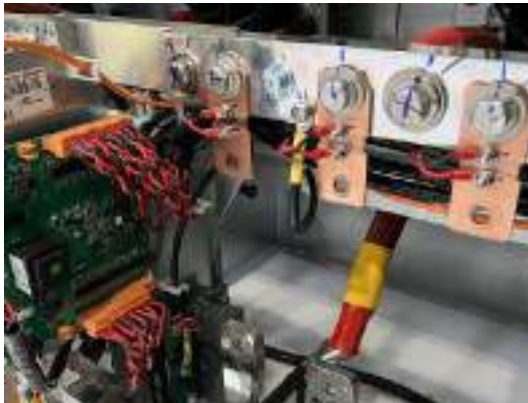
1. The fault setting is incorrect and fires when it is not due. If this occurs:
 - 1.1. Check by PowerCommsV the default value of the protection (parameter [DU Inv current prot] of group [G5: Protections]). If it is not the default value, check with the plant manager for the appropriate value. **INVG3_ST1B**.

Bus Plus and DCDC running with inverter stopped.

2. If the DCDC is kept running with the inverter stopped, current flow to the panels may occur. In this case:
 - 2.1. Check if the DCDC has been running while the inverter was stopped by comparing the SD of both inverters and if there is a plant log by the customer. If so, contact the plant manager to inform the customer that this status is incorrect. **INVG3_ST1B**.

Incorrectly configured channels.

3. If the channels of the inverters are not configured according to the corresponding hardware, it is possible that the fault is triggered by a value outside the full scale. If this occurs:
 - 3.1. Check that the input strings of the inverter that are connected to BBEP.63 or BBEP.64 and have the shunt connected to the E66 or E65 board. **INVKG3_ST2, INVMG3_ST5A**.



- 3.2. Configure by PowerCommsV the type of DU of the inverter (parameter [DU Fuses] of subgroup [SG10.6: DU]). **INVG3_ST1B**. It is possible to differentiate it visually as the option "Both Poles" of parameter [DU Fuses] requires that there are fuses in the positive busbar strings BBEP.63 or BBEP.64 as well as in the negative BBEN.03. In the case of positive pole configuration, the positive strings are connected to BBEP.63 or BBEP.64. **INVKG3_ST2, INVMG3_ST5A**.
- 3.3. Depending on the number of channels connected, this must be configured by PowerCommsV in the parameter [DU Number of Channels] of the subgroup [SG10.6: DU]. **INVG3_ST1B**.
- 3.4. If the configuration of the [DU Fuses] parameter does not match any of the options, in this case use the "Customisable" option and disable in the group [G10: DU] the channels that are not physically connected. **INVG3_ST1B**.

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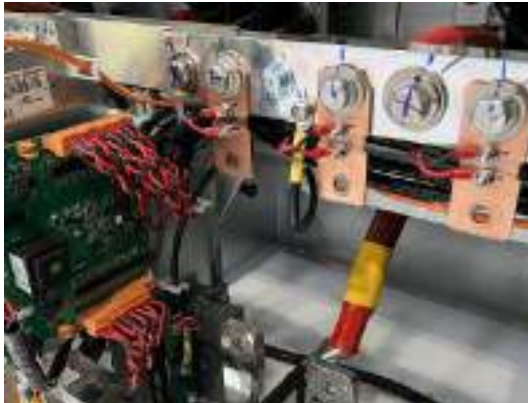
Note: Configure the channels of the inverter correctly to avoid false measurement failures.

Damaged wiring.

4. If the measurement wiring is damaged it is possible to trigger the fault. If this occurs:
 - 4.1. Check that the cables leading from the SHX measurement resistors to the E66 or E65 boards are correctly wired especially the connections on the resistor itself. **INVKG3_ST2, INVMG3_ST5A.**

Shunt connected backwards.

5. If the wiring to the shunt resistors is connected backwards, the current measurement will have the opposite sign to that expected. In this case:
 - 5.1. Check the wiring to the SHX shunts so that the red wiring is at the bottom and the black wiring is at the top. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.2. Check that the connections on the E65 and E66 boards are in the correct order and always alternate colours for each shunt. **INVKG3_ST2, INVMG3_ST5A.**



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Internal recirculation in the measurement board.

6. If none of the above cases apply, it is possible that the overcurrent is caused by a recirculation problem in the DU metering board. In this case:
 - 6.1. Check if the DU measurement boards are model E21801204BA or later. If this is not the case, replace these boards with this model **INVKG3_ST2, INVMG3_ST5A** and check the operation of the inverter by monitoring so that the fault is not repeated.



If it is repeated, check other causes. See document *INVG3IOP88849*.

Short-circuit in the channel.

7. It is possible that the overcurrent is due to a short circuit or shunt on the corresponding channel. If this is the case:
 - 7.1. Visually inspect the inside of the DU for damaged cables, blackened or bad busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU. **INVKG3_ST2, INVMG3_ST5A.**

- 7.2. With the strings / batteries disconnected, connect an isolation meter between positive and negative of each input of the inverter and make a measurement with a maximum of 1000Vdc. Note that previously it is necessary to disconnect the RFI boards and the isolation measurement boards.
 INVKG3_ST5B, INVMG3_ST5B.



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- 7.2.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables from each of the busbars and repeat the measurement.
 7.2.2. If the measurement exceeds $1\text{M}\Omega$ when any of the input cables to the inverter are disconnected, this means that this cable has a shunt between positive and negative. If this occurs, inform the customer so that this channel can be repaired.
 7.2.3. If the measurement does not improve after disconnecting all external cables, consult Power Electronics.

5.85. Fault code F214: IMD Comms

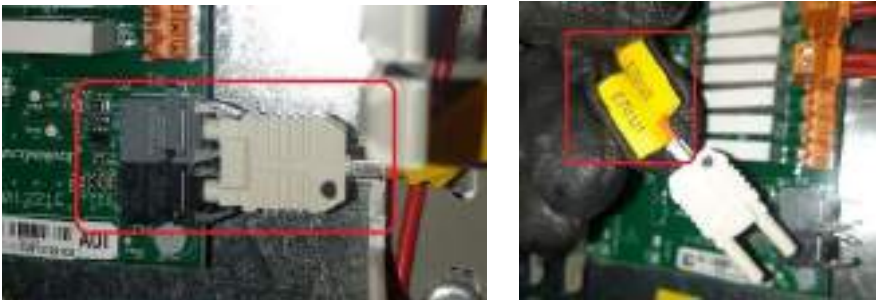
Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".	
Description	Failure of fibre-optic communications with the isolation monitor device	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

1. Check in PowerCommsV (Group [G6: Communications]), (Subgroup [G6.1: Modbus]), (Subgroup [G6.1.3: Master 3]), (Parameter [G6.1.3.7: Modbus Timeout 3]) if the value of Modbus Timeout 3 is different from that specified in the parameterisation provided, the reason for this difference must be consulted: **INVG3_ST1B**.
 2. The connection of the fibre optic cable (W160) between the central control board and the insulation monitor board shall be checked: **INVKG3_ST2, INVMG3_ST2**.
- E35:U12 (RX) - E70: U2 (TX) - Orange hose wire
 - E35:U13 (TX) - E70:U1 (RX) - Black hose wire



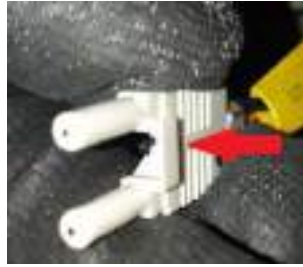
U12 and U13 of the board E35



U1 and U2 of the board E70

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Check that the connectors "click" when plugged into the boards. If you try to place the connector upside down, it will not click on the board because a notch in the aerial connectors collides with the connectors on the control board. **INVKG3_ST2, INVMG3_ST2.**



Check the connector of the fibres, the fibres must be flush with the connector and well-polished. If not, change the connector or the fibre (depending on availability). **INVKG3_ST2, INVMG3_ST2.**

3. Check that the light passes correctly from one optical fibre to another. To do this, shine a light source, either natural or artificial, from one end of the optical fibre and check that light is received at the other end. If not, replace the damaged optical fibre. **INVKG3_ST2, INVMG3_ST2.**



4. If the hose (W160) is found to be valid and also well connected, the problem is in the cards to which it is connected. The insulation monitor board E70 should be replaced first and if the fault persists, replace the central control board E35. **INVKG3_ST2, INVMG3_ST2.** See document *INVG3IOP88849*.

5.86. Fault code F215: Timeout GFDI

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	Close or open order has been sent to GFDI - the feedback does not match the action.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

The GFDI feedback does not correspond to the given command after a configured time.

In PowerCommsV the Timeout GFDI Retro must be checked (Subgroup [G9.3: Maneuver]), (Subgroup [G9.3.0: Manual Maneuver]), (Parameter [G9.3.2: Timeout GFDI Feedback]). If this value is different from the value specified in the parameterisation provided, the reason for this difference must be consulted ([INVKG3_ST1B](#); [INVMG3_ST1B](#)).

Check in PowerCommsV the status of the feedback (Group [SV9: IMD]), (Subgroup [SV9.3: Maneuver]), (Parameter [SV9.3.6: GFDI Feedback]). ([INVKG3_ST1B](#); [INVMG3_ST1B](#)). It must be checked in which state it should be in (Points below).

To find out in which status the feedback should be, the status of the GFDI (Group [SV9: IMD]), (Subgroup [SV9.3: Maneuver]), (Parameter [SV9.3.5: GFDI Output]). ([INVKG3_ST1B](#); [INVMG3_ST1B](#)).

If these statuses do not match, or match later than the defined time, the following checks must be performed.

1. De-energised checks: ([INVKG3_ST2](#); [INVMG3_ST5A](#))

1.1. The connections between the GFDI (Q12) and the insulation monitor board (E70) shall be checked.

1.1.1. To do this, detach the W315 hose from the GFDI (Q12) whose connector is XC556.1 and check its wiring.



Continuity must be checked using a 1500V multimeter in resistivity mode between the following pins:

- X29:1:2 - XC556.1:1 (WH)



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- X29:3:2 - XC556.1:6 (PK)



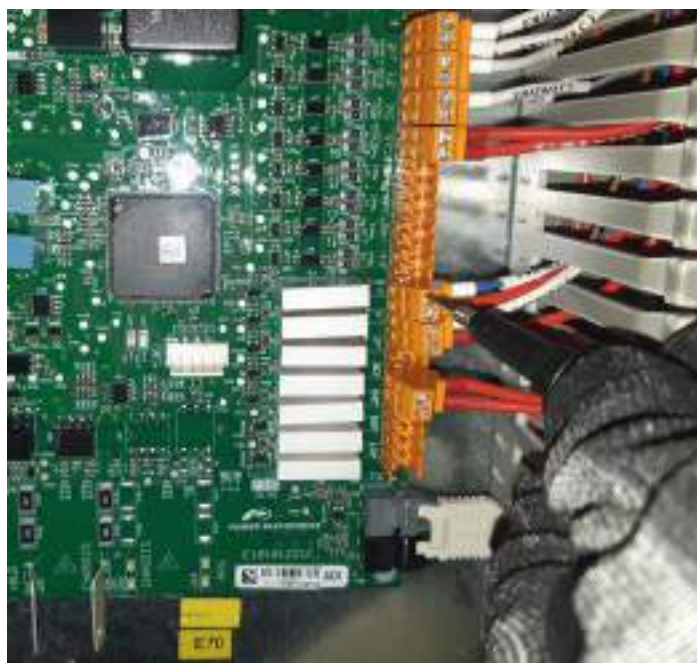
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- E70:J9:2 - XC556.1:2 (BN)



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- E70:J10:2 - XC556.1:7 (BU)



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- E70:J18:1 - XC556.1:8 (RD)



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- E70:J17:2 - XC556.1:4 (YE)



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- E70:J18:2 - XC556.1:10 (VI)



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- E70:J16:2 - XC556.1:5 (GY)



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- E70:J9:1 - X29:1:5



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- E70:J10:1 - X29:2:2



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- 1.1.2. Replace and/or modify if any of these cables do not give continuity (**INVKG3_ST2; INVMG3_ST5A**).

If all of them give continuity, the wiring is correct.

- 1.2. 24Vdc voltage shall be tested on the insulation monitor board (E70) and the central control board (E35) to verify operation (**INVKG3_ST2; INVMG3_ST5A + 24Vdc EXTERNAL POWER SUPPLY**).



G01 without external power supply



G01 with external power supply



G02 without external power supply



G02 with external power supply

- 1.2.1. If the fault persists, the fibre connector shall be checked. The fibres should be flush with the connector and well polished. If not, replace the connector or the fibre (depending on availability) (**INVKG3_ST2; INVMG3_ST5A**).

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- 1.2.2. Check that the light passes correctly from one optical fibre to another. To do this, shine a light source, either natural or artificial, from one end of the optical fibre and check that light is received at the other end. If not, replace the damaged optical fibre (INVKG3_ST2; INVMG3_ST5A).



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5.87. Fault code F216: Timeout IMI Meas

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The time set to wait for an earthed insulation measurement has expired.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	F229	

Analysis procedure

This fault is triggered when the IMD channel does not give correct insulation measurements after a time configured in PowerCommsV.

1. Consult in PowerCommsV the Measurement Timeout (Subgroup [G9.3: Maneuver]), (Subgroup [G9.3.0: Manual Maneuver]), (Parameter [G9.3.1: Timeout Measure]). If the value of this timeout is different to that specified in the parameterisation provided, the reason for this difference must be consulted (INVKG3_ST1B; INVMG3_ST1B).

This fault occurs in certain manoeuvres such as GFDI or diagnostics, not in a normal measurement.

2. Download via PowercommsV the SDs and check in Log SYS 2 the measurement time during the manoeuvre in which the fault occurred (INVKG3_ST1B; INVMG3_ST1B). If this time is less than the defined timeout and the fault continues to occur, replace the insulation monitor card (E70) (INVKG3_ST2; INVMG3_ST5A). See document INVG3/OP88849.
3. The AC and DC insulation resistor board must be checked.

The following checks are performed on the AC measurement channel board. If the fault is reported on a different channel, refer to the table for the equivalent component.

Measuring channel	Connector on E70 board	Resistor board
AC	J6	E75
DC1	J5	E76

Using a 1500V multimeter in resistivity mode, measure the resistance on the E75 resistor board between terminals J1-J3:1 and J2-J3:1. The resistance should be 240kOhms +-1%. If this value is not present, the resistor board must be replaced (INVKG3_ST2; INVMG3_ST5A).

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If the resistance is correct, check with a 1500V multimeter in continuity mode between the resistor board and the IMD board. Measure continuity between J3:1 of the E75 board and J6:2 of the E70 board, and between J3:2 of the E75 board and J6:1 of the E70 board. (INVKG3_ST2; INVMG3_ST5A)



4. Check the fibre connector. The fibres should be flush with the connector and well polished. If not, replace the connector or the fibre (depending on availability) (INVKG3_ST2; INVMG3_ST5A).

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5. Check that the light passes correctly from one optical fibre to another. To do this, shine a light source, either natural or artificial, from one end of the optical fibre and check that light is being received at the other end. If not, replace the damaged optical fibre ([INVKG3_ST2](#); [INVMG3_ST5A](#)).



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6. If the hose (W160) connecting the isolation board (E70) to the central control board (E35) is found to be valid and properly connected, the problem is in the boards to which it is connected. The insulation monitor board E70 should be replaced first, and if the fault persists, proceed to replace the central control board E35. (INVKG3_ST2; INVMG3_ST5A) See document INVG3/OP88849.

5.88. Fault code F229: Timeout IMI Channel

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting.			
	Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.			
Description				
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings				

Analysis procedure

One of the channels (AC, DC1, DC2, DC3, DC4) depending on the inverter type has not been measuring for more than a configured period of time.

- In PowerCommsV "Timeout No measurement" must be checked (Subgroup [G9.2: Channels]), (Subgroup [G9.2.1: Channel AC]), (Parameter [G9.2.1.5: No Measure Timeout]) and verify that this parameter is as indicated in the parameterisation: (INVKG3_ST1B; INVMG3_ST1B).
 - For HEM/k devices with GFDI= 3000 min.
 - For HEM/k equipment without GFDI= 120 min.
 - For MPCSM units with more than one busbar = 40 mins x no. of busbars.

- Via PowercommsV download the SDs and check in [Log SYS 2] the measurement time in the channel where the fault has occurred or if the measurement has not been taken (INVKG3_ST1B; INVMG3_ST1B).

If the parameterisation is correct and the measurement has not been carried out, the resistor board of the channel on which the fault occurred must be checked (INVKG3_ST2; INVMG3_ST5A).

- The following checks are performed on the AC measurement channel board. If the fault is reported on a different channel, refer to the table for the equivalent component.

Measuring channel	Connector on E70 board	Resistor card
AC	J6	E75
DC1	J5	E74
DC2	J4	E73
DC3	J3	E72
Dc4	J2	E71

Using a 1500V multimeter in resistivity mode, measure the resistance on the E75 resistor board between terminals J1-J3:1 and J2-J3:1. The resistance should be 240kOhms +-1%. If this value is not obtained, the resistor board must be replaced (INVKG3_ST2; INVMG3_ST5A). See document INVG3IOP88849.

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4. If the resistance is correct, check with a 1500V multimeter in continuity mode between the resistor board and the IMD board (E70). Measure continuity between J3:1 of the E75 board and J6:2 of the E70 board, and between J3:2 of the E75 board and J6:1 of the E70 board (**INVKG3_ST2; INVMG3_ST5A**).



If there is no continuity at any point, change wiring. (INVKG3_ST2; INVMG3_ST5A).

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5.89. Fault code F240: Du BP overcurrent

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Triggered when the sum of the measured currents in the DU channels exceeds the threshold and the device is not switching. Applies to inverters with optional Bus Plus.	
Applicable to	☒ HEM ☒ HEMK	☐ PCSM/MPCSM ☐ PCSK/MPCSK
Associated faults or warnings	Related to: F54: DU Overcurrent. F213: Negative DU current.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect parameterisation.

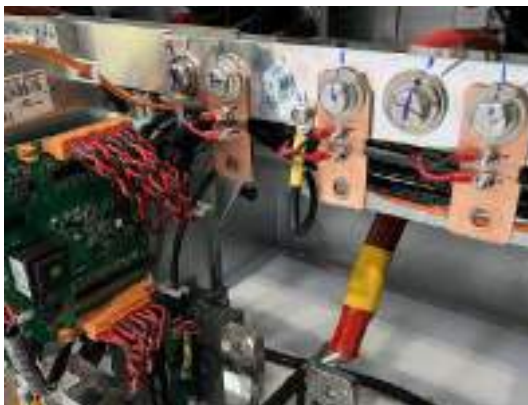
1. The fault setting is incorrect and fires when it is not due. If this occurs:
 - 1.1. Check by PowerCommsV the default value of the protection (parameter [DU BP overcurrent] of group [SG1.1: uP service]). If it is not the default value, check with the plant manager for the appropriate value. **INVKG3_ST1B**.

Bus Plus and DCDC running with inverter stopped.

2. If the DCDC is kept running with the inverter stopped, current flow to the panels may occur. In this case:
 - 2.1. Check if the DCDC has been running while the inverter was stopped by comparing the SD of both inverters and if there is a plant log by the customer. If so, contact the plant manager to inform the customer that this status is incorrect. **INVKG3_ST1B**.

Incorrectly configured channels.

3. If the channels of the inverters are not configured according to the corresponding hardware, it is possible that the fault is triggered by a value outside the full scale. If this occurs:
 - 3.1. Check that the input strings of the inverter that are connected to BBEP.63 or BBEP.64 and have the shunt connected to the E66 or E65 board. **INVKG3_ST2, INVMG3_ST5A**.



- 3.2. Configure by PowerCommsV the type of DU of the inverter (parameter [DU Fuses] of subgroup [SG10.6: DU]). **INVKG3_ST1B**. It is possible to differentiate it visually as the option "Both Poles" of parameter [DU Fuses] requires that there are fuses in the positive busbar strings BBEP.63 or BBEP.64 as well as in the negative BBEN.03. In the case of positive pole configuration, the positive strings are connected to BBEP.63 or BBEP.64. **INVKG3_ST2, INVMG3_ST5A**.

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- 3.3. Depending on the number of channels connected, this must be configured by PowerCommsV in the parameter [DU Number of Channels] of the subgroup [SG10.6: DU]. **INVG3_ST1B.**
- 3.4. If the configuration of the [DU Fuses] parameter does not match any of the options, in this case use the "Customisable" option and disable in the group [G10: DU] the channels that are not physically connected. **INVG3_ST1B.**

Note: Configure the channels of the inverter correctly to avoid false measurement failures.

Damaged wiring or shunt.

4. If the shunt resistors or measurement wiring is damaged it is possible to trigger the fault. If this occurs:
 - 4.1. Check that the SHX correspond to the current measurement shunt resistors. **INVKG3_ST2, INVMG3_ST5A.**
 - 4.2. Check that these resistors give continuity with a 1500V multimeter. **INVKG3_ST2, INVMG3_ST5A.**



- 4.3. Check that the cables that go from the SHX measurement resistors to the E66 or E65 boards are correctly wired, paying special attention to the connections on the resistor itself. **INVKG3_ST2, INVMG3_ST5A.**

String configuration.

5. If one of the inverter's input strings is poorly distributed, the amount of current that will enter through that measurement channel will be significantly higher and may cause a fault. This may be due to a bad distribution of the inputs by the wiring of the project. If this happens:
 - 5.1. Check that the DU strings are evenly distributed at the input of the inverter. If this is not the case, contact the plant manager so that he/she can check this wiring with the customer. **INVKG3_ST2, INVMG3_ST5A.**

Internal recirculation in the measurement board.

6. If none of the above cases apply, it is possible that the overcurrent is caused by a recirculation problem in the DU metering board. In this case:
 - 6.1. Check if the DU measurement boards are model E21801204BA or later. If this is not the case, replace these boards with this model **INVKG3_ST2, INVMG3_ST5A** and check the operation of the inverter by monitoring so that the fault is not repeated.



If it is repeated, check other causes. See document *INVKG3/OP88849*.

Short-circuit in the channel.

7. It is possible that the overcurrent is due to a short circuit or shunt on the corresponding channel. If this is the case:
 - 7.1. Visually inspect the inside of the DU for damaged cables, blackened or bad busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU.
INVKG3_ST2, INVMG3_ST5A.
 - 7.2. With the strings / batteries disconnected, connect an isolation meter between positive and negative of each input of the inverter and make a measurement with a maximum of 1000Vdc. Note that previously it is necessary to disconnect the RFI boards and the isolation measurement boards.
INVKG3_ST5B, INVMG3_ST5B.



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- 7.2.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables from each of the busbars and repeat the measurement.
- 7.2.2. If the measurement exceeds $1\text{M}\Omega$ when any of the input cables to the inverter are disconnected, this means that this cable has a shunt between positive and negative. If this occurs, inform the customer so that this channel can be repaired.
- 7.2.3. If the measurement does not improve after disconnecting all external cables, consult Power Electronics.

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5.90. Fault code F253: Imbalance V

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".		
Description	There is a voltage unbalance between busbars higher than the value set in the protection.		
Applicable to	☐ HEM	☒ PCSM/MPCSM	
	☐ HEMK	☒ PCSK/MPCSK	
Associated faults or warnings	N/A.		

Analysis procedure

This fault is reported when there is a voltage difference between busbars (Vdc1 - Vdc2) that exceeds the value set in **[Busbar V Threshold]** within group **[G5.13-Unbalanced Busbar]**, where Vdc1 and Vdc2 correspond to the voltage of each of the batteries.

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect configuration of the unbalance protection.

1. Check in PowerCommsV ([Faults history] within group [SV7.1-Faults]), that the device fault corresponds to the F253. **INV3_ST1B**.
2. Download the logs from the SD card and view on the SD card **(Log 4)** the DC voltages of the batteries at the time of the fault. To do so, check in LOG 4, the 'Emb' and 'Vpanels' columns, where the 'Emb' index corresponds to the busbar number and 'Vpanels' to the voltage of the battery in question. **INV3_ST1B**.
3. Check in PowerCommsV **[Busbar V Threshold]** within group **[G5.13-Unbalanced Busbar]**, the busbar unbalance protection setting. **INV3_ST1B**.
If the configured value is lower than the busbar unbalance present in the SD card, consult with the plant manager to know if this modification is justified for the protection of the installed batteries.
If not, change the parameter to its default value.

DC measurement failure.

4. Check in the equipment documentation ("Single Line Diagram" in the schematic) the number of DC inputs to know the number of busbars the equipment has.
5. Check in PowerCommsV **[General]** within group **[SG10-Drive select]**, if the inverter type is PCS / MPCS and the number of busbars matches the number of DC inputs available in the unit. **INV3_ST1B**.
6. Check in PowerCommsV within the group ([SV2-DC source]) the DC measurements of each busbar. Depending on the number of busbars, the measurements will be displayed in different parameters, refer to the following table: **INV3_ST1B**.

PARAMETER	NUMBER OF BUSBARS		
	2	3	4
[SV2.9-Dc Voltage extern 1]	X	X	X
[SV2.10- Dc Voltage extern 2]	X	X	X
[SV2.11- Dc Voltage extern 3]		X	X
[SV2.12- Dc Voltage extern 4]			X

If oscillation is detected in any of the measurements, proceed to the next check point.

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7. Check the wiring of the DC measurements by measuring with a 1500V multimeter the continuity between the E35 central control board and the DC busbar terminals at the following points:
INVKG3_ST2, INVMG3_ST5B.

- 7.1. Positive 1: between connector J1 of the central control board E35 and the BBEP:82 terminal, between terminal BBEP.82 and BBEP.81, between terminal BBIP.07 and BBIP.20.



E35:J1



BBEP.82

- 7.2. Negative 1: between connector J2 of the central control board E35 and the BBEN:67 terminal, between terminal BBEN.67 and BBEN.66, between terminal BBIN.07 and BBIN.20.



E35:J2



BBEN.67

- 7.3. Positive 2: between connector J1 of the busbar expansion board E36 and the BBEP:80 terminal, between terminal BBEP.80 and BBEP.79, between terminal BBIP.05 and BBIP.20.

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E36:J1



BBEP.80

- 7.4. Negative 2: between connector J2 of the busbar expansion board E36 and the BBEN:65 terminal, between terminal BBEN.65 and BBEN.64, between terminal BBIN.05 and BBIN.20.



E36:J2



BBEN.65

- 7.5. Positive 3: between connector J3 of the busbar expansion board E36 and the BBEP:78 terminal, between terminal BBEP.78 and BBEP.77, between terminal BBIP.03 and BBIP.20.



E36:J3



BBEN.78

- 7.6. Negative 3: between connector J4 of the busbar expansion board E36 and the BBEN:63 terminal, between terminal BBEN.63 and BBEN.62, between terminal BBIN.03 and BBIN.20.



E36:J4



BBEN.63

- 7.7. Positive 4: between connector J5 of the busbar expansion board E36 and the BBEP:76 terminal, between terminal BBEP.76 and BBEP.75, between terminal BBIP.01 and BBIP.20.



E36:J5



BBEP.76

- 7.8. Negative 4: between connector J6 of the busbar expansion board E36 and the BBEN:61 terminal, between terminal BBEN.61 and BBEN.60, between terminal BBIN.01 and BBIN.20.



E36:J6



BBEN.61

Carry out continuity measurements according to the number of busbars of the equipment, see the following table:

CHECK	NUMBER OF BUSBARS		
	2	3	4
Positive 1	X	X	X
Negative 1	X	X	X
Positive 2		X	X
Negative 2		X	X
Positive 3	X	X	X
Negative 3	X	X	X
Positive 4			X
Negative 4			X

If there is no continuity in any of the measured points or any cable is damaged, replace it. **INVKG3_ST2, INVMG3_ST5B.**

Damaged board.

If all the above checks have been satisfactory, replace the E35 central control board if the equipment has only one busbar. **INVKG3_ST2, INVMG3_ST5B.** See document *INVG3/OP88849*.

If the unit has 2, 3 or 4 busbars, replace the Busbar E36 expansion board. **INVKG3_ST2, INVMG3_ST5B.** See document *INVG3/OP88849*.

Check that the DC measurements are now correct and the fault disappears. **INVKG3_ST2, INVMG3_ST5B.**

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5.91. Fault code F255: BAT open detection

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	A sudden opening of the battery has been detected. This is a busbar fault.	
Applicable to	☐ HEM	☒ PCSM/MPCSM
	☐ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Incorrect fault parameterisation.

1. The fault configuration is incorrect and triggers when not applicable. If this occurs:
 - 1.1. Check by PowerCommsV the default value of the protection corresponding to battery opening (parameters of subgroup G5.15: Battery opening)]. If it is not the default value, check with the plant manager the appropriate value. **INVG3_ST1B.**

Incorrect battery parameterisation.

2. The battery configuration does not correspond to the plant hardware. In this case:
 - 2.1. Check by PowerCommsV with the plant manager that the values corresponding to the busbar start, stop and reconnection conditions are appropriate with respect to the installed batteries (subgroups [G2.1: Start], [G2.2: Stop] and [G2.3: Reconnection]). **INVG3_ST1B.**

Sudden battery opening caused by customer.

3. Customer has suddenly disabled the battery of an enabled busbar. If this occurs:
 - 3.1. Notify the plant manager not to open batteries without first disabling the busbar.
 - 3.2. If the battery has been disconnected due to a fault external to the inverter, it is possible that this fault has occurred on an exceptional event.
 - 3.3. If the customer has opened batteries in response to an inverter fault, (i.e. an abrupt or controlled stop), check the cause of this fault with the information indicated along this document. If the fault in question is configured with the default values, it is possible that the measure of opening batteries by the customer is excessive. In that case, check with the plant manager for the reason. **INVG3_ST1B.**

Faulty voltage measurements.

4. It is possible that either the central board (E35) or the expansion board (E36) is faulty. In this case:
 - 4.1. Exchange the connectors of the channel that is giving the fault with another one. **INVKG3_ST2, INVMG3_ST5A.** The measurement channels to be exchanged are two by two (positive and negative): on the central board (E35) J1 and J2, and on the expansion board (E36) J1 and J2, J3 and J4, J5 and J6. Check by PowerCommsV if the fault remains on the new channel that has been connected. **INVG3_ST1B.**
 - 4.2. If changing the channels does not fail, replace the corresponding board. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3IOP88849*.

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Failure in the measurement system.

- 5. It is possible that the difference caused by the central metering system is caused by a disconnection or failure of the metering wiring. To check this, it is necessary to check the module in which the fault has been triggered.

5.1. Check continuity between the central board (E35) and the corresponding bus. **INVKG3_ST2, INVMG3_ST5A.**

1 Busbar	2 Busbars	3 Busbars	4 Busbars
a y b	a, b, e y f	a, b, c, d, e y f	a, b, c, d, e, f, g y h

- a) Positive 1: check continuity from E35 terminal J1 to BBEP.82, from BBEP.82 to BBEP.81 and from BBIP.07 to BBIP.20.
- b) Negative 1: check continuity from E35 terminal J2 to BBEN.67, from BBEN.67 to BBEN.66 and from BBIN.07 to BBIN.20.



E35:J1



BBEP.82

EN



E35:J2



BBEN.67

- c) Positive 2: check continuity from E36 terminal J1 to BBEP.80, from BBEP.80 to BBEP.79 and from BBIP.05 to BBIP.20.
- d) Negative 2: check continuity from E36 terminal J2 to BBEN.65, from BBEN.65 to BBEN.64 and from BBIN.05 to BBIN.20.



E36:J1



BBEP.80



E36:J2



BBEN.65

- e) Positive 3: check continuity from E36 terminal J3 to BBEP.78, from BBEP.78 to BBEP.77 and from BBIP.03 to BBIP.20.
- f) Negative 3: check continuity from E36 terminal J4 to BBEN.63, from BBEN.63 to BBEN.62 and from BBIN.03 to BBIN.20.

EN



E36:J3



BBEP.78



E36:J4



BBEN.63

- g) Positive 4: check continuity from E36 terminal J5 to BBEP.76, from BBEP.76 to BBEP.75 and from BBIP.01 to BBIP.20.
- h) . Negative 4: check continuity from E36 terminal J6 to BBEN.61, from BBEN.61 to BBEN.60 and from BBIN.01 to BBIN.20.



E36:J5



BBEP.76

EN



E36:J6



BBEN.61

EN

- 5.2. Replace the affected cable section. Check tightness between plates and/or replace them. **INVKG3_ST2, INVMG3_ST5A.**

DC short-circuit.

6. It is possible that either the central board (E35) or the expansion board (E36) is faulty. In this case:
- 6.1. Exchange the connectors of the channel that is giving the fault. **INVKG3_ST2, INVMG3_ST5A.** The measurement channels to be exchanged are two by two (positive and negative): on the central board (E35) J1 and J2, and on the expansion board (E36) J1 and J2, J3 and J4, J5 and J6. Check by PowerCommsV if the fault remains on the new channel that has been connected. **INVG3_ST1B.**



- 6.2. If changing the channels does not fail, replace the corresponding board. **INVKG3_ST2, INVMG3_ST5A.** See document **INVG3IOP88849.**

Short-circuit on the outside of the busbar.

7. It is possible that the overcurrent is due to a short-circuit or shunt in the corresponding busbar. In this case:
 - 7.1. Perform a visual inspection inside the DU for damaged cables, blackened or bad busbars, bent busbars or any other foreign body that is making contact between positive and negative inside the DU. [INVKG3_ST2](#), [INVMG3_ST5A](#).
 - 7.2. With the batteries disconnected, connect an insulation meter between positive and negative of each input of the inverter and make a measurement with a maximum of 1000Vdc. Consider that previously it is necessary to disconnect the ground RFI cards and the ISOPRO measurement boards. [INVKG3_ST5B](#), [INVMG3_ST5B](#).



GND



- 7.2.1. If the measurement is less than $1\text{M}\Omega$, disconnect the input cables from each busbar and repeat the measurement.
- 7.2.2. If the measurement exceeds $1\text{M}\Omega$ when any of the input cables to the inverter are disconnected, this indicates that the cable has a shunt between positive and negative. If this occurs, inform the customer so that the channel can be repaired.
- 7.2.3. If the measurement does not improve after disconnecting all external cables, consult Power Electronics.

EN

5.92. Fault code F15: Synchronization

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The equipment has detected that it is connected to an AC mains when it is configured to generate AC voltage.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

Misconfiguration.

Check in PowerCommsV the parameter `[[Grid]]` within group `[SG4-Test mode]` and if its default value is "Yes", return to its default value. **INVG3_ST1B**.

If it defaults to "Yes" or does not reset the fault, contact Power Electronics.

EN

5.93. Fault code F33: Unable reconnect

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".			
Description	After a network event (voltage dip), when trying to activate the LVRT algorithm, the modules have tried to restart N times without success.			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

Analysis procedure

When a gap appears in the network (LVRT), the equipment stops the modules for a moment and restarts again with the conditions of the LVRT algorithm.

If the modules fail at startup, the restart is retry until the number of retries configured in `([Num max cons LVRT] within group [SG1.2-DSP service])` is reached in a time lower than `([max time cons LVRT] within group [SG1.2-DSP service])`, when F33 will be triggered.

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Verify the detected gaps.

1. Check in PowerCommsV ([Faults history] in group [SV7.1-Faults]), that the equipment fault corresponds to F33. **INV3_ST1B.**
2. Download in PowercommsV the logs from the SD card and view in `(Log Sys 2)` the states of the modules to verify the failure. **INV3_ST1B.**

It should be seen how at the moment of the gap in the network, the modules go to state 65 and then it will be seen how it performs different startups, where each step to state 6 will increase a count in the startup retries.

3. Check in PowercommsV that the number of times they enter state 6, matches the value set in `([Num max cons LVRT] within group [SG1.2-DSP service])` within the time set in `([max time cons LVRT] within group [SG1.2-DSP service])`, and then the equipment reports F33. If so, the protection has acted correctly. **INV3_ST1B.**

Incorrect fault configuration.

4. Check in PowerCommsV `([Num max cons LVRT] within group [SG1.2-DSP service])` if the fault setting is set to 0. **INV3_ST1B.**

If the configured value is different, consult with the plant manager to find out if this setting is justified.

If not, change the parameter to its default value.

5. Check in PowerCommsV if the parameters `([Sudden V drop] within group [SG1.2.a4-VRT Fault Avoid])` y `([Recovery] within group [SG1.2.a4-VRT Fault Avoid])` are set to their default values. **INV3_ST1B.**

If the configured values are different without a justification, modify the parameters to their default value. **INV3_ST1B.**

EN

Incorrect algorithm configuration.

6. Download in PowercommsV the logs from the SD card and view in (Log 1) the voltage value at the time of the grid gap (voltages Vrs, Vst, Vtr). **INVG3_ST1B**.
7. Check in PowerCommsV the parameter ([Threshold] within group [G4.3.1-LVRT General]). **INVG3_ST1B**.
8. Perform the following calculation to convert the parameter value to a voltage value.

$$(A*B) / 100$$

A is the parameter in PowerCommsV ([Nominal Vac] within group [SG10.1-Drive Select General]).

B is the parameter in PowerCommsV ([Threshold] within group [G4.3.1-LVRT General]).

9. Compare if the value seen in point 6 is higher than the value calculated in point 8, in that case the network is weak and has a low SCR, consult Power Electronics

EN

5.94. Fault code F104: PCB temperature

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	A temperature has been reached on the PCB of one of the modules higher than the fault threshold set in [G5.4.2.1.2: Umb fault temp mod].	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

This fault is reported when the PCB of any of the modules exceeds the configured overtemperature fault threshold.

1. Check the temperature reached by the module

1.1. From PowerCommsV, verify that the module fault corresponds to F104 (Parameter [SV7.4: Module events]). Check which module has reported the fault, or if there have been several of them (INVKG3_ST1B; INVMG3_ST1B).

1.2. Once the faulted modules are known, check in PowercommsV in the SD (LOG 3, columns 'mod_X PCB Temp') the T° reached by the module PCB at the time of the fault and verify that it is higher than the configured fault threshold (Subgroup [G5. 4.2: PCB]), (Subgroup [G5.4.2.1: Module]) (Parameter [G5.4.2.2.1.2-Umb fault temp mod]) (INVKG3_ST1B; INVMG3_ST1B). If so, the protection has acted correctly.

2. Incorrect setting of the fault threshold

Next, it should be checked if the fault threshold has been modified to a temperature that is easily reachable under normal conditions and therefore the fault jumps very easily.

2.1. From PowercommsV, check the configured fault threshold (Subgroup [G5.4.2.2: PCB]), (Subgroup [G5.4.2.1: Module]) (Parameter [G5.4.2.1.2-Umb fault temp mod]) (INVKG3_ST1B; INVMG3_ST1B). In case of detecting modified values without a justification for that particular case, modify the parameters to their default values (consult previously with the plant manager to know if this modification is justified). If the module temperatures previously checked in (Subgroup [SV12.X.3: Temperature Measurements]) (Parameter [SV12.X.3.1-Temperature Amb]) were lower than the default value of the protection (Parameter [G5.4.2.1.2-Umb fault temp mod]), the fault shall stop triggering for these conditions.

2.2. If it is the case that after the fault, it is not possible to reset it, although the T° of the modules has been reduced, it is possible that the hysteresis has been set to a very high value. To do this, check

in PowercommsV the hysteresis of the fault configured in (Subgroup [G5.4.2: PCB]), (Subgroup [G5.4.2.1: Module]) (Parameter [G5.4.2.1.3: Histère temp mod]). The module T° at which the fault can be reset shall be (Parameter [G5.4.2.2.1.2-Umb fault temp mod]) minus (Parameter [G5.4.2.1.3: Histère temp mod]). (INVKG3_ST1B; INVMG3_ST1B).

3. Ventilation problem

- 3.1. If the fault has occurred in most modules and their temperatures are excessively high, exceeding the default fault threshold of 75°C (Check in SD, LOG 3, columns 'mod_X PCB Temp'), it is a ventilation problem in the clean air circuit, on which the DU fans and LC filter depend. Check in PowerCommsV that the fans of the DU (Subgroup [G11.2.1: Fan 1]) (Subgroup [G11.2.2: Fan 2]) and LC filter (Subgroup [G11.2.7: Fan 7]) have correctly configured references, operating mode and fan type (Parameter [G11.2.1.1: Assigned Zone]) (Parameter [G11.2.1.2: Test]) (Parameter [G11.2.1.3: Type]) (INVKG3_ST1B; INVMG3_ST1B). If they are not at their default value, ask the customer if it is a special configuration for some reason and correct to solve the overtemperature faults.
- 3.2. Check the fan speed of the DU and LC filter. To do this, in PowercommsV you can check in the SD, LOG 3, the columns 'Set_Speed_Fan1', 'Set_Speed_Fan2', 'Set_Speed_Fan7', where the speed of these fans is indicated and must coincide with the speed setpoint sent by the control, shown in the columns 'Fb_Speed_Fan1', 'Fb_Speed_Fan2', 'Fb_Speed_Fan7' (INVKG3_ST1B; INVMG3_ST1B). You can also make this check from PowerCommsV, checking the setpoint and speeds of these fans, in the parameters indicated in the following table:

DU Fans		
Manufacturer	Default setting [G11.3.2]	Velocity [SV11.1.7 y SV11.2.7]
EBM-L7793B	80%	2324 rpm
TESOER-L8822	80%	2824 rpm
DELTA-9169	94%	2726 rpm

LC Filter Fans		
Manufacturer	Default setting [G11.6.2]	Velocity [SV11.7.7]
EBM-L7812A	80%	3184 rpm
TESOER-L8821	80%	3200 rpm
DELTA-9210	92%	2760 rpm
DELTA-L9587	80%	3360 rpm

- 3.3. If velocity oscillations are observed in any of them:

- 3.3.1. Check that there is no object obstructing the air intake of the LC and DU filter (INVKG3_ST2; INVMG3_ST5A).



Bad condition



Good condition

- 3.3.2. Check the correct rotation of the fan, making sure that there is no object obstructing its rotation or if there is any kind of friction. (INVKG3_ST2; INVMG3_ST5A).



Fan OK

EN

3.3.3. If the speed oscillation persists, replace the fan affected (INVKG3_ST2; INVMG3_ST5A).

4. Measurement error of the module PCB temperature sensor

If F104 has only tripped on one of the modules, while the other modules have lower temperatures, there may be a problem in the temperature measurement circuit of the module. To check this, do the following:

- 4.1. Compare in PowercommsV the measurement of all module boards in the SD (LOG 3, columns 'mod_X PCB Temp'). (INVKG3_ST1B; INVMG3_ST1B) If the Temperature read by the module that has reported the fault is very noisy and differs a lot from the temperatures of the rest of the modules, it is possible that the measurement sensor of that board is damaged or that an overheating is being generated inside that module for some reason.
- 4.2. With the equipment completely isolated from DC and AC and after checking that there is no accumulated energy in the module (DC voltage measurement between +Vbus: 'BBIP.20' and -Vbus: 'BBIN.20' with 1500V multimeter), remove the module and access the control card 'E02'. (INVKG3_ST2; INVMG3_ST5A)



EN

DC voltage measurement between +Vbus: 'BBIP.20' and -Vbus: 'BBIN.20'

- 4.3. Perform a visual inspection to check if there are any components in bad condition or signs of overheating ([INVKG3_ST2](#); [INVMG3_ST5A](#)).



Condition board OK

EN

- 4.4. If all is OK, replace module board 'E02' (**INVKG3_ST2**; **INVMG3_ST5A**) and then re-run the check in step 1 to ensure that all module boards are now reading a similar temperature value.

5.95. Fault code F171: Filter LC Current

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The equipment has detected that the current through filter C does not match to the theoretical current that should pass through the filter.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F172, W47.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Misconfiguration.

1. Check in PowerCommsV that the fault parameters configured in group **[G5.16-LC Filter]** are the default values. **INVKG3_ST1B**.
Otherwise, ask the plant manager why you have this configuration and if there is no cause, leave the default values of the fault.
2. Check in PowerCommsV (**[Benchs Number C]** within group **[SG10.5-Filter]**) that the number of capacitor banks of the equipment is set according to the technical specifications of the equipment depending on the size. **INVKG3_ST1B**.

Size	Number of banks
4	4
3	4
2	3

Otherwise, contact Power Electronics.

Capacitors in bad condition.

3. Check all capacitors of the capacitor filter (C25-C48). To do, measure with a 1500V multimeter the capacitance between: **INVKG3_ST2, INVMG3_ST5A**.
 - 3.1. Between L1 and L2 of the E20, E21, E22 and E23 discharge cards.
 - 3.2. Between L2 and L3 of the E20, E21, E22 and E23 discharge cards.
 - 3.3. Between L3 and L1 of the E20, E21, E22 and E23 discharge cards.
 The measured value must be $\approx 184\mu\text{F}$ (170-200 μF).

NOTE: The difference in the measured capacitance value between the points should not exceed 10 μF .

EN



L1-L2



L2-L3



L1-L3

If any measured value is not correct, replace the corresponding capacitor. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3IOP85367*.

Notch in bad condition (inductances L07, L08, L09).

4. Visually inspect the condition of the notch. Check that the plate insulators are in good condition (not broken, burned, ...). **INVKG3_ST2**, **INVMG3_ST5A**.

If the insulator is in bad condition, consult with Power Electronics about the possibility of replacing the notch. See document *INVG3IOP85367*.

5. Visually check the notch for excessive overheating. If the notch appears blackened or with overheating effects (blackened notch, melted stud insulators, notch sagging from its supports, ...). **INVKG3_ST2**, **INVMG3_ST5A**. Consult with Power Electronics about the possibility of replacing the notch. See document *INVG3IOP85367*.

EN



6. Check if the notch studs are short-circuited with the core and between the phases and the notch core. To do, perform the following checks: **INVKG3_ST2, INVMG3_ST5A.**
 - 6.1. Check with the 1500V multimeter that there is no continuity between the notch studs and the notch core (scrape lightly with the tip of the multimeter to make good contact). **INVKG3_ST2, INVMG3_ST5A.**



Espárrago y núcleo de L07

Repeat the measurement shown in the image on L08 and L09.

- 6.2. Check that there is no continuity between the core of the notch choke and each of the phases of the equipment. Measure between BBI1.12, BBI2.12 and BBI3.12 with the core of L07, L08 and L09. **INVKG3_ST2, INVMG3_ST5A.**



BBI1.12 and core L07

Repeat the measurement shown in the image between BBI2.12 and core L08, between BBI3.12 and core L09.

If there is continuity at any of the measured points, replace the damaged notch. See document *INVG3IOP85367*.

NOTE: It is recommended to replace all 3 notch chokes and not only the damaged one.

Hall sensors in bad condition.

7. If all the above checks have been successful, disable the fault and check in PowerCommsV ([C Filter] within group [SV3: Grid visualization]) that the current measurements in the LC filter are approximately the same, for this purpose the equipment must be started. **INVG3_ST1A**.
8. If any of the measurements differ by more than 10% from the rest, perform the following checks depending on which sensor is failing:

Phase	E35 board connector	RJ45 cable	Hall sensor + Board
R	J9	W056	HT05 + E24
S	J10	W057	HT06 + E25
T	J11	W058	HT07 + E26

- 8.1. Check the continuity of the RJ45 cables of the filter current HALL sensors (cables W056, W057 and W058) with the Ethernet cable tester. **INVKG3_ST2, INVMG3_ST5A**.



E35:J9



E24



E35:J10



E25

EN



E35:J11



E26

- 8.2. If there is no continuity, replace the cable. **INVKG3_ST2, INVMG3_ST5A.**
- 8.3. If the RJ45 cable does not fail, replace the board and the Hall sensor that is in bad condition, according to the table above. **INVKG3_ST2, INVMG3_ST5A.**
- 8.4. If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3IOP88849*.

Filter contactors in bad condition (KM01, KM02, KM03, KM04).

9. Remove the contactor front and perform the following checks on each contactor: **INVKG3_ST2, INVMG3_ST5A.**
 - 9.1. Internal contacts are in good condition and have no unusual, blackened spots.



- 9.2. Contacts are in correct position and none are bent or misplaced.

EN



- 9.3. The plate springs retain their elasticity and when the contactor is pressed, all the plates make good contact before the spring is fully compressed.



NOTE: The contactor must be properly mounted when it is returned to its normal state.

- 9.4. If any of the contactors have any of the problems described above, replace the corresponding contactor. See document *INV31OP85367*.

EN

5.96. Fault code F172: C LC Filter

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The capacity that has been calculated with the filter current and voltage does not correspond to the theoretical capacity of the equipment.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F171, W46.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Misconfiguration.

1. Check in PowerCommsV that the fault parameters configured in group **[G5.16-LC Filter]** are the default values. **INVKG3_ST1B**.
Otherwise, ask the plant manager why you have this configuration and if there is no cause, leave the default values of the fault.
2. Check in PowerCommsV (**[Benchs Number C]** within group **[SG10.5-Filter]**) that the number of capacitor banks of the equipment is set according to the technical specifications of the equipment depending on the size. **INVKG3_ST1B**.

Size	Number of banks
4	4
3	4
2	3

Otherwise, contact Power Electronics.

Capacitors in bad condition.

3. Check all capacitors of the capacitor filter (C25-C48). To do, measure with a 1500V multimeter the capacitance between: **INVKG3_ST2, INVKG3_ST5A**.
 - 3.1. Between L1 and L2 of the E20, E21, E22 and E23 discharge cards.
 - 3.2. Between L2 and L3 of the E20, E21, E22 and E23 discharge cards.
 - 3.3. Between L3 and L1 of the E20, E21, E22 and E23 discharge cards.
 The measured value must be $\approx 184\mu\text{F}$ (170-200 μF).

NOTE: The difference in the measured capacitance value between the points should not exceed 10 μF .

EN



L1-L2



L2-L3



L1-L3

If any measured value is not correct, replace the corresponding capacitor. **INVKG3_ST2**, **INVMG3_ST5A**. See document *INVG3IOP85367*.

Notch in bad condition (inductances L07, L08, L09).

4. Visually inspect the condition of the notch. Check that the plate insulators are in good condition (not broken, burned, ...). **INVKG3_ST2**, **INVMG3_ST5A**.

If the insulator is in bad condition, consult with Power Electronics about the possibility of replacing the notch. See document *INVG3IOP85367*.

5. Visually check the notch for excessive overheating. If the notch appears blackened or with overheating effects (blackened notch, melted stud insulators, notch sagging from its supports, ...). **INVKG3_ST2**, **INVMG3_ST5A**. Consult with Power Electronics about the possibility of replacing the notch. See document *INVG3IOP85367*.

EN



6. Check if the notch studs are short-circuited with the core and between the phases and the notch core. To do, perform the following checks: **INVKG3_ST2, INVMG3_ST5A.**
 - 6.1. Check with the 1500V multimeter that there is no continuity between the notch studs and the notch core (scrape lightly with the tip of the multimeter to make good contact). **INVKG3_ST2, INVMG3_ST5A.**



Espárrago y núcleo de L07

Repeat the measurement shown in the image on L08 and L09.

- 6.2. Check that there is no continuity between the core of the notch choke and each of the phases of the equipment. Measure between BBI1.12, BBI2.12 and BBI3.12 with the core of L07, L08 and L09. **INVKG3_ST2, INVMG3_ST5A.**



BBI1.12 and core L07

Repeat the measurement shown in the image between BBI2.12 and core L08, between BBI3.12 and core L09.

If there is continuity at any of the measured points, replace the damaged notch. See document *INVG3IOP85367*.

NOTE: it is recommended to replace all 3 notch chokes and not only the damaged one.

Hall sensors in bad condition.

7. If all the above checks have been successful, disable the fault and check in PowerCommsV ([C Filter] within group [SV3: Grid visualization]) that the current measurements in the LC filter are approximately the same, for this purpose the equipment must be started. **INVG3_ST1A**.
8. If any of the measurements differ by more than 10% from the rest, perform the following checks depending on which sensor is failing:

Phase	E35 board connector	RJ45 cable	Hall sensor + Board
R	J9	W056	HT05 + E24
S	J10	W057	HT06 + E25
T	J11	W058	HT07 + E26

- 8.1. Check the continuity of the RJ45 cables of the filter current HALL sensors (cables W056, W057 and W058) with the Ethernet cable tester. **INVKG3_ST2, INVMG3_ST5A**.



E35:J9



E24



E35:J10



E25



EN

E35:J11

E26

- 8.2. If there is no continuity, replace the cable. **INVKG3_ST2, INVMG3_ST5A.**
- 8.3. If the RJ45 cable does not fail, replace the board and the Hall sensor that is in bad condition, according to the table above. **INVKG3_ST2, INVMG3_ST5A.**
- 8.4. If the fault persists, replace the central control board (E35). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88849*.

Filter contactors in bad condition (KM01, KM02, KM03, KM04).

9. Remove the contactor front and perform the following checks on each contactor: **INVKG3_ST2, INVMG3_ST5A.**
 - 9.1. Internal contacts are in good condition and have no unusual, blackened spots.



- 9.2. Contacts are in correct position and none are bent or misplaced.



- 9.3. The plate springs retain their elasticity and when the contactor is pressed, all the plates make good contact before the spring is fully compressed.

EN



NOTE: The contactor must be properly mounted when it is returned to its normal state.

- 9.4. If any of the contactors have any of the problems described above, replace the corresponding contactor. See document *INV3IOP85367*.

EN

5.97. Fault code F211: Intern Isolation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.			
Description	The inverter has detected in the AC channel of the IMD an insulation resistance value lower than the configured fault threshold [G9.2.1.3-Fault Threshold] when the DU is open or the inverter is set to 'READY'.			
Applicable to	☒	HEM	☒	PCSM/MPCSM
	☒	HEMK	☒	PCSK/MPCSK
Associated faults or warnings	N/A			

Analysis procedure

This fault indicates that there is a derivation or loss of insulation in the AC part of the inverter. In order to detect the origin of this insulation loss, different checks will be carried out to narrow down the problem and detect the exact location.

1. Modified failure threshold

- 1.1. From PowercommsV, verify that the equipment fault corresponds to F211. To do this, check in [SV7.1.2: Fault history] and check that an F211 - Internal insulation fault has been recorded. (via PowerCommsV INVKG3_ST1B; INVMG3_ST1B).
- 1.2. Check the fault threshold configured on the channel that has reported the F211 [G9.2.1.2-Fault Threshold]. (via PowerCommsV INVKG3_ST1B; INVMG3_ST1B). Once the fault threshold is known, verify that the fault has been reported correctly by making an insulation measurement and comparing it to the fault threshold.. (INVKG3_ST5B; INVMG3_ST5B).
- 1.3. Check whether the failure threshold has been modified from its default value. If the configured threshold is different, consult with the plant manager to find out if this modification is justified. If not, modify the parameter to its default value. (Por medio de PowerCommsV INVKG3_ST1B; INVMG3_ST1B)

2. Detection of the loss insulation area.

It is necessary to identify where the earth fault is located. To do this, carry out the following actions:

- 2.1. Contact plant personnel and ensure that the inverter is isolated from AC and DC. (INVKG3_ST2; INVMG3_ST2).
- 2.2. It is necessary to perform different insulation measurements on the AC side of the inverter. For this purpose use a high voltage insulation measuring instrument (Megger). Set the meter to a voltage of 1000Vdc with a measurement time of 1 minute.. (INVKG3_ST2; INVMG3_ST2).

3. RFI board check (INVKG3_ST2; INVMG3_ST2)

- 3.1. Locate on the inverter the RFI board (E16) located in the LC filter cabinet.
- 3.2. Disconnect the earth cable (PEC36) from the RFI board (E16)

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- 3.3. Connect the earth cable of the insulation tester to the inverter chassis and the phase probe of the tester to phase L1 of the RFI board (E16).



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- 3.4. Set the meter to a voltage of 1000Vdc with a measurement time of 1 minute and perform the measurement.



- 3.5. Repeat the above steps in the phases L2 and L3.



L2



L3

- 3.6. If any of the measurements obtained is less than 100MOhms, the insulation is not correct and therefore it is necessary to progressively disconnect parts of the equipment that are connected to the AC busbar in order to find the point of failure. If a faulty component is detected, it is necessary to replace it. (INVKG3_ST2; INVMG3_ST2)

Below is the list of the most common components where this fault has occurred, together with the corresponding replacement procedure (if necessary)

- Inductance (L01). See document INVG3IOP85375.
- C Filter. See document INVG3IOP85367.
- Power stages. See document INVG3IOP88076.
- MV Transformer (T01). See document INVG3IOP85379.
- AASS transformer (T02). See document INVG3IOP85378.

5.98. Fault code F212: Extern Isolation

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “Introduction” .			
Description	The insulation monitoring has measured a resistance value on a DC channel lower than the fault threshold set on one of the DC channel of the IMD [G9.2.X.2] with the DU open.			
Applicable to	☒ HEM	☒ PCSM/MPCSM		
	☒ HEMK	☒ PCSK/MPCSK		
Associated faults or warnings	N/A			

Analysis procedure

This fault indicates that there is a shunt or loss of insulation in the DC part of the inverter or the field. To detect the origin of this loss of insulation, different checks will be carried out to narrow down the problem and detect the exact location. The following shows the available DC channels that the inverter will have depending on the number of busbars.

	Number of Busbars						
	1	2		4			
IMD Channel	1	1	3	1	2	3	4

1. Modified fault threshold

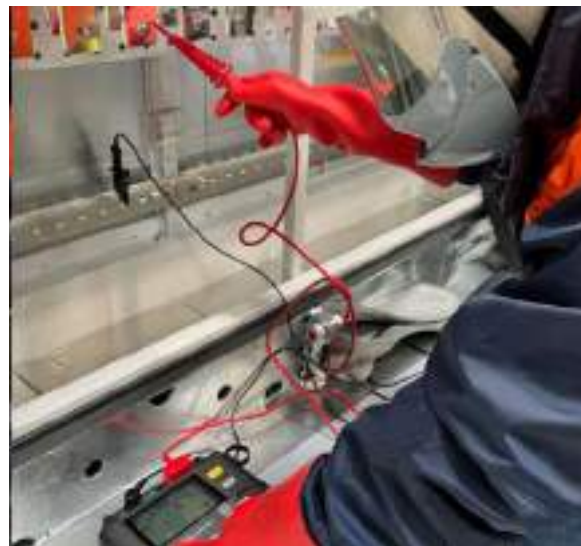
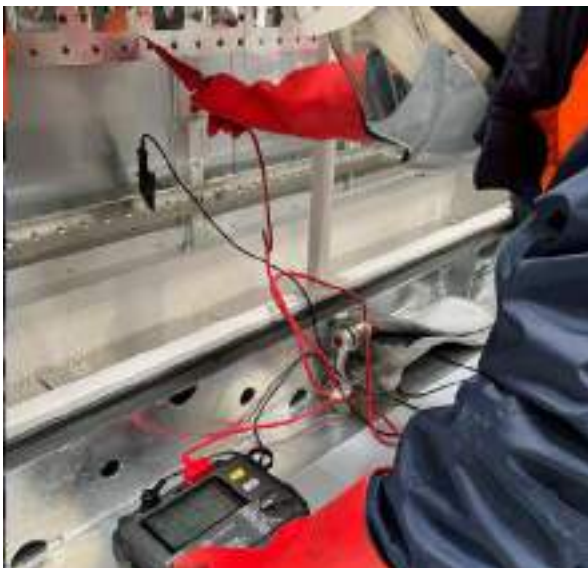
- 1.1. From PowercommsV, verify that the equipment fault corresponds to F212. To do this, check in [SV7.1.2: Fault history] and check which DC channel the fault corresponds to. This can be found by looking at the number shown in brackets for the fault, F212(X), where X is a number between 2-5, for channels 1-4 respectively. (INVKG3_ST1B; INVMG3_ST1B).
- 1.2. Check the fault threshold configured on the channel that has reported the F212 [G9.2.X.2-Fault Threshold]. (INVKG3_ST1B; INVMG3_ST1B). Once the fault threshold is known, verify that the fault has been reported correctly by making an insulation measurement and comparing it to the fault threshold as follows.
- 1.3. To carry out this measurement, it is necessary to have 100V external voltage, to have the AC and DC open (all disconnectors in OFF position), the parameter [SG10.6.1-DU Type = IMD or GFDI + IMD] depending on the equipment and [G9.1.1-Enabled =YES]. (INVKG3_ST3A; INVMG3_ST3A).
 - 1.3.1. For isolated poles inverter, with the above conditions, the measurement of each channel will start automatically, so wait for the measurements on the available DC channels [SV9.2.X] to be updated. This measurement must be lower than the threshold configured in G9.2.X.2. If so, the fault is jumping correctly. (INVKG3_ST3A; INVMG3_ST3A).
 - 1.3.2. For an equipment with ground pole, it will be necessary to carry out the manoeuvre manually, to do this, open the GFDI [G9.3.0.3-GFDI Output] and check its backfeed in [SV9.3.6-GFDI Retro], when it indicates that it is open, select one of the DC channels in [G9.3.0.2-Channel Measurement] according to the previous table and enable the manual measurement of the IMD in [G9.3.0.1-Enable]. In case of more than one busbar, repeat this measurement for all DC channels. Check the resistance measurement in [SV9.2.X.3] according to the DC channel selected in [G9.3.0.2-Channel Measurement]. It must be lower than the threshold configured in G9.2.X.2. If so, the fault is tripped properly. (INVKG3_ST3A; INVMG3_ST3A).
 - 1.3.3. Check if the fault threshold has been modified from its default value. (INVKG3_ST3A; INVMG3_ST3A). In the event that the configured threshold is different, consult with the plant manager to find out if this modification is justified. If not, modify the parameter to its default value.

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2. External inverter bypass check

To narrow down the problem, it is necessary to know whether the earth fault is present inside the equipment on the DC side or in the external wiring. To do this, carry out the following actions:

- 2.1. Contact the plant personnel and make sure that the DC input channels of the inverter are isolated (open strings) as well as the AC side.. (INVKG3_ST5B; INVMG3_ST5B).
- 2.2. Perform an insulation measurement at the entry point of the DC cables. Use a high voltage insulation measuring instrument (Megger) for this purpose. Set the meter to a voltage of 1000Vdc with a measurement time of 1 minute. Measure between BBEP.X and earth and between BBEN.X (X being 1 to 24) and earth to corroborate the insulation loss read on the equipment. (INVKG3_ST5B; INVMG3_ST5B). If the measurement is incorrect (lower than the fault) the problem can be found inside the inverter or in the sections from the string box to the inverter. To narrow down the problem, carry out the following step.



If possible disconnect the DC input wiring from the inverter or otherwise remove the fuses connected to the DC input and repeat the above measures. (INVKG3_ST5B; INVMG3_ST5B). If at this point a correct insulation measurement is obtained (higher than the fault), the problem lies in the field wiring. Otherwise the shunt will be inside the equipment and we must continue analysing the cause of the fault.



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3. IMD wiring derived from chassis

Check if the F212 is tripping due to a shunt in the wiring of the DC measurements connected to the IMD.

- 3.1. Perform a visual inspection of the wiring of the channel that is reporting the fault, to see if there is a visible earth fault. If a damaged cable is detected near an enclosure, replace it and route it so that it is contained in the ducts. (INVKG3_ST5B; INVMG3_ST5B). If no defects have been found in the wiring, proceed with the steps below..
- 3.2. Use an insulation tester (megometer) to take a measurement on the DC circuit breaker corresponding to the channel that is detecting the insulation fault. The following table shows the assignment of each channel to the DC circuit breaker according to the number of busbars. (INVKG3_ST2; INVMG3_ST5A).
- 3.3. Use an insulation tester (megometer) to take a measurement on the DC circuit breaker corresponding to the channel that is detecting the insulation fault. The following table shows the assignment of each channel to the DC circuit breaker according to the number of busbars. (INVKG3_ST2; INVMG3_ST5A).

	Number of busbars						
	1	2		4			
IMD channel	1	1	2	1	2	3	4
DC circuit breaker	PDC1	PDC1	PDC3	PDC1	PDC2	PDC3	PDC4
Measure points	P: BBEP.82 N: BBEP.67	P: BBEP.82 N: BBEP.67	P: BBEP.78 N: BBEP.63	P: BBEP.82 N: BBEP.67	P: BBEP.80 N: BBEP.65	P: BBEP.78 N: BBEP.63	P: BBEP.76 N: BBEP.61
Resistor board	E71	E71	E73	E71	E72	E73	E74
E70 IMD board: Connector	J5	J5	J3	J5	J4	J3	J2

- 3.4. Take a measurement between BBEP.X (X will be 82,80,78 or 76 depending on the IMD channel selected in the table above) and earth. In case of detecting an insulation measurement similar to the one read on the inverter previously [SV9.2.X.3], disconnect the BBEP.X:2. (INVKG3_ST2; INVMG3_ST5A).
- 3.5. Take a measurement between BBEP.X (X will be 67,65,63 or 61 depending on the IMD channel selected in the table above) and earth. In case of detecting an insulation measurement similar to

the one read on the equipment previously [SV9.2.X.3], disconnect the BBEP.X:2. (INVKG3_ST2; INVMG3_ST5A).



BBEP.82



BBEP.80

EN



BBEP.78



BBEP.76



EN

BBEN.67



BBEN.65



BBEN.63



EN

BBEN.61



GND

EN

- 3.6. Measure the insulation again at the same points as before. If the insulation measurement is now correct, then the fault is in the IMD measurement circuit. (INVKG3_ST2; INVMG3_ST5A).

**The variable Y appearing in the identifiers (e.g. E7Y:J1.1) shall be 1,2,3 or 4 depending on the IMD channel selected in the table above.*

**The variable Z appearing in the identifiers (e. g. E70:JZ.1) shall be 2,3,4 or 5 depending on the IMD channel selected in the table above..*

- 3.7. Disconnect connectors E7Y:J1.1 and E7Y:J1.2 and measure the earth insulation measurement on each of them. If the measurement on either wire is less than the fault, replace the wiring from BBEP.X:2 to E7Y:J1 or E7Y:J2. If a good insulation measurement is obtained (higher than the fault) continue with troubleshooting. (INVKG3_ST2; INVMG3_ST5A).

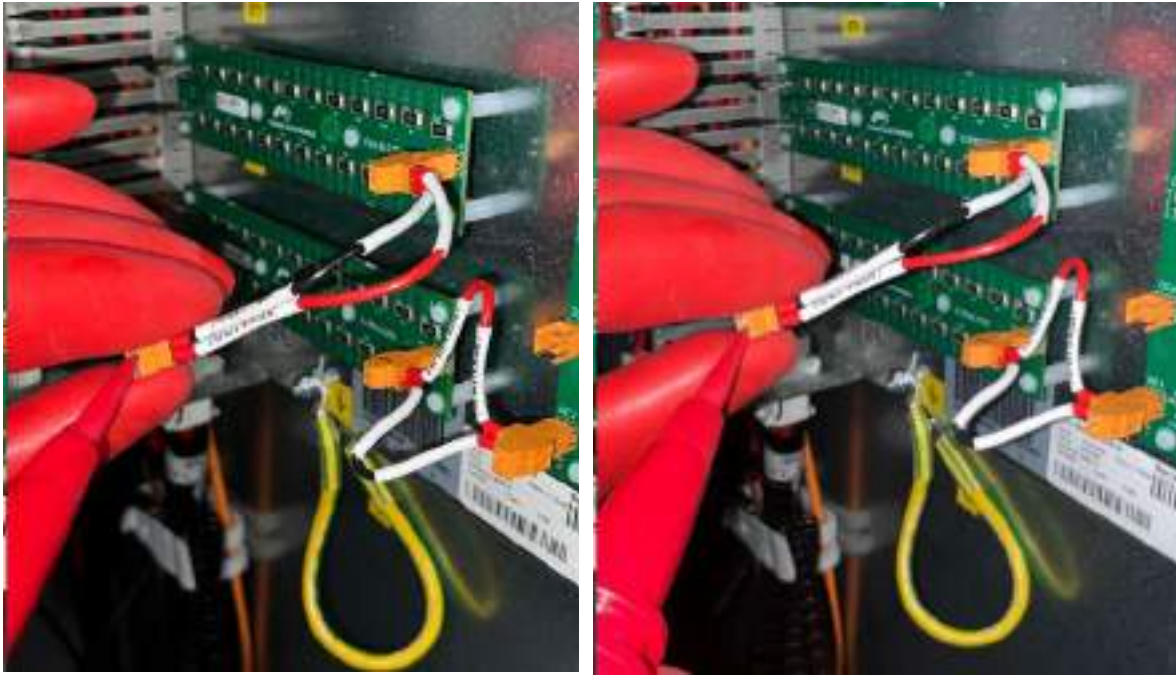




- 3.8. Reconnect J1 and J2 of E7Y. Disconnect wires E7Y:J3:1 and E7Y:J3:2 and measure from connector J3:1 and J3:2 to ground. In case of incorrect insulation measurement, replace the E7Y board. In case of good insulation measurement (above the fault) continue with troubleshooting.. (INVKG3_ST2; INVMG3_ST5A).



- 3.9. Reconnect E7Y:J3:1 and E7Y:J3:2. Disconnect wires E70:JZ:1 and E70:JZ:2 and measure between each wire and earth. In case of incorrect insulation measurement on either of them, replace the wiring from E7Y:J3:1 to E70:JZ:1 or E7Y:J3:2 to E70:JZ:2. (INVKG3_ST2; INVMG3_ST5A).



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3.10. If the insulation measurement above is correct, the problem lies in the insulation measurement board (E70), replace it and check if the problem has been solved. See document *INVG3/OP88849*. (*INVKG3_ST2*; *INVMG3_ST5A*).

5.99. Fault code F28: Passive Anti-island

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The passive island detection algorithm has detected a possible island condition in the AC part of the network.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	Related to: F28: Active Anti-island. F5: High Vac. F6: Low Vac. F7: High frequency. F8: Low frequency.	

Analysis procedure

The Anti-island fault occurs when the inverter detects a series of anomalous grid conditions and trips the inverter, stopping it and avoiding the option of it becoming an island. In other words, injecting power into the grid but without the grid being present (this indicates that the plant's head-end disconnect or part of the electrical distribution grid has been opened).

In this case it would occur when the grid frequency has a significant amplitude for a long time (order of magnitude 10-30 seconds approximately). This causes the inverter to fail and stop abruptly to avoid working under these conditions due to the existence of a possible island.

This fault is disabled by default. If the inverter were to fail due to these faults, then:

1. Check in PowercommsV that the inverter parameterisation is set to its default value (parameter [Mode] of subgroup [G5.3:Anti islanding]). In case the plant requires a special configuration for this algorithm, check that it is according to the requirements. **INVG3_ST1B**.
2. In case the Anti-island protection is activated, its parameterisation is as required and a sporadic fault has occurred due to a network event, it should not be problematic as it is the function of the algorithm itself. **INVG3_ST1B**.
3. If the Anti-island protection is activated, its parameterisation is as required, there is a sudden disconnection of the system from the mains and a device is left running, this indicates a dangerous and undesired situation. If this occurs:
 - 3.1. Check in PowercommsV that the configuration of the parameters corresponding to the "test mode" are in their default values (group SG4: Test mode). **INVG3_ST1B**.
4. If the Anti-Island protection is activated, its parameterisation is as required and there are frequent Anti-Island faults without apparent cause, **INVG3_ST1B** this is the case:
 - 4.1. Contact Power Electronics.

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5.100. Fault code F95: Source PCB

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	The module FPGA has detected that the 24Vdc voltage has dropped below 22Vdc or the 15Vdc voltage has dropped below 14Vdc.	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	F71-F84.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

15Vdc source failure.

1. Download the logs and the blackbox from the SD card after the failure and check if the 15Vdc voltage of the module has the correct value (15Vdc) and if this value remains above 14Vdc in the history. **INVKG3_ST3A, INVMG3_ST3A.**

If the voltage is below at any time, replace the module control board (E02). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88076*.

Short circuit in a GATE DRIVE.

2. If the fault persists after replacing the module control board (E02), replace the delivery board (E01). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88076*.
3. If not, replace the GATE DRIVE board (IGBT01-09) according to the module that is failing. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88076*.

24Vdc source failure.

4. Download the logs and blackbox from the SD card after the failure and check if the 24Vdc voltage is below 22Vdc at any time in the history. **INVKG3_ST3A, INVMG3_ST3A.**

In this case, carry out the following checks:

- 4.1. Measure with a 1500V multimeter the output voltage of the voltage source G01 between the positive and negative terminals. The measured value should be $\approx 24\text{Vdc}$ ($\pm 1\text{Vdc}$). **INVKG3_ST3A, INVMG3_ST3A.**

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- 4.2. If the measured value is not correct, adjust the output voltage of the G01 source by means of the regulation potentiometer. **INVKG3_ST3A, INVMG3_ST3A.**

If it cannot be adjusted correctly, replace the power supply G01. **INVKG3_ST2, INVMG3_ST5A.**

Power connector failure.

5. Check that the connector J15 of the module control board (E02) is in good condition.



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- 5.1. If the connector has play, replace it. **INVKG3_ST2, INVMG3_ST5A.**
- 5.2. If it is not properly connected, adjust it and reset the equipment. **INVKG3_ST2, INVMG3_ST5A.**
- 5.3. If the failure is recurrent and the connector is constantly out of adjustment, replace the connector. **INVKG3_ST2, INVMG3_ST5A.**

Wiring failure.

6. Lower the circuit breaker corresponding to the module that has failed. **INVKG3_ST2, INVMG3_ST5A.**

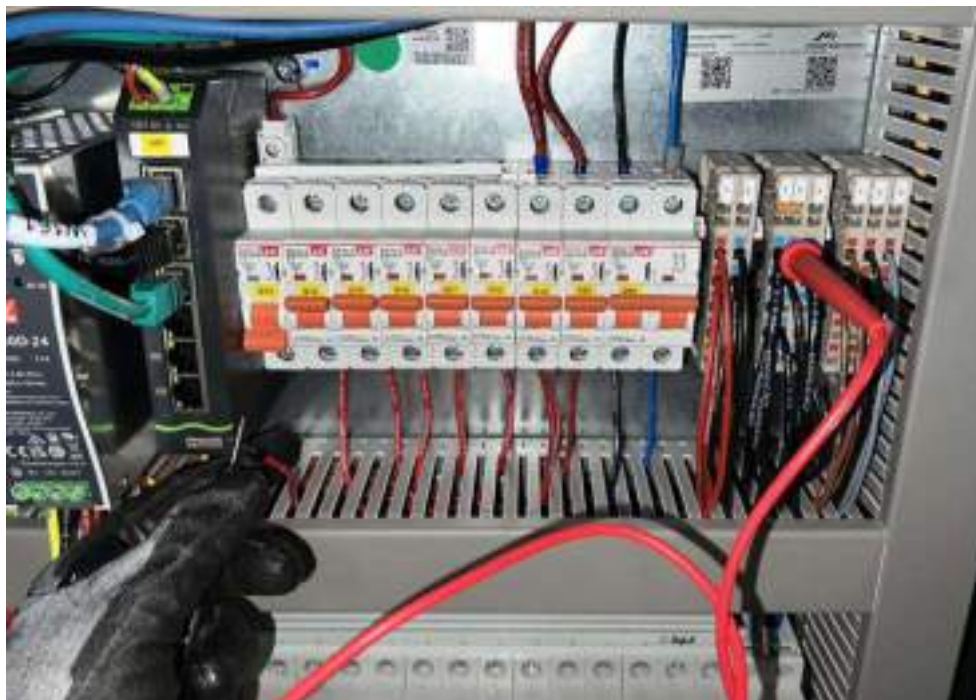
Module	24Vdc power supply circuit breaker
1	Q13
2	Q14
3	Q15
4	Q16

7. Make a short-circuit with a plug between positive and negative of the J15 connector of the module control board (E02). **INVKG3_ST2, INVMG3_ST5A.**



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8. Remove the wire from pin 2 of the MCB of the failed module and measure with a 1500V multimeter (magnitude $m\Omega$) between this wire and pin 1 or 2 of connector X11. **INVKG3_ST2, INVMG3_ST5A.**



If the measured value is $> 0.5\Omega$, replace the cable. **INVKG3_ST2, INVMG3_ST5A.**

24Vdc circuit breaker has tripped.

9. Reset the circuit breaker. Refer to the table above. **INVKG3_ST3A, INVMG3_ST3A.**
10. If it trips again, measure with the 1500V multimeter between pin 2 of the MCB and the negative pin of the 24Vdc power supply output (G01). **INVKG3_ST2, INVMG3_ST5A.**



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If the measured value is $< 6\Omega$, perform the following checks:

- 10.1. Disconnect connector J15 from the module control board (E02) and remeasure. **INVKG3_ST2, INVMG3_ST5A.**
- 10.2. If the measured value is $> 10M\Omega$, replace the module control board (E02). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP88076*.
- 10.3. If the measured value is $< 10M\Omega$, replace the wiring. **INVKG3_ST2, INVMG3_ST5A.**

5.101. Fault code F238: AEMO comms lost

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	Communications failure related to AEMO recovery ramps.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	Related to F51: PPC Comms.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

The PPC or SCADA does not send the Heartbeat script or does not reach the inverter because the IP is misconfigured.

1. Check via PowerCommsV that the inverter configuration and the correct IP are set according to the project configuration (subgroup [G6.2: Ethernet]). If the corresponding IP is not configured, change it according to the project/customer documentation. [INV3_ST1B](#).
2. Check via PowercommsV that the counter (parameter [PPC Heartbeat Reg] of subgroup [G6.5: PPC]) is constantly incremented. If it increments, this means that the inverter is well configured at the communications level and possibly a sporadic hardware fault. If it does not increment, go to the next point. [INV3_ST1B](#).

Rule out a possible problem with the configuration of access permissions to the inverter.

3. If the counter does not increment, check via PowercommsV that the IP filter is not enabled (parameter [Hab IP filter], within group [G6.5: PPC] and that the SCADA IP is not configured as "Black List" in the access configuration (parameter [IP User Level], within group [SG14: Access Config]). [INV3_ST1B](#).

The counter is not incremented. There may be a problem in wiring communications at the client software or central control board level.

4. To rule out a configuration problem, connect via PowercommsV an external computer via Ethernet to the inverter and send values other than 0 to the Modbus address corresponding to the [PPC Heartbeat Reg] parameter in subgroup [G6.5: PPC]. [INV3_ST1B](#).
 - 4.1. If the Heartbeat counter increments, it is a problem with the SCADA software or the communication installation. Discuss the problem with the customer to find a solution. [INV3_ST1B](#).
 - 4.2. If the counter does not increment, ping the central control board. If it is correct, via PowercommsV check the [G6.2: Ethernet] configuration and IP addresses. [INV3_ST1B](#). If the ping fails, change the central control board (E35). [INVKG3_ST5A](#), [INVMG3_ST5A](#). See document [INV3/OP88849](#).
 - 4.3. If the problem persists, contact Power Electronics.
5. Check customer's network RJ45 connectors for mis-crimped, bent or damaged connections. [INVKG3_ST5A](#), [INVMG3_ST5A](#). If a defect is found, report it to the customer to be fixed.

The counter is incremented but irregularly over time. If this happens, it is possible that there is a hardware communication problem with the SCADA.

6. Connect a computer and send different values via PowerCommsV from 0 to the computer's Heartbeat on a regular basis. [INV3_ST1B](#).
 - 6.1. If it increases regularly, the problem is with the external communications installation of the inverter. Talk to the customer to solve the problem.
7. If the communication is still irregular and Heartbeats are lost, the problem is the central control board (E35). In this case, replace. [INVKG3_ST5A](#), [INVMG3_ST5A](#). See document [INV3/OP88849](#).

8. If the problem persists after replacing the board, **INVKG3_ST5A**, **INVMG3_ST5A** contact Power Electronics.

If sporadic losses are detected, there could be a problem with the inverter's communications switch U01 where the customer cable runs.

9. Disconnect the customer cable from switch U01 and connect it directly to the central control board (E35). **INVKG3_ST5A**, **INVMG3_ST5A**. Observe if communications are re-established via PowerCommsV. **INVKG3_ST3A**, **INVMG3_ST3A**.



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- 9.1. If the Heartbeat counter increases regularly, the problem is the switch of the inverter. In this case, replace it. **INVKG3_ST5A**, **INVMG3_ST5A**. See document *INVG3IOP89282*.
- 9.2. Check if the problem has been solved by means of PowerCommsV. **INVKG3_ST3A**, **INVMG3_ST3A**

If the PPC Heartbeat Register is increased and regularly, it indicates that the communications are correct.

10. Check the PowerCommsV configuration of the communications timeout (parameter [AEMO lost detection], within group [G6.5: PPC]). **INVKG3_ST3A**, **INVMG3_ST3A**. This time must be much longer (4 times or more) than the increment time of the PPC Heartbeat Register (in case of doubt, set the default value of the timeout).
- 10.1. If the Heartbeat increments regularly but very slowly, talk to the customer to find out if this time is correct (there is no bug in the SCADA software) and if so, change the timeout configuration according to the customer's requirements via PowerCommsV. **INVKG3_ST3A**, **INVMG3_ST3A**.

11. Review the configuration by PowerCommsV (parameters [max losses] and [AEMO max time] of subgroup [G6.5: PPC]) to be in accordance with plant requirements and project/customer documentation. **INVKG3_ST3A, INVMG3_ST3A.**

Possible interference (noise) on the communication cables making it unable to communicate properly.

12. Review the configuration by PowerCommsV (parameters [max losses] and [AEMO max time] of subgroup [G6.5: PPC]) to be in accordance with plant requirements and project/customer documentation. **INVKG3_ST3A, INVMG3_ST3A.**

interference (noise) on the communication cables making it unable to communicate properly.

13. Instruct the customer to check the continuity of their RJ45 communications cables with a RJ45 tester. If there are any problems, they should fix them. **INVKG3_ST5A, INVMG3_ST5A.**

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5.102. Fault code F252: RFI OverCur Persistent

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction" .	
Description	The inverter has read a current for an RFI greater than the configured threshold or greater than 50A.	
Applicable to	☐ HEM ☐ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	Related to F253:Imbalance V.	

Analysis procedure

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Drive Select configuration.

1. Check by PowerCommsV that the busbar number and product type configurations match the actual hardware of the inverter (parameter [Number of busbar], [HW version number] in [SG10: General]). **INVKG3_ST1B**.
 - 1.1. If they do not match, configure according to the hardware of the inverter. **INVKG3_ST1B**. Note: this fault only applies to PCS inverters with more than 1 busbar. If it trips if it is not a PCS with more than one busbar, contact Power Electronics.

RFI protection configuration.

2. Check by PowerCommsV that the RFI current fault parameter settings are at their default value (subgroup [G5.13: Busbar unbalance]). If they are not at their default value, consult the plant manager if there is a reason why they are at that value, and if there is no reason, set to their default value. **INVKG3_ST3A, INVKG3_ST3A**.

Configuration of voltage unbalance protections.

3. Check that the DC voltage unbalance protection configuration is set to its default value (subgroup [G5.13: Busbar unbalance]). If it is not at its default value, ask the plant manager if there is a reason why it is at that value, and if there is no reason, set it to its default value. **INVKG3_ST3A, INVKG3_ST3A**.

Note: The voltage unbalance may not be set to a value greater than its default value without Power Electronics authorisation, if the customer has made this change, inform Power Electronics.

Limit unbalance power configuration.

4. Check by PowerCommsV that the power unbalance limit setting is at its default value (subgroup [G3.6: Reference Differences]). If it is not at its default value, ask the plant manager if there is a reason why it is at that value, and if there is no reason, set it to its default value. **INVKG3_ST3A, INVKG3_ST3A**.

Note: The reference difference limit cannot be set to a value higher than its default value without authorisation from Power Electronics. If the customer has made this change, inform Power Electronics.

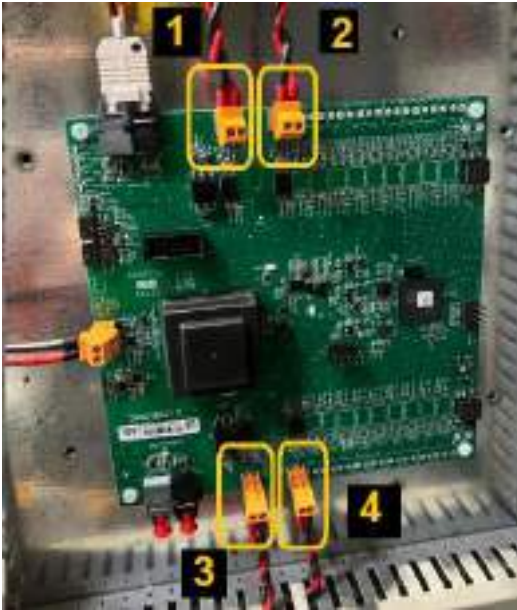
Measuring cable broken or disconnected.

5. Visually check that the board connectors E67 and J3 of the board are properly connected. If not, tighten the connectors.

Busbar failed	RFI board	Connector of the E67 board
1	E53	J29
2	E52	J28
3	E51	J11
4	E50	J10

EN

6. Check continuity between the following points: **INVKG3_ST2, INVMG3_ST5A.**
- Between connector Jxx:2 on the E67 board and pin J3:2 on the RFI Exx board.

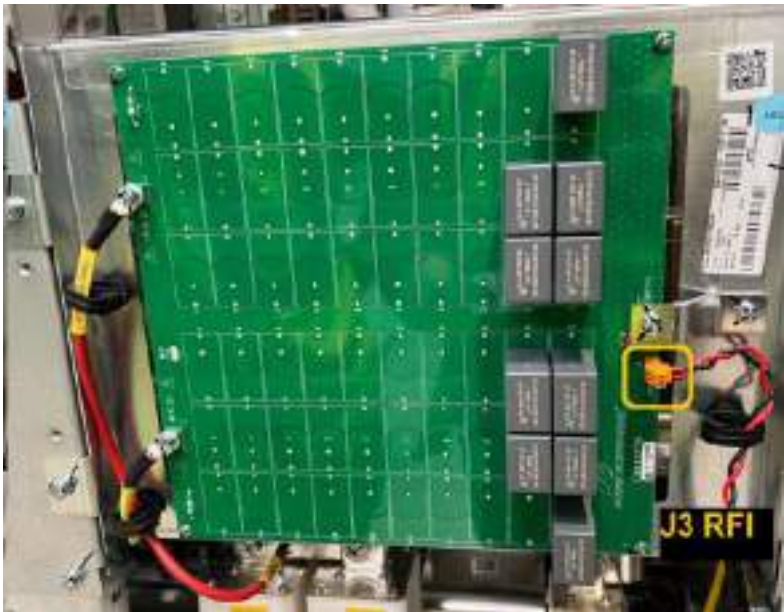


E67:JXX



E67:JXX:1 / E67:JXX:2

- Between connector Jxx:1 on the E67 board and pin J3:1 on the RFI Exx board.

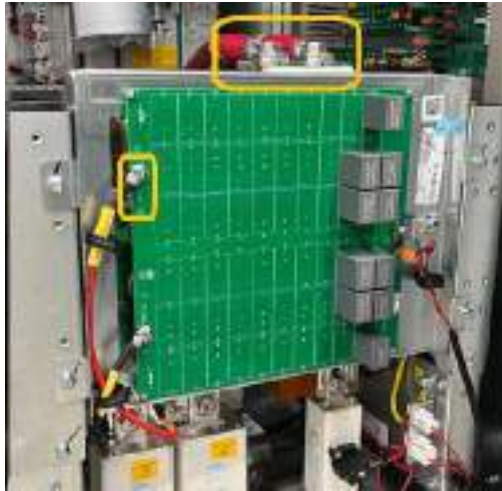


- If any of the points do not give continuity, replace the wiring.

Positive or negative cable disconnected.

7. Check that the cables connected to terminals J1 and J2 of the RFI board are connected to their corresponding busbar.

Busbar failed	Positive RFI board cable	Negative RFI board cable
1	J1 of the board E53 with BBEP.57	J2 of the board E53 with BBEN.57
2	J1 of the board E52 with BBEP.56	J2 of the board E52 with BBEN.56
3	J1 of the board E51 with BBEP.55	J2 of the board E51 with BBEN.55
4	J1 of the board E50 with BBEP.54	J2 of the board E50 with BBEN.54



BUS (+)



BUS (-)

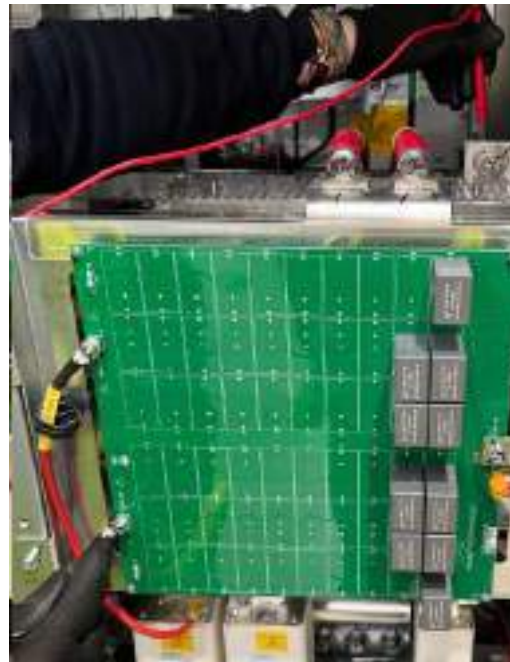


BUS (-)

8. Check continuity between the above points according to the busbar that has failed. **INVKG3_ST2**, **INVMG3_ST5A**.



BUS (+)



BUS (-)

EN

RFI board damaged.

9. Check the RFI board corresponding to the fault according to the fault instance and busbar according to the table. Disconnect the board cables from connector J1 and J2 and measure the capacitance with the 1500V multimeter between points J1-J4 and J2-J4. The capacitance in all cases should be approximately 5uF (between 5.5uF and 4.5uF). If the board is not between these values, replace the appropriate RFI board. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP88849*.

Busbar failed	RFI board reference
1	E53
2	E52
3	E51
4	E50





Damaged DU board (E67).

10. Connect a current probe with an oscilloscope between connector J4 of the faulty RFI board and ground **INVKG3_ST2; INVMG3_ST5A**.
11. Start the inverter totally enclosed by means of PowerCommsV. **INVG3_ST1A**. Check the current measured by RFI (parameters [RFI current x] of group [SV10: DU]).
12. If the current measured with the current probe does not correspond in RMS with the measurement displayed by PowerCommsV, replace the DU board (E67) **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88849*.

Damaged shock coils.

13. Dismantle all wires/plugs that connect to the coil depending on the busbar that has failed and check that between pins 2-6 and 4-8 with the LCR meter that the coil inductance is approximately 400uH (measure at 100Hz). **INVKG3_ST2, INVMG3_ST5A**.

Busbar failed	Coil
1	L03
2	L04
3	L05
4	L06

14. If the measurement does not correspond to approximately 400uH, consult Power Electronics about the possibility of changing the choke inductance.

EN

Module with damaged or non-firing IGBT.

15. Check the IGBT status of the affected busbar module(s). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP88076*.

Main choke damaged.

16. If all the checks have gone well, and the inverter has only one module due to busbar, exchange this power stage for another one that works properly. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3IOP85369*.

If the fault is still present in the busbar even after replacing the busbar with a power amplifier that works properly, this means that the main choke is damaged. Consult Power Electronics about the possibility of replacing the main choke (L01). See document *INVG3IOP85375*.

EN

5.103.Fault code F102: Unbalanced current

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	There is an unbalance of current between phases higher than the equipment protection set in [G5.2.1.2: Unbalance I].	
Applicable to	☒ HEM ☒ HEMK	☒ PCSM/MPCSM ☒ PCSK/MPCSK
Associated faults or warnings	N/A	

Analysis procedure

Possible causes of failure are:

1. Incorrect setting of the fault threshold.

It should be checked whether the fault threshold has been set to a minimum unbalance, the protection being very strict and triggering the fault with minimum unbalances. The fault setting is done by the following parameters:

- [G5.2.1.3- Unbalance I min]: Sets a current threshold above which the protection will operate, to avoid tripping at low power levels where there is less accuracy in the measurement.
- G5.2.1.2- Unbalance I]: Indicates the admissible error threshold between the module output currents ($I_{unbal} = I_{rst_max} - I_{rst_min} / 920$).
- G5.2.1.4-Delay Unbalance I]: Time that the threshold set in G5.2.1.2 must be exceeded for the fault to trip.

- 1.1. From PowerCommsV, verify that the module fault corresponds to F102. To do this, check which module has reported the fault, or if there have been several (Subgroup [SV7.4: Module events]) (Subgroup [SV7.4.X: Module X events]). (INVKG3_ST1B; INVMG3_ST1B).
- 1.2. Check in the SD the currents read at the time of the fault. To do this, check in LOG2, the currents lac_1, lac_2 and lac_3 of each of the modules indicated in the column 'module'. It should be seen that the rms values shown for the 3 phases differ in some of them. (INVKG3_ST1B; INVMG3_ST1B). If this is the case, the equipment has been correctly protected from the unbalance caused.
- 1.3. From PowerCommsV, check the fault threshold configured in (Subgroup [G5.2: Quality]) (Subgroup [G5.2.1: Unbalanced]) (Parameter [G5.2.1.2: Unbalance I]), (Parameter [G5.2.1.3: Unbalance I min]) and (Parameter [G5.2.1.4: Delay unbalance I]). (INVCG3_ST1B; INVMG3_ST1B). In case of detecting modified values without a justification for that particular case, modify the parameters to their default values (consult previously with the plant manager to know if this modification is justified).
2. Voltage unbalance

If we have a certain voltage unbalance, without triggering the voltage unbalance fault (F23), it could generate an unbalance in the output currents of the equipment. To check this, carry out the following checks:

- 2.1. Check in the SD the Vac voltages read at the time of F102. To do this, check in LOG1, the values of Vrs, Vst and Vtr. If an appreciable unbalance is seen in the AC voltages, it is possible that the current unbalance is a consequence of this. (INVKG3_ST1B; INVMG3_ST1B).

EN

- 2.2. Check in PowerCommsV the configuration of the voltage unbalance protection to check if it has been modified to a very high value and therefore the equipment allows a very large voltage unbalance (Subgroup [G5.2.1: Unbalanced]) (Parameter [G5.2.1.5: Unbalance V]). In case of detecting modified values without a justification for that particular case, modify the parameters to their default values (INVKG3_ST1B; INVMG3_ST1B) (consult previously with the plant manager to know if this modification is justified).

3. Erroneous module current reading

If one of the module current measurement hall sensors gives an erroneous measurement, this will cause the F102 to trip, as we will have a current measurement different from the others.

- 3.1. Check in the SD the currents read at the moment of the fault. To do this, check in LOG2, the currents lac_1, lac_2 and lac_3 of each of the modules indicated in the 'module' column (INVKG3_ST1B; INVMG3_ST1B).

- 3.2. Once you have detected which module and phase with different rms value has caused the fault, perform the following actions:

3.2.1. Make sure that the equipment is isolated from DC and AC. (INVKG3_ST2; INVMG3_ST5A).

3.2.2. If you have a Rogowski probe and oscilloscope, install the probe on the module output board of the detected phase. (INVKG3_ST2; INVMG3_ST5A). Otherwise skip to step 2.4.

3.2.3. After installing the probe, carry out the equipment start-up process. (With the equipment totally closed INVKG3_ST1A; INVMG3_ST1A). With the equipment running, compare whether the rms current read by the probe and the one read in PowerCommsV in SV12.X.2 Y coincide (X being 1, 2, 3 or 4, depending on the module detected and Y 1, 2 or 3 depending on the phase) (INVKG3_ST1A; INVMG3_ST1A). If the measurements do not coincide considering an admissible error of < 30A, it means that the HT0X measurement sensor (X being 1, 2 or 3 depending on the phase) is defective and must be replaced (INVKG3_ST2; INVMG3_ST5A).

- 3.2.4. If you do not have a current probe, compare from PowerCommsV if the currents read by the equipment at the LV output also show a difference in one of the phases (INVKG3_ST1A; INVMG3_ST1A). Depending on the phase you have previously detected, compare with the parameter indicated in the table.

lac_1	[SV3.2.4-Grid current I1 LV]
lac_2	[SV3.2.5- Grid current I2 LV]
lac_3	[SV3.2.6- Grid current I3 LV]

3.2.5. If after changing the hall sensor, the current measurement of that phase is still wrong, replace the sensor measurement processing board E0X (X being 4,5 or 6 depending on which phase has a bad reading) (INVKG3_ST2; INVMG3_ST5A).

4. Real unbalance

If after the previous checks no problem has been detected, then the fault is triggered by a real unbalance in the currents.

This unbalance may be generated by a possible recirculation of zero sequence between modules. Consult Power Electronics to analyse the problem in more detail.

5.104. Fault code F134: Module sensor comms

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".			
Description	The module FPGA cannot communicate with the temperature sensor integrated in the module FPGA board (E02).			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

Analysis procedure

Incompatible software.

If the failure occurs when updating the software, check with Power Electronics to see if the software version is compatible with the board version that has been installed. INVKG3_ST3A, INVMG3_ST3A.

NOTE: This fault can also occur if the module or module FPGA board (E02) is replaced.

Board in bad condition.

If the module FPGA board (E02) is in bad condition, replace the board. INVKG3_ST2, INVMG3_ST5A. See document INV3IOP88076.

If the failure is recurrent, inform Power Electronics.

EN

5.105.Fault code F200: Overvoltage Lim DC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section “ Introduction ”.			
	Description			
Applicable to	Overvoltage has been detected in the DC voltage limiter.			
	☒ HEM	☒ PCSM/MPCSM		
Associated faults or warnings	☒ HEMK	☒ PCSK/MPCSK		
	N/A.			

Analysis procedure

This protection can be configured as a fault or warning in PowercommsV ([Over VDC Fault Type] within group [G5.5-I/O Protections]).

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Problem in the surge protector.

1. Check in PowerCommsV ([Digital Inputs] within group [SV9.1-IMD: General]) for DC overvoltage (bit 1: DC overvoltage). INV~~G~~3_ST1B.
2. Visually check whether the front indicators of the surge arrester indicate failure. INV~~K~~G3_ST3A, INV~~M~~G3_ST3A.



In this case, the surge arrester cartridge must be replaced. **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3/OP90116*.

If the fault no longer appears, restart the unit.

3. If the fault occurs repeatedly, analyze with the inverter data the possible overvoltages detected, for this purpose via PowercommsV download the logs from the SD card after the fault and view on the SD card (Log 4) the value of the DC voltage to verify if a real overvoltage has occurred (measured value > 1500Vdc). **INVG3 ST1B.**

If the digital input that we have checked in point 1 is still at the value "X", continue to the next point.
INVKG3 ST2, INVMG3 ST5A.

Problem in the surge protector wiring.

If the measurements are correct, continue with the following checks:

HEM / HEMK.

4. Check the status of the surge protector (P05) by measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

- 4.1. between terminals P05:11 and P05:12 there must be continuity.



- 4.2. between terminals P05:11 and P05:14 there must be no continuity.



If the checks are not correct, replace the surge protector (P05). **INVKG3_ST2, INVMG3_ST5A.** See document *INVG3IOP90116*.

5. Check that the wiring of the surge protector (P05) located on the DU protection panel (PPD) is correct: **INVKG3_ST2, INVMG3_ST5A.**

- 5.1. verify that two 10mm² section power cables are connected to the PV+ and PV- terminals and that the ground cable is connected. **INVKG3_ST2, INVMG3_ST5A.**

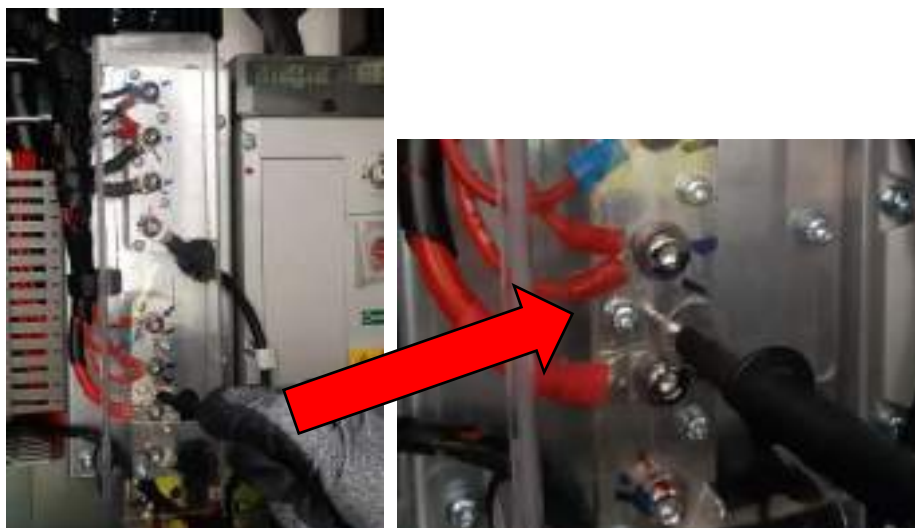
EN



EN

5.2. measuring continuity with a 1500V multimeter: **INVKG3_ST2**, **INVMG3_ST5A**.

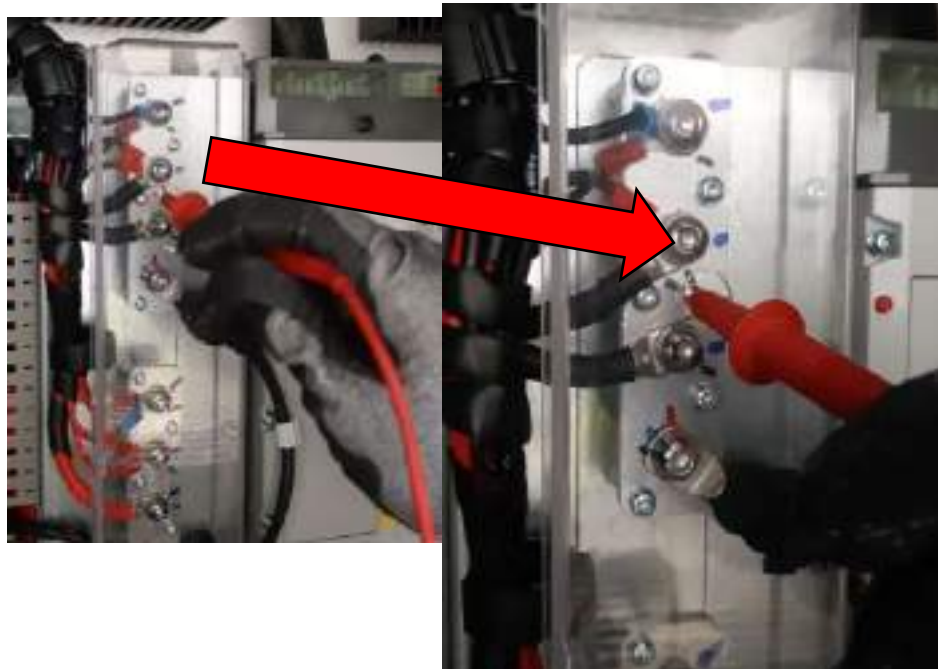
5.2.1. between BBEP.82:3 and P05:DC+/-.



5.2.2. between BBEN.67:3 and P05:DC+/-.



EN



5.2.3. between P05:PE and PE75.



5.2.4. between connector J15:1 of the E70 isolation monitor board and P05:11.



J15:1 de la tarjeta E70



P05:11

5.2.5. between connector J15:2 of the E70 isolation monitor board and P05:12.





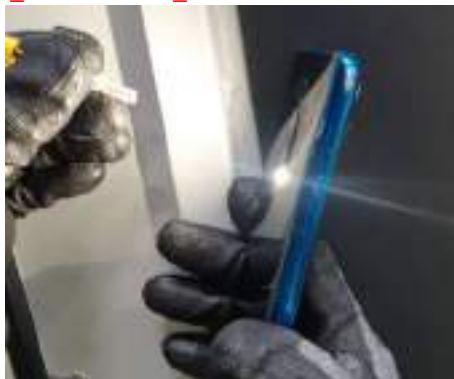
J15:2 de la tarjeta E70



P05:12

If there is no continuity at any of the measured points, replace the corresponding wiring or connector. **INVKG3_ST2, INVMG3_ST5A.**

6. Check the communication between the E70 isolation monitor board and the E35 central control board.
 - 6.1. Check the fiber optic connectors to ensure that they are flush and polished. If not, replace the connector or the fiber. **INVKG3_ST2, INVMG3_ST5A.**
 - 6.2. Check that the light passes correctly from one optical fiber to another. To do, shine a light source, either natural or artificial, from one end of the optical fiber and check that light is received at the other end. If this is not the case, replace the damaged optical fiber (depending on availability). **INVKG3_ST2, INVMG3_ST5A.**



PCSM / PCSK

In these models of equipment, we can have different number of panels depending on the number of busbars of the equipment and the size.

NOTE: These integrate a fuse connected upstream of the surge arrester. The feedback of each surge arrester is connected in series with the feedback of the fuse. They are all connected in series to the same digital input of the insulation monitor (E70:J15) in analogy to a HEM/HEMK.

The components and connections according to the configuration of the equipment are shown below:

	MODULE 1	MODULE 2	MODULE 3	MODULE 4
LOCATION	PIND1	PIND2	PIND3	PIND4
SURGE PROTECTOR	P13	P12	P11	P10
+ BUSBAR	BBEP.61:2	BBEP.60:2	BBEP.59:2	BBEP.58:2
-BUSBAR	BBEN.57:6	BBEN.56:6	BBEN.55:6	BBEN.54:6
GND CONNECTION	PE60	PE61	PE62	PE63
FUSE	F114	F111	F108	F105

The surge protectors (P10-P13) are located on the DU choke panel (PIND1-4).

7. Visually check that the fuse has not blown to be sure that the feedback is reaching the surge protector. **INVKG3_ST2, INVMG3_ST5A.**

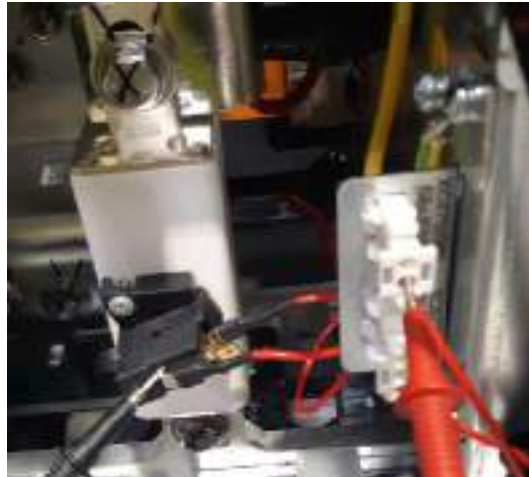
If it is blown, replace the fuse. **INVKG3_ST2, INVMG3_ST5A.**

8. Perform the following checks on each module that is failing according to the instance referenced by the failure in PowerCommsV.

NOTE: The following indications are for module 1. Refer to the table above for the rest of the modules.

- 8.1. Check the status of the surge protector (P13) by measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

8.1.1. between terminals P13:11 and P13:12 there must be continuity.



8.1.2. between terminals P13:11 and P13:14 there must be no continuity.



P13:14



P13:11

If the checks are not correct, replace the surge protector (P13). **INVKG3_ST2, INVMG3_ST5A.**
See document *INVG3/OP90116*.

- 8.2. Check that the surge arrester (P13) wiring is correct: **INVKG3_ST2, INVMG3_ST5A.**
 - 8.2.1. verify that two 10mm² section power cables are connected to the DC+ and DC- terminals and that the ground cable is connected. **INVKG3_ST2, INVMG3_ST5A.**



8.2.2. measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**
 8.2.2.1. between BBEP.61:2 and P13:DC+.



BBEP.61:2



P13:DC+

8.2.2.2. between BBEN.57:6 and P13:DC-.



BBEN.57:6



P13:DC-

8.2.2.3. between P13:PE and PE60.

EN



8.2.2.4. between terminals 1 and 2 of connector J15 of the E70 isolation monitor board.



If there is no continuity at any of the measured points, replace the corresponding wiring or connector. **INVKG3_ST2, INVMG3_ST5A.**

9. Check the communication between the E70 isolation monitor cards and the E35 central control board.
 - 9.1. Check the fiber optic connectors to ensure that they are flush and polished. If not, replace the connector or the fiber. **INVKG3_ST2, INVMG3_ST5A.**
 - 9.2. Check that the light passes correctly from one optical fiber to another. To do, shine a light source, either natural or artificial, from one end of the optical fiber and check that light is received at the other end. If this is not the case, replace the damaged optical fiber (depending on availability). **INVKG3_ST2, INVMG3_ST5A.**



5.106. Fault code F219: Overvoltage Lim AC

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section " Introduction ".	
Description	Overvoltage has been detected in the AC voltage limiter.	
Applicable to	☒ HEM	☒ PCSM/MPCSM
	☒ HEMK	☒ PCSK/MPCSK
Associated faults or warnings	N/A.	

Analysis procedure

This protection can be configured as a fault or warning in PowercommsV ([Over VAC Fault Type] within group [G5.5-I/OP Protections]).

There are different causes that can trigger this failure. Follow the steps below to identify the possible cause and resolve it.

Problem in the surge protector.

1. Check in PowerCommsV ([Status DSP RJ18] within group [SV1.5-Digital I/O status]) if there is AC surge protector feedback (bit 4: Feedback sdp AC). [INVKG3_ST1B](#).
2. Visually check whether the front indicators of the surge arrester indicate failure. [INVKG3_ST3A](#), [INVMG3_ST3A](#).



In this case, the surge arrester cartridge must be replaced. [INVKG3_ST2](#), [INVMG3_ST5A](#). See document [INVKG3IOP90116](#).

If the fault no longer appears, restart the unit.

3. If the fault occurs repeatedly, analyze with the inverter data the possible overvoltages detected, for this purpose download the logs from the SD card after the fault and view on the SD card ([Log 1](#)) the value of the AC voltage to verify if a real overvoltage has occurred (measured value > 1000Vac). [INVKG3_ST1B](#).

EN

If the digital input that we have checked in point 1 is still at the value "X", continue to the next point.
INVKG3_ST2, INVMG3_ST5A.

Problem in the surge protector wiring.

If the measurements are correct, continue with the following checks:

4. Check the status of the surge protectors (P01, P02 and P03) by measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**

- 4.1. between terminals P01:11 and P01:12 there must be continuity.



- 4.2. between terminals P01:11 and P01:14 there must be no continuity.



- 4.3. between terminals P02:11 and P02:12 there must be continuity.



4.4. between terminals P02:11 and P02:14 there must be no continuity.



4.5. between terminals P03:11 and P03:12 there must be continuity.



4.6. between terminals P03:11 and P03:14 there must be no continuity.

EN



If the checks are not correct, replace the corresponding surge protector. **INVKG3_ST2, INVMG3_ST5A.**
See document *INVG3IOP90116*.

EN

5. Check that the wiring of the surge protectors (P01, P02 and P03) located on the main protection panel (MPP) is correct. Each protector has 3 modules, one for each phase L1, L2 and L3. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.1. verify that each surge protector has 3 power cables (L1, L2 and L3) of section 16mm² connected and that the ground cable is connected. **INVKG3_ST2, INVMG3_ST5A.**
 - 5.2. measuring continuity with a 1500V multimeter: **INVKG3_ST2, INVMG3_ST5A.**
 - 5.2.1. between F100:1 y P01:L/N



- 5.2.2. between F100:3 y P02:L/N

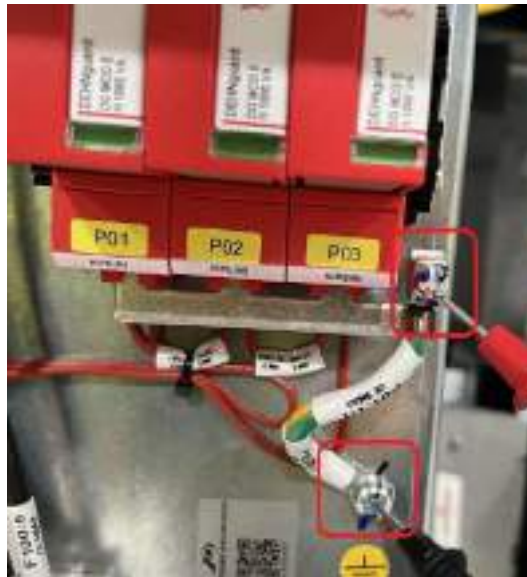


5.2.3. between F100:5 y P03:L/N



5.2.4. between P01.1:PE y PE05.

EN



5.2.5. between terminals 3 and 4 of connector J4 on the E19 input/output board.



If there is no continuity at any of the measured points, replace the corresponding wiring or connector.
INVKG3_ST2, INVMG3_ST5A.

6. Check the communication between the input/output board E19 and the central control board E35.
 - 6.1. Disconnect the RJ45 cable (W060) connecting connector J2 on the E19 input board to connector J18 on the E35 central control board and check for continuity with the network cable continuity tester. **INVKG3_ST2, INVMG3_ST5A.**

EN



- 6.2. If there is no continuity, replace the cable. *INVKG3_ST2, INVMG3_ST5A.*
- 6.3. If the RJ45 cable does not fail, replace the E19 input/output board. *INVKG3_ST2, INVMG3_ST5A.*
See document *INVG3IOP88849.*

If the fault persists, replace the central control board (E35). *INVKG3_ST2, INVMG3_ST5A.* See document *INVG3IOP88849.*

EN

5.107. Fault code F173: CTs measure

Mandatory	The user must check the Safety Instructions for Operating, Troubleshooting and Maintenance before starting. Each action described in the sequence has a defined status. There are no general statuses. User must read and understand the statuses described in the document mentioned in section "Introduction".			
Description	The inverter has detected a current error higher than the configured threshold between the current CTs (current transformers) and the current summation of the modules.			
Applicable to	☒ HEM		☒ PCSM/MPCSM	
	☒ HEMK		☒ PCSK/MPCSK	
Associated faults or warnings	N/A.			

Analysis procedure

Problem with configuration of default values.

- Check in PowercommsV that the parameters corresponding to the CTs are in their default value (parameters [Enable CTs prot] and [Unbalance threshold] of the subgroup [G5.2: Quality]). **INV3_ST1B**.
 - If they are not, consult with the plant manager if there is still a reason why they are not at their default value. If there is no reason, set to default value. **INV3_ST1B**.

Central control board is damaged.

- Install three current probes, connected to an oscilloscope, on the red wires leading into the central control board from the current CTs. See table. **INV3_ST2, INV3_ST5A**.

Note: the de-energised probes **INV3_ST2, INV3_ST5A** must be installed. Subsequently, close all the doors of the inverter to start it up and measure.

Current transformer (CT)	Positive connector on E35 board (red)	Negative connector on the E35 board (black)	Plates HEMK/PCSK	Plates HEM/PCSM
CT01	J6:2	J6:1	BBS3.01	BBI1.02
CT02	J7:2	J7:1	BBS2.01	BBI2.02
CT03	J8:2	J8:1	BBS1.01	BBI3.02

- Start up the unit (with the INV3_ST1A unit completely closed). Use PowerCommsV to compare the AC output current measurements of the inverter (group SV3: Grid visualization) with the current measurements on the oscilloscope. Note: the oscilloscope currents must be multiplied by 800 to correspond with the measurements of the inverter. **INV3_ST1A**.
- If the measurement shown by the oscilloscope (multiplied by 800) differs by more than 2% from that shown by the inverter, it indicates that the central control board is damaged. In this case:
 - Replace the central control board (E35). **INV3_ST2, INV3_ST5A**. See document **INV3/OP88849**.

Note: The measurement must be made at least at 50% of the inverter's active power reference to be valid, if the test is made at less reference, the error may be greater.

The CTs are in poor condition.

- Install three current probes, connected to an oscilloscope, on the low voltage PCBs of the inverter right next to the current CTs. See table above.

Note: the de-energised probes INV3_ST2, INV3_ST5A must be installed. Subsequently, close all the doors of the equipment to start it and measure.
- Start up the unit (with the INV3_ST1A unit completely closed). Use PowerCommsV to compare the AC output current measurements of the inverter (group SV3: Network display) with the current measurements on the oscilloscope. **INV3_ST1A**.

7. If the measurement shown by the oscilloscope differs by more than 2% from that shown by the inverter, this indicates that the current transformer (CT) is damaged. In this case:

- 7.1. Replace the corresponding CT (according to the table above). **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP86981*.

Note: the measurement must be done at least at 50% of the inverter's active power reference to be valid, if the test is done at less reference, the error can be higher.

Wiring is damaged.

8. Disconnect the current transformer (CT) connector from the central control board (E35) and measure with 1500V multimeter between its pins (positive and negative, according to the table) if there is continuity between the two points both on the central control board and between the wires of the disconnected connector. **INVKG3_ST2, INVMG3_ST5A**.



- 8.1. If there is no continuity between the pins of the central control board (E35). Replace the central control board (E35) and any CTs connected to the connectors that do not give continuity. **INVKG3_ST2, INVMG3_ST5A**. See documents *INVG3/OP88849* and *INVG3/OP86981*.
- 8.2. If there is no continuity between the pins of the connector already disconnected, check continuity with 1500V multimeter between pins S1 and S2 of the CT itself. If there is still no continuity, the CT is damaged. In this case, replace the corresponding CT. **INVKG3_ST2, INVMG3_ST5A**. See document *INVG3/OP86981*.

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8.2.1. If continuity is obtained in the second measurement, replace the CT and its corresponding wiring. [INVKG3_ST2](#), [INVMG3_ST5A](#). See document [INVG3/OP86981](#).

